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Chapter 5

GELFAND-P000335

[source number: 1]

Gelfand sem.

1/III — 71

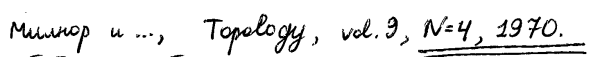
Gabrielov on the art. of Griffiths (Griffits).

On the periods of certain rational integrals

Ann. Math., v. 90, $N = 3$.

I. Bernstein.

Milnor and ..., Topology, vol. 9, $N = 4$, 1970.



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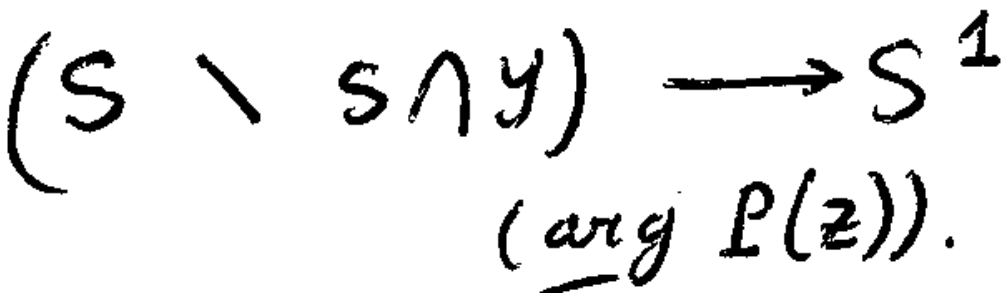
Manuscript notation. Three dots follow "Milnor and".

Generalization of Brieskorn's work.

$P(z)$ — a polynomial, $z \in \mathbb{C}^n$.

0 — a sing. point (isol.) of the surf. $\{P(z) = 0\} = Y$.

Topology of $S \setminus S \cap Y$ (S — a sphere of small rad.).



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Diagram and typed description.

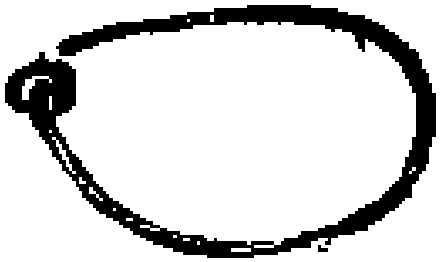
$$(S \setminus S \cap Y) \longrightarrow S^1$$

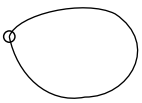
$$(\arg P(z)).$$

We study the fibres and the circuit.

(homeomorphism of the fibre: $T : F \longrightarrow F$).

Diagram and typed description.



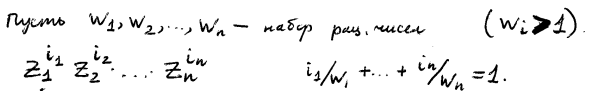


Original source details: GELFAND-P000335

Fibre F: $P(z) = \varepsilon$, ε — a small complex number.

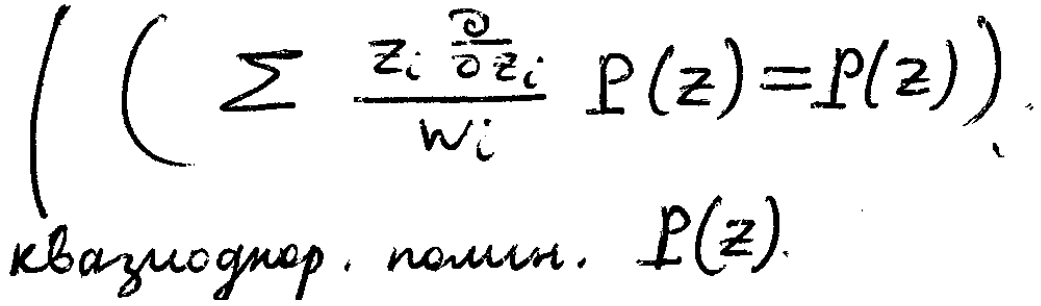
Let $w_1, w_2, \ldots, w_n$ — a collection of rational numbers ($w_i > 1$).

$$z_1^{i_1} z_2^{i_2} \dots z_n^{i_n}; \qquad \frac{i_1}{w_1} + \dots + \frac{i_n}{w_n} = 1.$$



Original source details: GELFAND-P000335

Uncertain reading. Before 1 is a heavy angular mark, provisionally $>$. The letter w in the list and denominators does not clearly distinguish upper from lower case.

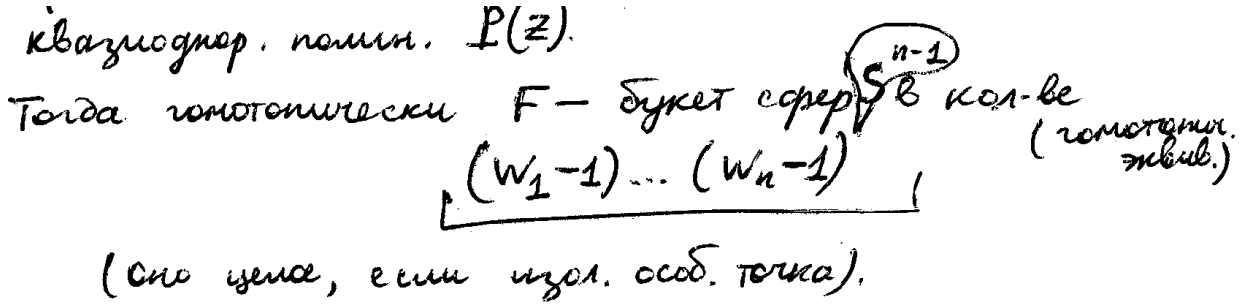


Original source details: GELFAND-P000335

Diagram and typed description.

$$\left(\sum \frac{z_i \frac{\partial}{\partial z_i}}{w_i} P(z) = P(z) \right)$$

quasi-homog. polyn. $P(z)$.



Original source details: GELFAND-P000335

Diagram and typed description.

Then homotopically F — a bouquet of spheres S^{n-1} in the number

$$\underbrace{(w_1 - 1) \dots (w_n - 1)}$$

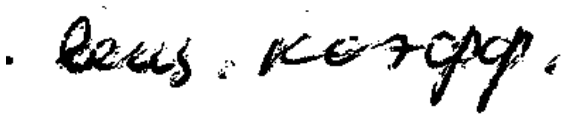
(homotopy equiv.)

(It is an integer if an isol. sing. point).

GELFAND-P000336

The characterist. polynomial of the transfor. T has been computed

(Example: $z_1^2 + \ldots + z_n^2 = \varepsilon$. — one sphere — real coord. and everything retracts onto it).



Uncertain reading. After the abbreviation for "real", the next word reads "coord." with two loops and a dot. Both words are abbreviated.

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Brieskorn's case:

$$z_1^{i_1} + \ldots + z_n^{i_n} = \varepsilon.$$

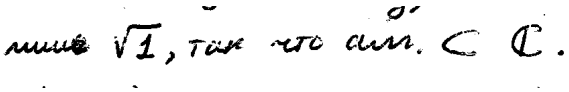
(I. Bernstein: the poles and residues of P^λ are connected with these cycles.)

$$H^{n-1}(F) = \mathbb{Z}^{(w_1-1)\ldots(w_n-1)}.$$

Consider the group ring $\mathbb{Q}(\mathbb{C}^*)$.

$$\sum r_i \langle z_i \rangle, \qquad \langle z_i \rangle \cdot \langle z_j \rangle = \langle z_i z_j \rangle.$$

Instead of $\mathbb{C}^*$ only $\sqrt{1}$ occurs, so alg. $\subset \mathbb{C}$.



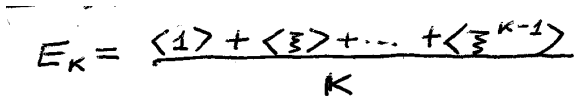
Uncertain reading. Before $\subset$ the abbreviation is provisionally "alg.". The written radical $\sqrt{1}$ has no index.

Original source details: GELFAND-P000336

$$\operatorname{div}(t - d_1)\ldots(t - d_k) = \langle d_1 \rangle + \ldots + \langle d_k \rangle \quad \text{(divisor)}.$$

Divisor of the char. polynomial $\chi(T)$

$$E_k = \frac{\langle 1 \rangle + \langle \xi \rangle + \ldots + \langle \xi^{k-1} \rangle}{k},$$



Uncertain reading. The angular denominator is larger than the subscript on E_k. The reading is provisionally k. Its case is unresolved.

Original source details: GELFAND-P000336

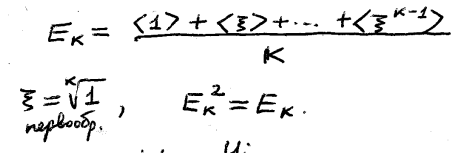


Diagram and typed description.

$\xi \underset{\text{prim.}}{=} \sqrt[k]{1}, \qquad E_k^2 = E_k.$

Original source details: GELFAND-P000336

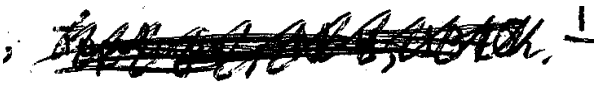
Let $w_i = \dfrac{u_i}{v_i}$ — irreduc.

$$\operatorname{div} \chi(T) = (w_1 E_{u_1} - 1)(w_2 E_{u_2} - 1)\ldots(w_n E_{u_n} - 1).$$

This gives an integral posit. divisor.

2nd work: Connection between Reidemeister tors. and ζ-funct. of the corresp. Laplace oper. on a Riem. manif.

Singer and someone else;



Original source details: GELFAND-P000336

Undeciphered passage. The crossed-out line after “Singer and someone else;” and before “Preprint”: overlapping letters and long strokes; the complete text has not been recovered.

Preprint.

GELFAND-P000337

[source number: 2]

I. M. Gelfand.

$$\alpha = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \qquad A, B, C, D \text{ — matrices.}$$

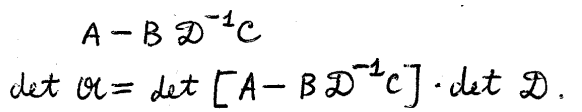
$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} \text{ — ?}$$

AD cannot be mult., in diff. spaces, AB, AC do not occur in det.

$$A - B \cdot C. \qquad D \text{ only in the middle.}$$

$A - BD^{-1}C$.

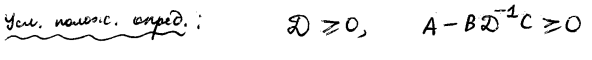
$$\det \alpha = \det[A - BD^{-1}c] \cdot \det D. \qquad (?)$$



Original source details: GELFAND-P000337

Uncertain reading. In the formula for $\det \alpha$, the letter c is shorter than the C on the line above. A separate $(?)$ stands at the right.

Cond. for posit. definiteness: $D \geq 0, \quad A - BD^{-1}C \geq 0.$



Original source details: GELFAND-P000337

Manuscript notation. Both comparisons have a lower stroke under $>$.

$A - BD^{-1}C$.

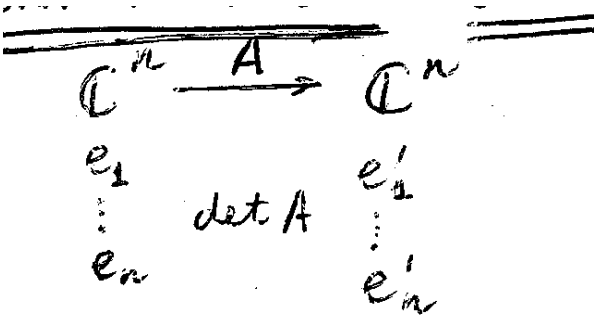
S. P. Novikov. On det



Original source details: GELFAND-P000337

Undeciphered passage. A short word between det and “(commut.)” is crossed out; the earlier letters overlap and have not been recovered.

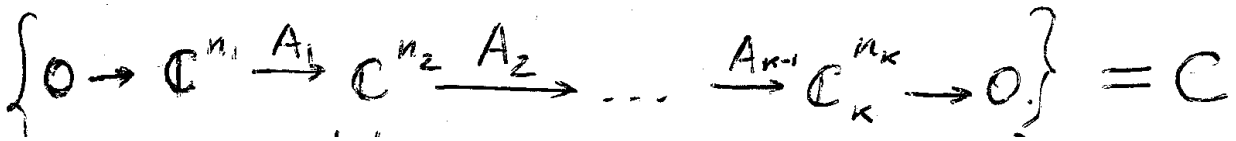
(commut.).



Original source details: GELFAND-P000337

Diagram and typed description.

$$\begin{array}{ccc} \mathbb{C}^n & \xrightarrow{\quad A \quad} & \mathbb{C}^n \\ e_1 & & e'_1 \\ \vdots & \det A & \vdots \\ e_n & & e'_n \end{array}$$



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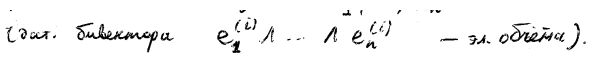
Uncertain reading. The last $\mathbb{C}$ has n_k above and a separate k-shaped sign below. Both are retained; the meaning of the lower sign is not established. Braces enclose the chain, with $= C$ at the right.

$$\left\{0 \longrightarrow \mathbb{C}^{n_1} \xrightarrow{A_1} \mathbb{C}^{n_2} \xrightarrow{A_2} \dots \xrightarrow{A_{k-1}} \mathbb{C}^{n_k}_k \longrightarrow 0.\right\} = C.$$

Classical det is defined for exact sequences.

In all $\mathbb{C}^{n_i}$ their own bases $e_1^{(i)}, \dots, e_n^{(i)}$.

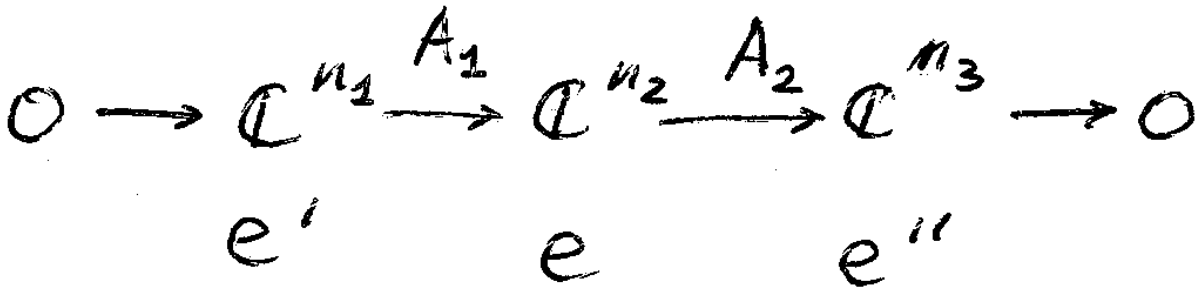
(dat. bivector $e_1^{(i)} \wedge \dots \wedge e_n^{(i)}$ — vol. elem.).



Original source details: GELFAND-P000337

Uncertain reading. The compressed first word is provisionally shown as “dat.” (Russian “дат.”); the ending of “бивектора” also permits “бивекторы”. The last $e_n^{(i)}$ has only n below, without an additional i. These provisional words and this index are retained.

$$0 \longrightarrow \mathbb{C}^n \xrightarrow{A} \mathbb{C}^n \longrightarrow 0. \qquad \det A.$$



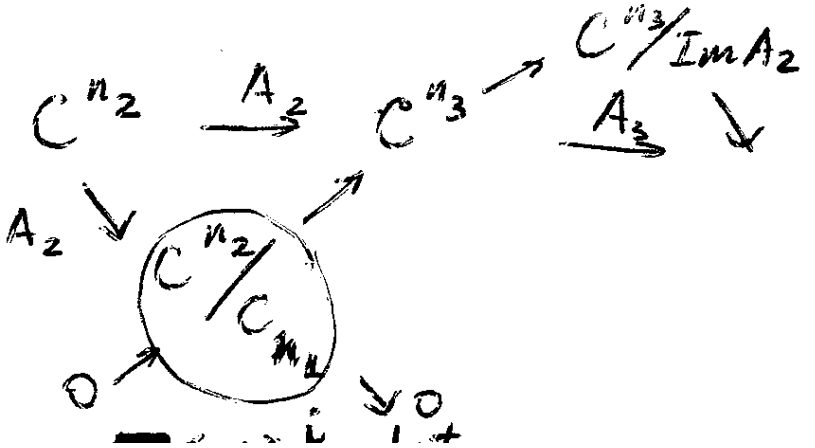
Original source details: GELFAND-P000337

Diagram and typed description.

$$0 \longrightarrow \mathbb{C}^{n_1} \xrightarrow{A_1} \mathbb{C}^{n_2} \xrightarrow{A_2} \mathbb{C}^{n_3} \longrightarrow 0$$

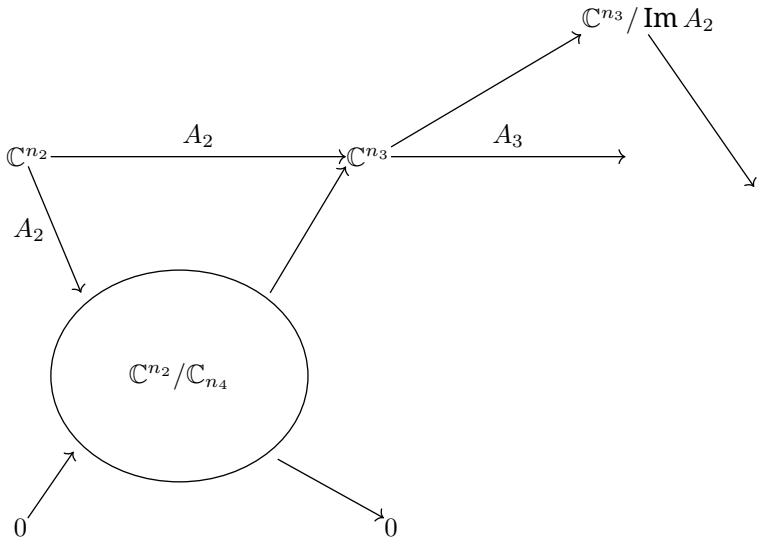
$$e' \qquad\qquad e \qquad\qquad e''$$

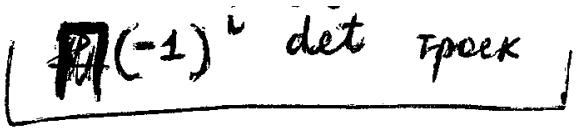
Compute the det of the transition from e to $(A_1 e', A_2^{-1} e'')$. This is $\det C$.



Original source details: GELFAND-P000338

Uncertain reading. The circled object's denominator has a dense subscript; it is provisionally shown as n_4, with n_1 also possible. Two arrows at the right end without labelled objects. The written Im A_2, both labels A_2, the label A_3, the oval and both zeros are retained; the exact index and connections are not established.





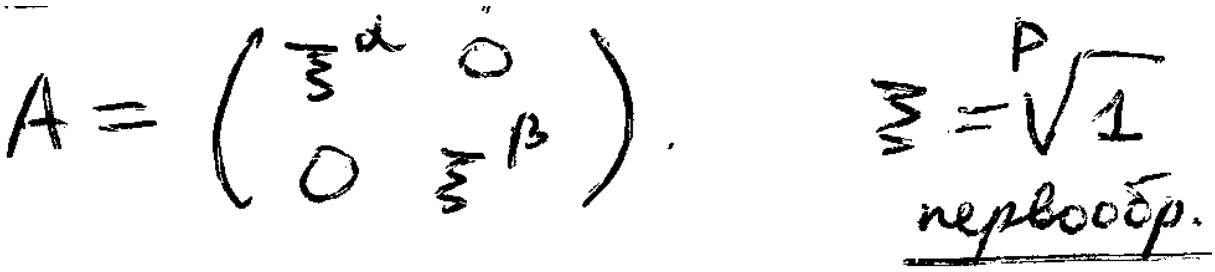
Original source details: GELFAND-P000338

Uncertain reading. The crossed-out sign before $(-1)^i$ is provisionally shown as $\prod$; two uprights and an upper stroke are visible, but the whole sign is not established. A three-sided bracket is below the entire expression.

$\not\prod (-1)^i$ det of triples

does not depend on the choice of bases in the intermediate spaces.

Lens spaces.



Original source details: GELFAND-P000338

Diagram and typed description.

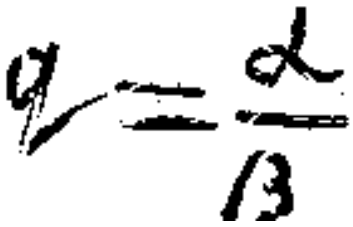
$$A = \begin{pmatrix} \xi^{\alpha} & 0 \\ 0 & \xi^{\beta} \end{pmatrix}, \qquad \xi = \sqrt[p]{1}.$$

prim.

Without fixed points.

$$q = \frac{\alpha}{\beta};$$

$A_{1,\beta}$.

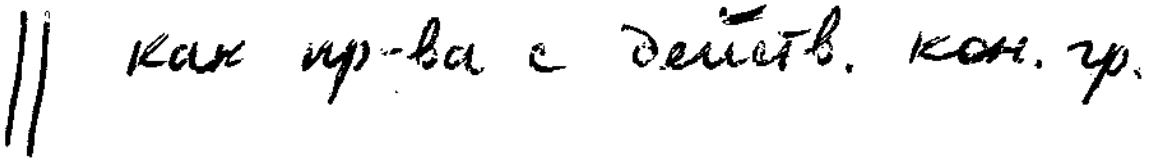


Uncertain reading. To the left of α/β is a small loop with a long downstroke and an oblique stroke, provisionally q. A separate $A_{1,\beta}$ is double-underlined below.

Original source details: GELFAND-P000338

Quotient of the sphere — lens space.

$$L^3_p = S^3/A_{1,\beta}.$$



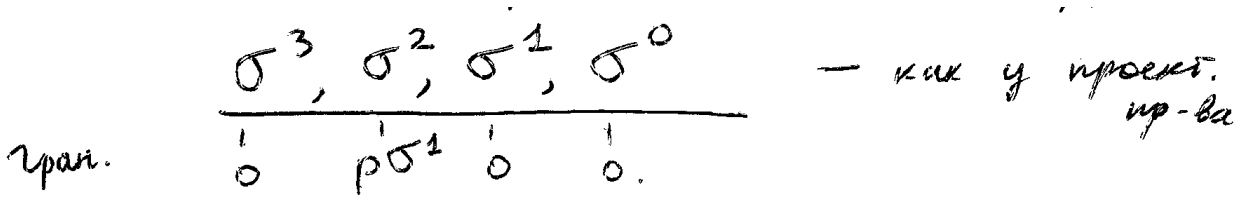
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Diagram and typed description.

|| as spaces with an action of a finite group.

Are they homeom. or not? The homologies are the same.

$\xi^p = 1$. The fund. gr. of a lens manif. is a cyclic group.



Original source details: GELFAND-P000338

Uncertain reading. The coefficient below σ^2 is provisionally shown as p; the form also permits ρ. A horizontal line separates the rows, with short vertical strokes above the four lower entries. The comparison with project. space is written at the right.

	σ^3,	σ^2,	σ^1,	σ^0
Bound.	0	$p\sigma^1$	0	0.

— as for project. space.

Univ. cover.

$$\xi\sigma^3$$
$$\xi^2\sigma^3$$
$$\vdots$$

Module

Over the group ring $\mathbb{Z}[\pi_1]$, $\pi_1 = \mathbb{Z}_p$.



Uncertain reading. The last $\mathbb{Z}$ has a subscript provisionally shown as p; its looped form also permits ρ.

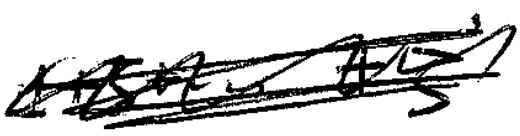
Original source details: GELFAND-P000338

GELFAND-P000339

[source number: 3]

The cochain complex is exact.

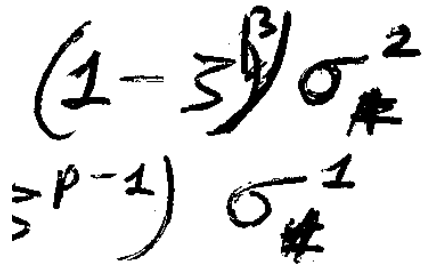
$$\sigma^3 \xrightarrow{\partial} \text{[unreadable crossed-out expression]} \quad (1-\xi^\beta)\sigma^2_{\not k}$$



Undeciphered passage. A whole expression after the first arrow and before $(1-\xi^\beta)\sigma^2$ is crossed out. Overlapping loops, sloping strokes and long horizontal strokes do not permit recovery of the entire expression.

Original source details: GELFAND-P000339

$$\sigma^2 \xrightarrow{\partial} (1+\xi+\ldots+\xi^{p-1})\sigma^1_{\not k}.$$



Original source details: GELFAND-P000339

Uncertain reading. The first ξ has an exponent with a tall stem and two right-hand loops, provisionally β. The cancelled lower marks at both occurrences of σ are provisionally shown as crossed k.

$\partial\sigma^1 = (1-\xi\ )\sigma_0$. exact.



Original source details: GELFAND-P000339

Undeciphered passage. The exponent of ξ inside $(1-\xi\ )$ is a solid dark cancelled sign. The blank preserves its position; the earlier exponent has not been recovered.

Exact sequence.

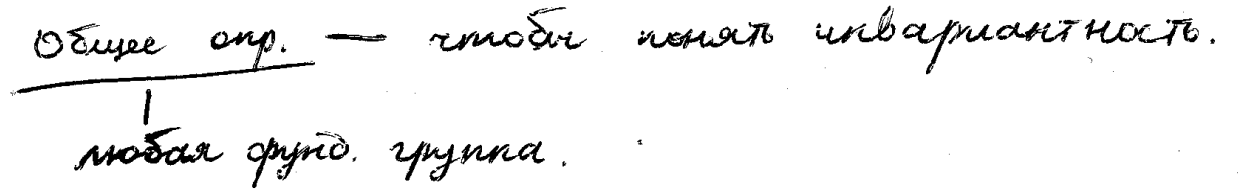
In the middle — the zero operator.



Original source details: GELFAND-P000339

Manuscript notation. The line begins «Посредине» (“In the middle”) and separately has «0-й».

$$\det = (1-\xi)(1-\xi^{\beta}).$$



Original source details: GELFAND-P000339

Diagram and typed description.

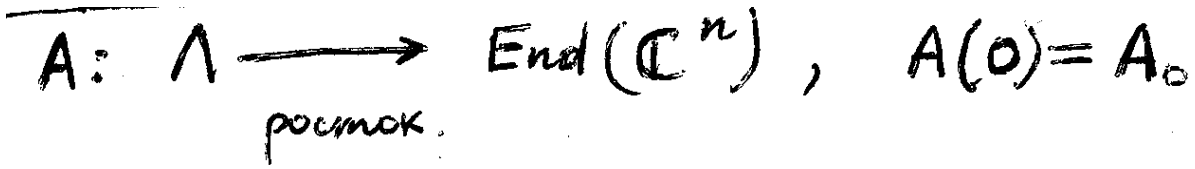
General def. — to understand invariance.

|

any fund. group.

A repres. such that the complex on the covering is exact.

V. I. Arnold. $\Lambda = \mathbb{C}^k$.



Original source details: GELFAND-P000339

Diagram and typed description.

$A : \Lambda \longrightarrow \mathrm{End}(\mathbb{C}^n)$ $\qquad A(0) = A_0$.

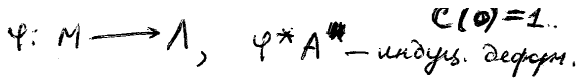
germ.

They are a category. The final object is a vers. deform.

$C(\lambda)$ — a deform. of the identity.

$$C(\lambda)A(\lambda)(C(\lambda))^{-1} = B(\lambda), \qquad C(0) = 1.$$

$\varphi : M \longrightarrow \Lambda$, $\varphi^* A$ — induced deform.



Original source details: GELFAND-P000339

Uncertain reading. The map sign is looped, with a descending stroke; it is provisionally shown as φ, with ψ also possible. The signs before the star and inside $A(\varphi(\mu))$ are also looped; the same provisional candidate is retained at all three sites.



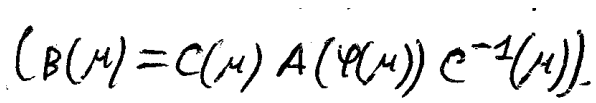
Original source details: GELFAND-P000339

Undeciphered passage. A small solid cancelled mark is above and after A. The blank preserves its position; the mark's content has not been recovered.

A versal deform. A — every deform. B of the matr. $A_0 \sim$ induced.

A.

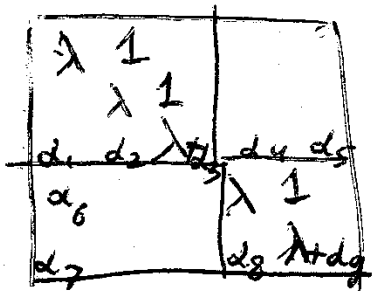
$$(B(\mu) = C(\mu)A(\varphi(\mu))c^{-1}(\mu)).$$



Original source details: GELFAND-P000339

Uncertain reading. The first letter in the final inverse factor is smaller than the neighbouring capital C and open, with a small middle curl. The provisional $c^{-1}(\mu)$ is retained; e and C are also possible.

$\exists$ a vers. fam. with 9 params.



Original source details: GELFAND-P000339

Uncertain reading. The rectangular boundary, internal vertical and horizontal lines, and the written λ, 1 and $\alpha_1, \ldots, \alpha_9$ are encoded. The label near $\lambda + \alpha_3$ intersects the lines; the exact cells and overlaps are not established. Lower labels are close to the border. Empty places remain empty, without supplied zeros.

λ	1			
	λ	1		
α_1	α_2	$\lambda + \alpha_3$	α_4	α_5
α_6			λ	1
α_7			α_8	$\lambda + \alpha_9$

α_i — free parameters.

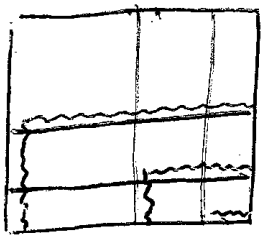
GELFAND-P000340

Dim. of the space of variations:

$$\sum_{\lambda} (n_1 + 3n_2 + 5n_3 + \ldots)$$

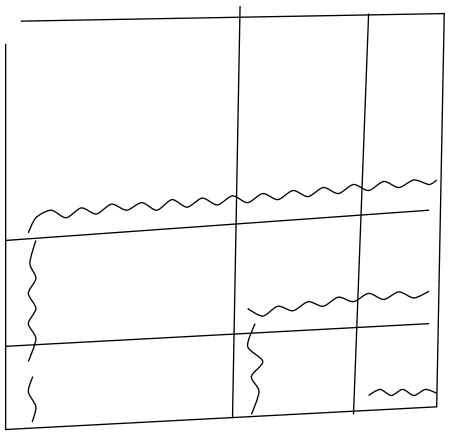
$$n_1(\lambda) \geq n_2(\lambda) \geq \ldots$$

Take a manif. through A_0 transverse to the orbit.
Deformations of the identity — n^2 params.

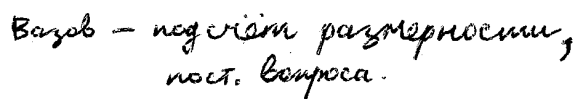


Original source details: GELFAND-P000340

Uncertain reading. An unlabelled rectangular sketch with a broad left region and two narrower right regions. Below are two horizontal divisions and several wavy bands; the outer boundary has a small gap at the upper left. The visible band positions are encoded, but the exact cells, block order and meaning of the waves are not established; no labels or entries are supplied.



Wasow [reading provisional] — dimension count, formul. of the question.

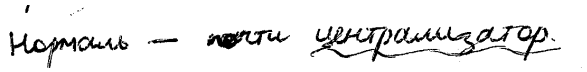


Original source details: GELFAND-P000340

Uncertain reading. The name before the dash is provisionally read «Вазов». In context it likely refers to Wolfgang Wasow.

Versality ⇔ transversality (to the orbit) —

— a normal to the orbit.
Normal — almost centralizer.



Original source details: GELFAND-P000340

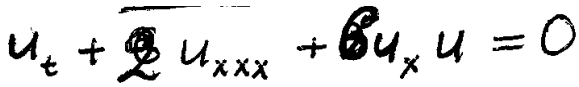
Uncertain reading. The word is provisionally read as “almost” (Russian “почти”): its beginning is overwritten by dense strokes, and the ending “ти” is readable. The earlier initial letters are not established. “Centralizer” has a wavy underline.

For quadr. forms — the canon. form of quadr. forms.

GELFAND-P000341

15/III L. Faddeev.

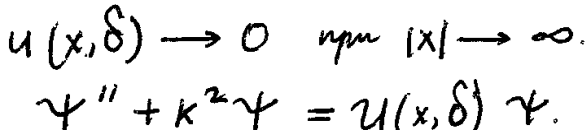
$$u_t + 2u_{xxx} + 6u_x u = 0.$$



Original source details: GELFAND-P000341

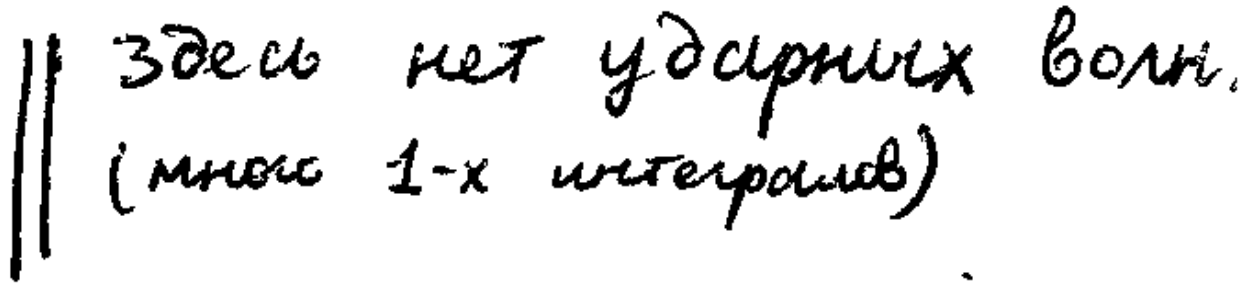
Uncertain reading. Before u_{xxx} is a visible 2 with a dense earlier looped stroke above it; before $u_x u$ is 6 with an overwritten top. The candidates 2 and 6 are retained; the earlier coefficients are not established. Both written signs before these terms are pluses.

$u(x, \delta) \longrightarrow 0$ as $|x| \longrightarrow \infty$.



Original source details: GELFAND-P000341

Uncertain reading. The second argument in both lines is rendered δ. Its upper hook and lower loop differ from the t on the lines below. The value of the argument is unspecified.



Original source details: GELFAND-P000341

Diagram and typed description.

There are no shock waves here.
(many 1st integrals).

Consider $\psi'' + k^2\psi = u(x, \delta)\psi$.

Spec. solution:

$$\psi|_{x\to-\infty} = e^{ikx} + a(k)e^{-ikx}$$
$$\psi|_{x\to+\infty} = e^{ikx} \cdot b(k).$$

One derives $|a(k)|^2 + |b(k)|^2 = 1$.

Known fact: $a(k)$ determines $u(x)$

$u(x) \longleftrightarrow a(k)$ (up to the discr. spectrum).

Recipe: $|a(k,t)| = |a(k)|$

$\arg a(k,t) = \arg a(k) + ck^3t$.



Uncertain reading. The coefficient of k^3t is provisionally read as c, with a small upper curl. A corresponding constant occurs in the formula for the phase derivative on the next leaf.

Original source details: GELFAND-P000341

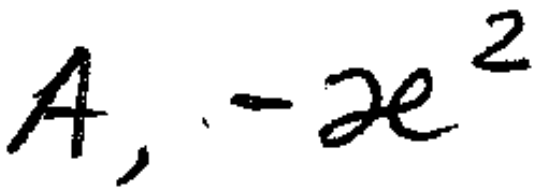
$u(x,t) \longleftrightarrow a(k,t)$.

And with a discr. spectrum A, $-æ^2$ — these are solitons,



Uncertain reading. The initial sign is provisionally read as the Russian conjunction «И» (“and”). The letter u, as on the preceding line, is also possible.

Original source details: GELFAND-P000341



Uncertain reading. After A, the squared symbol made of two connected loops is rendered æ and may be handwritten $\varkappa$.

Original source details: GELFAND-P000341

when $a(k) \equiv 0$.



Uncertain reading. The letter before $(k) \equiv 0$ is dense, with an overlapping stroke; the candidate a is retained, and the earlier form is not established.

Original source details: GELFAND-P000341

GELFAND-P000342

Conclusion: resembles a Hamilton. eq. in action — angle variables.

$a(k,t)$ — a solution of a classical Hamiltonian system



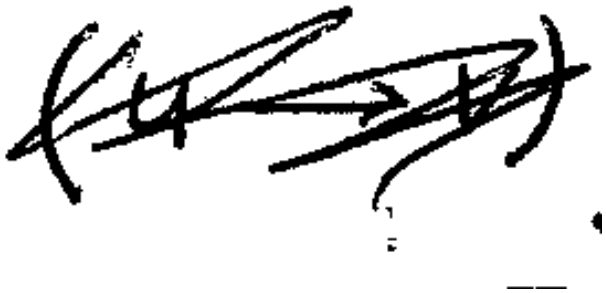
Original source details: GELFAND-P000342

Uncertain reading. At the right is a crossed-out entry with an H-shaped sign and a small k-shaped lower stroke. The crossed H_k is provisional; the exact earlier expression is not established.

$$\begin{cases} \dfrac{\partial |a|}{\partial t} = 0 \\ \dfrac{\partial \varphi}{\partial t} = Ck^3. \end{cases} \qquad \cancel{H_k}$$

Assertion: — the equation is an ∞-dimensional completely integr. Hamilton. eq. The solution of the scattering problem is its canonical transfor. to an explicitly solvable form.

Symplectic structure in the space of functions $u(x)$.



Original source details: GELFAND-P000342

Uncertain reading. Between the minus sign and the second integral are crossed parentheses, u, v and a horizontal sign. The crossed $(u \to v)$ is provisional; $(u \leftrightarrow v)$ is also possible, with the direction obscured by cancellation.

$$\langle u|v\rangle = \int_{-\infty}^{\infty} u(x)\,dx \int_{-\infty}^{x} v(y)\,dy - \cancel{(u \to v)} \int_{-\infty}^{\infty} v(x)\,dx \int_{-\infty}^{x} u(y)\,dy.$$

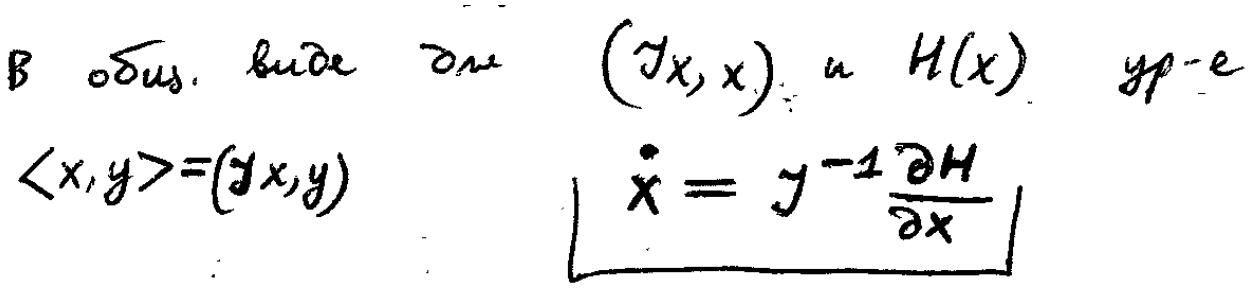
a nondeg. skew-symm. form.

Hamiltonian:

$$Q(u) = \int_{-\infty}^{\infty} (u^3 + u'^2)\,dx$$

$$\frac{\partial u}{\partial t} = \left(\frac{d}{dx}\right)\left(\frac{\delta Q}{\delta u(x)}\right).$$

In general form, for (Jx, x) and $H(x)$ the eq.

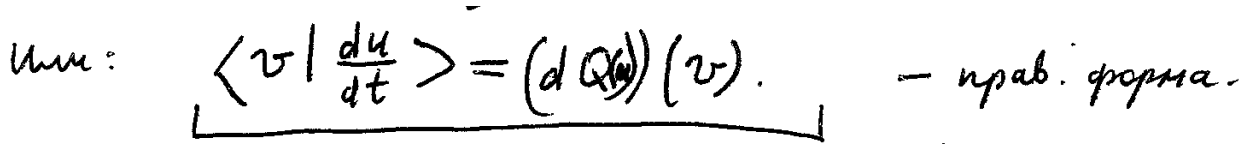


Original source details: GELFAND-P000342

Uncertain reading. Both letters in the introductory pair resemble x: (Jx, x). The next pair is (Jx, y); the difference is retained. Below the equation with J^{-1} is a line with upright sides; no upper side is visible.

$$\langle x, y\rangle = (Jx, y); \qquad \dot{x} = J^{-1}\frac{\partial H}{\partial x}.$$

Or:

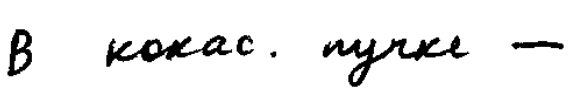


Original source details: GELFAND-P000342

Diagram and typed description.

$$\left\langle v \middle| \frac{du}{dt} \right\rangle = (dQ(u))(v).$$ — correct form.

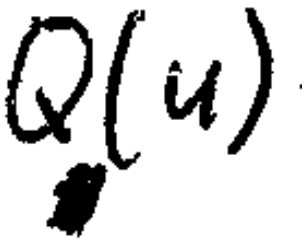
In the cotang. bundle [reading provisional] — 2 eqs. for p and q.



Uncertain reading. The manuscript abbreviation «кокас.» before “bundle” likely stands for “cotangent.”

Original source details: GELFAND-P000342

Pass from u to $a(k)$.

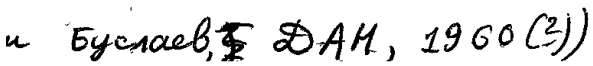


Original source details: GELFAND-P000342

Uncertain reading. Below Q in the final equality is a separate short, dense, sloping stroke. Its position is encoded; its reading and meaning are not established.

$$Q(u) = \int_{-\infty}^{\infty} (u^3 + u'^2)\,dx = \int_{-\infty}^{\infty} \ln(1 - |a|^2)k^4\,dk.$$

(Trace f-la: L. D. Faddeev and Buslaev,



Original source details: GELFAND-P000342

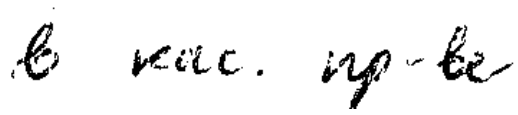
Undeciphered passage. A small sign with remnants of a loop is densely crossed out between “Buslaev,” and “DAN”; the earlier entry has not been recovered.

DAN, 1960 (2)).

GELFAND-P000343

[source number: 6]

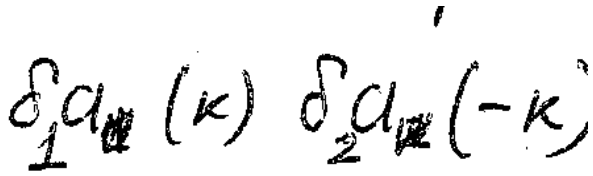
Express any $\int$ of this type (a local functional) through moments of the logarithm $\ln(1-|a|^2)$. Image of the differential form $\longrightarrow$ in the tang. space (the transition is nonlinear).



Original source details: GELFAND-P000343

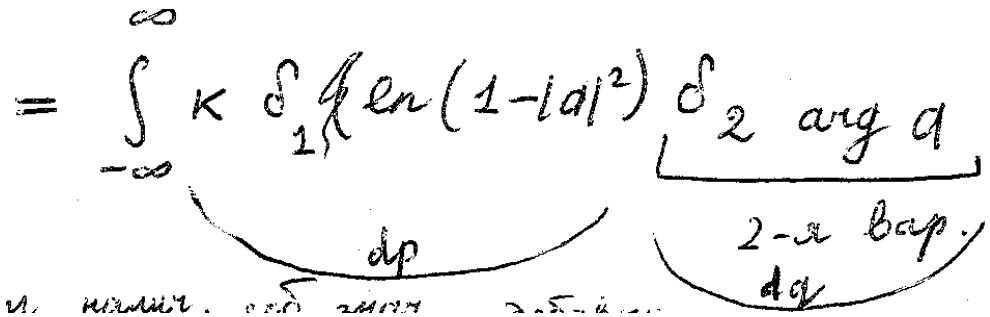
Uncertain reading. The abbreviation is provisionally read «кас.», likely short for “tangent.”

$$\langle \delta u_1, \delta u_2 \rangle_u = i \int_{-\infty}^{\infty} \frac{k}{1-|a|^2} \delta_1 a_k(k)\, \delta_2 a_k(-k)\, dk =$$



Original source details: GELFAND-P000343

Uncertain reading. Both occurrences of a have densely canceled lower signs. The overwritten sign may have been k, and the reading with no operative subscript is also possible.



Original source details: GELFAND-P000343

Uncertain reading. Before $\ln(1-|a|^2)$ is an opening brace; no closing brace or trailing dk is visible in this row. An arc labelled dp is below the left part; below $\delta_2 \arg a$ are a three-sided bracket labelled “2nd var.” and a lower arc labelled dq. Unseen completions are not supplied.

$$= \int_{-\infty}^{\infty} k\delta_1\{\ln(1-|a|^2) \qquad \delta_2 \arg a$$

dp 2nd var. dq

When eigenvalues are present, there are additional terms.

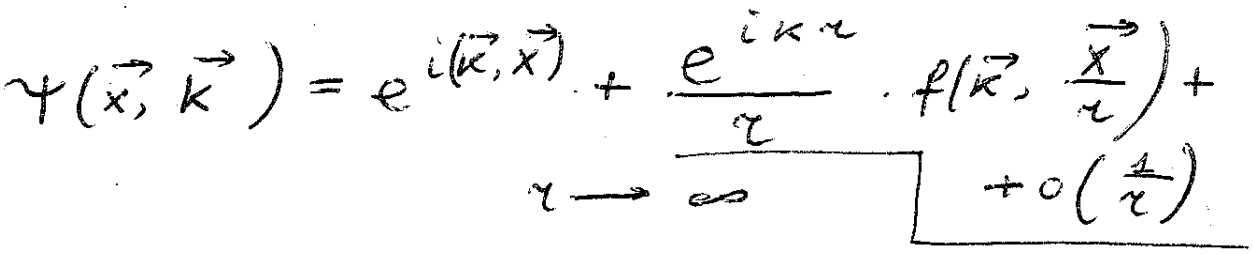
The three-dimensional inverse scattering problem.

$$-\Delta\psi + v(\vec{x})\psi = k^2\psi, \qquad \vec{x} \in \mathbb{R}^3,$$

$v(\vec{x})$ decays sufficiently rapidly as $|\vec{x}| \longrightarrow \infty$

Scattering amplitude.

The natural solution:



Original source details: GELFAND-P000343

Uncertain reading. A plus is written after f at the end of the first row, and another before $o(1/r)$ in the next row. Both are retained, and o is lowercase. A stepped line is below the fraction and formula ending, with $r \longrightarrow \infty$ at the lower left.

$$\psi(\vec{x},\vec{k}) = e^{i(\vec{k},\vec{x})} + \frac{e^{ikr}}{r}\cdot f\left(\vec{k},\frac{\vec{x}}{r}\right) +$$

$$r \longrightarrow \infty \qquad +o(1/r).$$

f — the scattering amplitude, a function of 5 variables (since $\vec{x}/r$).

Relations on the scattering amplitude have been found, eliminating 2 parameters.



Original source details: GELFAND-P000343

Uncertain reading. The crossed beginning of the line contains a tall f-shaped form and (k). The crossed $f(k)$ is provisional; the dense initial form is not definitively established.

~~$f(k)$~~

$$f\left(k\left|\frac{\vec{k}}{k},\frac{\vec{x}}{r}\right.\right) \longrightarrow f(k|\vec{\alpha},\vec{\beta})$$

in $L_2(S^2)$ a unitary operator! — this is a condition;

analogous.

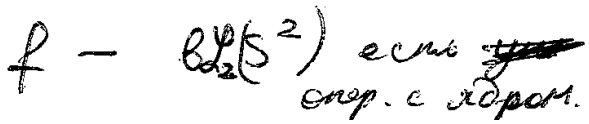


Original source details: GELFAND-P000343

Undeciphered passage. A short expression between "analogous" and $a^2+b^2=1$ is crossed out; the overlapping letters have not been recovered.

$a^2 + b^2 = 1$.

$S(k) = I + f$, f — $\mathcal{B}L_2(S^2)$ if

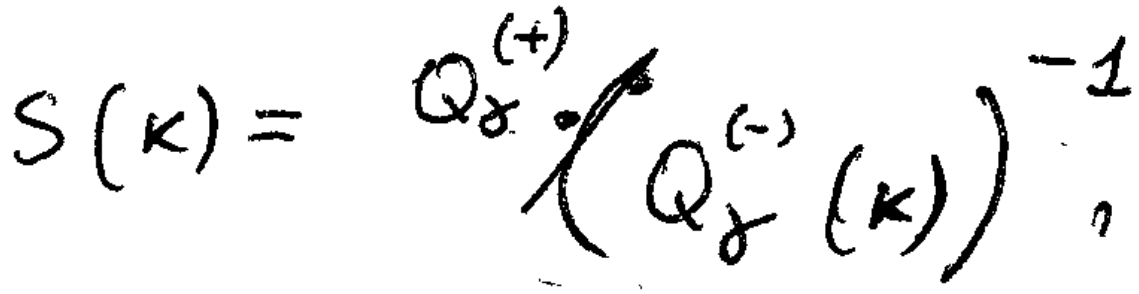


Original source details: GELFAND-P000343

Undeciphered passage. After "if" and before "operator with a kernel", a word is crossed out by overlapping strokes; the word has not been recovered. Before this is a B-shaped sign directly before the space notation, without a supplied membership sign or new parentheses.

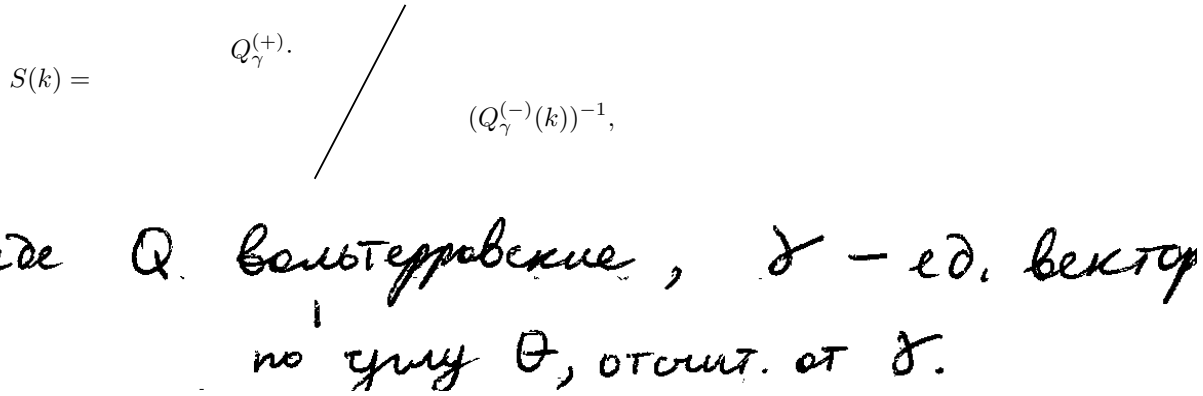
an operator with a kernel.

GELFAND-P000344



Original source details: GELFAND-P000344

Uncertain reading. After $Q_\gamma^{(+)}$ are a dot and a long slant near the opening of the next parenthesis. Both strokes retain their positions. A product is only a candidate; the slant's role and possible earlier division are not established. No argument k is visible on the first Q, and none is supplied.



Original source details: GELFAND-P000344

Diagram and typed description.

where Q are Volterra, γ — a unit vector.

in the angle θ, measured from γ.

$$h_\gamma(k;\alpha,\beta)\Theta[\pm(\alpha-\beta,\gamma)] = f\left(k\left|\frac{\vec{k}}{k},\frac{\vec{x}}{r}\right.\right)$$

When $(\alpha,\gamma) = (\beta,\gamma)$ there is a function analytic in k in the upper half-plane.

The Green function and factorization were written down (several years ago).

There was no inverse problem because of the discrete spectrum.

A. Gabrielov.

Griffiths's work on periods of rational integrals. Annals of Math.

$$\omega = \frac{P(z)}{Q(z)}\, dz_1 \wedge \ldots \wedge dz_n, \qquad P, Q \text{ — polynomials in } \mathbb{C}^n.$$

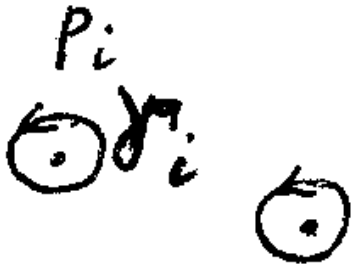
V — the zeros of $Q(z)$.

Γ — n-dimensional cycles in $\mathbb{C}^n \setminus V$.

$\int_\Gamma \omega$ — the periods.

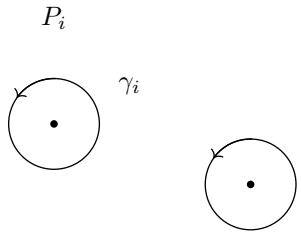
1 variable: $\omega = \dfrac{P(z)}{Q(z)}\, dz$

Γ reduces to a standard collection of paths



Original source details: GELFAND-P000344

Uncertain reading. Two contours enclose points, with both upper arrowheads pointing left. The label over the first is provisionally read P_i and may be lowercase p_i. At its right is γ_i.



$$\Gamma = \sum \mu_i \gamma_i; \qquad \int_\Gamma \omega = \sum \mu_i \int_{\gamma_i} \omega.$$

Reduction to 1st order — subtract an exact differential.

And now $\sum$ of residues.

Regard the form as defined on $\mathbb{CP}^n$.

$P_n \supset V$ — the set of poles.

There exists Q — a homogeneous form of degree m, minimal.

GELFAND-P000345

[source number: 7]

$$\Gamma \subset P_n \setminus V, \qquad \Gamma \in H_n(P_n \setminus V, \mathbb{Z}).$$

Заметим, что Γ в P_n есть 0;

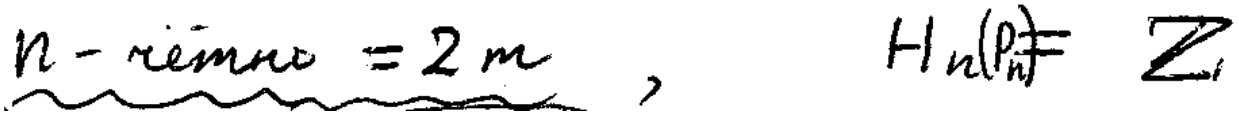
Original source details: GELFAND-P000345

Diagram and typed description.

Notice that Γ in P_n is 0;

n — odd $\Rightarrow H_n = 0$.

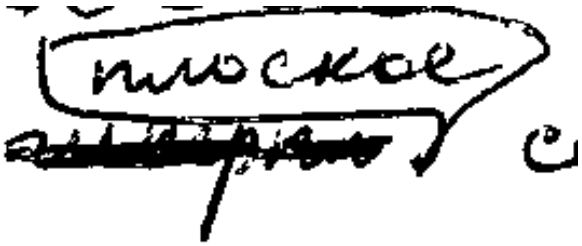
$n = 2k+1 \Rightarrow H_n(P_n) = 0$.



Original source details: GELFAND-P000345

Diagram and typed description.

n — even $= 2m$, $H_n(P_n) = \mathbb{Z}$



Original source details: GELFAND-P000345

Undeciphered passage. The earlier word after "Then" is crossed out by dense horizontal strokes; the whole word has not been recovered. The inserted "plane" is enclosed, with a downward arrow at its right towards the earlier entry. An initial loop and a long lower stroke are visible; "alg." remains only a candidate.

Then plane ~~[crossed out]~~

section P_m — the only generator. It suffices to compute the intersection index of Γ with P_m.
Consider $W = P_{m+1} \cdot V = q \cdot P_m$, q — the degree of the polyn. defining V.
This follows from $W \cdot P_m = V \cdot P_1$ (Bézout's thm.),

$$(z_1^2 + z_2^2 + z_3^2 = 0 \sim 2P_1).$$

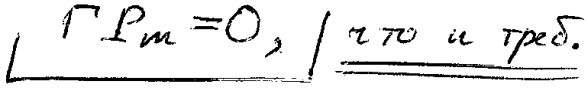


Original source details: GELFAND-P000345

Uncertain reading. After ΓP_m and before the comma is a raised small crossed sign with a loop and a lower stroke. A crossed 2 is provisional; the earlier sign is not established. Its position is not expanded as a secure exponent.

$0 = \Gamma \cdot W = q\Gamma P_m$ ~~2~~ ,

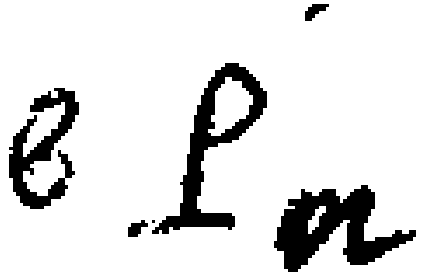
hence



Original source details: GELFAND-P000345

Diagram and typed description.

$\Gamma P_m = 0$, as required



Original source details: GELFAND-P000345

Uncertain reading. Below R is the insertion "in P_m". Its lower letter is dense, with joined strokes; m is provisionally retained, with n also possible. The insertion stays below R.

There is a chain R, such that $\partial R = \Gamma$.

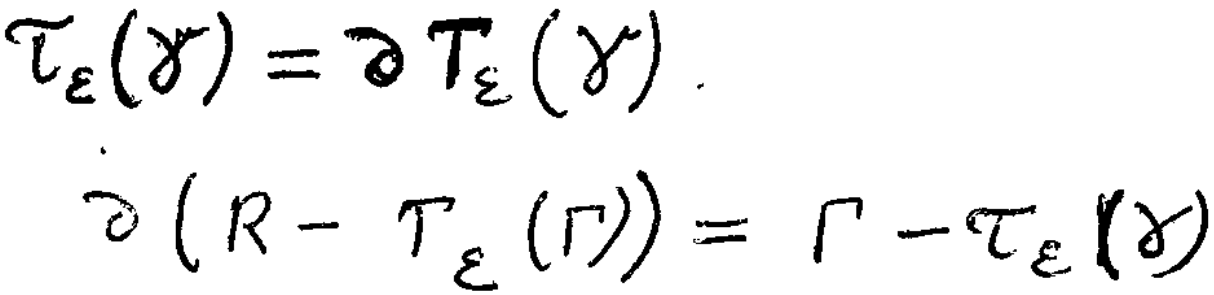
in P_m

Assume R and V in general position, $\gamma = R \cdot V$, $\gamma \in H_{n-1}(V, \mathbb{Z})$.
Take $T_\varepsilon(\gamma)$ — an ε-neighbourhood of the cycle γ in P_n.



Uncertain reading. In the first argument of T_ε the letter with a lower loop is strongly reinforced and provisionally shown as γ; earlier strokes cannot be securely separated.

Original source details: GELFAND-P000345



Original source details: GELFAND-P000345

Diagram and typed description.

$$\tau_\varepsilon(\gamma) = \partial T_\varepsilon(\gamma).$$
$$\partial(R - T_\varepsilon(\Gamma)) = \Gamma - \tau_\varepsilon(\gamma)$$

Hence $\tau_\varepsilon(\gamma) \sim \Gamma$. We obtain:

$$\tau : H_{n-1}(V, \mathbb{Z}) \longrightarrow H_n(P_n \setminus V, \mathbb{Z}).$$

Manuscript notation. The upper wavy stroke corresponds to the crossbar of the handwritten τ_ε on this leaf. No separate tilde is legible.



Original source details: GELFAND-P000345

This gives a map [crossed out] of homologies.



Original source details: GELFAND-P000345

Undeciphered passage. Between “map” and “of homologies” a word is covered by several horizontal strokes. Joined loops are visible beneath them; the whole earlier word has not been recovered, including the candidate Russian “когомол.”.

$$\tau_\varepsilon(\partial\gamma) = \partial(\tau_\varepsilon(\gamma)).$$

$$\tau : H_k(V, \mathbb{Z}) \longrightarrow H_{k+1}(P_n \setminus V, \mathbb{Z}).$$

Manuscript notation. The upper wavy stroke corresponds to the crossbar of the handwritten τ_ε on this leaf. No separate tilde is legible.



Original source details: GELFAND-P000345



Original source details: GELFAND-P000345

Diagram and typed description.



GELFAND-P000346

Arnold. Let $P_n(x, y) = 0$ be a plane nonsingular alg. curve.



Uncertain reading. The subscript on the first P is strongly reinforced; n is provisionally retained. Dense overlapping strokes do not securely separate the earlier form.

Original source details: GELFAND-P000346

$$n = 2k.$$

Harnack's thm.: ovals $\leq g + 1$, $g = \dfrac{(n-1)(n-2)}{2}$.

4th degree.

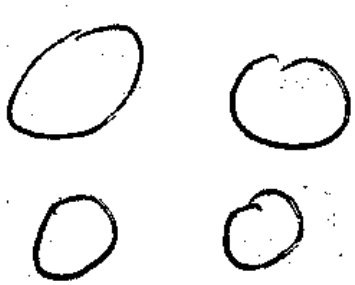
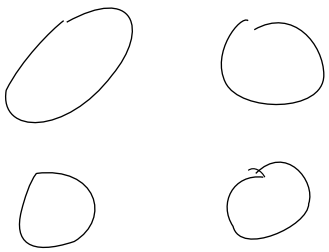


Diagram and typed description.



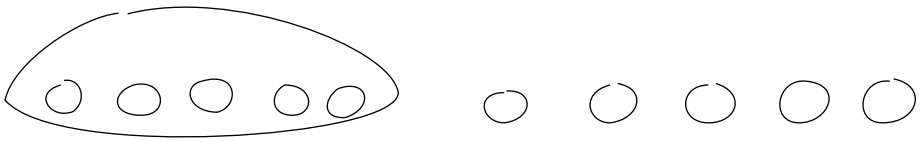
Original source details: GELFAND-P000346

6th degree. — 50 cases (Gudkov). Let the max. number of ovals be 11 (for $n = 6$).



Original source details: GELFAND-P000346

Diagram and typed description.



and another 2 cases,

Posit. ovals — lie in an even number of others.

Petrovsky: $|2(p - m) - 1| \leq 3k^2 - 3k + 1$. Hence not all 11 are outside one another.

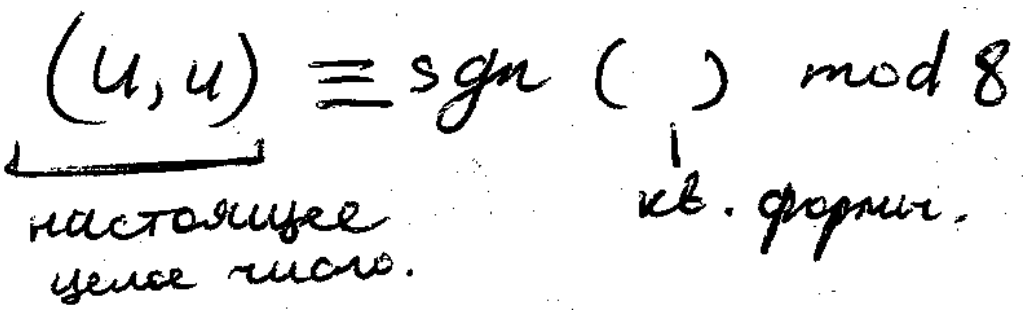
Gudkov: $p - m \equiv k^2 \pmod 8$.

Arnold and Varchenko: proof mod 4 because of integral forms.

$$(x, x) = (u, x) \pmod 2$$

— the quad. form is linear mod 2.

Lemma.



Original source details: GELFAND-P000346

Diagram and typed description.

$(u,u) \equiv \operatorname{sgn}(\quad) \mod 8$

an actual integer.

quad. form.

Proof:

$$(u+2x, u+2x) = u^2 + 4ux + 4x^2,$$

and ux and x^2 have the same parity by the formula.

$z^2 = f(x,y)$, z, x, y; $(p-m)$ — the Euler char. of the set of smaller values.

GELFAND-P000347

[source number: 8]

22/III. A. Gabrielov.

$$\int_\Gamma \frac{P(z)}{Q(z)}\, dz_1 \wedge \cdots \wedge dz_n, \qquad \Gamma \text{ — a cycle in } P^n \setminus V.$$

$$V = \{Q(\xi) = 0\}, \qquad \deg Q = q.$$

The cycle Γ in P_n is homologous to zero.

$$\Gamma \in H_n(P_n \setminus V, \mathbb{Z}).$$

The map $H_n(P_n \setminus V, \mathbb{Z}) \longrightarrow H_n(P_n, \mathbb{Z})$ is zero.
Let $\Gamma \in H_n(P_n \setminus V, \mathbb{Z})$, $n = 2m$.



Uncertain reading. Before V in the repeated group are dense crossed oblique strokes. The resulting complement is retained as \; the earlier sign is not recovered.

Original source details: GELFAND-P000347

Intersec. no.

$\langle P_m, P_m \rangle = 1$ — direct check.

$$\langle P_{m+1} \cdot V, P_m \rangle = q.$$

$$\langle P_{m+1} \cdot V; P_m \rangle = \langle V, P_{m+1} \cdot P_m \rangle = V \cdot P_1 = q.$$

(Bézout's thm.).

$$0 = \Gamma \cdot (P_{m+1} \cdot V) = q(\Gamma, P_m), \qquad \text{hence } (\Gamma, P_m) = 0.$$

Thus, $\Gamma = \partial R$, $R \cdot V = \gamma \in H_{n-1}(V, \mathbb{Z})$.

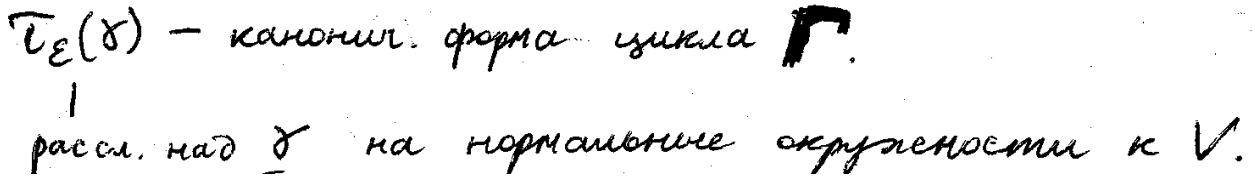


Uncertain reading. The subscript before -1 is heavily reinforced. It is provisionally read as n, with the earlier strokes inseparable.

Original source details: GELFAND-P000347

$T_\varepsilon(\gamma)$ — fibred into small circles over γ.

$$\partial T_\varepsilon(\gamma) = \tau_\varepsilon(\gamma).$$
$$\partial(R - T_\varepsilon(\gamma)) = \Gamma - \tau_\varepsilon(\gamma).$$



Original source details: GELFAND-P000347

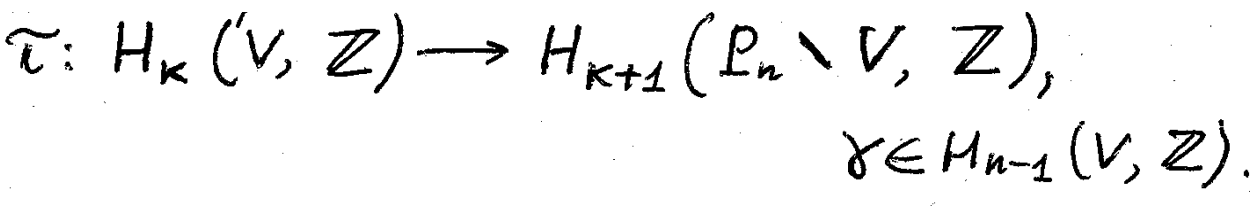
Diagram and typed description.

$\tau_\varepsilon(\gamma)$ — the canon. form of the cycle Γ.

|

fibred over γ into circles normal to V.

A map has been constructed:



Original source details: GELFAND-P000347

Uncertain reading. The map symbol has an upper wavy stroke separated from its lower part. τ is provisionally retained; a separate tilde, $\widetilde{\tau}$, is also possible.

$\tau : H_k(V, \mathbb{Z}) \longrightarrow H_{k+1}(P_n \setminus V, \mathbb{Z}),$

$\gamma \in H_{n-1}(V, \mathbb{Z}).$

How ambiguous is this?

$$\operatorname{Ker} \tau\big|_{H_k} - ?$$

Manuscript notation. The handwritten τ has a wavy crossbar, as in the τ_ε above.



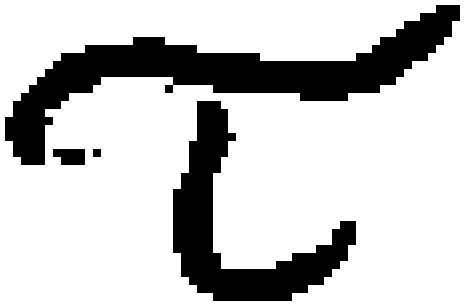
Original source details: GELFAND-P000347

GELFAND-P000348

This is part of the exact seq. of the pair:

$$H_{n+1}(P_n, \mathbb{Z}) \xrightarrow{\cdot V} H_{n-1}(V, \mathbb{Z}) \xrightarrow{\tau} H_n(P_n \setminus V, \mathbb{Z}) \xrightarrow{0} H_n(P_n, \mathbb{Z}).$$

Manuscript notation. The wavy crossbar of τ resembles that of τ_ε on the preceding leaves.



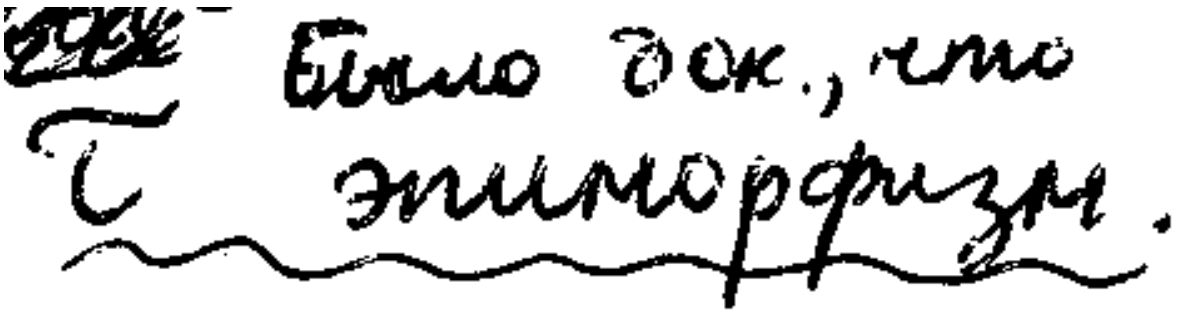
Original source details: GELFAND-P000348

Uncertain reading. A word before “It is exact” is crossed out. “otsyuda” (“hence”) remains a possible reading beneath the diagonal strokes.



Original source details: GELFAND-P000348

It is exact. Usually mod torsion, but here everything is fine.



Original source details: GELFAND-P000348

Diagram and typed description.

It was proved that

τ is an epimorphism.

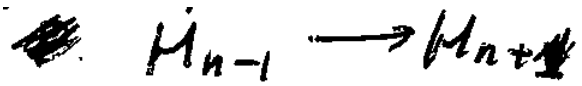
Hence exactness at that term.

From exactness: Ker τ is described.

$$\operatorname{Ker}\tau = H_{n+1}(P_n,\mathbb{Z}),\quad \begin{cases} n \text{ — even — this is } 0,\\ n \text{ — odd — } \mathbb{Z} \text{ with gen. } P_{m+1}. \end{cases}$$

Exactness is proved in the standard way.

Alternatively: the pair P_n, P_n, — a tubul. neighb. of V. Apply to the pair and the Thom isom.:

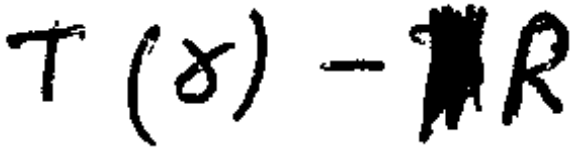


Original source details: GELFAND-P000348

Uncertain reading. A small sign after "Thom isom." is densely crossed out. In the right subscript the last digit is reinforced over other strokes; $n+1$ is provisionally retained.

$$H_{n-1} \longrightarrow H_{n+1}.$$

Proof of exactness: Let $\tau(\gamma) = \partial R$, $\partial(T(\gamma) - R) = 0$ and hence $T(\gamma) - R$ is the bdry. of something, intersect with V and precisely γ.



Original source details: GELFAND-P000348

Uncertain reading. Before the final R in $T(\gamma) - R$ is a dense correction with a vertical stroke and diagonals. The earlier sign is not recovered; R is readable.

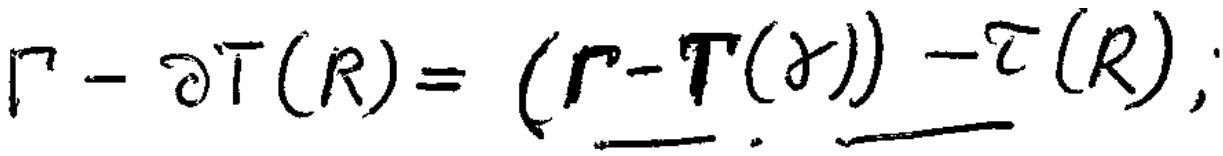
[In the left margin: $\Gamma \cdot V = \gamma = \partial R$;]

Manuscript notation. A semicolon follows R.



Original source details: GELFAND-P000348

$$\partial T(R) = \tau(R) + T(\gamma);$$



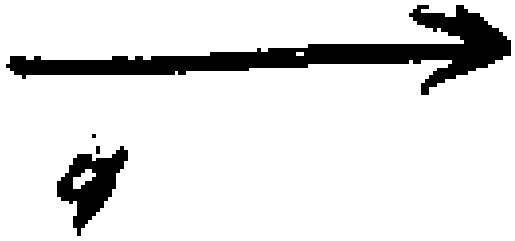
Original source details: GELFAND-P000348

Diagram and typed description.

$\Gamma - \partial T(R) = (\mathbf{\Gamma} - T(\gamma)) \ -\tau(R);$

Let there be a rat. diff. form on P_n. In homogeneous coord.

$$\mathbb{C}^{n+1} \setminus \{0\} \xrightarrow[\varphi]{} P_n.$$

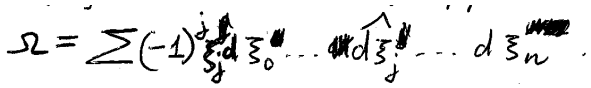


Uncertain reading. Below the arrow is a small looped sign with a descending oblique leg. φ is provisionally retained; π remains a historical candidate.

Original source details: GELFAND-P000348

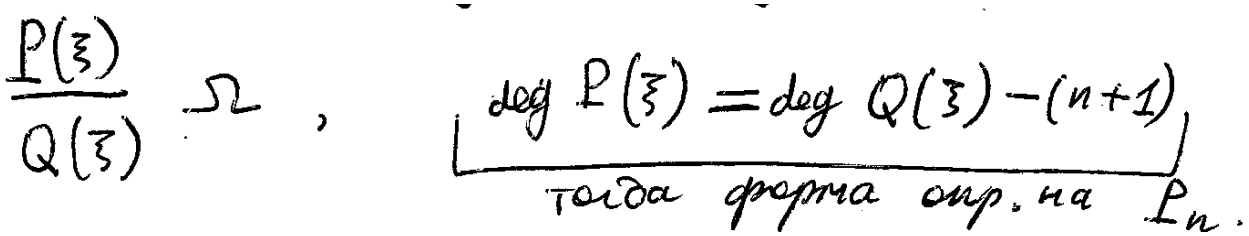
The induced form

$$\Omega = \sum (-1)^j \xi_j \, d\xi_0 \cdots \widehat{d\xi_j} \cdots d\xi_n.$$



Uncertain reading. Earlier signs above ξ_j, after $d\xi_0$, at $\widehat{d\xi_j}$ and above the final $d\xi_n$ are reinforced or crossed out. The earlier indices are not recovered.

Original source details: GELFAND-P000348



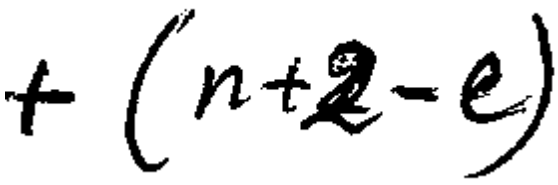
Original source details: GELFAND-P000348

Diagram and typed description.

$\frac{P(\xi)}{Q(\xi)}\Omega,$ $\quad \underline{\deg P(\xi) = \deg Q(\xi) - (n+1)}$

then the form is defined on P_n.

$$\varphi = \frac{1}{B(\xi)} \left\{ \sum_j (-1)^{|j|} \left[\sum_{\alpha=1}^{\ell} (-1)^{\alpha} \xi_{j_\alpha} A_{j_1 \cdots \widehat{j_\alpha} \cdots j_\ell}(\xi) \right] \right.$$
$$\left. (d\xi_0 \wedge \cdots \widehat{d\xi_{j_1}} \cdots \wedge \widehat{d\xi_{j_\ell}} \wedge \cdots \wedge d\xi_n) \right\}; \qquad \deg B = \deg A + (n + \mathbf{2} - \ell).$$



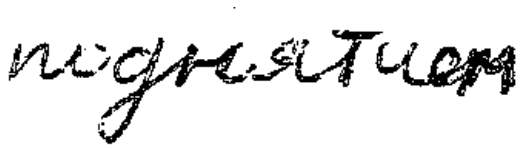
Uncertain reading. The digit before $-\ell$ is written over earlier strokes. The upper reading is provisionally 2.

Original source details: GELFAND-P000348

GELFAND-P000349

[source number: 9]

Everything is nonsingular.
This is the type of forms that are pullbacks from P_n.



Uncertain reading. The reinforced ending is provisionally read «-ием». The earlier ending is inseparable.

Original source details: GELFAND-P000349

$$\frac{P(\xi)}{Q^{\ell}(\xi)}\Omega.$$

We lower the order of poles by subtracting coboundaries.

$$\frac{1}{Q(\xi)^{\ell}}\left\{\sum_{i<j}(-1)^{i+j}[\xi_i A_j(\xi)-\xi_j A_i(\xi)]\right.$$
$$\left.(d\xi_0\wedge\cdots\wedge\widehat{d\xi_i}\wedge\cdots\wedge\widehat{d\xi_j}\cdots\wedge d\xi_n)\right\}.$$

$$P=\ell\left(\sum A_j\frac{\partial Q}{\partial\xi_j}\right)-Q\left(\sum\frac{\partial A_j}{\partial\xi_j}\right),$$
$$\deg P(\xi)=\deg Q(\xi)-(n+1).$$

Annals, v. 90, n. 3, 1969.

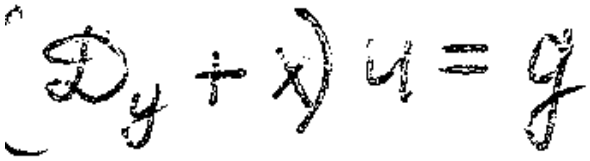
Palamodov.

On the characteristics of systems of lin. eqns.

$$\text{system}\begin{cases}a_1u=f_1\\ \vdots\\ a_tu=f_t\end{cases}\qquad a_1,\dots,a_t\text{ — diff. operators}$$

Example

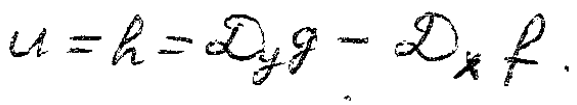
$$\begin{cases}D_xu=f,\\(D_y+x)u=g.\end{cases}\qquad\text{princ. part — ?}$$



Uncertain reading. The sign after the plus is provisionally read x. Its form also permits λ.

Original source details: GELFAND-P000349

Not $\begin{pmatrix}D_x\\D_y\end{pmatrix}$, since a consequence of the system is: $u=h=D_yg-D_xf$.



Manuscript notation. The line reads D_yg-D_xf.

Original source details: GELFAND-P000349

One must write out all consequences and deal with them.

———— . ————

Char. — ? $D_x u + D_y v = f$.

Holmgren's thm. does not apply in any direction, whereas the thm. of **C**auchy-**K**oval. in any direction.

GELFAND-P000350

29/III. Rolewicz.

A — a lin. operator. in a fin.-dim. space.

$\{x, Ax, A^2x, \ldots\}$ — closure.

Limit points: 0, $+\infty$ or

Uncertain reading. The heavily reinforced mark before ∞ is provisionally read as $+$.



Original source details: GELFAND-P000350

Undeciphered passage. An entire crossed-out word between "or" and "a compact set" is unreadable.



Original source details: GELFAND-P000350

a compact set not containing zero.

Let the oper. be in a Hilbert space, ℓ^p or c^0. It can be everywhere dense. — a shift with multiplication.

$$S(x_1, \ldots, x_n, \ldots) = (x_2, \ldots, x_n, \ldots).$$

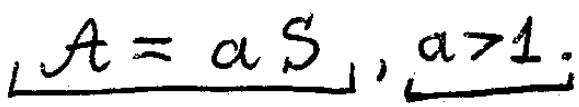
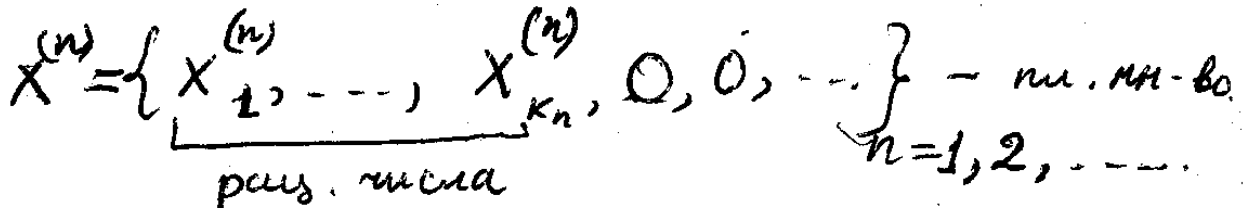


Diagram and typed description.

$A = aS$, $a > 1$.

Original source details: GELFAND-P000350

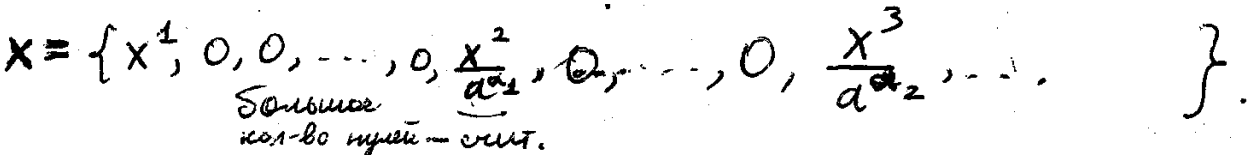


Original source details: GELFAND-P000350

Diagram and typed description.

$x^{(n)} = \{x_1^{(n)}, \dots, x_{k_n}^{(n)}, 0, 0, \dots\}$ — a lin. set

rat. numbers $n = 1, 2, \dots$.



Original source details: GELFAND-P000350

Uncertain reading. After “large no. of zeros —” is a short abbreviation. “schit.” is provisionally retained; “ochit.” remains a candidate. A short arc is below the first denominator.

$$\boldsymbol{x} = \{x^1, 0, 0, \dots, 0, \underbrace{\frac{x^2}{a^{\alpha_1}}}, 0, \dots, 0, \frac{x^3}{a^{\alpha_2}}, \dots\}.$$

large
no. of zeros — schit.

$$x^{(1)} = (x_1^{(1)}, x_2^{(1)}, \dots, x_{k_1}^{(1)}).$$

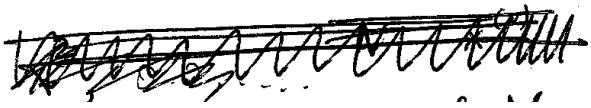
Topol. equiv. of lin. opers.: $\exists$ a homeom. U: $A = UBU^{-1}$.

There exist inequivalent opers. — density of an orbit — a topol. prop.



Uncertain reading. Before “opers. not equiv.” a word is crossed out, with “Exist.” inserted above. The earlier word is not recovered; “Eti” (“These”) remains a candidate.

Original source details: GELFAND-P000350



Undeciphered passage. An entire line between “topol. prop.” and “One can construct” is crossed out with horizontal lines and repeated loops. Its original words are unreadable.

Original source details: GELFAND-P000350

One can construct an oper. unbounded in $\forall$ norm.

Arnold.

Milnor — the theory of unimod. quad. integral forms.

$$\varphi : \mathbb{Z}^n \times \mathbb{Z}^n \longrightarrow \mathbb{Z}.$$

Unimod.: $\tilde{\varphi} : \mathbb{Z}^n \longrightarrow \mathbb{Z}^{n*}$ is an isomorphism.

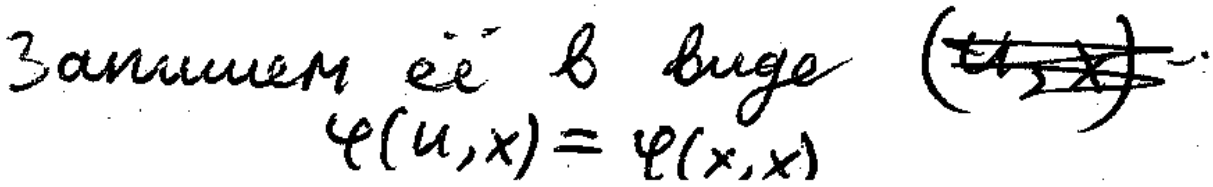
Invariants: r — rank; sgn — signature; type, I, II.



Uncertain reading. Before "rank" is a sign with a curved upper part and a lower hook. Latin r is provisionally retained; the form also resembles τ.

Original source details: GELFAND-P000350

If the form is not definite, then the system is complete.
Consider the quad. form $\hat{\varphi}_2 : \mathbb{Z}_2^n \longrightarrow \mathbb{Z}_2$, reduced mod 2. It is linear.



Original source details: GELFAND-P000350

Diagram and typed description.

Write it in the form $\varphi(u,x)$

$$\varphi(u,x) = \varphi(x,x).$$

GELFAND-P000351

[source number: 10]

or $\varphi(x,x) \equiv \varphi(u,x) \bmod 2 \quad \forall x$
Type I — $\exists$ an elt. whose square is odd.
Lemma: 1) $\varphi(u,u) \bmod 8$ does not depend on u. (u is defined mod $2\mathbb{Z}^n$).
2) $\varphi(u,u) \equiv \operatorname{sgn}\varphi \bmod 8$.
(from Fresnel's $\int$).
Classif.: Not definite.

$$\text{odd} \quad \sum x_i^2 - \sum y_i^2$$



Uncertain reading. After $\sum x_i^2 - \sum y_i^2$ a small term is densely crossed out with oblique and horizontal strokes. Its notation is not recovered.

Original source details: GELFAND-P000351

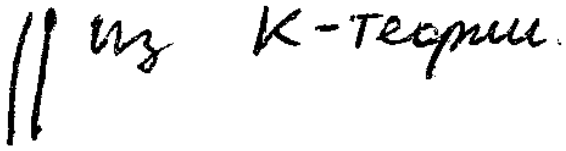


Diagram and typed description.

from K-theory.

Original source details: GELFAND-P000351

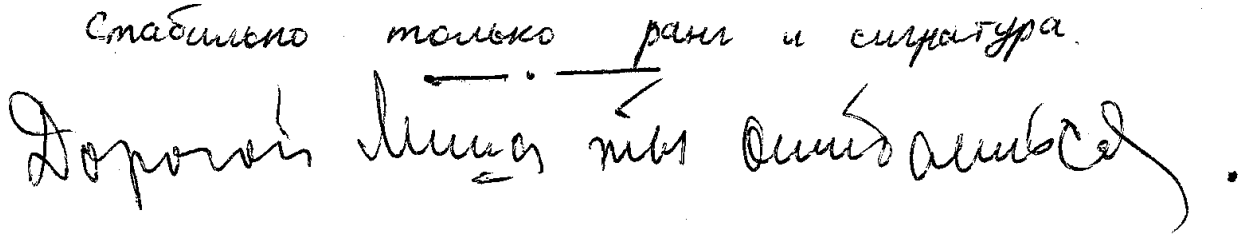
even $\quad kU \oplus \ell\Gamma_8, \qquad \Gamma_8$ — the Zolotarev form,

related to E_8.

$$U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Posit. def. — a finite number in $\forall$ dimension.

Serre — a textbook of arithmetic.



Original source details: GELFAND-P000351

Diagram and typed description.

Stably only rank and signature.

Dear Misha you are wrong.

GELFAND-P000352

[source number: 12]

5/IV Kirillov Representation according to Geronimus.



Uncertain reading. The last letter of «Представление» is written over earlier strokes and is provisionally read e.

Original source details: GELFAND-P000352

Based on the notion of a Grothendieck topology (“Topó”).

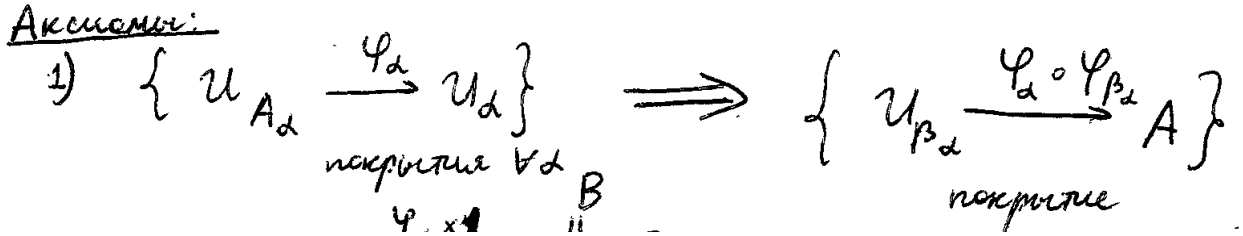
C — a category, for $\forall A \in C$ covers are specified $\{U_\alpha \xrightarrow{\varphi_\alpha} A\}$.



Uncertain reading. The loop and subscript above the arrow are heavily reinforced. The provisional reading is φ_α.

Original source details: GELFAND-P000352

Axioms:

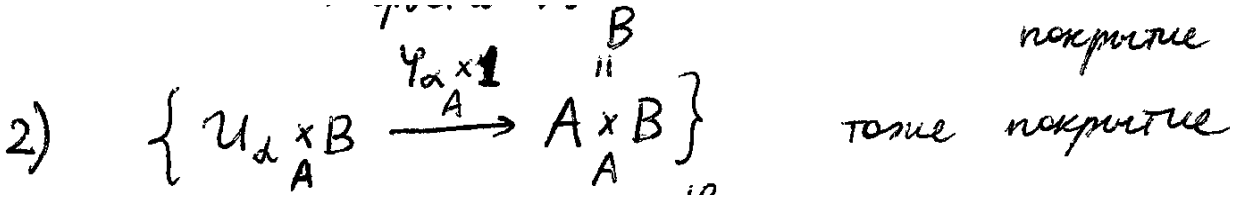


Original source details: GELFAND-P000352

Diagram and typed description.

1) $\{U_{A_\alpha} \xrightarrow{\varphi_\alpha} U_\alpha\} \quad \Longrightarrow \quad \{U_{\beta_\alpha} \xrightarrow{\varphi_\alpha \circ \varphi_{\beta_\alpha}} A\}$

covers $\forall \alpha$ a cover



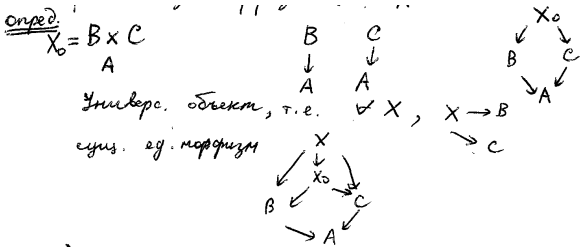
Original source details: GELFAND-P000352

Diagram and typed description.

$$\begin{array}{ccc} & & B \\ & & \| \\ \{U_\alpha \times_A B & \xrightarrow{\varphi_\alpha \times_A 1} & A \times_A B\} \end{array}$$

2) also a cover

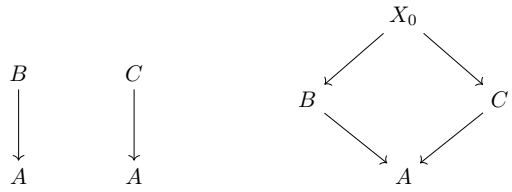
A morphism $B \xrightarrow{\varphi} A$ is given here.



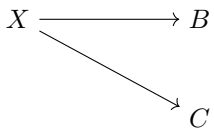
Original source details: GELFAND-P000352

Diagram and typed description.

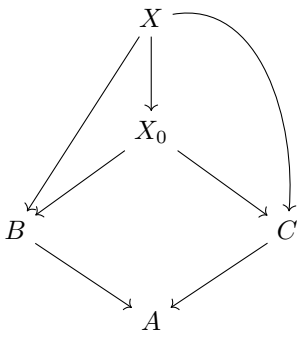
Def. $X_0 = B \times_A C$.



A universal object, i.e. $\forall X$,



there exists a unique morphism



8) $U \longrightarrow U$ a cover.



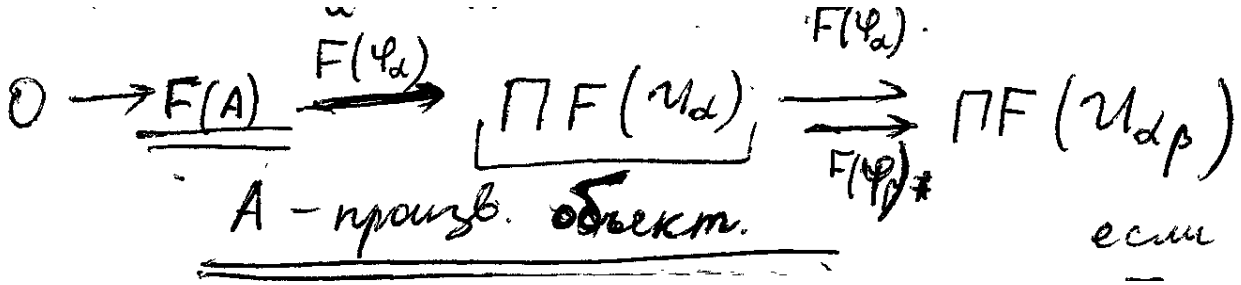
Original source details: GELFAND-P000352

Uncertain reading. Before $U \to U$ is a densely reinforced number with two loops. The visible 8 is provisionally retained; 3 remains a candidate, and the earlier number is not recovered.

(see **Mumford**.)

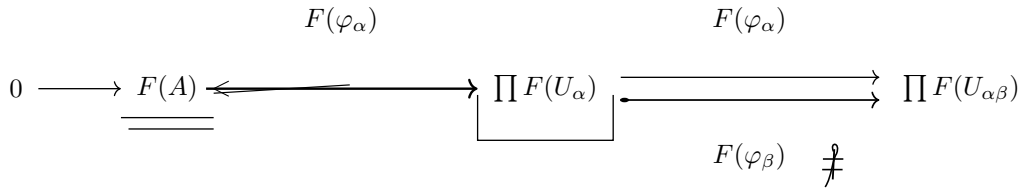
A sheaf — a functor $F : C \underset{\text{contravar.}}{\longrightarrow} K$ — another category. And the sheaf ax. holds.

$$U_\alpha \xrightarrow{\varphi_\alpha} A.$$



Original source details: GELFAND-P000352

Diagram and typed description. The first arrow from $F(A)$ is reinforced; beneath its strokes at the left is a remnant of a left-facing arrowhead, while the right arrowhead is clear. The order of alteration is not established. At the left of the lower parallel arrow are thickened strokes. After $F(\varphi_\beta)$ is a dense sign with two crossbars and a loop; its meaning is not established. Below $F(A)$ are two lines; below $\prod F(U_\alpha)$ is a bracket.

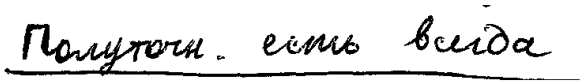


A — an arbitrary object.

If the images agree, then

$$A \longleftarrow U_\alpha \longleftarrow U_\alpha \times U_\beta.$$

Half-exactn. exists everywhere.



Original source details: GELFAND-P000352

Uncertain reading. The final abbreviation is provisionally read «Полуточн.» (“half-exactn.”).

GELFAND-P000353

$\mathcal{U} = \{U_\alpha \to A\}$. • Cohomology $H(\mathcal{U}, F)$ (Čech).

Nontriv. theorem. (!) One can construct a sheaf from a presheaf.



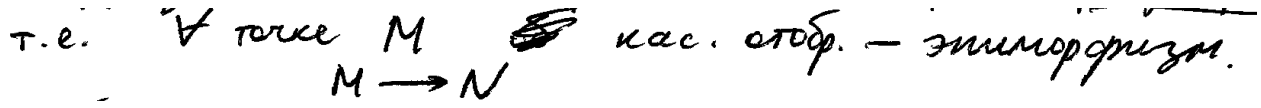
Original source details: GELFAND-P000353

Undeciphered passage. The entire crossed-out heading to the left of “from a group” is not read in full. “Predstavleniya” (“Representations”) remains an uncertain candidate.

from a group G construct “Topó”. and from a repr. — a sheaf over this “topó”.

Geronimus’s idea — cohomology and its relation to the old reprs.

G — a (Lie) group. C — the G-manifolds. Morphisms — G-maps, and moreover regular,



Original source details: GELFAND-P000353

Uncertain reading. Between M and "tang. map" a small sign is crossed out with several oblique strokes. Neither its letter nor its direction is established. Below M is written $M \to N$.

i.e. at $\forall$ point M the tang. map is an epimorphism.

$$M \longrightarrow N$$

2nd ex.

G — a fin. group — categ. — arbitrary G-sets. (Finite). Cat. K — the category of lin. spaces. Sheaves $\leftrightarrow$ representations. Let a repr. T_g in E be given.

[Crossed out: ~~Repr.~~ $\longrightarrow$]



Undeciphered passage. The crossed-out word at the right after the readable "Repr." and arrow is unreadable.

Original source details: GELFAND-P000353

commuting with the action of G.



Original source details: GELFAND-P000353

Diagram and typed description.

If X is a G-space, then one must consider maps from X to E,

of 5 points.

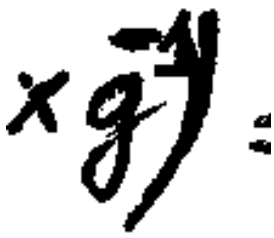
$$\xi_{g(i)} = T_g \xi_i.$$

The five-tuples — a lin. space.

———— · ————

From X and T_g construct a space of fns. $\xi(x)$.

$$T_g \xi(xg^{-1}) = \xi(x), \qquad \xi(x) \in E.$$



Uncertain reading. The exponent on g is reinforced and is provisionally read -1.

Original source details: GELFAND-P000353

There are sheaves that are not obtained from representations.

Elementary-particle theory



Uncertain reading. Before “not constructed” is a short crossed-out word with an e-shaped first loop and following shch-shaped strokes. “eshchyo” (“yet”) and “eshch.” remain uncertain candidates; the dots over yo and the ending are not established.

Original source details: GELFAND-P000353

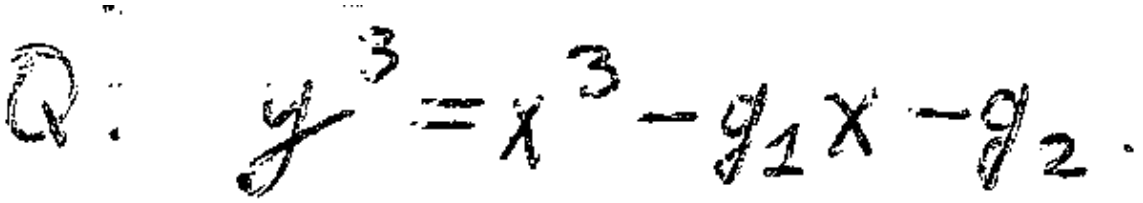
is not constructed, since not all reprs. are known. (cf. spinors and electrons.) (For a fin. group everything is obtained from here).

Sometimes one obtains the cohom. of a topol. space and cohom. as in Gelfand–Fuchs.

GELFAND-P000354

[source number: 13]

I. M. Gelfand on the Weil conjecture.



Original source details: GELFAND-P000354

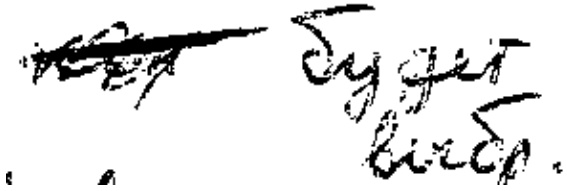
Diagram and typed description.

$$Q: \quad y^3 = x^3 - g_1 x - g_2.$$

g_1, g_2 — rat. numbers. — an ell. curve T. Over the compl. numbers.

There is a discr. group of matrices $\begin{pmatrix} \alpha & \beta \\ A\gamma & \delta \end{pmatrix}$, $\alpha, \beta, \gamma, \delta$ — integers, det $= 1$.

$A \in \mathbb{Z}$,



Uncertain reading. Between $A \in \mathbb{Z}$, and “will be chosen” a small word is crossed out with a horizontal stroke and diagonals. “pozd.” (“later”) remains an uncertain candidate.

Original source details: GELFAND-P000354

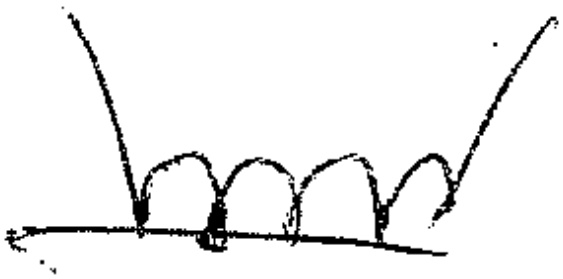
will be chosen. Group mod A.

They define fractional-lin. transf. of the upper half-plane into itself.

$$z' = \frac{\alpha z + A\gamma}{\beta z + \delta}.$$

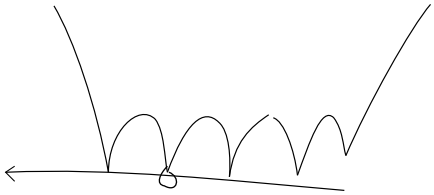
Fund. domain — the modular manifold X_A.

The group $\Gamma_0(A) = \begin{pmatrix} \alpha & \beta \\ A\gamma & \delta \end{pmatrix}$.



Original source details: GELFAND-P000354

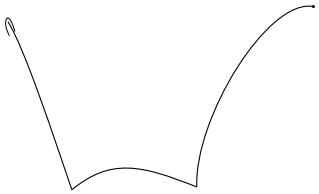
Diagram and typed description. Four arcs above a slanted nearly horizontal line with a small left arrowhead. The two outer branches rise; the third arc has an upper gap, and the first inner foot has a loop. The loop is not identified as a label.



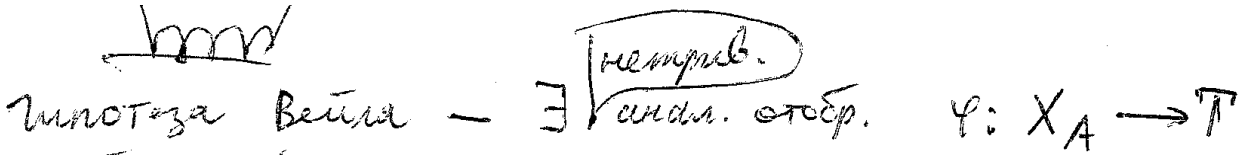


Original source details: GELFAND-P000354

Diagram and typed description.



$A = 1.$

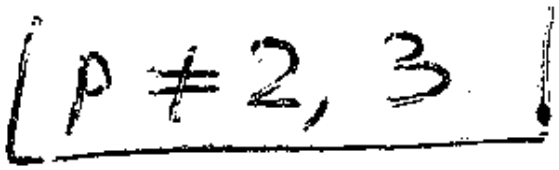


Original source details: GELFAND-P000354

Diagram and typed description.

The Weil conjecture — ∃ nontriv. an anal. map $\varphi : X_A \longrightarrow T$

(There are finitely many maps; from ∀ curve to a torus)



Original source details: GELFAND-P000354

Diagram and typed description.

$p \neq 2, 3$

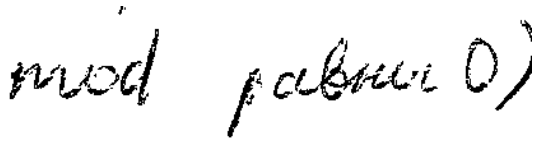
Replace T mod p. A curve over $\mathbb{F}_p$ results. It



Uncertain reading. After “It” at the start of the next line before “can” is a dense overlap of letters resembling “on” (“he/it”). It is not established whether this is a cancelled repetition of the pronoun.

Original source details: GELFAND-P000354

can become singular.
(I.e. the partial derivs. mod equal 0).



Original source details: GELFAND-P000354

Manuscript notation. After “mod” the manuscript goes directly to “equal,” with the modulus unstated.

1) No sing. points.

2) Sing. points with distinct tangents.

2a — real



Original source details: GELFAND-P000354

Undeciphered passage. An entire crossed-out word between “real” and “tang.” is unreadable.

tang.

2b — imaginary tangents.

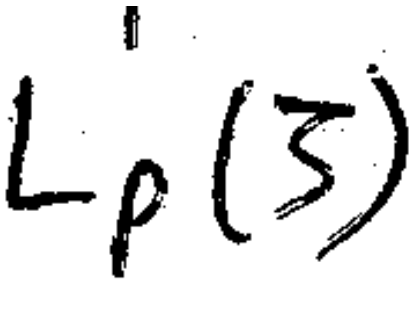
3) The tangents coincide.

$$\begin{array}{lll} A_p = 1 & p & \text{type 1} \\ A_p = p & p & \text{type 2} \\ A_p = p^2 & p & \text{type 3} \end{array}$$

GELFAND-P000355

$A = \prod_p A_p$ — the conductor of the curve.

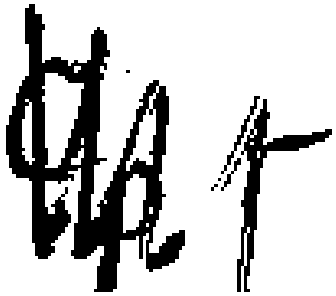
$$L_p(\zeta) = (1 - c_p p^{-s} + p^{1-2s})^{-1}, \qquad p \text{ — of type 1.}$$



Original source details: GELFAND-P000355

Manuscript notation. The first argument reads ζ, as in $\zeta(s)$ below.

[Crossed out before the sign “+”:]



Original source details: GELFAND-P000355

Undeciphered passage. The initial entry before “+ number of points of the curve” is densely overwritten and crossed out. The whole earlier sign or short entry has not been recovered.

$+$ number of points of the curve:

$$1 - c_p + p.$$

$$L_p(s) = (1 - \varepsilon_p p^{-s})^{-1}, \qquad \text{case 2)}; \qquad \begin{matrix} \varepsilon_p = +1 & \text{2a}, \\ \varepsilon_p = -1 & \text{2b}. \end{matrix}$$

$L_p(s) = 1$, case 3.

$$\zeta(s) = \prod_p L_p(s).$$

Expand $\zeta(s)$ in a series:

$$\zeta(s) = \sum_1^\infty \frac{c_n}{n_s}.$$

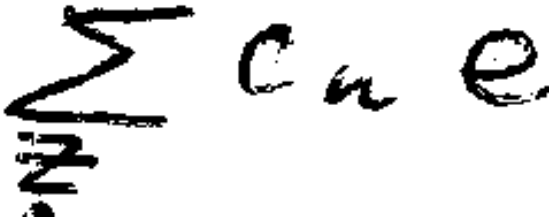


Uncertain reading. The small s in the denominator is below n. The visible n_s is retained without replacing it by a power n^s.

Original source details: GELFAND-P000355

Now consider

$$\sum_z c_n e^{inz} = \omega(z).$$

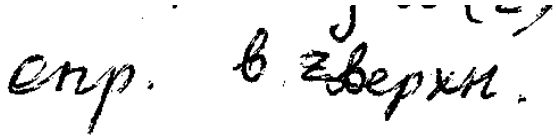


Uncertain reading. The lower summation sign has two horizontal strokes and an oblique middle, resembling z. The coefficient and exponent use n; the lower z is provisionally retained.

Original source details: GELFAND-P000355

Assertion: $\int \omega(z)\, dz$ — the sought fn..

It is str. in the upper half-plane. (and analyt.)



Uncertain reading. The abbreviation reads “str.”. Overlapping earlier strokes before “upper” are crossed out.

Original source details: GELFAND-P000355

Maps the upper half-plane into the compl. line.

The Weil conjecture:

I. $\exists$ a lattice $\Gamma \subset \mathbb{C}^1$ such that $\forall g \in \Gamma_0(A)$

$$\int_z^{gz} \omega(z)\, dz \in \Gamma.$$

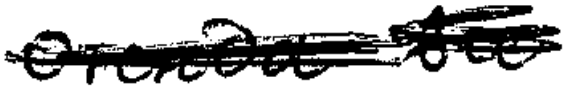
[Crossed out at lower left:]



Original source details: GELFAND-P000355

Undeciphered passage. The short entry in the lower left margin is covered by horizontal and oblique strokes. A complete reading is not established.

[Crossed-out line at the bottom:]

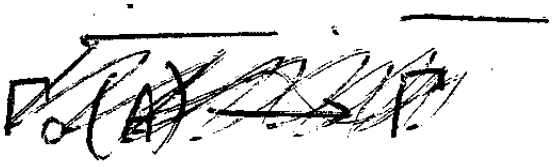


Original source details: GELFAND-P000355

Uncertain reading. The crossed line is provisionally read “otsyuda vse” (“from here all”); “vsyo” (“everything”) is also possible. The last ending and possible dots over yo are obscured by horizontal strokes.

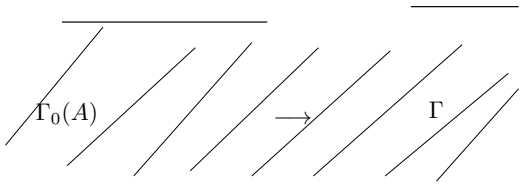
~~from here all~~

[Crossed out above the continuation of this line:]



Original source details: GELFAND-P000355

Uncertain reading. The crossed entry $\Gamma_0(A) \to \Gamma$ is shown provisionally. The letter in parentheses is reinforced over other strokes; A remains a candidate, and the earlier argument has not been recovered. The oblique strokes and two upper horizontal lines are rendered typographically.

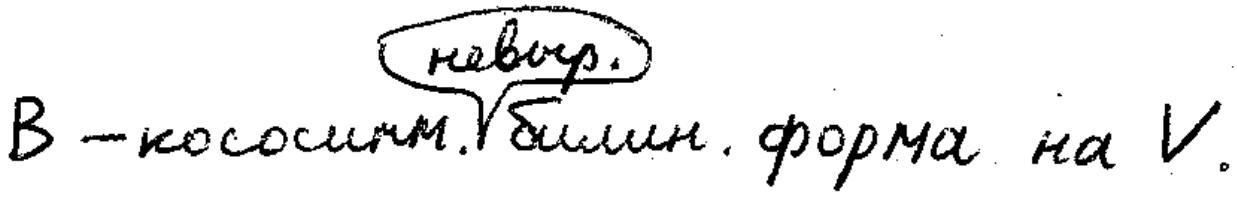


GELFAND-P000356

[source number: 14]

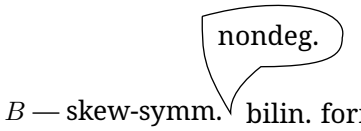
12/IV. Perelomov.

V — a $2N$-dim. real space,



Original source details: GELFAND-P000356

Diagram and typed description.



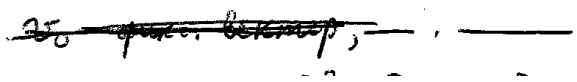
A B-lattice $L \subset V$,



Undeciphered passage. The short crossed-out entry between $L \subset V$, and the following dash has not been read in full.

Original source details: GELFAND-P000356

— on L, B takes integer values. $J^2 = -1$, J is compatible with B: $J^T B J = B$.



Uncertain reading. The crossed row is provisionally read v_0 fix. vector. Middle strokes of "fix." are covered by horizontals; the abbreviation is not expanded. After the word, a dash, comma and long horizontal stroke are retained.

~~v_0 fix. vector~~ —, ———————

Original source details: GELFAND-P000356

Gelfand on Pauli's book:

$$P^2 = P, \qquad \lambda^2 = \lambda, \qquad \lambda = 0, +1, -1.$$

-1 discarded for phys. reasons.



Diagram and typed description.

——— , ———————

Original source details: GELFAND-P000356

Volume of the fund. parallelotope $= 1$. (principal lattice). Let $BJ > 0$.

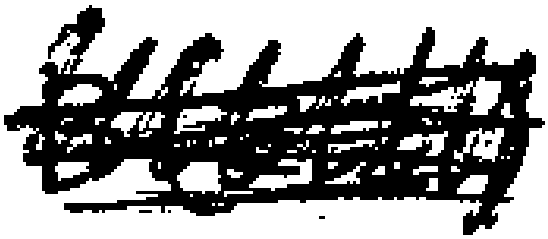


Uncertain reading. The sign after BJ is reinforced over earlier strokes, with an additional horizontal line. $>$ is shown provisionally; $\geq$ remains a possible reading.

Original source details: GELFAND-P000356

A compl. structure on V. $z \in \mathbb{C}^N$.

$H(z, z') = B(z, Jz') + iB(z, z')$ — a Hermitian form.



Undeciphered passage. Between the equals sign and the current $B(z, Jz')$, an earlier formula with a B-like beginning is covered by repeated cancellation. Its arguments and signs are unreadable.

Original source details: GELFAND-P000356

Consider a fn. F: — integer-valued.

$$F(\alpha+\beta) \equiv F(\alpha) + F(\beta) + B(\alpha,\beta) \pmod 2.$$

A lin. form may be added to F. Choose a basis in L.

$$v \in L \mapsto \kappa_1, \dots, \kappa_N, \ell_1, \dots, \ell_N.$$

Assume:

$$B = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}.$$



Undeciphered passage. The short crossed-out word before $F(\alpha)$ has not been read in full. An initial loop and a lower stroke are visible, but a complete reading is not established.

Original source details: GELFAND-P000356

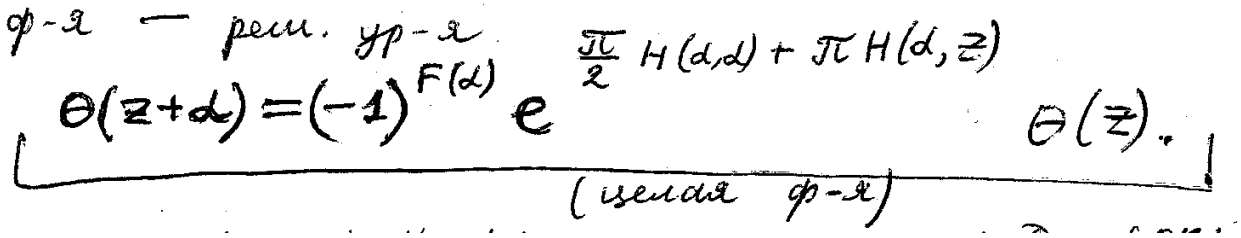
$$F(\alpha) = \sum \kappa_i \ell_i + \sum \varepsilon_i \kappa_i + \sum \tilde{\varepsilon}'_i \ell_i \qquad \text{(there exist } 4^N \text{ solutions)}.$$



Uncertain reading. Above the last coefficient is a separate oblique stroke below the right part of the tilde. It is provisionally rendered as an additional prime on $\tilde{\varepsilon}_i$′; its exact position remains uncertain.

Original source details: GELFAND-P000356

A Θ-fn. — a solution of the eqn.



Original source details: GELFAND-P000356

Uncertain reading. The current $H(\alpha, z)$ has no separate upper bar on z; below it is a remnant of an overlapped curve, not separated into a secure sign. Below the whole formula is a three-sided bracket without a top, with “(entire fn.)” underneath.

$$\Theta(z+\alpha) = (-1)^{F(\alpha)} e^{\frac{\pi}{2} H(\alpha,\alpha) + \pi H(\alpha,z)} \Theta(z).$$

(entire fn.).

There exists only one solution of the functional eqn. up to const. (It is important that the lattice is principal — $\det B = 1$). This is written in Cartier (although known before him).

GELFAND-P000357

Phys. problem (see von Neumann). Let a — annihil. oper., $a|\alpha\rangle = \alpha|\alpha\rangle$ — eigenvectors.

$$e^{-\frac{(x-\lambda)^2}{2}}.$$

Spectrum — the whole compl. plane. The system is overcomplete. Problem: select a complete system.

~~Result~~



Manuscript notation. The crossed-out word reads “Result.”

Original source details: GELFAND-P000357

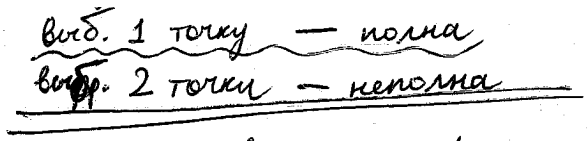
Overcomplete — if discarding $\forall$ finite number leaves a complete system. (if λ has an accumulation point, then ...) For which seq. λ_n is this a complete system? Let λ — a lattice on the compl. plane C.

cell area $> \pi$ — incomplete,
— ″ — $< \pi$ — overcomplete.

Let the cell area $= \pi$.

vyb. 1 point — complete.

vybr. 2 points — incomplete.

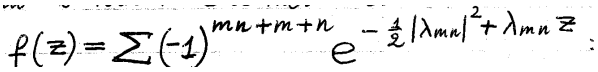


Uncertain reading. Before “2 points”, “vybr.” is shown provisionally; “vyb.” remains a candidate. After the b there is a loop and a descending r-like stroke, but the earlier entry is not securely separated. The abbreviation is left unexpanded. The first line has a wavy underline, the second two horizontal underlines.

Original source details: GELFAND-P000357

I.e., there is exactly 1 relation.

$$f(z) = \sum (-1)^{mn+m+n} e^{-\frac{1}{2}|\lambda_{mn}|^2 + \lambda_{mn} z} = 0.$$



Manuscript notation. The exponent ends in z with a crossing stroke, as in $f(z)$.

Original source details: GELFAND-P000357

$\lambda_{mn} = m\omega_1 + n\omega_2$. (already in F_2) — everything is divisible. or

$$\sum (-1)^{m\boldsymbol{n}+m+n} e^{-\frac{(x-\lambda_{mn})^2}{2}}.$$

Diagram and typed description. In the initial mn of the sign exponent, the second letter is thickened and overwritten. A separate earlier reading is not established.



Original source details: GELFAND-P000357

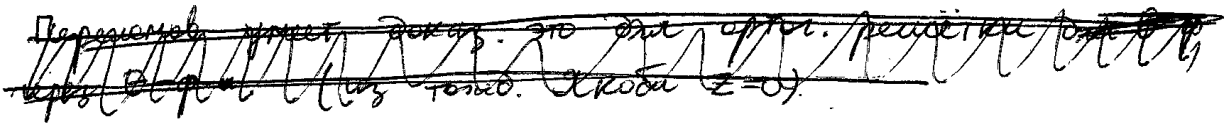
$$|\langle\alpha|\beta\rangle|^2 = e^{-|\alpha-\beta|^2}.$$

(Neumann and Schoenberg — pos. def. fns.). Von Neumann's question: consider a lattice with area π and fns. $|\alpha_{mn}\rangle$. In this state determine the variances of X and p. $\delta_x\,\delta_p$ usually grows for an orthonorm. system.

GELFAND-P000358

[source number: 15]

Von Neumann's question. Will $\delta_x\,\delta_p$ after orthogonalization of these vectors be bounded on this system? Gelf. considers this unlikely. The gen. question for an orthog. system is unclear.



Original source details: GELFAND-P000358

Undeciphered passage. Legible fragments of the two crossed-out lines are retained. The end of the first line and beginning of the second, including part of a parenthesis, are unrecovered.

~~Perelomov can prove this for an orthog. lattice [?]~~

~~[?] (from the Jacobi ident. $(z = 0)$).~~

π The gen. ident. is not deduced from the Jacobi ident..

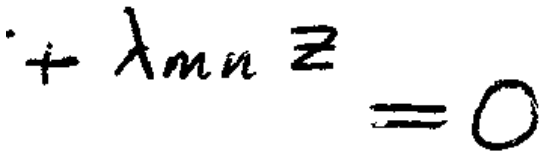
Uncertain reading. The current preposition is provisionally read "from"; it is rewritten in dark strokes. The earlier entry underneath is not securely separated.



Original source details: GELFAND-P000358

This carries over to the N-dim. case. Obtain $2^{N-1}(2^N - 1)$ identities. Thus, an elem. problem:

$$\sum (-1)^{mn+n+m} e^{-\frac{1}{2}|\lambda_{mn}|^2 + \lambda_{mn} z} = 0$$



Manuscript notation. The final z in the exponent has a stroke through the middle of the letter.

Original source details: GELFAND-P000358

if the lattice area $= \pi$. Proof: consider a Hilb. space with

$$\int \bar{f}(z) g(z) e^{-|z|^2} \, dz \, d\bar{z}.$$

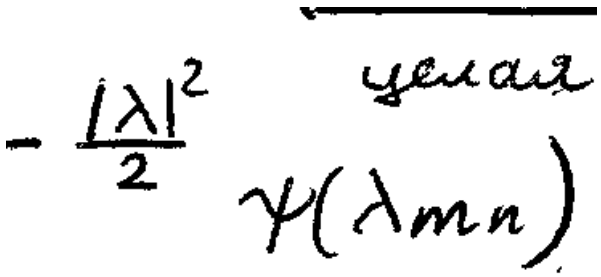
$|\boldsymbol{\lambda}_{mn}\rangle$
Consider $|\lambda\rangle$ and $|\psi\rangle \in \mathcal{H}$.

$$\langle \psi | \lambda \rangle = e^{-\frac{|\lambda|^2}{2}} \, \psi(\lambda) \cdot$$

entire analytic.

Orthog.

$$\langle \psi | \lambda_{mn} \rangle = e^{-\frac{|\lambda|^2}{2}} \psi(\lambda_{mn}).$$



Manuscript notation. The norm has λ without mn. The ket and the argument of ψ have λ_{mn}.

Original source details: GELFAND-P000358

Entire with 0s at the prescribed set of points. Need to prove that it does not exist in F_2. (see Levin, standard theorem).

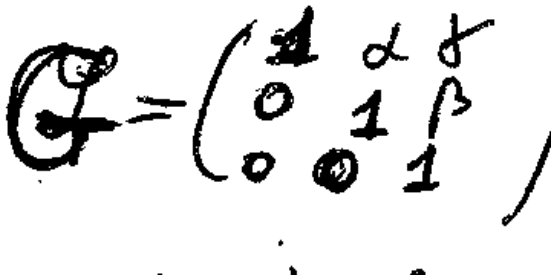


Undeciphered passage. The isolated crossed-out mark before G is not recovered in full.

Original source details: GELFAND-P000358

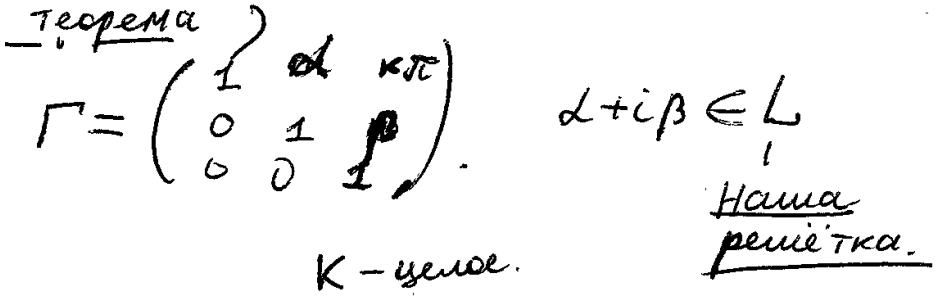
$$G = \begin{pmatrix} \mathbf{1} & \alpha & \delta \\ 0 & 1 & \beta \\ 0 & \mathbf{0} & 1 \end{pmatrix},$$

α, β, δ — real.



Manuscript notation. The upper left one and the middle zero in the bottom row are written over in heavier strokes.

Original source details: GELFAND-P000358



Original source details: GELFAND-P000358

Diagram and typed description. The current α and β in Γ are heavily overwritten; earlier strokes are unrecovered. The upper right entry is provisionally shown as $k\pi$. The note below has a capital K. Below L is a short detached vertical stroke towards the two underlined words “Our lattice”; its exact connection is not established.

$$\Gamma = \begin{pmatrix} 1 & \boldsymbol{\alpha} & k\pi \\ 0 & 1 & \boldsymbol{\beta} \\ 0 & 0 & 1 \end{pmatrix}.$$

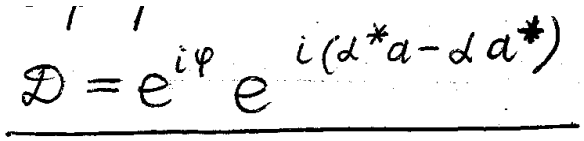
K — integer.

$\alpha + i\beta \in L$
|
Our lattice.

GELFAND-P000359

Consider the operators.

$\mathcal{D} = e^{i\varphi} e^{i(\alpha^* a - \alpha a^*)}$ — a 3-param. group.



Manuscript notation. The exponent of the second exponential has an i before the parenthesis.

Original source details: GELFAND-P000359

Lie alg. — a, a^+ and 1; $[a, a^+] = 1$, other commutators 0. Γ — the α-points of the lattice; φ — 0 and π. Then the repr. induced by the repr. $(-1)^{mn+m+n}$ of the discrete group Γ is irreducible. There is an extension to an arbitrary group.

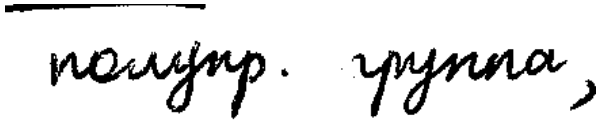
Kazhdan.



Uncertain reading. “Kazhdan.” is shown provisionally; compressed middle strokes also permit the literal “Kamdan.”.

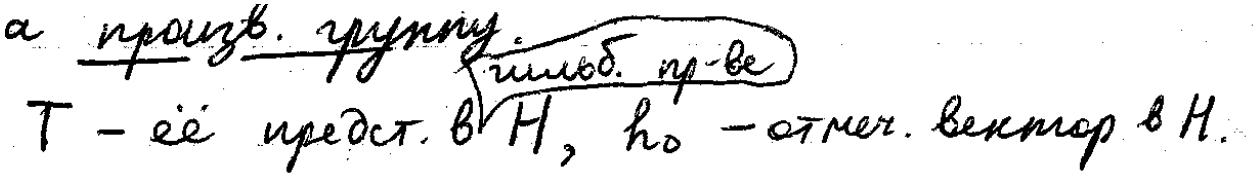
Original source details: GELFAND-P000359

G — a polupr. group,



Uncertain reading. After G — “polupr.” is shown provisionally; “polugr.” remains a candidate. The abbreviation is not expanded.

Original source details: GELFAND-P000359



Original source details: GELFAND-P000359

Diagram and typed description.

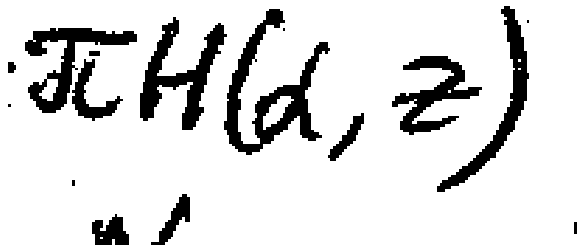
Hilb. space

T — its repr. in H,

h_0 — a distinguished vector in H. $T(g)h_0 \in H$. $\Gamma \subset G$ a discrete subgroup, $T(\gamma)h_0 \in H$. For a reasonable subgroup Γ the system is complete and satisfies a finite number of relations. Here — a unipotent group.

$$f(z) = \sum_{\alpha} (-1)^{F(\alpha)} e^{-\frac{\pi}{2}H(\alpha,\alpha)+\pi H(\alpha,z)} = 0 = c\Theta(z).$$

Manuscript notation. The z in $H(\alpha, z)$ has a crossing stroke through the letter.

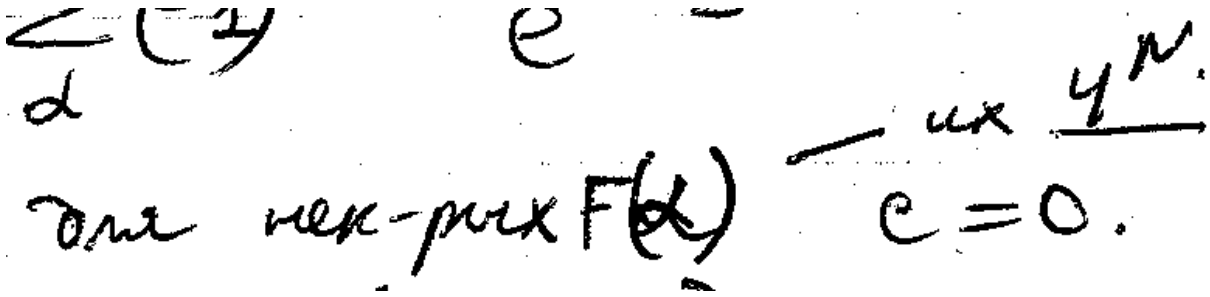


Original source details: GELFAND-P000359

Uncertain reading. The source literally writes $= 0 = c\Theta(z)$. The last argument of Θ is repeatedly retraced, with overlapping upper and lower strokes. z is shown provisionally; the earlier sign and any separate upper bar are not established.



Original source details: GELFAND-P000359



Original source details: GELFAND-P000359

Diagram and typed description.

— of them $\underline{4^N}$

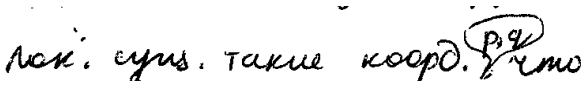
for some $F(\alpha)$ $\quad c = 0.$

$c \neq 0$ — A new repr. of the Θ-fn. For half of the $F(\alpha)$ this is 0. $\Theta(z)$ — the standard fn., $\neq 0$. Next meeting 1st Mon. of September.

GELFAND-P000360

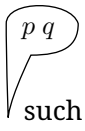
[source number: 22]

Kirillov on quantization. (Kostant etc.) Trivial part. A symplectic manifold — X^{2n} — even-dimensional with a nondeg. closed 2-form ω ($d\omega = 0$).



Original source details: GELFAND-P000360

Diagram and typed description.

Locally there exist coords.  such that

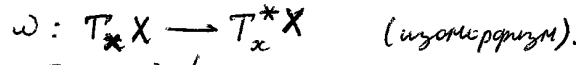
$$\omega = \sum_{i=1}^{n} dp_i \wedge dq_i.$$

(see Rashevskii, “Geom. theory of diff. eqns.”, Sternberg, “Lectures on diff. geom.”). Volume

$$\Omega = \frac{1}{n!}\omega \wedge \omega \wedge \cdots \wedge \omega = \frac{1}{n!}\omega^n \quad \text{(exterior power)}.$$

Identify tang. space with cotang. using the form ω.

$$\omega(\xi, \eta) \text{:—} \quad \omega : T_x X \to T_x^* X \quad \text{(isomorphism)}.$$



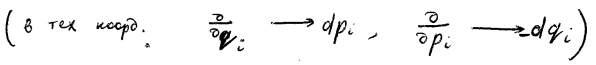
Original source details: GELFAND-P000360

Uncertain reading. The lower index of the first T is heavily overwritten. x with overlapping upper strokes is shown provisionally; $*$ remains a candidate.

$$\xi \to \alpha_\xi;$$

$$\alpha_\xi(\eta) = \omega(\xi, \eta).$$

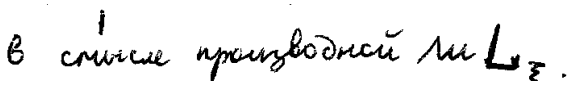
$$\left(\text{in those coords. } \frac{\partial}{\partial q_i} \to dp_i, \quad \frac{\partial}{\partial p_i} \to -dq_i\right).$$



Original source details: GELFAND-P000360

Manuscript notation. The coordinate line gives dp_i and $-dq_i$.

Closed 1-forms $\leftrightarrow$ vector fields preserving the form ω.



Original source details: GELFAND-P000360

Diagram and typed description.

|

in the sense of the Lie derivative L_ξ.

L_ξ acts on all geom. objects on the manifold. φ_t — the corresponding one-param. group, corresponding to the field ξ. ($\dot{x} = \xi(x)$).

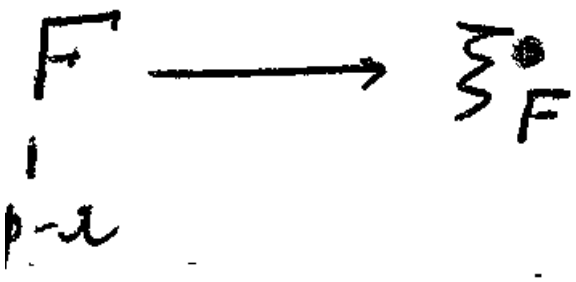
$$\left.\frac{d}{dt}\right|_{t=0} = L_\xi.$$

It commutes with differentiation of diff. forms: $L_\xi d = dL_\xi$. $L_\xi f = \xi f$, $L_\xi \eta = [\xi, \eta]$.

$$\boxed{L_\xi \omega = 0 \;\Leftrightarrow\; d\alpha_\xi = 0.}$$ — a short formulation.

Such fields — Hamiltonian, forms — gradient. ξ — loc. Hamiltonian $\Leftrightarrow \alpha_\xi = df$. f is defined up to const and is multivalued.

GELFAND-P000361



Uncertain reading. Beside the first ξ_F is a dense round overwritten mark. It is shown as a round mark without assigning a mathematical meaning; the exact earlier sign is not established.

$F \longrightarrow \xi_F^{\bullet}$

|

fn.

Original source details: GELFAND-P000361

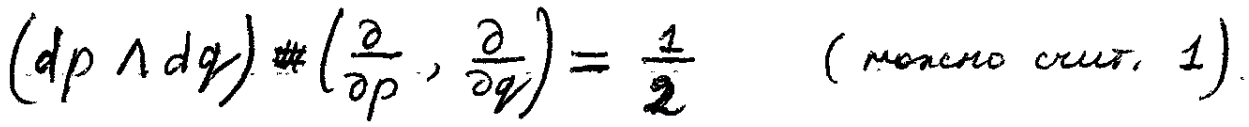
dF — a form $\longleftrightarrow$ Hamiltonian field; $dF = 2\alpha_{\xi_F}$. Poisson bracket.

$$\{F_1, F_2\} = \xi_{F_1} F_2 = -\xi_{F_2} F_1 = 2\omega(\xi_{F_1}, \xi_{F_2}).$$

Identity:

$$\xi_{\{F_1,F_2\}} = [\xi_{F_1}, \xi_{F_2}].$$

Remark:



Original source details: GELFAND-P000361

Uncertain reading. Between $(dp \wedge dq)$ and the parenthesis with vectors is a dense crossed hash-like mark. Its strokes are shown without a mathematical interpretation.

$(dp \wedge dq)$ ## $\left(\frac{\partial}{\partial p}, \frac{\partial}{\partial q}\right) = \frac{1}{2}$ (one may take 1).

$$\frac{1}{2}[\alpha_1(\xi)\alpha_2(\eta) - \alpha_2(\xi)\alpha_1(\eta)] = \alpha_1 \wedge \alpha_2.$$

A new object is constr. when $[\omega] \in H^2(X, Z)$. Constr. a complex 1-dim. bundle and a map $\mathbb{C}$.

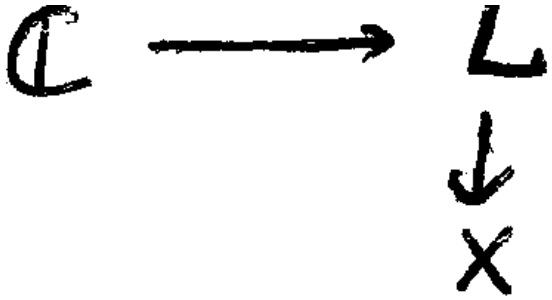
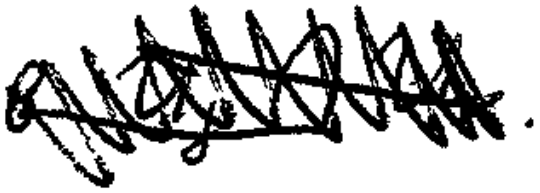


Diagram and typed description.

$$\begin{array}{ccc} \mathbb{C} & \longrightarrow & L \\ & & \downarrow \\ & & X \end{array}$$

Original source details: GELFAND-P000361

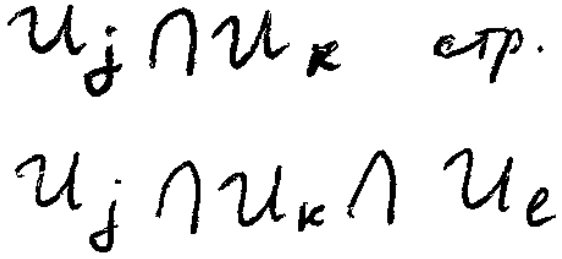
$\{U_j\}$ — a covering, $\omega = d\alpha_j$ in U_j. On $U_j \cap U_k$ constr. $\alpha_j - \alpha_k = df_{jk}$. ~~$\alpha$~~[?]



Undeciphered passage. After df_{jk}, the beginning of the crossed entry is provisionally shown as α. Small signs, subscripts and the continuation are not recovered in full.

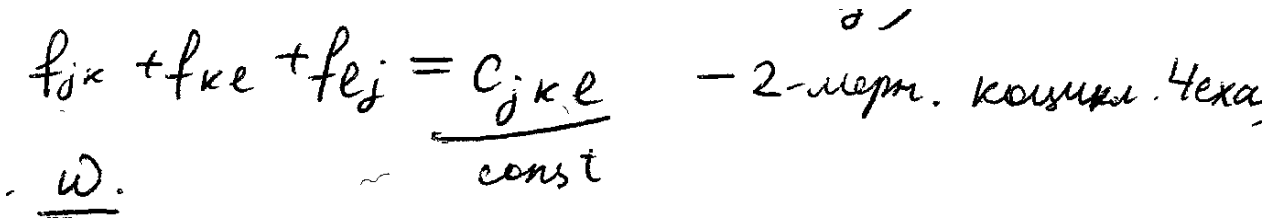
Original source details: GELFAND-P000361

on $U_j \cap U_k \wedge U_\ell$ we have $d(f_{jk} + f_{k\ell} + f_{\ell j}) = 0$ and hence



Uncertain reading. The second sign between the sets is pointed: $\wedge$ is shown provisionally, without replacing it by the usual intersection.

Original source details: GELFAND-P000361



Original source details: GELFAND-P000361

Diagram and typed description.

$f_{jk} + f_{k\ell} + f_{\ell j} = \underset{\text{const}}{C_{jk\ell}}$ — a 2-dim. Čech cocycle,

corresponding to the form $\underline{\omega}$. Integer-valued — $\exists \alpha_j, f_{jk}$ such that $C_{jk\ell}$ — integer-valued.

$g_{jk} = e^{2\pi i f_{jk}}$ — a set of bundle transition fns. (1-dim.).

$$g_{jk} \cdot g_{k\ell} \cdot g_{\ell j} = 1.$$

GELFAND-P000362

[source number: 23]

Choose other α, [?] ~~const [?]~~



Original source details: GELFAND-P000362

Undeciphered passage. In the cancelled continuation a const-like fragment is shown provisionally. Other words and signs among long looping cancellation strokes are not read in full.

The equiv. class of the constr. bundle depends on choosing an element of

$$H^1(X, \mathbb{C}^*).$$

$H^1(X, \mathbb{C}^*)$ acts simply transitively on the set of bundles. ($\mathbb{C}^*$, rather than



Undeciphered passage. The small crossed-out token before “sheaf” is wholly unreadable.

Original source details: GELFAND-P000362

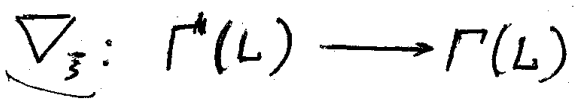
the sheaf of continuous fns.) s_j in U_j, $s_j = g_{jk}s_k$ — a section of the bundle.



Original source details: GELFAND-P000362

Diagram and typed description.

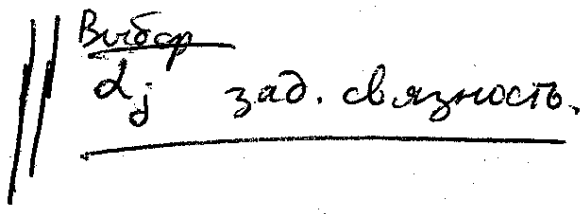
Connection — derivative of sections along a vector field.



Original source details: GELFAND-P000362

Uncertain reading. The small upper mark on the first Γ is thickened. $\Gamma^1(L)$ is shown provisionally; $\Gamma\prime(L)$ remains a candidate. Below ∇_ξ is a curved underline.

$$\nabla_\xi : \quad \Gamma^1(L) \to \Gamma(L).$$



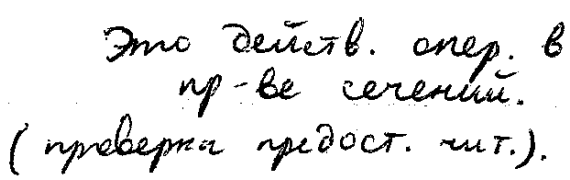
Original source details: GELFAND-P000362

Diagram and typed description.

Choice of α_j defines the connection.

$$(\nabla_\xi s)_j = \xi s_j - 2\pi i \alpha_j(\xi) s_j.$$

This acts as an oper. on the space of sections. (checking left to the reader).



Original source details: GELFAND-P000362

Uncertain reading. The first word is provisionally read “checking”; “to check” remains a candidate. The following Russian abbreviations are “predost. chit.”, with a separate o-like loop after d. The abbreviations remain unexpanded in the Russian text.

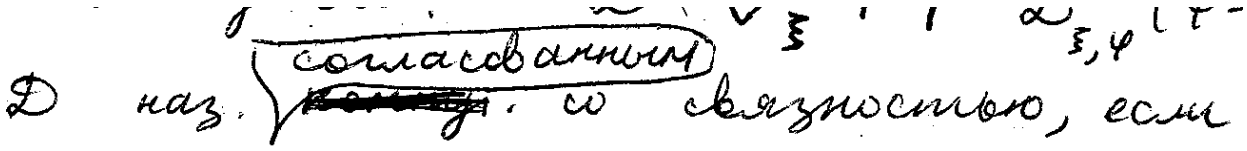
Check:

$$(\nabla_\xi s)_j = g_{jk}(\nabla_\xi s)_k.$$

Define a Hermitian bundle with a connection and a generalized Fourier transform in it. Consider an arbitrary 1st-order diff. oper. in the bundle L.

$$\mathcal{D} : \Gamma(L) \to \Gamma(L) \quad \text{(ordinary in local coords.)}.$$

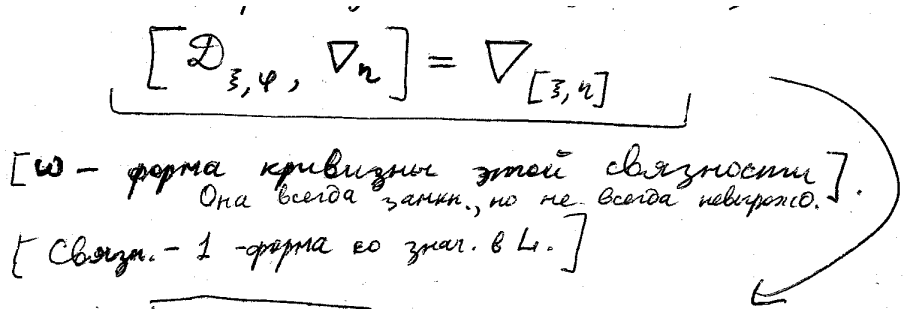
Its gen. form: $\mathcal{D} = \nabla_\xi + \varphi = \mathcal{D}_{\xi,\varphi}$ (φ — multiplication by a fn.).



Original source details: GELFAND-P000362

Undeciphered passage. The earlier adjective is crossed out and not recovered in full. Above it is the legible inserted and outlined word “compatible”.

$\mathcal{D}$ is called ~~[?]~~ compatible with the connection if



Original source details: GELFAND-P000362

Diagram and typed description. An arc starts at the right of the commutator, bows to the right beside the remarks in square brackets, and ends down-left above the lower frame. No additional mathematical relation is assigned to the arc.

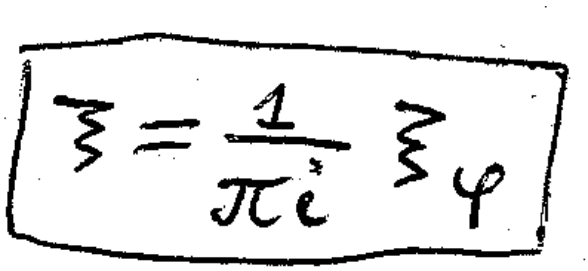
$$[\mathcal{D}_{\xi,\varphi}, \nabla_\eta] = \nabla_{[\xi,\eta]}.$$

[ω — the curvature form of this connection.
It is always closed, but not always nondeg.]

[Connection — a 1-form with values in L.]

Or

$$\boxed{\xi = \tfrac{1}{\pi i}\xi_\varphi}$$



Manuscript notation. The box reads $\xi = \frac{1}{\pi i}\xi_\varphi$.

Original source details: GELFAND-P000362

(in particular, ξ is actually Hamiltonian.)

GELFAND-P000363

~~$\mathcal{D}_\varphi$~~



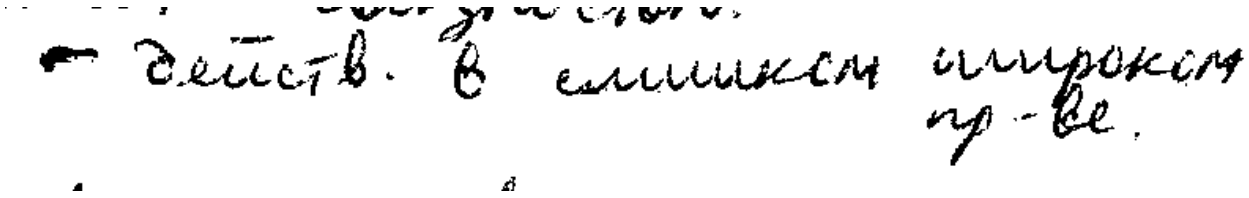
Uncertain reading. The crossed initial sign is provisionally shown as $\mathcal{D}_\varphi$; overlapping strokes do not permit a secure reading.

Original source details: GELFAND-P000363

The operator $\mathcal{D}_\varphi = \nabla_{\xi_\varphi} + \pi i\varphi : \Gamma(L) \to \Gamma(L)$. Fund. theorem:

$$[\mathcal{D}_\varphi, \mathcal{D}_\psi] = \mathcal{D}_{\{\varphi,\psi\}}.$$

(obtain a repr. of the Lie algebra) compatible with the connection. (Van Hove repr.: fns. $\to$ opers. on bundle sections) compatible with the connection. This is called prequantization.

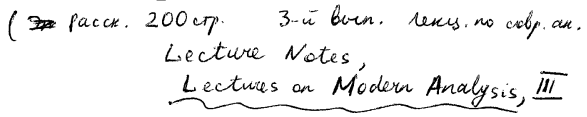


Original source details: GELFAND-P000363

Diagram and typed description. Before the note at right is a short left arrow with a shaft on its right. Its exact endpoint is not established; no additional logical connection is assigned.

$\leftarrow$ acts on too large a space.

(~~This.~~ consider p. 200 of the 3rd issue of Lectures on Modern An., Lecture Notes, Lectures on Modern Analysis, III).



Uncertain reading. Before "consider," the crossed-out word provisionally reads "This." The source puts "200" before the page abbreviation.

Original source details: GELFAND-P000363

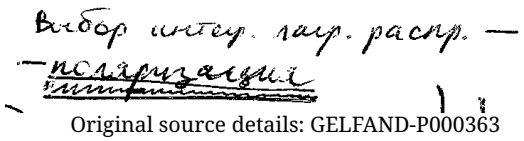
A Lagrangian distribution (isotropic space) — — at $\forall$ point a Lagr. subspace is given. $\frac{1}{2}$ the

dimension.

$$F : F_x \subset T_* X, \qquad \omega|_{F_x} = 0, \qquad \dim F_x = n = \frac{1}{2} \dim X.$$

Complex distributions will be considered:

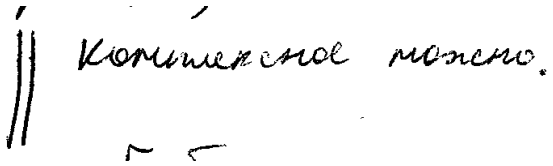
$$F_x \subset T_x^C X, \qquad \omega|_{F_x} = 0, \qquad \dim_C F_x = n = \frac{1}{2} \dim X.$$



Original source details: GELFAND-P000363

Diagram and typed description. Written vertically in the left margin.
The choice of an integr. Lagr. distrib. —
— polarization

It is considered integrable (Frobenius cond.) — see Dieudonné. (Locally exists — because of $\sum dp \wedge dq$) take the p-subspace.

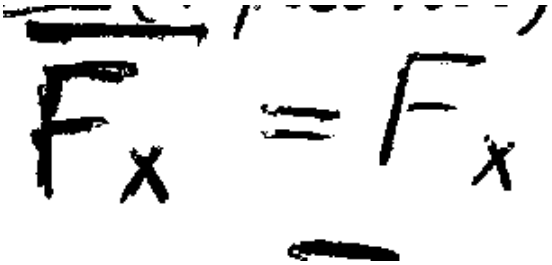


Original source details: GELFAND-P000363

Diagram and typed description.
The complex case is possible.

Example: a sphere with its usual volume. Globally impossible. Special cases (extreme).

1) $\overline{F}_x = F_x$ — a real distribution. (real polarization)



Original source details: GELFAND-P000363

Uncertain reading. The first F and its bar in the first equality are heavily thickened or rewritten. The current reading is provisionally shown as $\bar{F}_x = F_x$; earlier strokes are not separated.

2) $F_x \cap \bar{F}_x = 0$. (Kahler polarization).

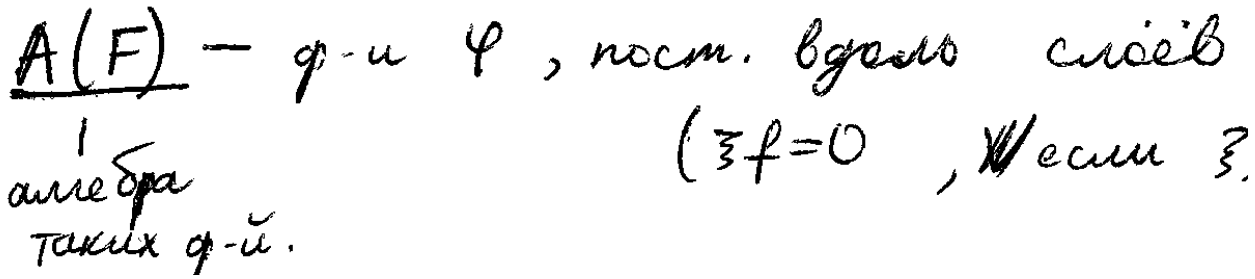
$$\dim(F_x + \bar{F}_x) = k \quad (= \text{const}) \quad (\text{and an integr. bundle}).$$

There is a mixed-mfd. structure there (a bundle with a real base and complex fibres; transitions — holom. transformations of the fibres).

GELFAND-P000364

[source number: 24]

F — a distrib., $\Gamma(L,F)$ — sections constant along the leaves, i.e., $\nabla_\xi s = 0$ if $\xi_x \in F_x$. Now consider the action of opers. on $\Gamma(L,F)$.



Original source details: GELFAND-P000364

Diagram and typed description.

$A(F)$ — fns. φ constant along the leaves.

|

algebra of
such fns.

($\xi f = 0, \forall$



Uncertain reading. The crossed sign is provisionally shown as $\forall$; overlapping diagonal strokes cross it.

Original source details: GELFAND-P000364

if $\xi_x \in F_x$). $\varphi \in A(F) \Rightarrow \mathcal{D}_\varphi$ preserves the space of sections ($\mathcal{D}_\varphi$ acts on $\Gamma(L,F)$ as multiplication by φ). $A(F)$ — a max. commutative subalg. with the Poisson bracket (locally). Now consider operators (also preserving $\Gamma(L,F)$)

$$A_1(F) = \text{normalizer of } A(F) \text{ in the Poisson-bracket alg.} =$$
$$= \{\varphi : \{\varphi, A(F)\} \subset A(F)\}.$$

Thm. Operators $\mathcal{D}_\varphi$ with $\varphi \in A_1(F)$ preserve $\Gamma(L,F)$ and act there as 1st-order opers. This is already quantization, and the fns. — quantizing. (Other quant. are already nonunique). There are many of them.



Uncertain reading. The current word “ikh” (“of them”) is readable, but its first letter is rewritten. Earlier strokes are not separated.

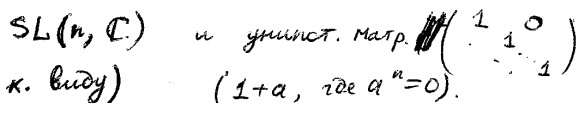
Original source details: GELFAND-P000364

p, q; $dq = 0$ — subspace, fns. — of q only. Opers. — fns. of q — multiplication opers.,
a linear fn. of p — a 1st-order oper.

Next time: Kirillov.

GELFAND-P000365

13/IX-71. I. M. Gelfand on the work of Springer (comput. char. for semisimple groups over a finite field). 1. Consider the group $SL(n, \mathbb{C})$ and unipot. matrices. [?]

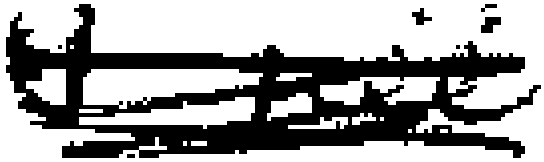


Original source details: GELFAND-P000365

Uncertain reading. A separate dark overwritten sign precedes the first matrix. Neither U nor an equals sign has been established securely.

$$\begin{pmatrix} 1 & & & 0 \\ & 1 & & \\ & & \ddots & \\ \dots & & & 1 \end{pmatrix}$$

~~U~~[?]



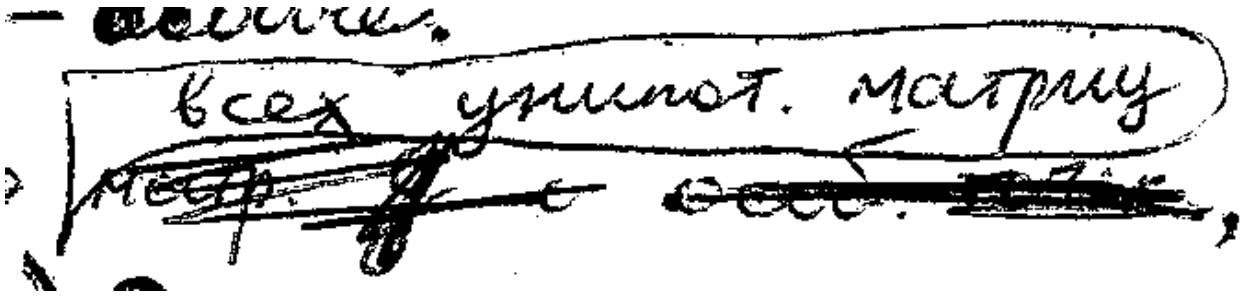
Original source details: GELFAND-P000365

Undeciphered passage. The beginning of the short cancelled entry is provisionally shown as a U-like sign. The remaining signs under horizontal strokes have not been read in full.

(reduced to this form). $(1 + a,$ where $a^n = 0)$. At 1 — a sing. point.

$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $ad - bc = 1$, unipot. $a + d = 2$, in $\mathbb{R}^3$ a cone, 1 — a singularity.

Nonsingular — one Jordan block. Several Jordan blocks — **singular**. Answer: R — the set

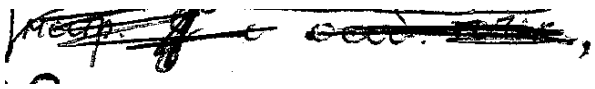


Original source details: GELFAND-P000365

Undeciphered passage. Beneath the outlined insertion “all unipot. matrices”, the earlier phrase is cancelled. Its remainder between and after the provisionally legible fragments has not been recovered.

all unipot. matrices

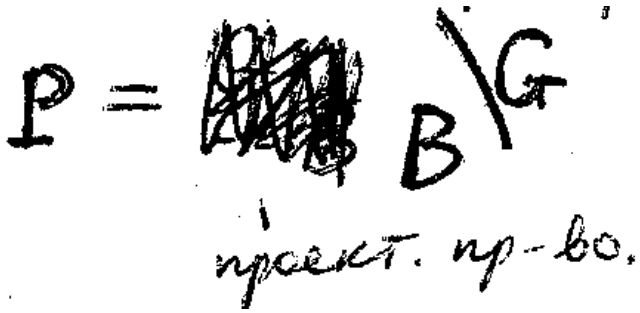
~~matr.[?] [?] sing.[?] [?],~~



Original source details: GELFAND-P000365

Uncertain reading. The contractions “matr.” and “sing.” are provisionally rendered in the cancelled line. Endings and intervening writing are covered by strokes; the contractions are not expanded.

$g \in R$.



Original source details: GELFAND-P000365

Uncertain reading. Between $P =$ and $B\backslash G$ a sign is heavily cancelled, provisionally P. The exact earlier sign has not been recovered. Beneath it is a short vertical stroke and the caption “project. space.”.

$P =$ ~~P~~[?] $B\backslash G$

project. space.

B — triangular matrices

$$\begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ \dots & & \lambda_n \end{pmatrix}.$$

Consider pairs (p, g) with conditions

$$pgp^{-1} \in U = \left\{ \begin{pmatrix} 1 & & 0 \\ & 1 & \\ & & \ddots \\ \dots & & 1 \end{pmatrix} \right\}, \qquad p \in P.$$

(This does not depend on the chosen representative elem. from p.) $\bar{R}$ — the set of pairs.

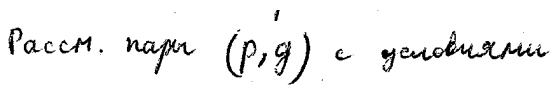
$$(p, g) \longrightarrow g.$$

This is $\bar{R} \to R$, a resolution of singularities (all). Over a point in gen. position one element. $\bar{R}$ — a nonsingular variety. (This is for an arbitrary reductive group over $\forall$ field.)

Uncertain reading. The Russian word “nad” (“over”) is readable, but its last letter is heavily rewritten, with a separate arc above it. Earlier strokes are not separated.



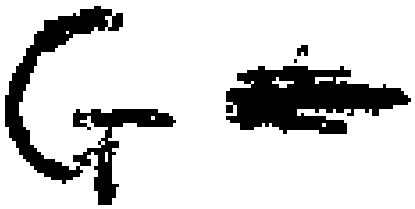
Original source details: GELFAND-P000365



Original source details: GELFAND-P000365

Manuscript notation. The words before the conditions on the pair read “with conditions.”

2. Geom. interpretation of Hall polynomials. Consider G [?]

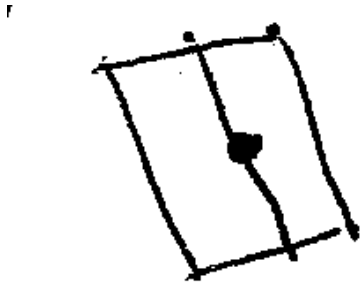


Original source details: GELFAND-P000365

Undeciphered passage. After G is a short overwritten and cancelled group of strokes. An equals sign remains a candidate; no secure earlier reading has been established.

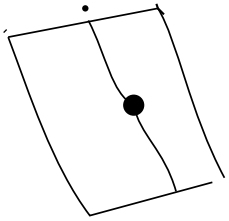
— the group of matrices, B — the group of triangular matrices.

$$G = GL(n, F_q), \qquad B\backslash G \text{ — flags.}$$



Original source details: GELFAND-P000365

Diagram and typed description. A sloping four-sided outline has a roughly middle bent line with a large dot on it. The small detached dot above its upper middle join, short trace at upper left and thickened upper right join are encoded without assigning additional nodes or mathematical meaning.



The number of flags — a polynomial in q, whose degree depends on n.

GELFAND-P000366

[source number: 25]

A coset class is determined by a partition $n = n_1 + \cdots + n_k$, $u \in U$ — an elem. of this class.

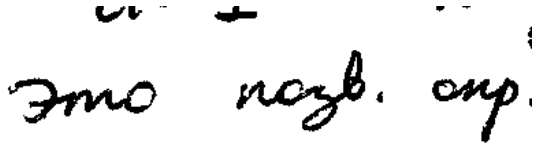
$$|P^u| = P_{n_1,\ldots,n_k}(q) \text{ — a Hall polynomial}$$

(the number of fixed points of this element). $u = 1$ — obtain the Poincaré polynomial. This allows one to determine the Poincaré polynomial for any split group.



Original source details: GELFAND-P000366

Manuscript notation. The opening reads “coset class.” The Russian wording is abbreviated.



Original source details: GELFAND-P000366

Manuscript notation. The abbreviation after “This” reads “pozv.” (“allows”).

J. A. Green, Trans. A. M. S., 1955 — recommended.

There are bounds on the dim. of the variety lying over a point here.

$$\pi : \bar{R} \longrightarrow R,$$

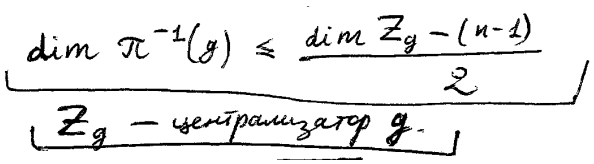


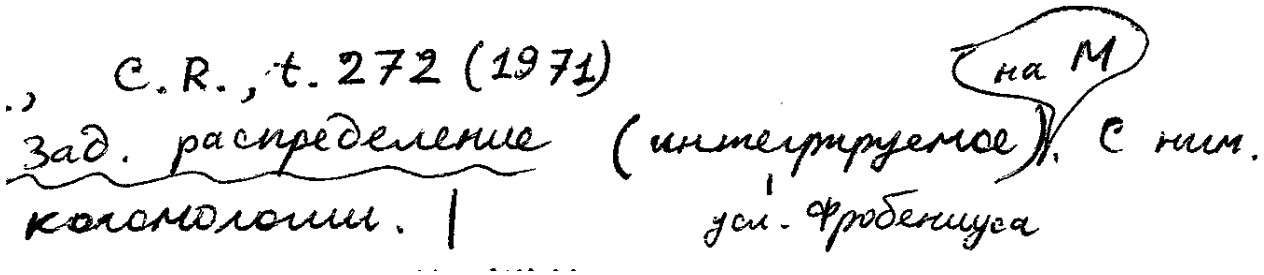
Diagram and typed description.

$\dim \pi^{-1}(g) \leq \dfrac{\dim Z_g - (n-1)}{2}$

Z_g — the centralizer of g.

Original source details: GELFAND-P000366

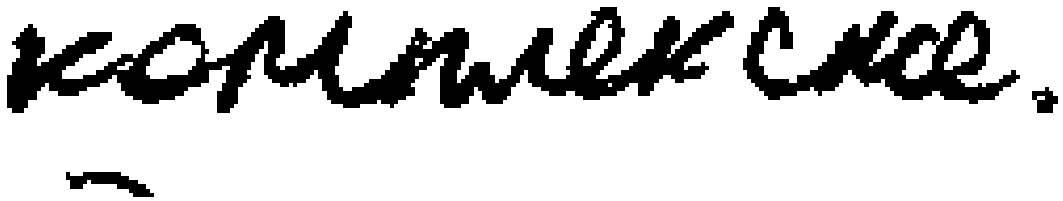
B. Rosenfeld. On notes by Rego, C. R., t. 272 (1971). ~~In the tang. space.~~



Original source details: GELFAND-P000366

Diagram and typed description.

Given a distribution (integrable) on M with it Frobenius cond. are assoc. certain cohomologies.



Original source details: GELFAND-P000366

Diagram and typed description.

Every diff. of the graded ring complex. diff. forms on M has the form

$$d_{L_1} + i_{L_2}$$

(i_{L_2} — interior multiplication by a vector-form).

– – – – – – – – – – – – – – – – – –

$\omega(X_1, \ldots, X_q)$ — a scalar form.
$L(X_1, \ldots, X_p)$ — a form (vector-valued).
$(i_L\omega)(X_1, \ldots, X_{p+q-1})$ substitute and antisymmetrize.
$d_L\omega = (d \circ i_L + (-1)^p i_L \circ d)\omega.$

L — a form of degree 0 (vector field) — d_L — the Lie derivative. ~~$d_L^2 = 0$~~[?]



Uncertain reading. Before "integrability" a short formula is cancelled. $d_L^2 = 0$ is provisionally shown; the index and exponent on d are hidden by overlapping strokes and not established securely.

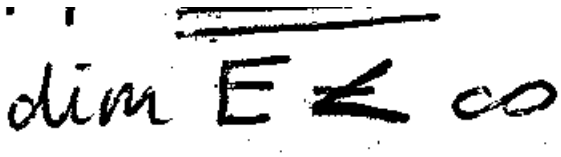
Original source details: GELFAND-P000366

Integrability. (square $= 0$)

GELFAND-P000367

Prof. Pták.

$$\dim E \leq \infty, \qquad E \text{ with a norm.}$$



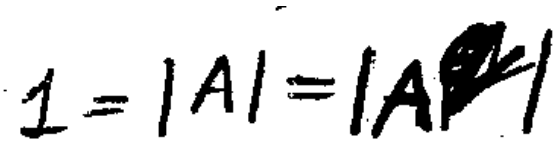
Uncertain reading. Between dim E and ∞ several oblique and horizontal strokes overlap. $\leq$ is provisionally shown; $<$ and $\neq$ remain candidates.

Original source details: GELFAND-P000367

q — an exponent (crit.) if

$$1^\circ. \quad A \in L(E), \quad 1 = |A| = |A^q| \Rightarrow |A|_\sigma = 1$$

(sp. radius)



Uncertain reading. The exponent on the second A is heavily thickened or rewritten. q is shown provisionally; the earlier exponent is unestablished.

Original source details: GELFAND-P000367

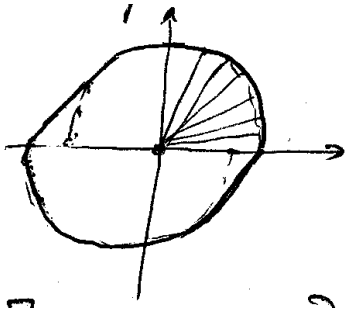
2°. $\exists B$, $\quad 1 = |B| = |B^{q-1}|$, $\quad$ but $|B|_\sigma < 1$.

$\ell^\infty \quad |x| = \max |x_i|,$

$\ell_1 \quad \sum |x_i|, \qquad q(n) = n^2 - n + 1.$

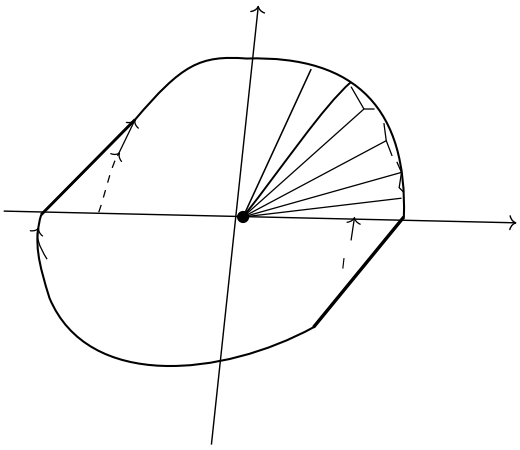
$\ell_2 \qquad q(n) = n.$

There exists a 2-dim. space in which $q = \infty$. (The norm gives poor estimates of convergence; eigenvalues are needed).



Original source details: GELFAND-P000367

Diagram and typed description.



A — rotation by a small angle (Danzer's example). ℓ_p — unknown whether it exists, except for $p = 2^m$. A useful problem for computational mathematics.

$M(n) =$ min over all ban. $\leq n$.

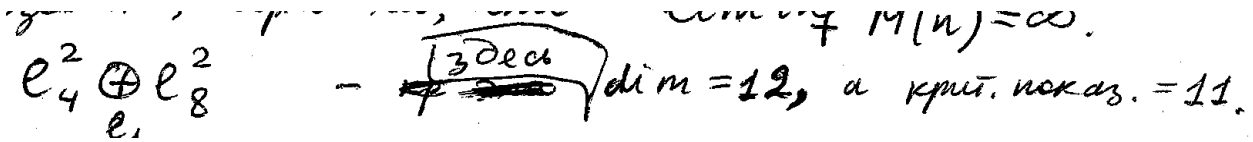


Original source details: GELFAND-P000367

Uncertain reading. The abbreviation after "over all" is provisionally shown as "ban.". Its middle letters are heavily thickened and rewritten; earlier strokes are not separated. The abbreviation is not expanded.

$\exists$ those for which $< n$! There exists a seq. $M(n_k) \sim \sqrt{n_k}$. It is unknown whether $\liminf M(n) = \infty$ is true.

$$\ell_4^2 \underset{\ell_1}{\oplus} \ell_8^2$$



Original source details: GELFAND-P000367

Uncertain reading. Above the cancelled earlier word is the legible outlined insertion "here". The earlier "eto" ("this") is shown provisionally; the covered letters do not establish its complete reading.

— here ~~this~~[?] dim $= 12$, and crit. exponent $= 11$.

$C(E, r, \rho) = \max\{|T^r|,\ |T| \leq 1,\ |T|_\sigma \leq \rho\}.$



Original source details: GELFAND-P000367

Diagram and typed description.

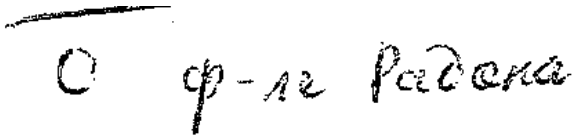
$0 < \rho < 1.$

GELFAND-P000368

[source number: 26]

I. M. Gelfand.

On the Radon formula over a finite field.



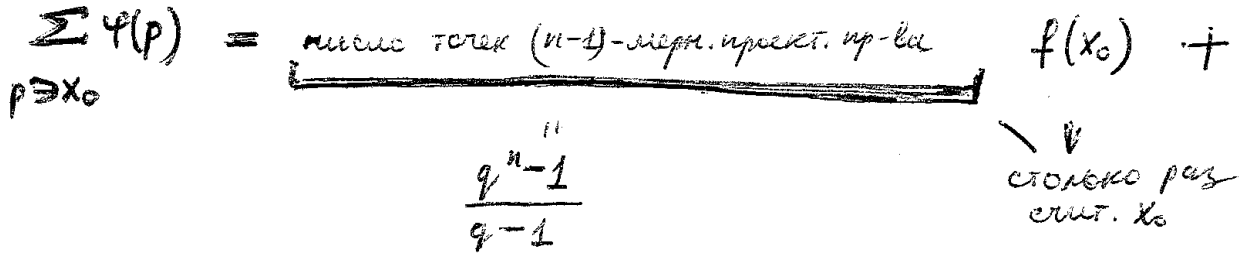
Uncertain reading. The initial glyph has a gap on the right. It is provisionally read as Russian О; its shape also allows C.

Original source details: GELFAND-P000368

Over F_q an n-dim. space. $(x_1, \ldots, x_n)$. q^n points.
Integrals over all planes are known.

$$\varphi(p) = \int_{x \in p} f(x) \quad \text{(sum)}.$$

p — a plane.

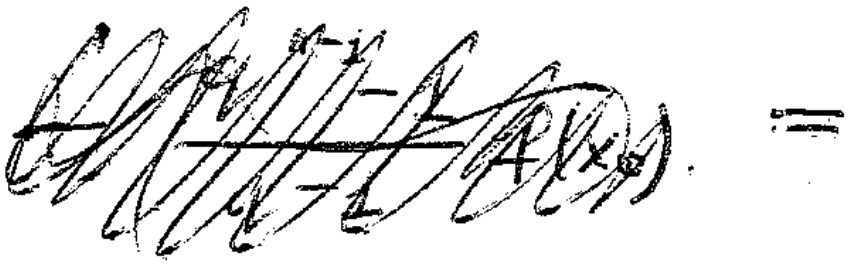


Original source details: GELFAND-P000368

Diagram and typed description.

$$\sum_{p \ni x_0} \varphi(p) = \underbrace{\text{number of points of } (n-1)\text{-dim. project. space}}_{\frac{q^n - 1}{q - 1}} \; f(x_0) + \; \downarrow \text{this many times counted } x_0$$

$$+\frac{q^{n-1}-1}{q-1}\sum_{x\neq x_0} f(x).$$

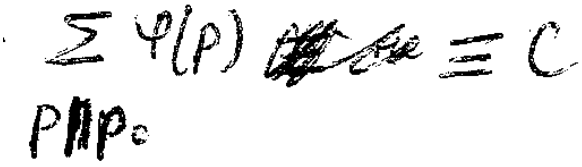


Original source details: GELFAND-P000368

Uncertain reading. In the cancelled term the angular sign before (x_0) is provisionally shown as A. f remains a candidate; the crossing cancellation strokes prevent a unique reading. The entire term is cancelled; an equals sign follows it.

~~$+\frac{q^{n-1}-1}{q-1}A(x_0)$ [?] $=$~~

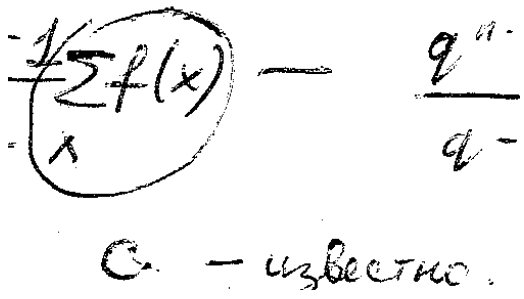
$$\sum_x f(x) = \sum_{p\|p_0} \varphi(p) \quad \text{[?]} \quad \equiv C.$$



Undeciphered passage. Between the sum over $p \parallel p_0$ and $\equiv C$ several signs are cancelled. Their reading is unestablished.

Original source details: GELFAND-P000368

(does not depend on p_0) — a cond. on $\varphi(p)$ — an analogue of the Paley–Wiener theorem.



Original source details: GELFAND-P000368

Diagram and typed description. The sum $\sum_x f(x)$ is circled; beneath it "C — known." is underlined. The lower index has overlapping strokes and a small loop; x is retained provisionally.

$$= \frac{q^n-1}{q-1}f(x_0) + \frac{q^{n-1}-1}{q-1}\left(\sum_x f(x)\right) - \frac{q^{n-1}-1}{q-1}f(x_0).$$

C — known.

Rota — as $q \to 1$ one obtains a passage to simple theorems. $C_k^n(q)$ — the number of k-dim.

subspaces of an n-dim. space.

$$\lim_{q\to 1} C_k^n(q) = C_k^n \text{ — combinations.}$$

GELFAND-P000369

Kirillov.

Consider an arbitrary generalized function a on a manifold M (e.g. $\mathbb{R}^n$). For it a wave front is defined

$$WF(A) \subset T^*M.$$

A smooth function — zero wave front (empty). Let $\xi \in T^*M$.

$$\xi \overline{\in} WF(A), \quad \text{if}$$
$$\langle A, \varphi e^{it\psi}\rangle = o(t^{-n}), \quad \forall n \quad \text{as } t \to +\infty.$$

Manuscript notation. The first line uses lowercase a.



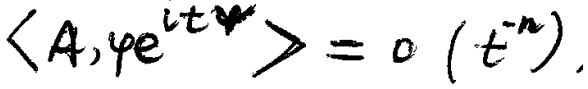
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Manuscript notation. A separate horizontal stroke appears above $\in$.



Original source details: GELFAND-P000369

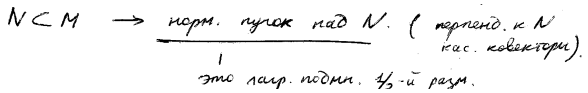
Uncertain reading. The phase sign after it is heavily thickened or rewritten. ψ is shown provisionally; the earlier strokes are not separated.



Original source details: GELFAND-P000369

(here $d\psi = \xi$ at the point π_ξ and $\varphi \in C_0^\infty$ is arbitrary).

It is convenient to study functions for which the wave front $\in$ a Lagr. manifold.



Original source details: GELFAND-P000369

Diagram and typed description.

$N \subset M \longrightarrow$ norm. bundle over N

|

(tangent covectors perpend. to N).

This is a Lagr. subman. of $1/2$ dimension.

~~[?]~~ P. D. O. — a generalized function on $M \times M$ (kernel) with a wave front constructed in this way from the diagonal.



Original source details: GELFAND-P000369

Uncertain reading. Before “P. D. O.” a short word is heavily cancelled. “Napr.” (“e.g.”) remains a candidate; the complete reading is unestablished.

~~$A(p,q) =$~~

$$\langle A, [?]u \rangle = \int e^{i\psi(x,\xi)} a(x,\xi) u(x)\, dx\, d\xi.$$



Uncertain reading. A comma is visible after A, followed by a separate thickened vertical mark before u. The mark is undeciphered. The right angle bracket is retraced.

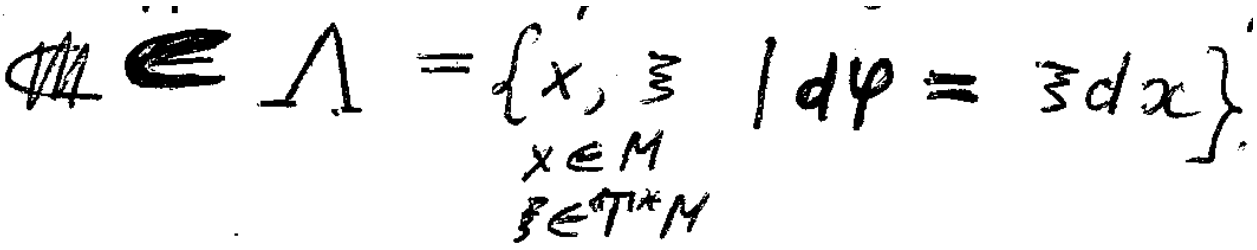
Original source details: GELFAND-P000369

(Value of A on the test function u).
Every A ~~[?]~~ with wave front defined this way is given by this formula.



Undeciphered passage. After “Every A” a separate sign is heavily cancelled. Its reading is unestablished.

Original source details: GELFAND-P000369



Original source details: GELFAND-P000369

Uncertain reading. After “Wave front” a rounded sign is cancelled. φ is shown provisionally; another reading is not excluded. The following $\in \Lambda$ is retained separately from the cancellation.

$$\text{Wave front } \varphi[?] \in \Lambda = \left\{ \begin{matrix} x, \xi \\ x \in M \\ \xi \in T^* M \end{matrix} \;\middle|\; d\psi = \xi\, dx \right\}.$$

GELFAND-P000370

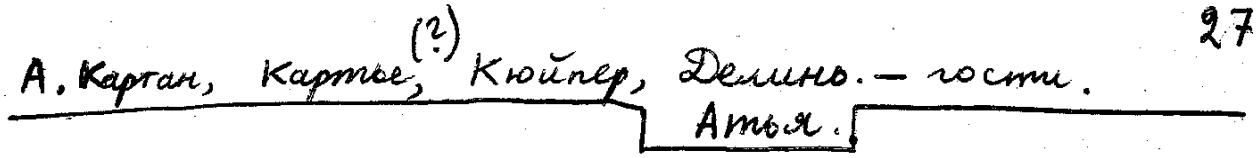
[source number: 27]

20/IX

Uncertain reading. To the right of 20/IX is a separate short vertical 1-like stroke. Its meaning is unestablished.



Original source details: GELFAND-P000370



Original source details: GELFAND-P000370

Diagram and typed description.

A. Cartan, Cartier (?), Kuiper, Deligne. — guests.

Atiyah.

I. Bernstein. on work of Godbillon, Vey. Comptes rendus. M — a compact manifold. On M is given a form ω (a foliation S of codim. 1), which is integrable.

$$d\omega = \omega \wedge \omega_1.$$

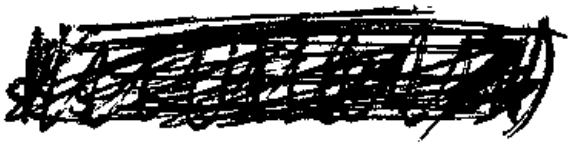
— ~~[?]~~



Undeciphered passage. After the formula an earlier word is heavily cancelled; its reading is unestablished.

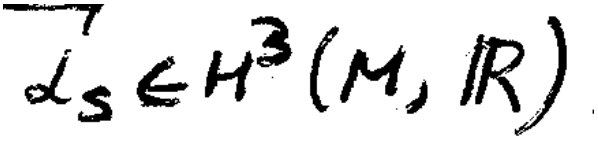
Original source details: GELFAND-P000370

~~[?]~~ To it is associated $\alpha_S \in H^3(M, \mathbb{R})$, it



Undeciphered passage. Before “To it is associated” an entire phrase is heavily cancelled; its reading is unestablished.

Original source details: GELFAND-P000370



Manuscript notation. The subscript S repeats the foliation label used above.

Original source details: GELFAND-P000370

~~[?]~~ $= 0$ if this is a fibration and is stable under deformations,



Original source details: GELFAND-P000370

Undeciphered passage. After “it” and before $= 0$ a short word is heavily cancelled. No confident reading of its letters is established.

~~[?]~~ $\neq 0$.

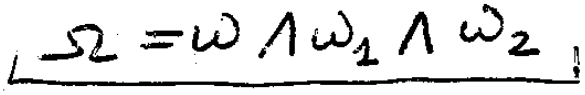


Original source details: GELFAND-P000370

Undeciphered passage. Between “under deformations” and $\neq 0$ a short word is crossed out; its reading is unestablished.

Construction: there exist forms ω_1, ω_2:

$$d\omega = \omega \wedge \omega_1;$$
$$d\omega_1 = \omega \wedge \omega_2.$$



Original source details: GELFAND-P000370

Diagram and typed description.

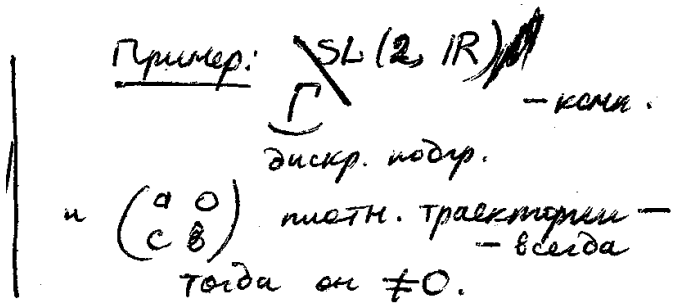
Consider $\Omega = \omega \wedge \omega_1 \wedge \omega_2$

— closed, and the cohom. class does not depend on the choice of ω_1 and ω_2. There is a connection with the cohomology of algebras.



Original source details: GELFAND-P000370

Uncertain reading. After $SL(2, R)$ is a reinforced M-like sign. M is shown provisionally; the overlapping strokes are unestablished.



Original source details: GELFAND-P000370

Diagram and typed description.

Example: $SL(2,R)\,M$[?]

Γ — compact.

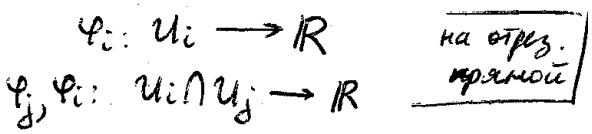
discr. subgr.

and $\begin{pmatrix} a & 0 \\ c & b \end{pmatrix}$ dense trajectories —

— always.

Then it is $\neq 0$.

Foliation:



Original source details: GELFAND-P000370

Diagram and typed description.

$\varphi_i : U_i \longrightarrow \mathbb{R}$, $\varphi_j\varphi_i : U_i \cap U_j \longrightarrow \mathbb{R}$. — on an interval of the line

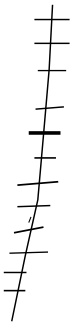
Since the leaves are the same, one obtains $\Psi_{ij} : \mathbb{R} \to \mathbb{R}$ (an interval). If $\Psi_{ij} \in$ a group, then a cocycle with coeff. in the group $\longrightarrow$ a sheaf and from it cohomology classes. But this is a pseudogroup. But consider a Lie algebra isom. to W_1 — formal vector fields on the line, $f(x)\frac{d}{dx}$. They are $= 0$, except there exists $H^3(W_1, \mathbb{R}) = \mathbb{R}$. This is precisely that class.

Atiyah. On the spectrum of a selfadj. elliptic operator. X — a compact manifold, D — an elliptic diff. operator, selfadj. $D^* = D$. Discr. spectrum; eigenvalues are divided into positive and negative. Let $D = D_t$, t — a param.; the spectrum changes with t.

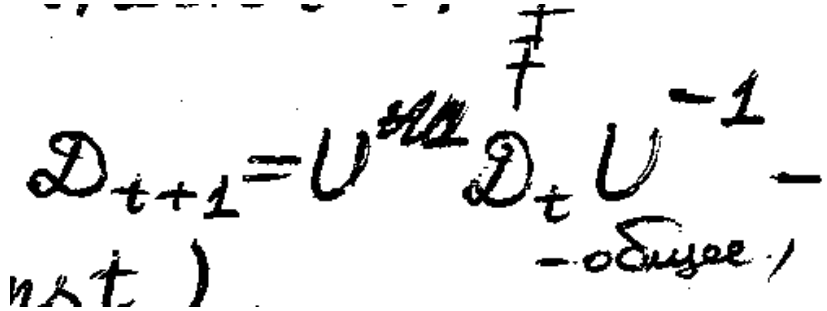


Original source details: GELFAND-P000370

Diagram and typed description. An inclined nearly vertical line with twelve distinguishable transverse strokes of different lengths and short overlapping strokes. One stroke in the middle is thickened. Exact intervals and separation of all overlapping strokes are unestablished; there are no numerical labels.



Let $D_{t+1} = D_t$ — periodicity, or



Original source details: GELFAND-P000370

Uncertain reading. The first U stands on the main line; a small cancelled earlier sign is above and to its right. “na” remains a candidate; the exact reading is unestablished. The order of insertion is undetermined.

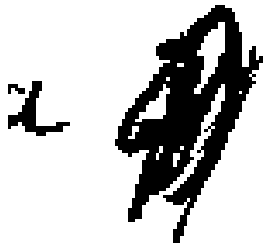
$$D_{t+1} = U \overset{\text{[?]}}{} D_t U^{-1}$$

— general case,

($U = \text{const}$).

$$\operatorname{Spec} D_{t+1} = \operatorname{Spec} D_t.$$

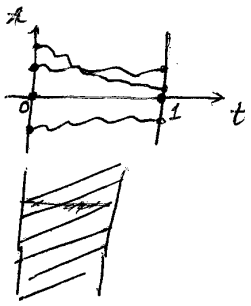
The spectrum moves (eigenvalues may shift by q[?]).



Original source details: GELFAND-P000371

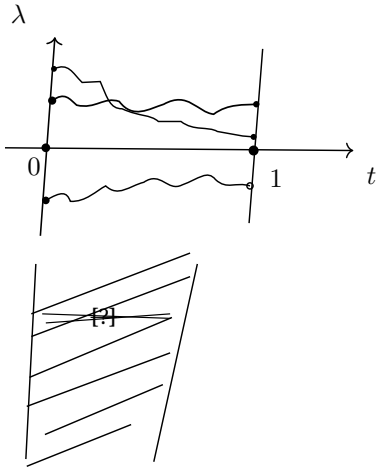
Uncertain reading. After “by” is a reinforced q-like sign. q is shown provisionally; a question mark remains a candidate.

One can define an integer characterizing this shift. (Place an observer at 0 and see what mass has passed to the left and to the right.)



Original source details: GELFAND-P000371

Diagram and typed description. Upper drawing: axes λ, t, marks $0, 1$, two verticals and three wavy curves. Thick dots occur at endpoints and on the axis; the lower right dot has a light centre. Below are six distinguishable inclined segments between nearly vertical lines. Exact separation of overlapping strokes and joins is unestablished.





Original source details: GELFAND-P000371

Undeciphered passage. In the middle of the lower drawing a short caption is cancelled, distinct from the inclined lines. No complete word or formula has been recovered.

Example. Spectral flow $\in \mathbb{Z}$.

$$X \text{ — a circle}, \qquad D_t = -i\frac{d}{dx} + t, \qquad Uf = e^{ix} f.$$

Flow $= \pm 1$ $(n \to n+1)$. This is a top. invariant under deformations of the family. Problem: compute spectral flow from the geom. data of the problem: manifold, operator, U. Connected with the index. Let the operator D be of first order (a system). Consider

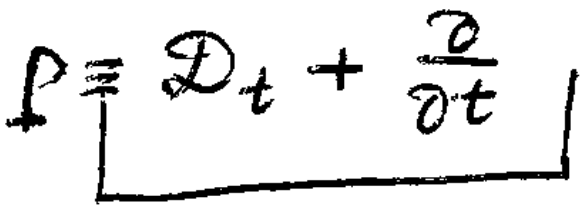
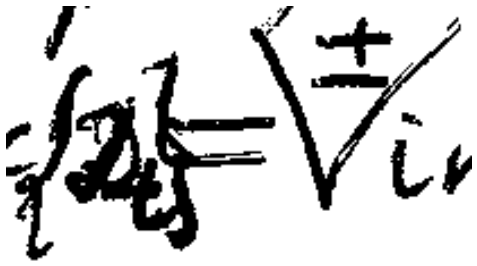


Diagram and typed description.

$$P \equiv D_t + \frac{\partial}{\partial t}$$

Original source details: GELFAND-P000371

on functions $f(x,t)$, $f(x,t+1) = f(x,t)$, $x \in X$, $(U = 1)$. (if $U \neq 1$, adjust $U \otimes 1$). This is an elliptic operator on $X \times S^1$ (or in a bundle, base S^1, fibre X). (D_t — Hermitian, $\frac{\partial}{\partial t}$ — skew-Hermitian). Theorem.



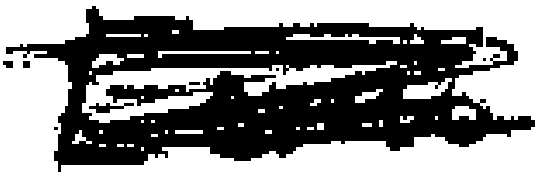
Original source details: GELFAND-P000371

Uncertain reading. Near the right family brace an upper 1-like stroke and a lower 0-like loop join the reinforced brace strokes. Possible limits 1 and 0 are not confidently separated; no definite interval is assigned.

Spectral flow $\{D_t\} = \pm$ index P

$=$ the index formula.

A direct proof is difficult. It is easier to express spectral flow by a formula. Heuristic considerations. Let D_t depend real-analytically on t. [?]



Undeciphered passage. After “real-analytic” a short word is entirely crossed out. Its reading is unestablished.

Original source details: GELFAND-P000371

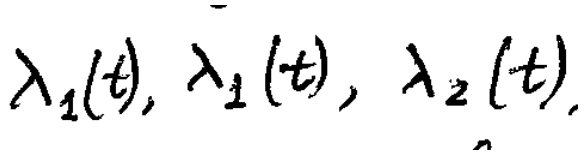
[?]



Uncertain reading. At left an entry is cancelled: an initial t-like form and at right a t_0- or 0-like ending. $t \dots t_0$, $t = t_0$, $t = 0$, $t \dots 0$ remain candidates; the relation sign is unestablished.

Original source details: GELFAND-P000371

$$\lambda_1(t),\ \lambda_1(t),\ \lambda_2(t), \dots$$



Original source details: GELFAND-P000371

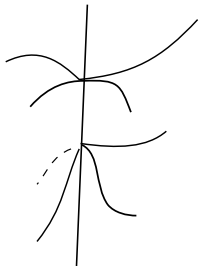
Manuscript notation. The first two subscripts both read 1.

— analytic functions!, defined for all t. (Roots of char. polynomials; real $\Rightarrow$ no branching).



Original source details: GELFAND-P000371

Diagram and typed description. A nearly vertical line and two groups of branches at different levels: upper crossing curves, below a right rising branch and a downward-bent branch, and oblique strokes at left. Exact joins of thick strokes and endpoints are unestablished.



GELFAND-P000372

[source number: 28]

Periodicity $\Rightarrow$ a permutation σ of them.

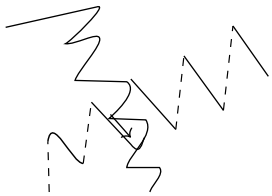
$$-i\frac{d}{dx} + t, \qquad \sigma(n) = n + 1.$$

This σ is not arbitrary. Orbits of σ — 4 classes are possible. 1) finite — an eigenvalue that returns. 2) bounded on one side (from $-\infty$ to $-\infty$ or from $+\infty$ to $+\infty$). 3) unbounded on either side.



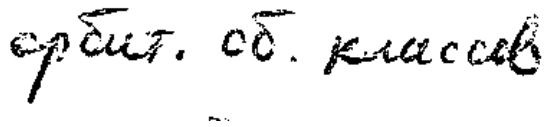
Original source details: GELFAND-P000372

Diagram and typed description. To the right of case3 is a descending zigzag and several wavy or angular strokes with solid and interrupted parts; in the middle is a downward arrow-like stroke. Exact joins and geometric meaning are unestablished.



$(|\lambda'(t)| < C|\lambda(t)|.)$

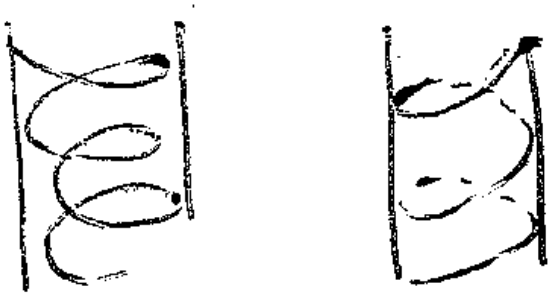
Increasing and decreasing — the numbers of orbits of ob.[?] classes p and q are finite.



Original source details: GELFAND-P000372

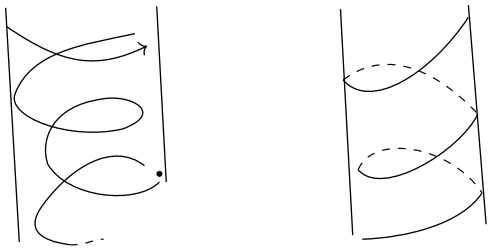
Uncertain reading. The abbreviation after “orbits” is provisionally shown as “ob.”. The first letter has a gap; “sb.” remains a candidate. The abbreviation is not expanded.

Functions $\underline{\underline{\lambda(t)}}$ on the cylinder. 3) — spirals go upward or downward.

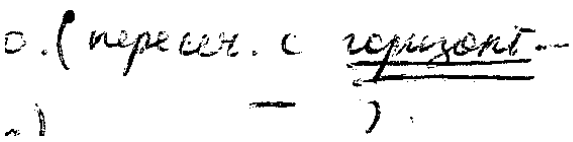


Original source details: GELFAND-P000372

Diagram and typed description. Two drawings between verticals: at left overlapping loops, solid and interrupted parts, an upper rightward arrow and a lower dot at right. At right are several arcs with solid and interrupted parts. Additional directions and hidden joins are unestablished.

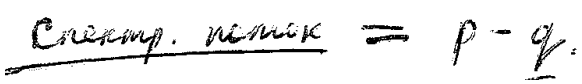


The number of orbits of class3) is finite. (intersection with a horizontal —) Spectral flow $= p - q$. (obvious from this).



Original source details: GELFAND-P000372

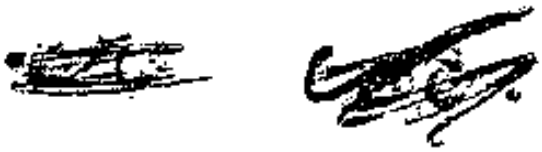
Manuscript notation. A separate horizontal stroke is drawn below the end of the line, before the closing parenthesis.



Original source details: GELFAND-P000372

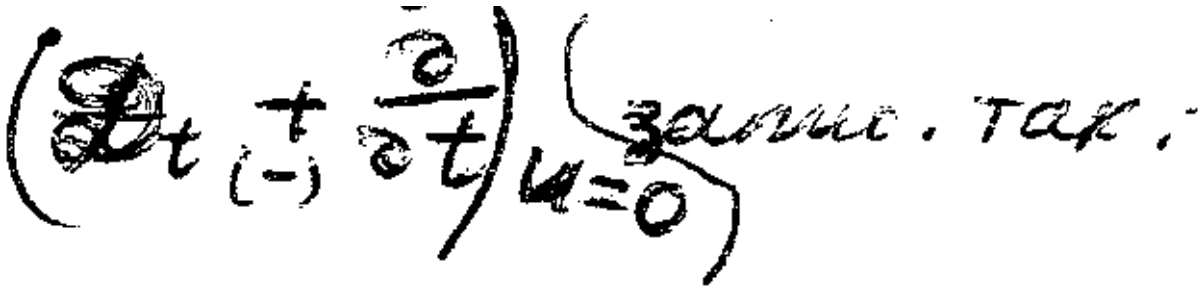
Manuscript notation. The two strokes before $p - q$ form an equals sign.

Assumption: let the eigenvectors be independent of t. Then ~~[?]~~ ~~[?]~~ exact solution



Original source details: GELFAND-P000372

Undeciphered passage. After “Then” and before “exact solution” two short words are crossed out. Neither reading is established.



Original source details: GELFAND-P000372

Uncertain reading. The first operator before the plus is reinforced over an earlier sign. D_t is shown provisionally; earlier strokes with a possible ∂ are not separated. Below the plus is a separate $(-)$; this insertion does not replace the main plus.

$\left(D_t[?] + \underset{(-)}{\frac{\partial}{\partial t}}\right) u = 0$ write it thus:

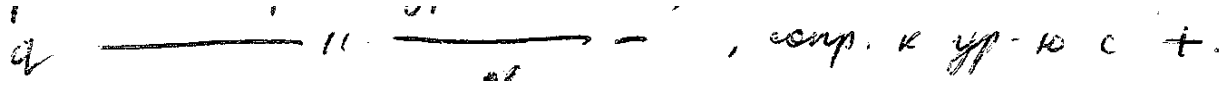
$$u_n = C_n \exp\left(-\int \lambda_n(t)\,dt\right), \qquad u = \sum u_n e_n, \quad e_n \text{ — an eigenvector.}$$

$$\lambda_n(t+1) = \lambda_{\sigma(n)}(t).$$

$$\sigma(n) \longrightarrow \infty \iff \text{finite } n \text{ and large } t.$$

$$u_n(t) = C_n \exp\left(-\int_0^t \lambda_n(s)\,ds\right).$$

p — the number of solutions of the equation with $+$,



Original source details: GELFAND-P000372

Diagram and typed description.

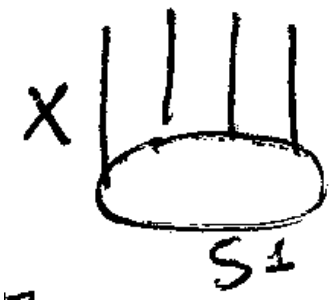
q ——————— // ——————→ $-$, adjoint to the equation with $+$.

Finite orbits $\int_0^N \lambda_n(t)\,dt = 0$, N — the period.

Index $= p - q$. In the general case (vectors move), the equation $D_t + \frac{\partial}{\partial t}$ contains an additional term expressing the motion of eigenvectors. A direct analytic proof cannot be given.

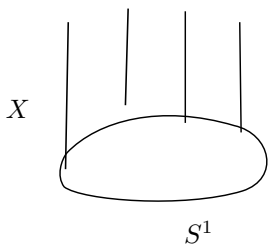
GELFAND-P000373

Examples. ①. $T : X \longrightarrow X$ — a diffeom. Define a bundle.



Original source details: GELFAND-P000373

Diagram and typed description.



Let there be a functorial situation (X, g, D), g — a Riem. metric, D — selfadjoint. (The case of a spectrum unbounded on both sides is interesting). One then obtains:

$$X_t, g_t, D_t$$



Original source details: GELFAND-P000373

Uncertain reading. After "when" a separate sign is cancelled; A remains a candidate. Above it "this" is readable. The earlier sign is not confidently recovered.

this
(E.g.: $d + \delta$, when ~~[?]~~ is selfadj.).

Spectral flow does not depend on the choice of Riem. metric. If T is an isometry, then $D_t = \text{const}$ and spectral flow $= 0$. The invariant is an obstruction to the existence of a metric preserved by T. ②. $X \times S^1$, the operator $(d + \delta)_t$ on forms. Twisted using this map. Suppose given

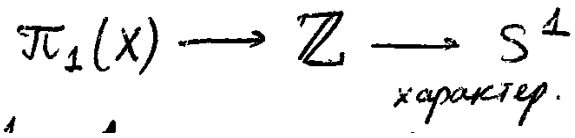


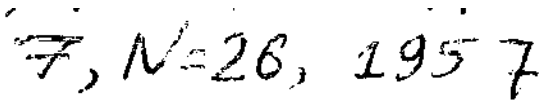
Diagram and typed description.

$\pi_1(X) \longrightarrow \mathbb{Z} \longrightarrow S^1$
character.

Original source details: GELFAND-P000373

$$X^{4k+1} \times S^1$$

— a Novikov invariant using the Hirzebruch L-polynomial. Deligne, in Manin's translation. On the theory of motives. Different cohomology theories of alg. varieties. 25/X̄-71 | S. Gelfand. Schubert cells. Horrocks, Proc. Lond. Math. Soc., 7, $N = 26$, 1957.



Manuscript notation. After 7 the manuscript reads $N = 26$.

Original source details: GELFAND-P000373

$G(N,n)$ — the complex Grassm.

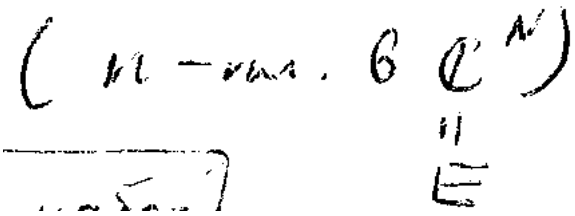
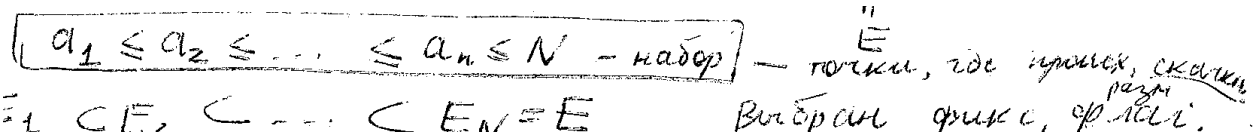


Diagram and typed description.

(n-planes in $\mathbb{C}^N$)
||
E

Original source details: GELFAND-P000373



Original source details: GELFAND-P000373

Diagram and typed description. The tuple is framed. At right a line leads to "points where jumps occur[?]"; below is the insertion "dim." joined by an oblique stroke to the end of the caption. Its exact attachment to the lower line is unestablished.

$a_1 \le a_2 \le \cdots \le a_n \le N$ — tuple — — points where jumps occur[?]

dim.

A fixed flag is chosen.

$$E_1 \subset E_2 \subset \cdots \subset E_N = E,$$
$$\dim E_i = i.$$
$$F \in \Omega(a_1, \ldots, a_n), \quad \text{[?]}$$



Undeciphered passage. After $F \in \Omega(a_1, \ldots, a_n)$ an entire word is cancelled by several thick horizontal strokes. Its reading is unestablished.

Original source details: GELFAND-P000373

$$\Updownarrow$$
$$\dim(F \cap E_{a_i+i}) = i.$$

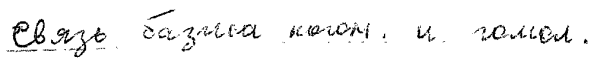
$$\varphi(i) = \dim(F \cap E_i).$$

$\underline{\underline{a_i + i}}$ — the growth points of this function.

GELFAND-P000374

[source number: 29]

This is a cell decomposition of the Grassm. manifold. Problem. Find all invariant diff. forms (Chern classes). Connection between the basis of cohom. and homol.



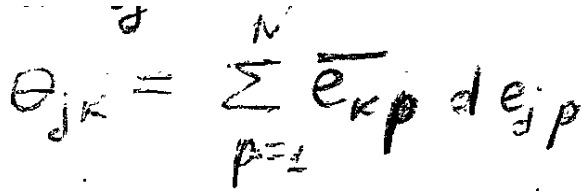
Manuscript notation. The last abbreviation reads “homol.”

Original source details: GELFAND-P000374

$$(*) \quad U(N)/(U(n)\times U(N-n)) = G(N,n) = G/F.$$

The form can be lifted to $U(N)$. Construction of forms: de_{ij}. $U=(e_{ij})$, (e_{ij}) — a matrix-element function on $U(N)$.

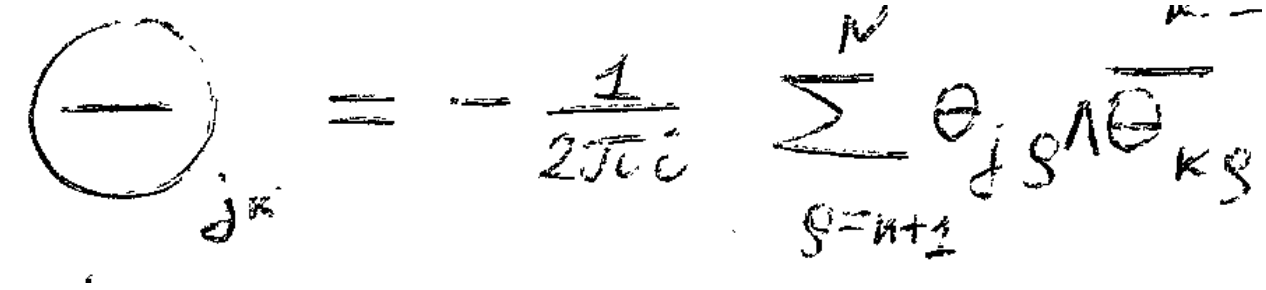
$$\theta_{jk} = \sum_{p=1}^{N} \bar{e}_{kp}\, de_{jp}, \qquad j=1,\dots,n, \quad k=n+1,\dots,N.$$



Uncertain reading. In the index on $\bar{e}$, after k, is a heavily reinforced p-like sign. p is shown provisionally; ρ remains a candidate.

Original source details: GELFAND-P000374

$j,k=1,\dots,n$.

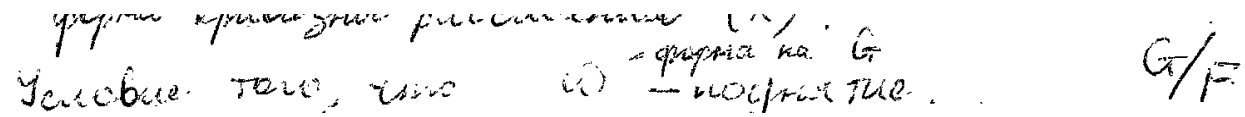


Original source details: GELFAND-P000374

Uncertain reading. The two factors use signs of the same size, smaller than Θ at left. The barred sign is provisionally read as θ; its case is uncertain.

$$\Theta_{jk} = -\frac{1}{2\pi i}\sum_{s=n+1}^{N} \theta_{js}\wedge\bar{\theta}_{ks}$$

curvature form of the bundle $(*)$.



Original source details: GELFAND-P000374

Diagram and typed description. “Form on G” is written above ω, with G/F separately at right.

ω — form on G

The condition that ω — a pullback. G/F

1) $\omega(X_1,\dots,X_k)=0$, $X_i\in T(F)$ — tangent to F. 2) ω — invariant under F. All Θ_{jk} satisfy condition 1). Invariant under $U(N-n)$; under $U(n)$ it transforms by conjugation (adjoint action).



Original source details: GELFAND-P000374

Diagram and typed description.

$$GL(n) \ni (x_{ik})_{i,k=1}^{n}.$$

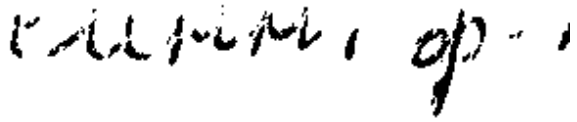
Construct polynomials in the matrix elements with the property

$$p(x) = p(gxg^{-1}).$$

Coefficients of the char. polynomial — elementary symm. functions. A basis of the linear space is needed. Let $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0$ be integers. x — a matrix, $\alpha_1, \ldots, \alpha_n$ — eigenvalues. One must take

GELFAND-P000375

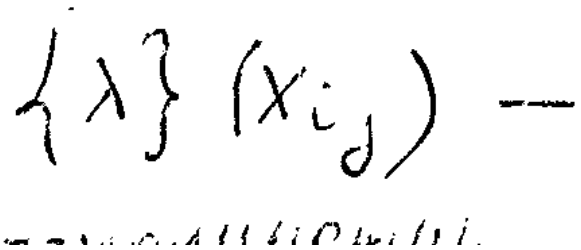
symm. fns.



Original source details: GELFAND-P000375

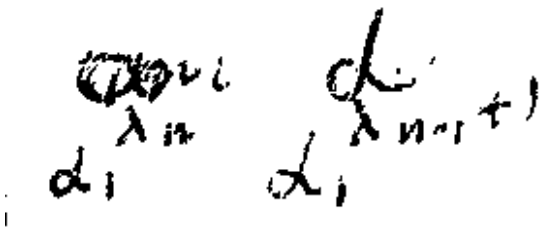
Uncertain reading. The small faint note above the first determinant is provisionally rendered "symm. fns."; a longer final contraction remains possible.

$$\{\lambda\} = \frac{\begin{vmatrix} \alpha_1^{\lambda_n} & \alpha_1^{\lambda_{n-1}+1} & \cdots & \alpha_1^{\lambda_1+n-1} \\ \vdots & \vdots & & \vdots \\ \alpha_n^{\lambda_n} & \alpha_n^{\lambda_{n-1}+1} & \cdots & \alpha_n^{\lambda_1+n-1} \end{vmatrix}}{\begin{vmatrix} 1 & \alpha_1 & \cdots & \alpha_1^{n-1} \\ \vdots & \vdots & & \vdots \\ 1 & \alpha_n & \cdots & \alpha_n^{n-1} \end{vmatrix}} = \{\lambda\}(X_{ij}) \text{ — a polynomial.}$$



Original source details: GELFAND-P000375

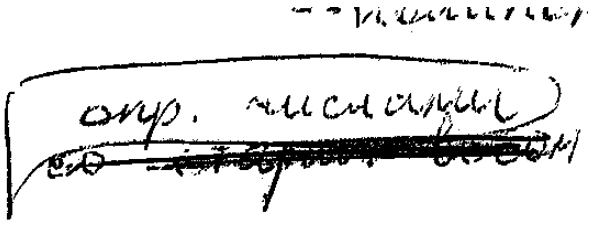
Manuscript notation. Two crossing strokes form the argument X_{ij}.



Original source details: GELFAND-P000375

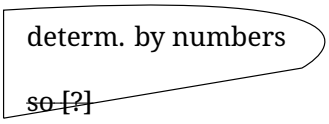
Uncertain reading. Above the first numerator entry is a separate heavily overwritten entry. An α-like form and neighboring short strokes are visible; the earlier index is not securely recovered.

This is the character of a representation.



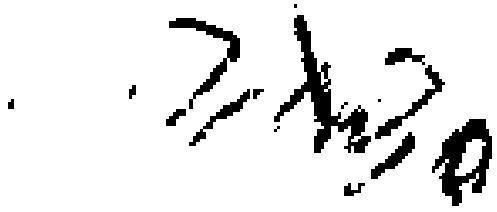
Original source details: GELFAND-P000375

Undeciphered passage. The outlined insertion “determ. by numbers” lies above the cancelled earlier phrase. Its initial Russian “so” and separate final strokes are visible; the words under the dense horizontal strokes are not established in full.



$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0[?].$$

Uncertain reading. The last small oval has an attached lower stroke. 0 is provisionally shown; a remains a candidate.



Original source details: GELFAND-P000375

$\{\lambda, \circleddash\} = \circleddash^{\{\lambda\}}$ — notation.



Uncertain reading. The elevated entry after the enclosed letter contains λ and two curly strokes. Braces $\{\lambda\}$ are provisionally rendered; not all overlapping strokes have been separated.

Original source details: GELFAND-P000375

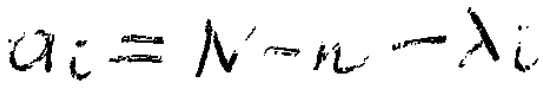
$\circleddash$ — a matrix of diff. forms (they have degree 2 and commute).



Uncertain reading. The circle encloses a short crossbar and two side strokes. Its reading as a letter is uncertain.

Original source details: GELFAND-P000375

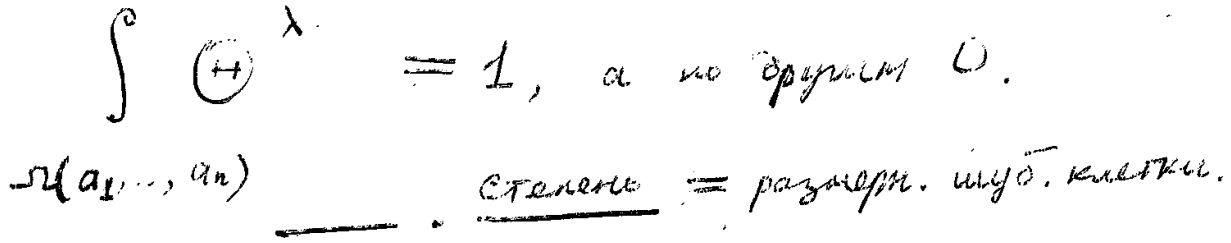
Put $a_i = N' - n - \lambda_i$.



Uncertain reading. Between N and the minus is a separate thin rising stroke above the line. N' is provisionally rendered; the meaning of the stroke is not established.

Original source details: GELFAND-P000375

Then



Original source details: GELFAND-P000375

Diagram and typed description.

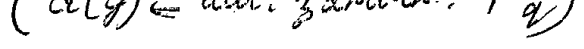
$\int \Theta^{\lambda} = 1$, and on the others 0.

$\Omega(a_1, \ldots, a_n)$ degree = dim. of the Schubert cell.

Problem (I. M. Gelfand)

$$g \in SL(2, F_q).$$

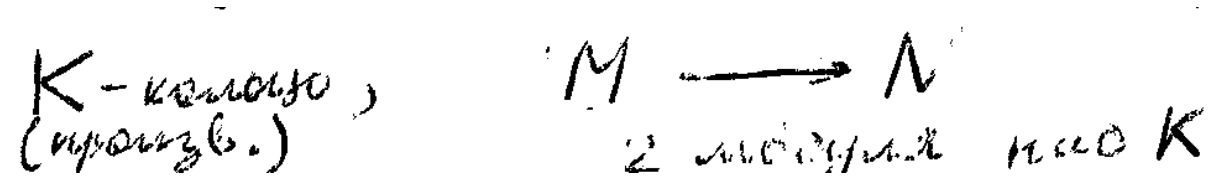
Group ring $\sum_g a(g)g$. ($a(g) \in$ alg. closure F_q).



Original source details: GELFAND-P000375

Uncertain reading. The sign after $a(g)$ is provisionally rendered as $\in$. Its curve and crossbar resemble the sign after g above, in a more compact form.

Describe it. A. Vinogradov. Global aim — to expel analysis from the theory of linear diff. operators.



Original source details: GELFAND-P000375

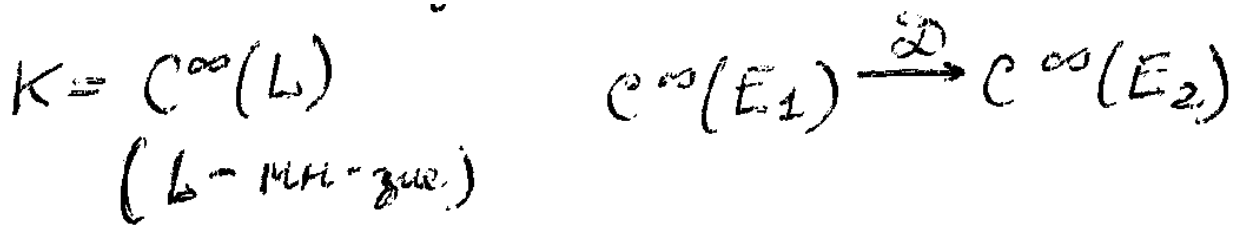
Diagram and typed description.

K — a ring, (arbitr.)

$M \longrightarrow N$

2 modules over K

Classical situation



Original source details: GELFAND-P000375

Diagram and typed description.

$K = C^{\infty}(L)$ (L — manifold)

$C^{\infty}(E_1) \xrightarrow{\mathcal{D}} C^{\infty}(E_2)$

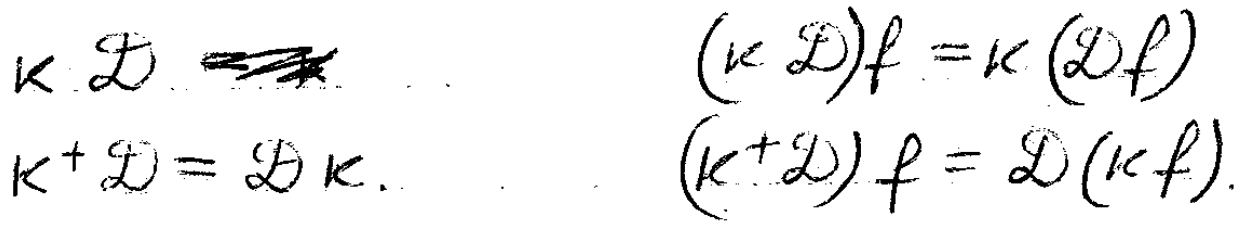
K may be noncommutative.

GELFAND-P000376

[source number: 30]

$$M \xrightarrow{\mathcal{D}} N.$$

$k \in K$,



Original source details: GELFAND-P000376

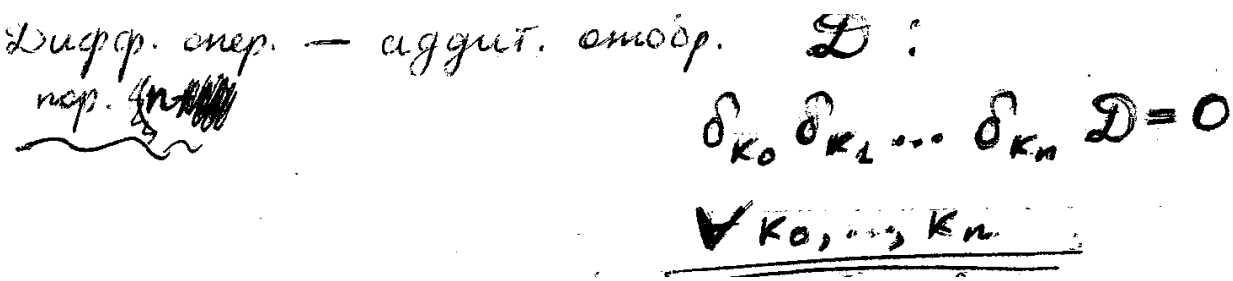
Undeciphered passage. The right-hand side after $k\mathcal{D} =$ is crossed by horizontal and oblique strokes. Its earlier signs have not been read in full.

$k\mathcal{D} =$ [?] $\qquad (k\mathcal{D})f = k(\mathcal{D}f)$

$k^{+}\mathcal{D} = \mathcal{D}k.$ $\qquad (k^{+}\mathcal{D})f = \mathcal{D}(kf).$

$$\underline{\delta_k = k^{+} - k}.$$

Basic definition. A diff. operator — an additive map



Original source details: GELFAND-P000376

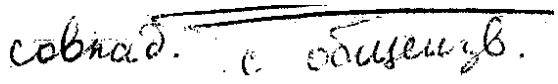
Undeciphered passage. After the visible n an earlier sign is densely cancelled. Curly strokes lie to the left of n, with a wavy line below "ord.". The complete earlier reading is not established.

$\mathcal{D}$:

ord. n [?] $\qquad \delta_{k_0}\delta_{k_1}\cdots\delta_{k_n}\mathcal{D} = 0$

$\underline{\forall k_0, \ldots, k_n}$

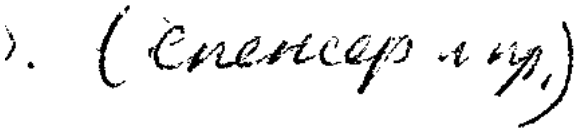
In the classical case this agrees with the customary definition.



Manuscript notation. The last word is abbreviated in the manuscript.

Original source details: GELFAND-P000376

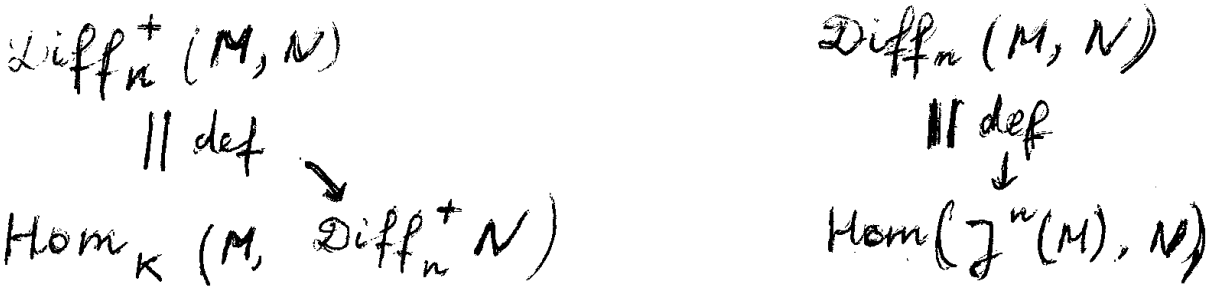
In this way one algebraically constructs the theory of de Rham homology and so on. (Spencer and oth.)



Manuscript notation. The parenthesis reads "Spencer and others," with the last word abbreviated.

Original source details: GELFAND-P000376

$\mathrm{Diff}_n^+(M,N)$ — all these operators. a module with multiplication $\underline{\underline{k^+}}$.



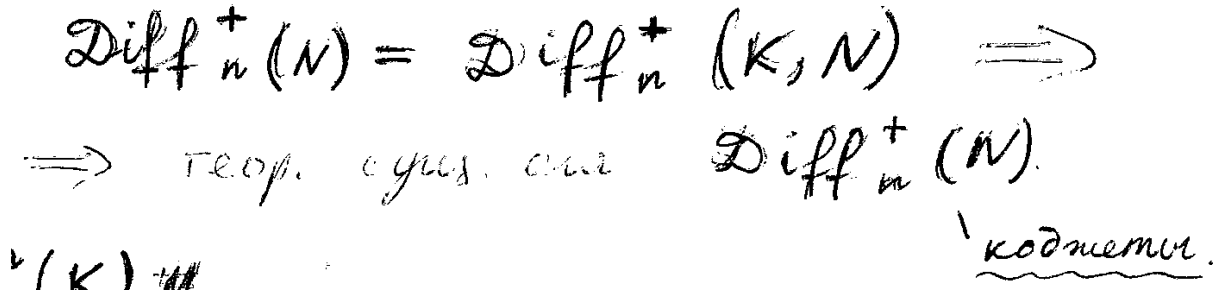
Original source details: GELFAND-P000376

Diagram and typed description.

$\mathrm{Diff}_n^+(M,N)$ $\quad$ $\mathrm{Diff}_n(M,N)$

$\|$ def $\searrow$ $\quad$ $\|$ def $\downarrow$

$\mathrm{Hom}_K(M,\ \mathrm{Diff}_n^+ N\ )$ $\quad$ $\mathrm{Hom}(J^n(M),N)$



Original source details: GELFAND-P000376

Diagram and typed description.

$\mathrm{Diff}_n^+(N) = \mathrm{Diff}_n^+(K,N) \quad \Longrightarrow$

$\Longrightarrow$ existence theorem for $\mathrm{Diff}_n^+(N)$.

$\backslash$

cojets.

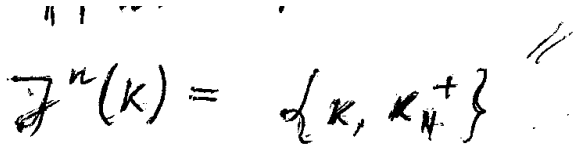
$J^n(K)$ ~~[?]~~ — an existence theorem also exists.



Undeciphered passage. After $J^n(K)$ is a separate small sign with overlapping cancellation strokes. Its whole reading is not recovered.

Original source details: GELFAND-P000376

In $\mathrm{Diff}_n K$ consider the algebra of operators, generated by k, k^+.



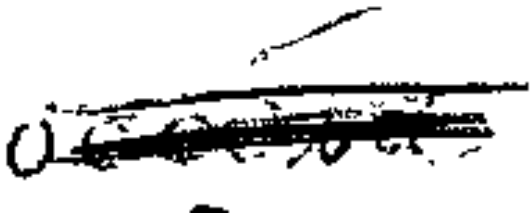
Original source details: GELFAND-P000376

Uncertain reading. Beneath the last k is a separate lower N/n-like sign. N is provisionally rendered; lowercase n and a reading without an independent index remain candidates. The upper plus is retained separately.

$$J^n(K) = \{k, k_N^+\}$$

GELFAND-P000377

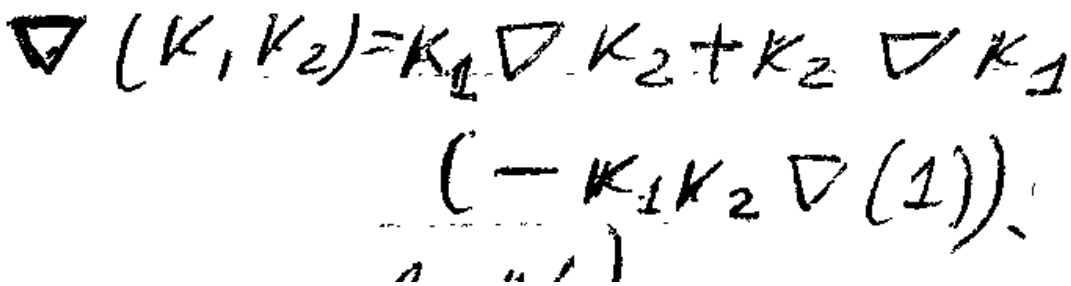
D_i — ~~[?]~~ functors $D_i(M)$ (de Rham homology).



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Undeciphered passage. The word before "functors" is cancelled by several horizontal strokes. It has not been recovered in full.

Define inductively. $D = D_1$ — derivations of K into N — denoted $D(N)$. $K \xrightarrow{\nabla} N$ is a derivation if



Original source details: GELFAND-P000377

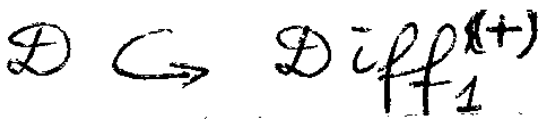
Uncertain reading. The parentheses contain k, k_2. The first low stroke reads as a comma.

$$\nabla(k, k_2) = k_1 \nabla k_2 + k_2 \nabla k_1$$

This is a diff. op. of order 1. $(-k_1 k_2 \nabla(1))$.

This is $D(N)$ (vector fields with values in N). D — ordinary vector fields.

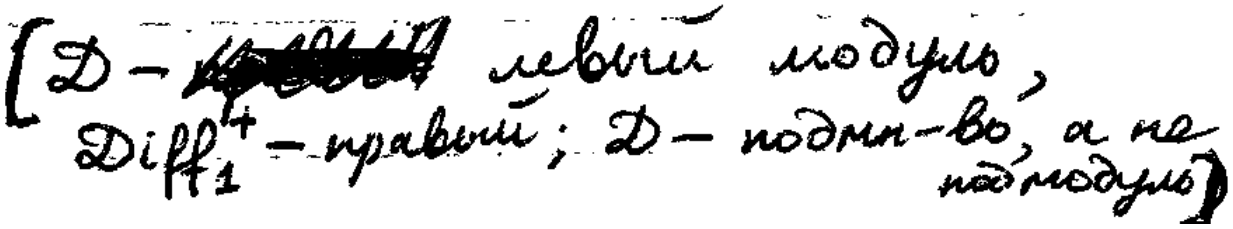
$$D_2(N) = D\{D(N) \subset \mathrm{Diff}_1^+ N\}.$$



Original source details: GELFAND-P000377

Diagram and typed description.

$D \hookrightarrow \mathrm{Diff}_1^{(+)}$



Original source details: GELFAND-P000377

Uncertain reading. The old adjective before “left” is densely cancelled. “right” is rendered as a candidate; the overlapping strokes do not yield an indisputable whole reading.

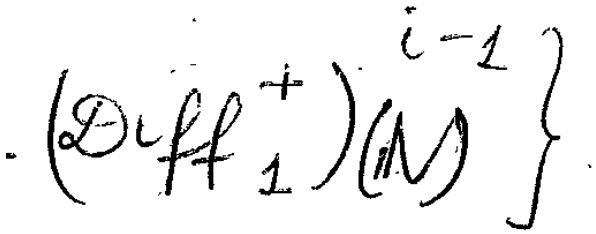
D — ~~right[?]~~ left module,

Diff_1^+ — right; D — a subset, and not a submodule

In $\mathrm{Diff}_1^{(+)}$ it is not a submodule, and the inclusion — a diff. operator of order 1. $D_2(N)$ — derivations of K into $\mathrm{Diff}_1^+ N$ which go into $D(N)$.

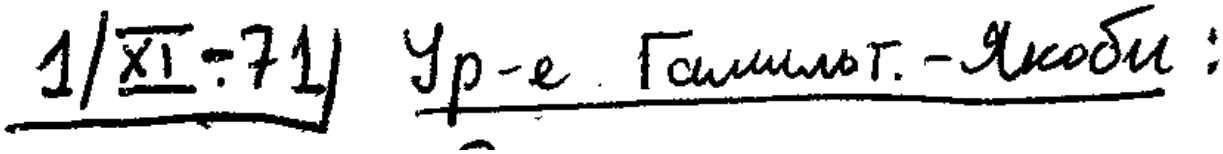
$$D_i(N) = D\{D_{i-1}(N) \subset (\mathrm{Diff}_1^+)(N)^{i-1}\}.$$

(by induction).



Manuscript notation. The exponent $i-1$ is written above (N).

Original source details: GELFAND-P000377

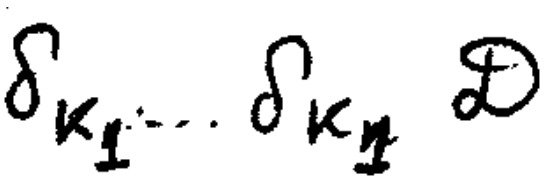


Original source details: GELFAND-P000377

Diagram and typed description.

1/XI-71 Hamilton-Jacobi equation:

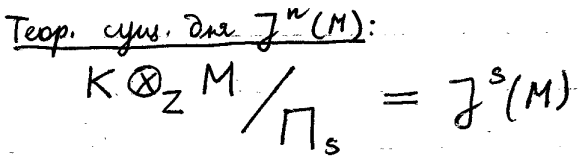
$\delta_{k_1} \cdots \delta_{k_1} \mathcal{D}$ — left side of the Hamilton-Jacobi eq.



Uncertain reading. The last lower index contains thick overlapping strokes. k_1 is provisionally rendered; k_n is retained among the candidates.

Original source details: GELFAND-P000377

(calculate) (k_1 — the same everywhere). Existence theorem for $J^n(M)$:



Uncertain reading. The heading has n; the following quotient and Π_s have s. Both source indices are retained.

Original source details: GELFAND-P000377

$$K \otimes_{\mathbb{Z}} M/\Pi_s = J^s(M).$$

Π_s — the submodule in $K \otimes_{\mathbb{Z}} M$ generated by the required relations.

$$k(k' \otimes m) = kk' \otimes m,$$
$$\bar{k}(k' \otimes m) = k' \otimes km.$$

GELFAND-P000378

[source number: 31]

$$\Delta_{k_0 \ldots k_s} = (\bar{k}_0 - k_0) \cdots (\bar{k}_s - k_s).$$

Consider the submodule generated by these Δ.

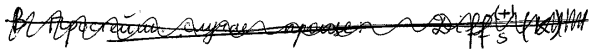
$$K \otimes_{\mathbb{Z}} M = A \supset \Delta(A) = \Pi_s \quad \text{(the usual definition of jets).}$$



Diagram and typed description.

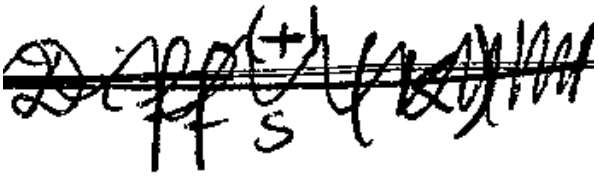
Original source details: GELFAND-P000378

~~In the simplest[?] case simply Diff$_s^{(+)}(K, N[?])$ [?]~~



Undeciphered passage. The whole line is cancelled. “In the simplest[?] case simply” and formula strokes can be distinguished. The adjective ending and final overlapping signs are not recovered in full.

Original source details: GELFAND-P000378



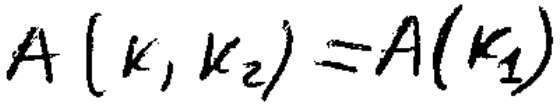
Uncertain reading. In the cancelled Diff$_s^{(+)}(K, N[?])$ the target sign is N/M-like. N is provisionally rendered; M remains a candidate.

Original source details: GELFAND-P000378

$J^s(K)$ is simpler than in the general case.

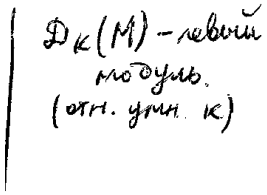
$D_1(M)$ — diff. operators without a constant term. $D_1(M) = D(M)$ (derivations of K with values in M). $D(M) \subset \mathrm{Diff}_1 M$.

$$A \in D_1(M) \overset{def}{\iff} A(k, k_2) = A(k_1)k_2 + k_1 A(k_2).$$



Manuscript notation. The argument of A reads k, k_2, as in the related formula on the previous frame.

Original source details: GELFAND-P000378



Original source details: GELFAND-P000378

Diagram and typed description.

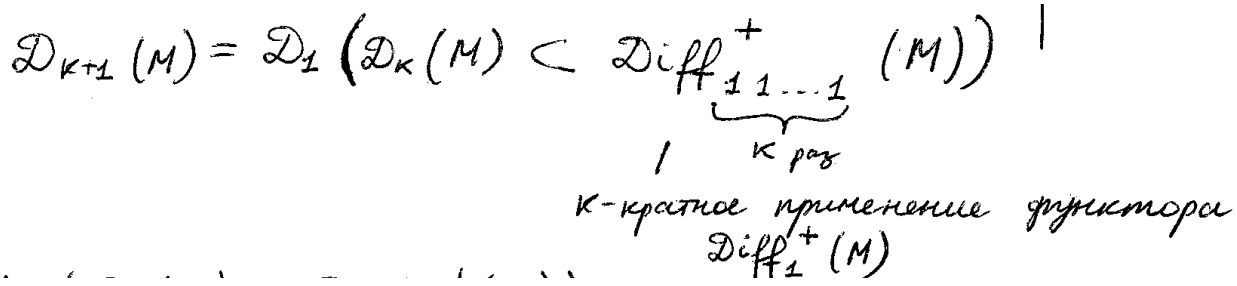
$D_0(M) = M$;

$D_k(M)$ — a left

module.

(w.r.t. multiplication by k).

$$D_2(M) = D_1(D_1(M) \subset \mathrm{Diff}_1^+(M)).$$



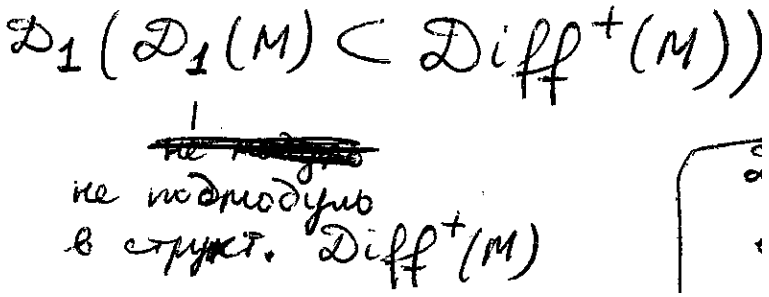
Original source details: GELFAND-P000378

Diagram and typed description.

$D_{k+1}(M) = D_1(D_k(M) \subset \mathrm{Diff}^+_{\underbrace{1\,1\ldots1}_{k \text{ times}}}(M))$

/

the k-fold application of the functor $\mathrm{Diff}_1^+(M)$.



Original source details: GELFAND-P000378

Undeciphered passage. Above the uncancelled “not a submodule” is a cancelled entry with visible parts “not” and “submod…”. The other letters are not recovered in full. Above the entry is a separate short vertical stroke.

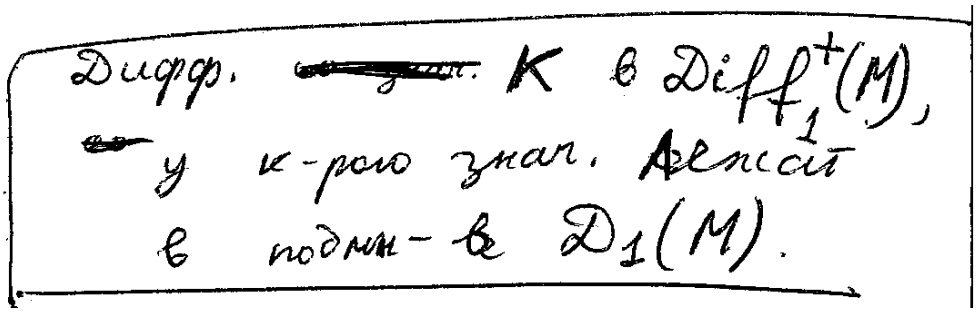
$D_1(D_1(M) \subset \text{Diff}^+(M))$.

|

~~not a submod[?]~~

not a submodule

in the structure $\text{Diff}^+(M)$.



Original source details: GELFAND-P000378

Undeciphered passage. After “Diff.” before K a word is densely cancelled. It has not been read in full.

Diff. ~~[?]~~ of K into $\text{Diff}^+_1(M)$,

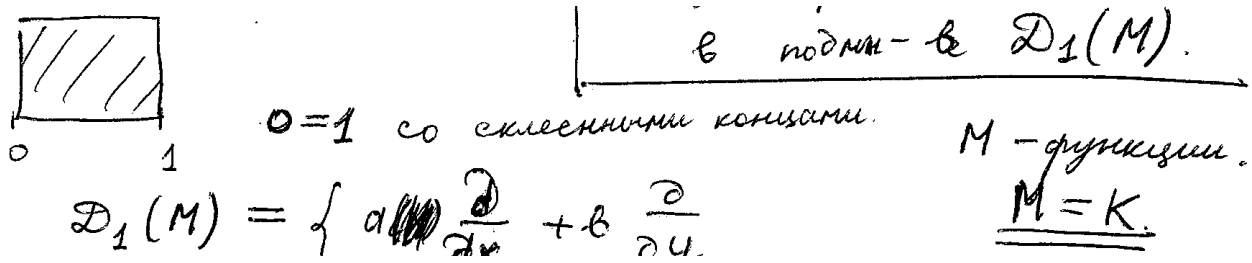
~~[?]~~ whose values lie

in the subset $D_1(M)$.



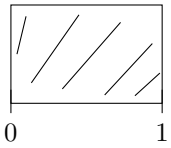
Undeciphered passage. Before “whose” is a separate short fragment beneath cancellation strokes. It has not been deciphered in full.

Original source details: GELFAND-P000378



Original source details: GELFAND-P000378

Diagram and typed description.

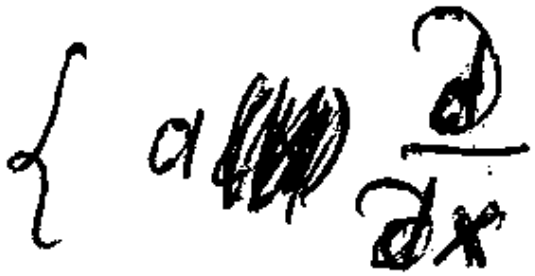


$0 = 1$ with ends identified.

M — functions.

$\underline{\underline{M = K}}$.

$$D_1(M) = \left\{ a \,[?]\, \frac{\partial}{\partial x} + b \frac{\partial}{\partial y} \right. .$$



Undeciphered passage. After the legible a its argument is densely overwritten and cancelled. It has not been deciphered in full. An opening curly brace is visible; no closing brace has been added.

Original source details: GELFAND-P000378

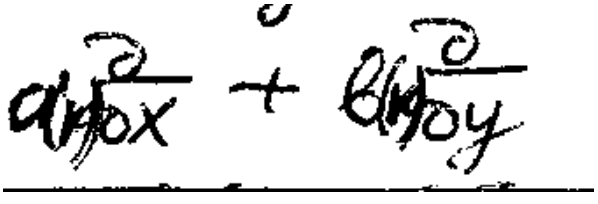
$$\mathrm{Diff}_1^+ : a \frac{\partial}{\partial x} + b \frac{\partial}{\partial y} + c.$$

$$K(x) \longrightarrow [?]\, a_k \frac{\partial}{\partial x} + b_k \frac{\partial}{\partial y}.$$

Undeciphered passage. Immediately after the arrow a short prefix is cancelled. It cannot be read in full. The line lies at the lower edge of the frame.



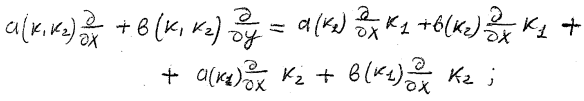
Original source details: GELFAND-P000378



Uncertain reading. Beside a and b are compact k-like strokes. Lower indices a_k, b_k are provisionally rendered; $a(k), b(k)$ remain candidates. Complete parentheses are not recovered.

Original source details: GELFAND-P000378

$$a(k,k_2)\frac{\partial}{\partial x}+b(k,k_2)\frac{\partial}{\partial y}=a(k_2)\frac{\partial}{\partial x}k_1+b(k_2)\frac{\partial}{\partial x}k_1+$$
$$+a(k_1)\frac{\partial}{\partial x}k_2+b(k_1)\frac{\partial}{\partial x}k_2;$$

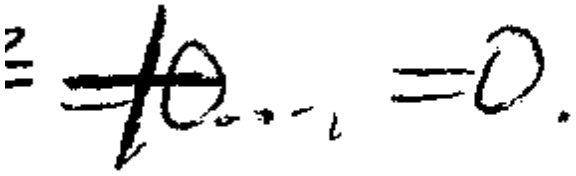


Manuscript notation. On the left, the arguments of a and b read k, k_2. All four denominators on the right read ∂x, including the terms with b.

Original source details: GELFAND-P000379

Hence $a(k,k_2)=a(k_2)k_1+a(k_1)k_2$ and b — a derivation, and also the condition:

$$a(k_2)\frac{\partial k_1}{\partial x}+a(k_1)\frac{\partial k_2}{\partial x}$$
$$+b(k_2)\frac{\partial k_1}{\partial x}+b(k_1)\frac{\partial k_2}{\partial x}\quad =0\ldots=0.$$



Uncertain reading. After the sum, $=0$ is crossed by a separate oblique stroke. Separate dots and a faint lower stroke follow; then there is an uncancelled $=0$.

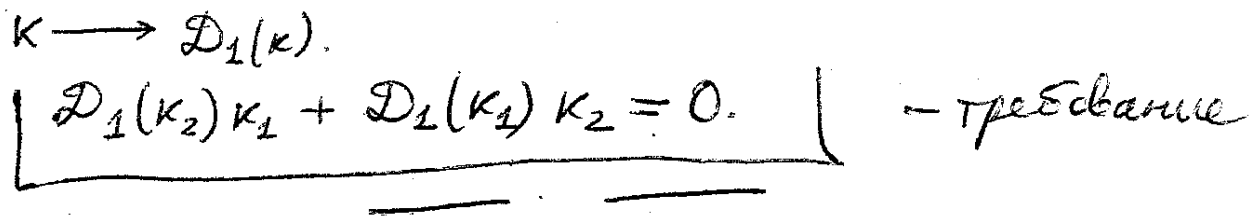
Original source details: GELFAND-P000379

$$\left(a(k_2)\frac{\partial}{\partial x}+b(k_2)\frac{\partial}{\partial y}\right)k_1+\sim=0.$$
$$D_1(k_2)k_1+D_1(k_1)k_2=0.$$



Uncertain reading. The first lower 1 is densely overwritten. D_1 is provisionally rendered; the earlier overlapping strokes are not recovered in full.

Original source details: GELFAND-P000379



Original source details: GELFAND-P000379

Diagram and typed description.

$k \longrightarrow D_1(k)$.

$D_1(k_2)k_1 + D_1(k_1)k_2 = 0.$ — requirement.



Original source details: GELFAND-P000379

Undeciphered passage. At the left margin before $D_k(M)$ a separate sign is cancelled. It has not been deciphered in full.

~~[?]~~ $D_k(M)$ —k-forms on the ~~[?]~~ manifold with values in M.

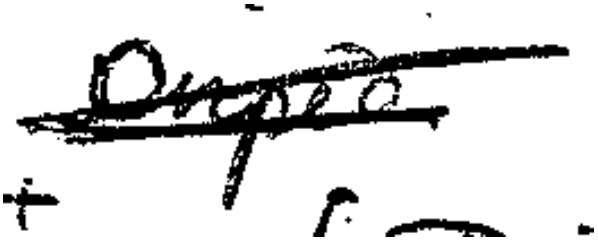
~~[?]~~

~~Def.-e.[?]~~



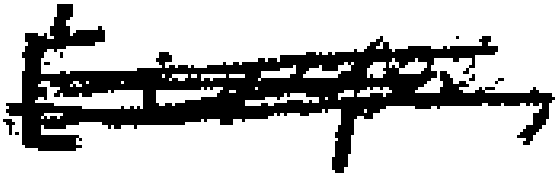
Undeciphered passage. Between “on” and “manifold” a sign is densely cancelled. It has not been recovered in full.

Original source details: GELFAND-P000379



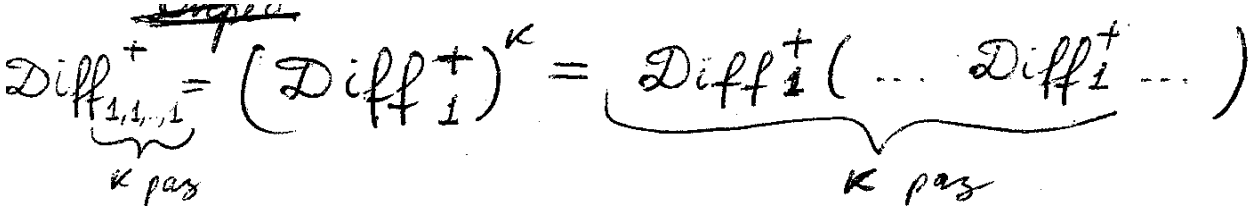
Uncertain reading. Below the line at left is a cancelled contraction, provisionally “Def.-e.”; “Def.-o.” and “Def.” are retained as alternatives. The ending is not expanded.

Original source details: GELFAND-P000379



Original source details: GELFAND-P000379

Undeciphered passage. Below “forms” is a separate cancelled entry. It has not been recovered in full.



Original source details: GELFAND-P000379

Diagram and typed description.

$$\underbrace{\mathrm{Diff}^+_{1,1,\dots,1}}_{k\text{ times}} = (\mathrm{Diff}^+_1)^k = \underbrace{\mathrm{Diff}^+_1(\cdots \mathrm{Diff}^+_1 \cdots)}_{k\text{ times}}.$$

$$(\mathrm{Diff}^+_1)^2(M) = \mathrm{Diff}_1(K, \mathrm{Diff}^+_1 M).$$

Definition of exterior differential forms. (the de Rham complex over the ring K)

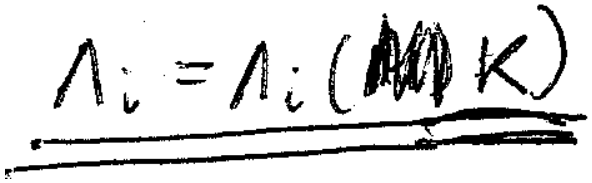


Original source details: GELFAND-P000379

Diagram and typed description.

$$D_i(M) \overset{def}{=} \mathrm{Hom}_K(\Lambda_i, M);$$

$$\Lambda_i = \Lambda_i(\cancel{M}\,[?]\,K).$$



Original source details: GELFAND-P000379

Uncertain reading. In the argument before the legible K an M/N-like sign is cancelled. M is provisionally rendered; N remains a candidate.

Theorem: a representing object exists



Original source details: GELFAND-P000379

Manuscript notation. The manuscript uses the singular “a representing object exists.”

$K = \mathbb{Z}_2$ will have $D_2 \neq 0$! (the 2-dimensional de Rham complex).

GELFAND-P000380

[source number: 32]

Consider the group ring $\Gamma_2(\mathbb{Z}_2)$ (with coefficients in $\mathbb{Z}_2$).



Manuscript notation. The first letter has a vertical and a top stroke, Γ, with no middle crossbar.

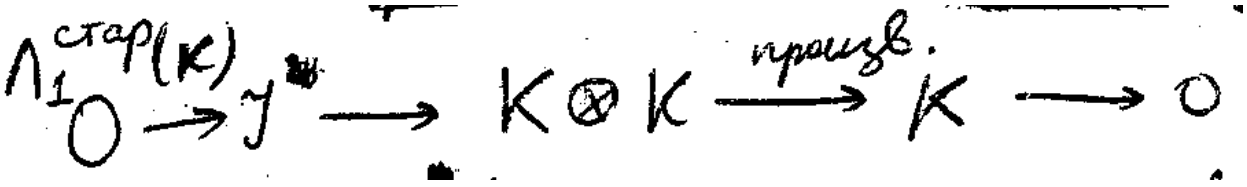
Original source details: GELFAND-P000380

All de Rham cohomology groups — $\mathbb{Z}_2$. For group rings, de Rham cohom. — group cohom.!



Diagram and typed description.

Original source details: GELFAND-P000380



Original source details: GELFAND-P000380

Uncertain reading. At the upper right of J in the sequence is a separate dense sign with overlapping strokes. Its whole reading and meaning are not established; it is rendered as [?].

$\Lambda_1^{\text{old}}(K)$

$$0 \longrightarrow J^{[?]} \longrightarrow K \otimes K \xrightarrow{\text{mult.}} K \longrightarrow 0$$

$$J^1/J^2 = \Lambda_1, \quad \text{the rest — exterior powers.}$$



Uncertain reading. At the upper right of the numerator are dense overwritten strokes. J^1 is provisionally rendered; J without an independent power and J with an unidentified earlier power remain candidates. The denominator J^2 is legible.

Original source details: GELFAND-P000380

For a field these cohomologies will be 0, and the Vinogradov ones $\neq 0$. This definition is unnatural in the categorical sense. They are not defined for noncommutative rings. This is the only natural definition in the category of rings. The differential d.



Original source details: GELFAND-P000380

Undeciphered passage. Below the map is a separate small sign with overlapping cancellation strokes. It has not been recovered in full.

$D_k \longrightarrow D_{k-1}(\mathrm{Diff}_1^+)$ morphism of functors.

(in other terms, d).

i.e. $D_k(M) \longrightarrow D_{k-1}(\mathrm{Diff}_1^+(M))$. natural map.

~~[?]~~

$$D_1 = D \underset{\text{incl.}}{\longrightarrow} \mathrm{Diff}_1^+ \quad (= D_0(\mathrm{Diff}_1^+)).$$



Original source details: GELFAND-P000380

Uncertain reading. The first lower index is densely overwritten, with a loop-like component. 1 is provisionally rendered; 0 remains a visual candidate.

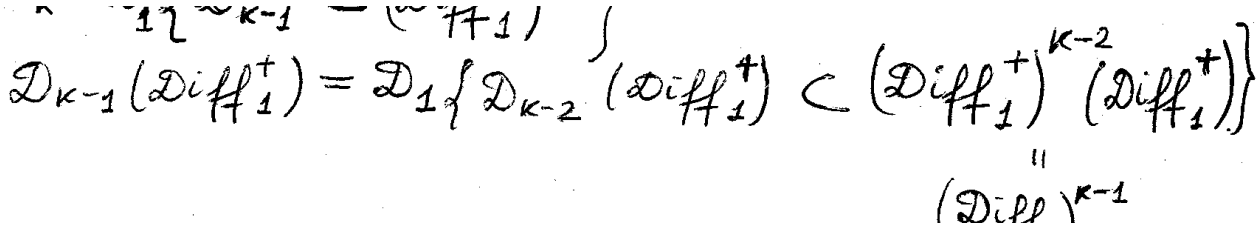
$$D_1 \longrightarrow D_0(\mathrm{Diff}_1^+).$$

(not a module inclusion, but a differential operator of order 1).

$\mathrm{Diff}_n \longrightarrow \mathrm{Diff}_n^+$
(a differential operator of order n).

Induction:

$$D_k = D_1\{D_{k-1} \subset (\mathrm{Diff}_1^+)^{k-1}\}.$$



Original source details: GELFAND-P000380

Diagram and typed description.

$$D_{k-1}(\mathrm{Diff}_1^+) = D_1\{$$

$$D_{k-2}(\mathrm{Diff}_1^+) \subset (\mathrm{Diff}_1^+)^{k-2}(\mathrm{Diff}_1^+)\ \}.$$
$$\| \|$$
$$(\mathrm{Diff}_1)^{k-1}$$

Assume inductively that $D_{k-1} \subset D_{k-2}(\mathrm{Diff}_1^+)$, and then we obtain:

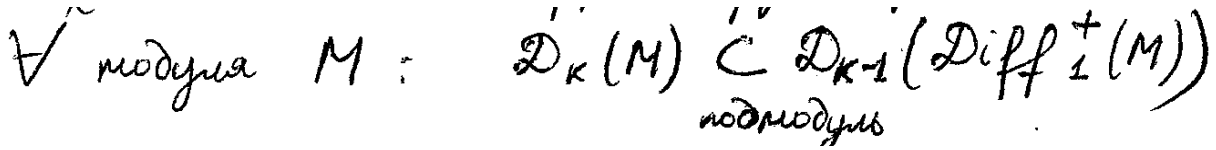
GELFAND-P000381

$D_k \subset D_{k-1}(\mathrm{Diff}_1^+)$. This is an [?] injective morphism of functors, that is



Undeciphered passage. Before “injective” a short sign or fragment is cancelled. It has not been deciphered in full.

Original source details: GELFAND-P000381

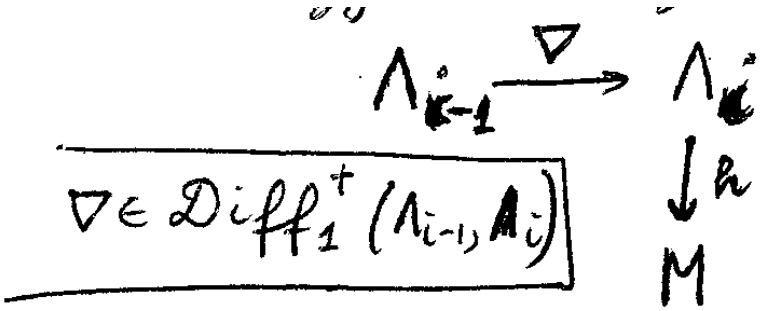


Original source details: GELFAND-P000381

Diagram and typed description.

$\forall$ module M: $D_k(M) \subset D_{k-1}(\mathrm{Diff}_1^+(M))$
submodule

Express this in the language of differential forms:



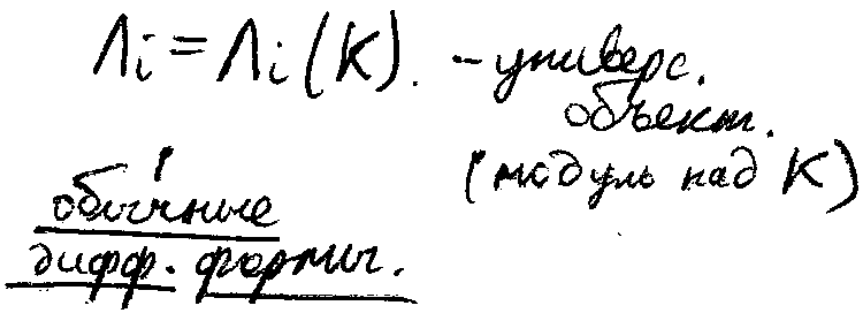
Original source details: GELFAND-P000381

Uncertain reading. In the diagram the lower $i-1$ at left and i at right are written over reinforced strokes. The current pair is provisionally rendered; the earlier lower signs are not recovered in full.

$$\begin{array}{ccc} \Lambda_{i-1} & \xrightarrow{\nabla} & \Lambda_i \\ & & \downarrow h \\ & & M \end{array}$$

$\nabla \in \mathrm{Diff}_1^+(\Lambda_{i-1}, \Lambda_i)$

$$D_i(M) = \mathrm{Hom}(\Lambda_i, M).$$



Original source details: GELFAND-P000381

Uncertain reading. Above “ordinary” is a separate thick, nearly vertical stroke. An independent lower dot is not established; no complete exclamation mark has been supplied.

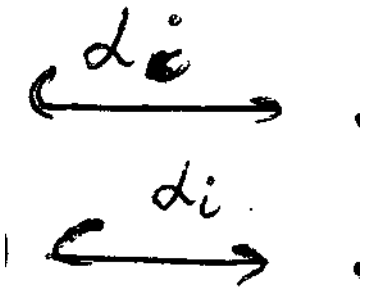
$\Lambda_i = \Lambda_i(K)$. — universal object.

(a module over K).

ordinary diff. forms.

$$D_i(M) \overset{\alpha_i}{\hookrightarrow} D_{i-1}(\mathrm{Diff}_1^+(M)).$$

$$h \in \mathrm{Hom}(\Lambda_i, M) \overset{\alpha_i}{\hookrightarrow} [?]\,\mathrm{Hom}(\Lambda_{i-1}, \mathrm{Diff}_1^+(M)).$$



Manuscript notation. Both arrows have a hook at left and an arrowhead at right.

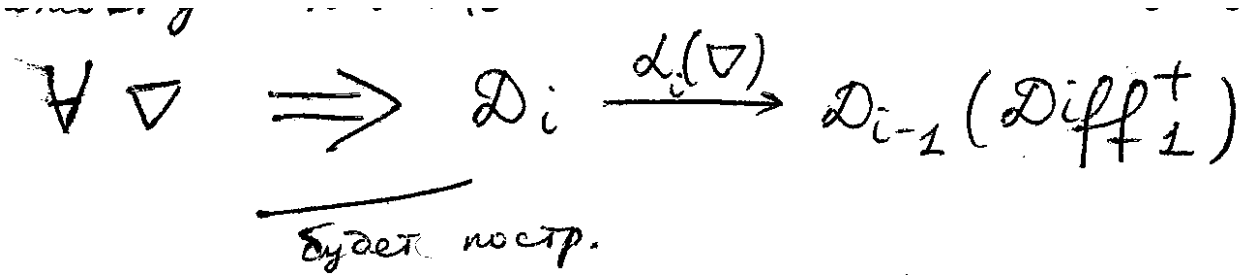
Original source details: GELFAND-P000381



Undeciphered passage. Before the right-hand Hom a short fragment is densely cancelled. It has not been recovered in full.

Original source details: GELFAND-P000381

Let a differential operator of order 1 be given, $\Lambda_{i-1} \overset{\nabla}{\longrightarrow} \Lambda_i$. Then to each $h : \Lambda_i \longrightarrow M$ assign the composition $h \circ \nabla \in \mathrm{Diff}_1^+(\Lambda_{i-1}, M)$.

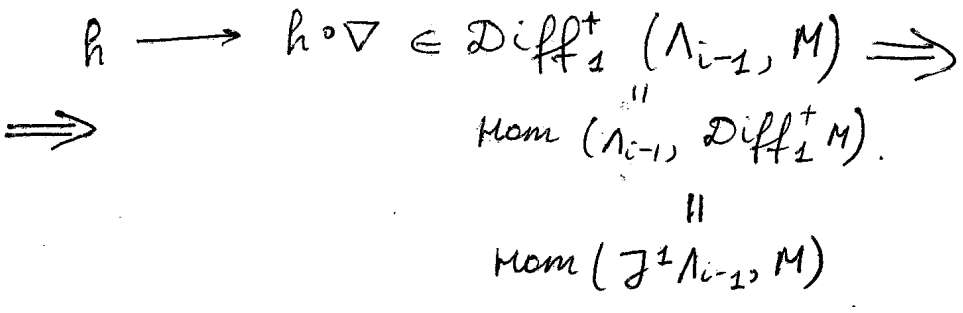


Original source details: GELFAND-P000381

Diagram and typed description.

$$\forall \nabla \Longrightarrow D_i \overset{\alpha_i(\nabla)}{\longrightarrow} D_{i-1}(\mathrm{Diff}_1^+)$$

will be constructed.



Original source details: GELFAND-P000381

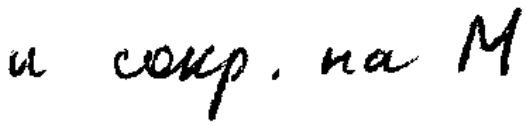
Diagram and typed description.

$$h \longrightarrow h \circ \nabla \in \mathrm{Diff}_1^+(\Lambda_{i-1}, M) \quad \Longrightarrow$$

$$\Longrightarrow \qquad \mathrm{Hom}(\Lambda_{i-1}, \mathrm{Diff}_1^+ M)$$

$$\mathrm{Hom}(J^1\Lambda_{i-1}, M)$$

And adj. on M



Original source details: GELFAND-P000381

Manuscript notation. The abbreviation is left unexpanded in the manuscript.



Original source details: GELFAND-P000381

Undeciphered passage. The first cancelled entry lies under the strokes of a reinforced rightward arrow; its whole reading is not recovered. At right, “diff.” is separately cancelled. Both parts and the arrow are retained.

~~[?]~~ ⟶ ~~diff.~~

a homomorphism $J^1\Lambda_{i-1} \longrightarrow \Lambda_i$, i.e. a differential operator $\Lambda_{i-1} \longrightarrow \Lambda_i$. Taking $M = \Lambda_i$ and the identity map gives what is needed.

GELFAND-P000382

[source number: 33]

$$\mathrm{Hom}(\Lambda_i, M) \overset{\alpha_i}{\hookrightarrow} \mathrm{Hom}(\Lambda_{i-1}, \mathrm{Diff}_1^+(M)).$$



Manuscript notation. The α_i arrow has a hook at left and an arrowhead at right.

Original source details: GELFAND-P000382

$M = \Lambda_i$ and id on the left. An element of $\mathrm{Hom}(\Lambda_{i-1}, \mathrm{Diff}_1^+(M))$ is by definition a differential operator of order 1 from Λ_{i-1} to M. (We find ∇, which exists and is unique). We must prove that

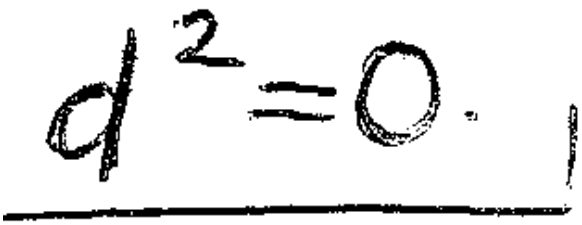
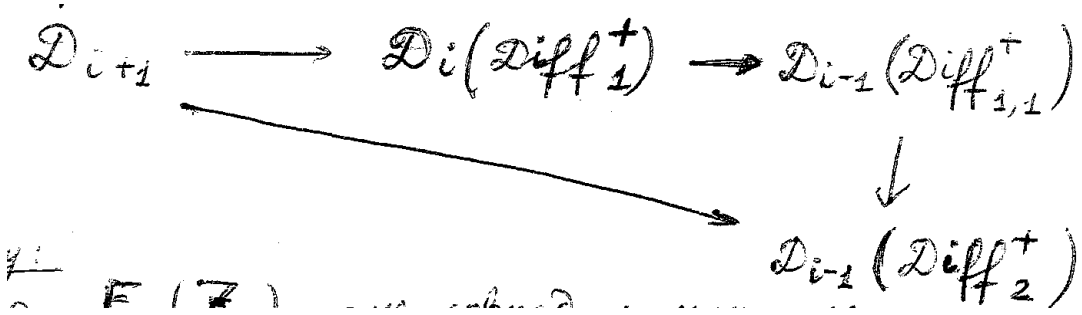


Diagram and typed description.

$$d^2 = 0.$$

Original source details: GELFAND-P000382

Consider

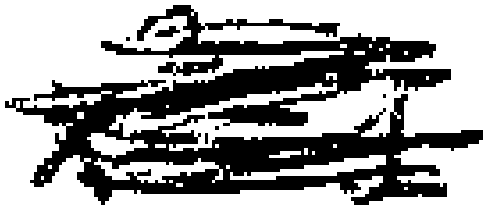


Original source details: GELFAND-P000382

Diagram and typed description.

$$\begin{array}{ccccc} D_{i+1} & \longrightarrow & D_i(\mathrm{Diff}_1^+) & \longrightarrow & D_{i-1}(\mathrm{Diff}_{1,1}^+) \\ & \searrow & & & \downarrow \\ & & & & D_{i-1}(\mathrm{Diff}_2^+) \end{array}$$

An aside: The ring $F_2(\mathbb{Z}_2)$—does not coincide with the classical ones. $0, 1, x$, ~~[?]~~, $1 + x$, $(x^2 = 1)$.



Undeciphered passage. After x and before $1 + x$ a separate element is cancelled. Its whole reading has not been recovered.

Original source details: GELFAND-P000382

$y = x + 1$, then $y^2 = 0$. $0, 1, y, 1 + y$, where $y^2 = 0$.

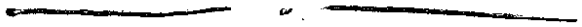
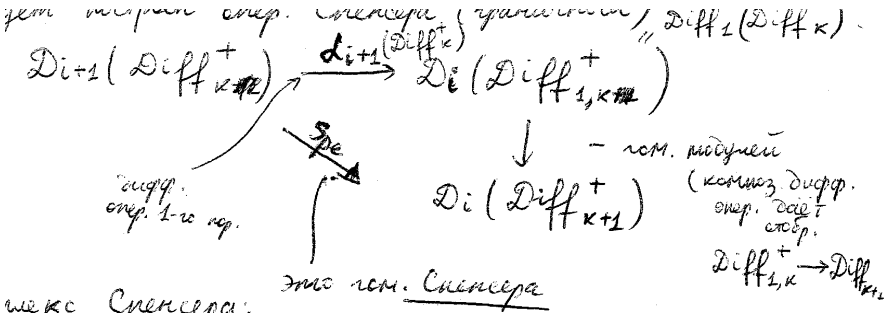


Diagram and typed description.

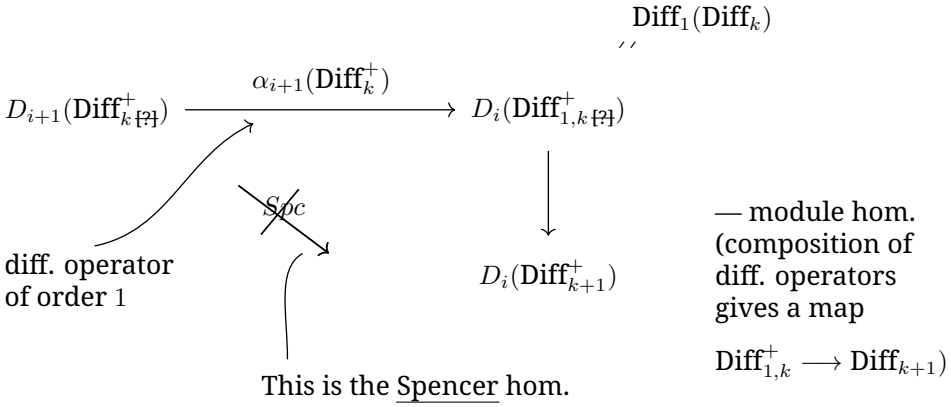
Original source details: GELFAND-P000382

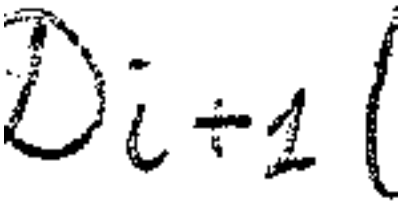
The Spencer operator (boundary) will be constructed.



Original source details: GELFAND-P000382

Diagram and typed description. At upper left is $D_{i+1}(\mathrm{Diff}_k^+)$, at upper right $D_i(\mathrm{Diff}_{1,k}^+)$, and below at right $D_i(\mathrm{Diff}_{k+1}^+)$. A horizontal arrow bears $\alpha_{i+1}(\mathrm{Diff}_k^+)$; at right is a downward arrow. A short diagonal down-right arrow stops before the lower expression; its Spc label is crossed out. A curved pointer from below at left approaches the start of the upper horizontal arrow. Under the diagonal is a curved vertical bracket to "This is the Spencer hom."; "Spencer" is underlined. Above the right expression are $\mathrm{Diff}_1(\mathrm{Diff}_k)$ and two short slanted strokes; the composition annotation is at right.





Uncertain reading. The sign after i has discontinuous vertical portions. Provisionally D_{i+1}; possibly D_{i-1}.

Original source details: GELFAND-P000382



Uncertain reading. After the left k a linking mark and a 2-like sign are cancelled. The earlier reading may be -2, $+2$, or another index ending in 2.

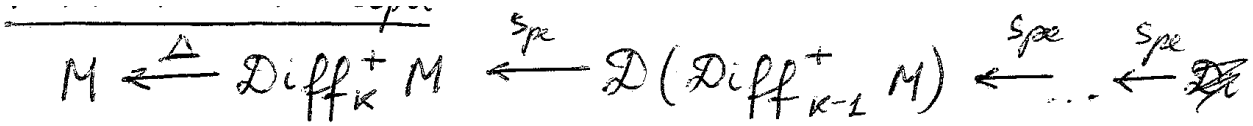
Original source details: GELFAND-P000382

Undeciphered passage. After the right k an additional index fragment is cancelled. Its whole reading has not been recovered.



Original source details: GELFAND-P000382

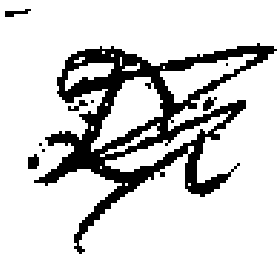
The Spencer complex:



Original source details: GELFAND-P000382

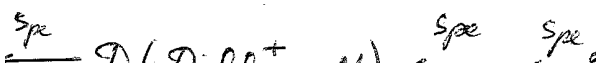
Diagram and typed description.

$$M \xleftarrow{\Delta} \mathrm{Diff}^{+}_{k} M \xleftarrow{Spc} D(\mathrm{Diff}^{+}_{k-1} M) \xleftarrow{Spc} \cdots \xleftarrow{Spc} \not{D}_{\bar{i}}$$



Uncertain reading. The cancelled expression at the far right is provisionally D_i; the lower hook also admits D_1.

Original source details: GELFAND-P000382



Original source details: GELFAND-P000382

Uncertain reading. All four labels have a third sign resembling c/e or a lower e-like sign. Provisionally Spc; possibly Spe. The diagonal label is crossed out; the three lower labels are current.

GELFAND-P000383

$$\longleftarrow D_i(\mathrm{Diff}^+_{k-i} M) \longleftarrow \cdots$$

This is the Spencer resolution (coresol. Sm.)



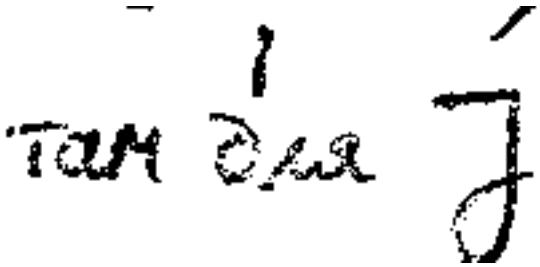
Original source details: GELFAND-P000383

Manuscript notation. A separate dot is visible above the last sign in $k-i$.



Original source details: GELFAND-P000383

Uncertain reading. In parentheses, “coresol. Sm.”; the final two joined Cyrillic letters admit “См.” or “Сп.”. The abbreviation is left unexpanded.

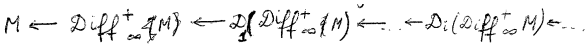


Original source details: GELFAND-P000383

Diagram and typed description. The small note reads “there for J”. A separate vertical stroke is above “for”.

there for J.

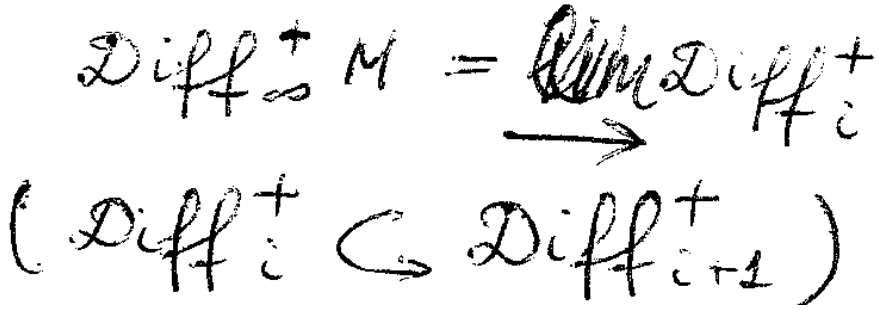
$$M \longleftarrow \mathrm{Diff}^+_\infty\{M\} \longleftarrow D_1(\mathrm{Diff}^+_\infty\{M\})$$
$$\longleftarrow \cdots \longleftarrow D_i(\mathrm{Diff}^+_\infty M) \longleftarrow \cdots$$



Original source details: GELFAND-P000383

Uncertain reading. The first M has braces with reinforced strokes. Before the second M is a brace-like sign, and after it a closing round parenthesis. Provisionally $D_1(\mathrm{Diff}^+_\infty\{M)$; the earlier form of the reinforced strokes is not established in full. All arrows point left.

$$\mathrm{Diff}^+_\infty M = \varinjlim \mathrm{Diff}^+_i .$$



Original source details: GELFAND-P000383

Diagram and typed description.

$$(\mathrm{Diff}^+_i \hookrightarrow \mathrm{Diff}^+_{i+1})$$

This is a complex. And a map (not yet defined)

$$\mathrm{Diff}^+_k M \longrightarrow M.$$

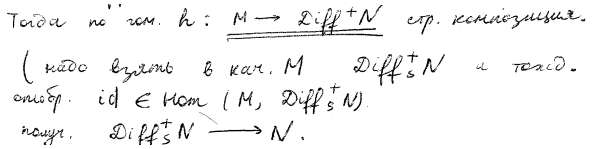
$$\mathrm{Diff}^+_s(M,N) \stackrel{def}{=} \mathrm{Hom}_K(M, \mathrm{Diff}^+_s N).$$

Let $\mathrm{Diff}^+ N \xrightarrow{\Delta} N$; Δ — an operator of order s. Then from the hom. $h : M \longrightarrow \mathrm{Diff}^+ N$ construct the composition. (we must take $\mathrm{Diff}^+_s N$ as M and the identity map

$$id \in \mathrm{Hom}(M, \mathrm{Diff}^+_s N).$$

we obtain $\mathrm{Diff}^+_s N \longrightarrow N$.

This defines the first morphism.



Original source details: GELFAND-P000383

Diagram and typed description. The map $M \longrightarrow \mathrm{Diff}^+ N$ following h is doubly underlined. Below Hom a separate thin diagonal stroke runs down to the right, without an arrowhead, towards the start of the next arrow. Before "we must" is an opening round parenthesis; no separate overall closing parenthesis is visible after the lower N.

$$\Delta \in \mathrm{Diff}^+_s(K,M) \longrightarrow \underline{\underline{\Delta(1)}}.$$

Headings: All Spencer resolutions in non-coordinate form (a distinction from Spencer). The index over any ring. The Hamiltonian formalism for $\forall$ ring. Next time—continuation.

GELFAND-P000384

[source number: 34]

[ ]

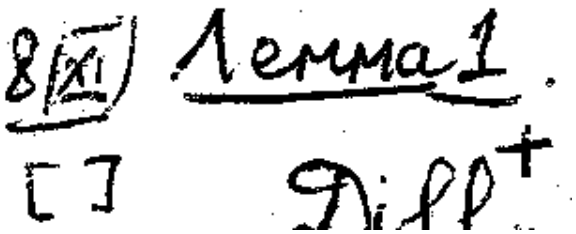


Diagram and typed description.

8 XI Lemma 1.

Original source details: GELFAND-P000384

$\mathrm{Diff}_n^+(M,N)$ for fixed N —a functor of M (contravar.).

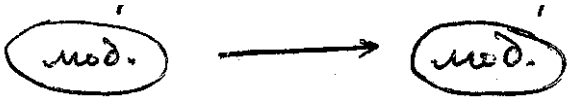


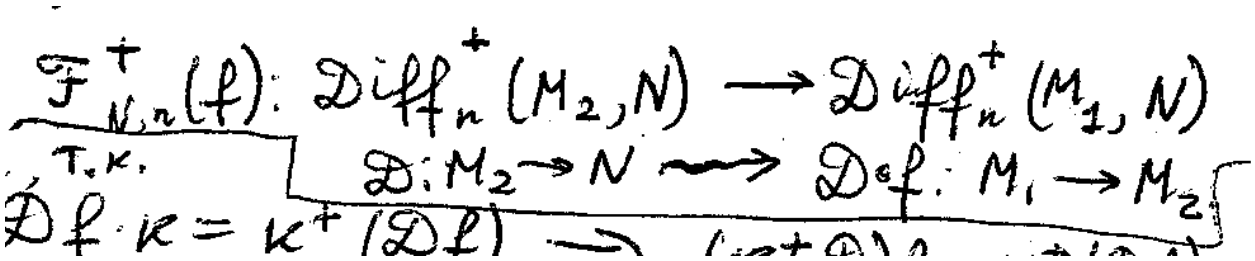
Diagram and typed description.

mod. ⟶ mod.

Original source details: GELFAND-P000384

(and if we take $\mathrm{Diff}_n(M,N)$, then this is a functor of N for fixed M (covar.)). Proof: $\mathcal{F}^+_{N,n}(M) = \mathrm{Diff}_n^+(M,N)$. If $f : M_1 \longrightarrow M_2$ a module hom. $\Longrightarrow$

$$\mathcal{F}^+_{N,n}(f) : \mathrm{Diff}_n^+(M_2,N) \longrightarrow \mathrm{Diff}_n^+(M_1,N).$$

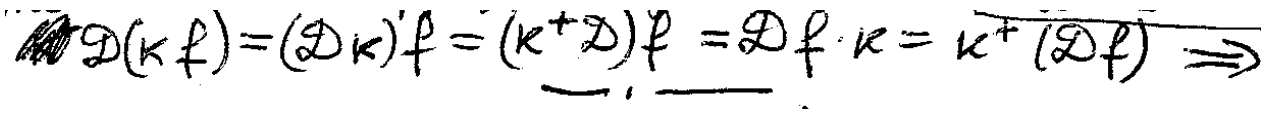


Original source details: GELFAND-P000384

Uncertain reading. The right-hand inset reads $D \circ f : M_1 \longrightarrow M_2$; the final M_2 is retained. Its frame is open above, with a long lower stroke and short side bends.

$D : M_2 \longrightarrow N \rightsquigarrow D \circ f : M_1 \longrightarrow M_2$

And this is a module homomorphism, since



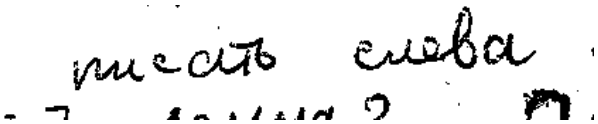
Original source details: GELFAND-P000384

Undeciphered passage. Before $D(kf)$ is a separate short, densely cancelled fragment. Its whole reading has not been recovered.

[?] $D(kf) = (Dk)f = (k^+D)f = Dfk = k^+(Df)$

$$\Longrightarrow (k^+D)f = k^+(Df).$$

Similarly, if we consider $\mathrm{Diff}_n(M,N)$ as a functor of N, then f must be written on the left, and this also gives a module homomorphism.



Original source details: GELFAND-P000384

Manuscript notation. The manuscript reads "be written on the left".

[ ] Lemma 2. Let $A \in \mathrm{Diff}_1(K,M)$. Then

$$A(k_1k_2) = k_1A(k_2) + k_2A(k_1) - k_1k_2A(1). \qquad (1)$$

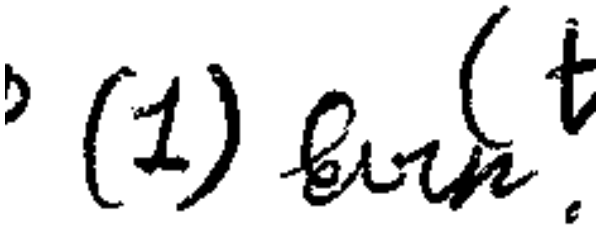
and conversely, if $A : K \longrightarrow M$ —an additive map satisfying (1), then $A \in \mathrm{Diff}_1(K,M)$.
Proof: The general situation $A \in \mathrm{Diff}_1(N,M) \Longleftrightarrow$

$$Ak_0k_1 = k_0Ak_1 + k_1Ak_0 - k_0k_1A \quad (\forall k_0, k_1). \qquad (2)$$

For $N = K$ this means that

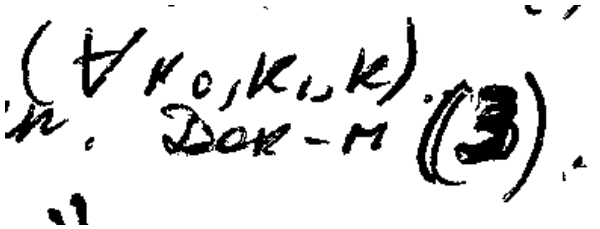
$$A(k_0k_1k) = k_0A(k_1k) + k_1A(k_0k) - k_0k_1A(k). \qquad (3)$$

Setting $k = 1$ gives (1). Conversely, let (1) hold.



Original source details: GELFAND-P000384

Manuscript notation. The manuscript abbreviates "holds" after "let (1)".

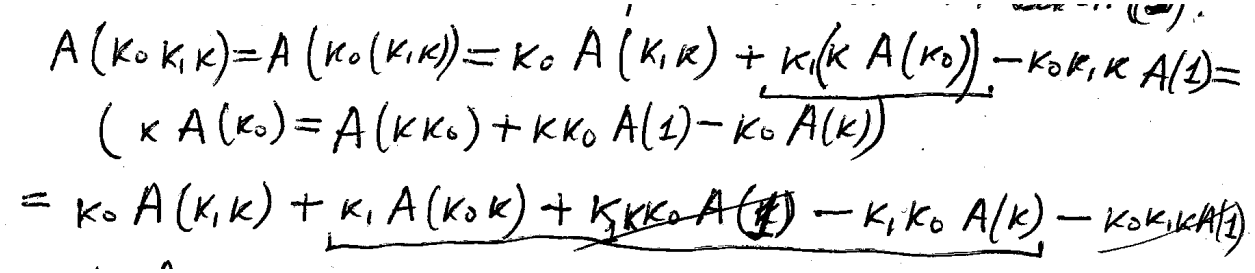


Original source details: GELFAND-P000384

Uncertain reading. The digit 3 is densely reinforced over earlier ink; an overwritten 2 is possible. At left are two close opening parentheses; at right one closing parenthesis. Above is the small insertion $(\forall k_0, k_1, k)$.

$(\forall k_0, k_1, k)$

We prove $((3)$.



Original source details: GELFAND-P000384

Uncertain reading. The whole $k_1kk_0A(1)$ is crossed out; the final argument is reinforced and also admits $A(k)$. The final term $-k_0k_1kA(1)$ is crossed out separately.

$$A(k_0k_1k) = A(k_0(k_1k)) = k_0A(k_1k)+k_1(kA(k_0))-k_0k_1kA(1) =$$

$$\big(kA(k_0) = A(kk_0) + kk_0A(1) - k_0A(k)\big)$$

$$= k_0A(k_1k)+k_1A(k_0k) + \cancel{k_1kk_0A(1)} - k_1k_0A(k)\cancel{-k_0k_1kA(1)}.$$

$$= k_0Ak_1k + k_1Ak_0k - k_0k_1Ak,$$

as required.

Corollary: $A \in \mathrm{Diff}_1(K, M) \Longrightarrow A - A(1)$ —a derivation from K to M. Lemma 3. Let $A_1 \in \mathrm{Diff}_m(M_1, M_2)$, $B \in \mathrm{Diff}_n(M_2, M_3)$.



Uncertain reading. In the hypothesis, the first A has an extra reinforced 1-like lower index. Provisionally A_1; the earlier lower sign is not established.

Original source details: GELFAND-P000384

Then $B \circ A \in \mathrm{Diff}_{m+n}(M_1, M_3)$. Proof: induction on $m+n$. For $m+n = 0 (m = n = 0)$ everything is evident. Assume it proved for $m+n \le d-1$. Let $m+n = d$.

$$\delta_k(B \circ A) = kBA - BAk = (kB - Bk)A + B(kA - Ak)$$
$$= (\delta_k B)A + B(\delta_k A) \quad \blacksquare$$



Manuscript notation. The order of the terms at the start of the line is $kBA - BAk$.

Original source details: GELFAND-P000384

GELFAND-P000385

Lemma 4. There exists a canon. diff. operator of ord. n:

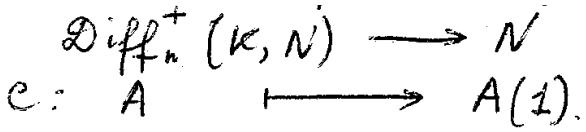


Diagram and typed description.

$$\mathrm{Diff}_n^+(K, N) \longrightarrow N$$

$$C: \quad A \longmapsto A(1)$$

Original source details: GELFAND-P000385

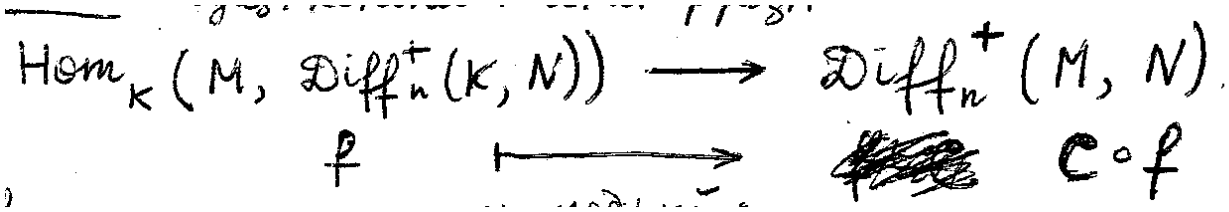
Proof.

$$(\delta_k C)(A) = (kC - Ck)(A) = kA(1) - CAk$$
$$= kA(1) - Ak(1) = (\delta_k A)(1).$$

hence

$$(\delta_{k_0}\delta_{k_1}\dots\delta_{k_n}C)(A) = (\delta_{k_0}\delta_{k_1}\dots\delta_{k_n}A)(1) = 0$$

and C is a diff. op. Consequence: a canon. homomorphism exists



Original source details: GELFAND-P000385

Diagram and typed description.

$$\mathrm{Hom}_K(M, \mathrm{Diff}_n^+(K,N)) \longrightarrow \mathrm{Diff}_n^+(M,N)$$

$$f \longmapsto [?] \quad C \circ f$$

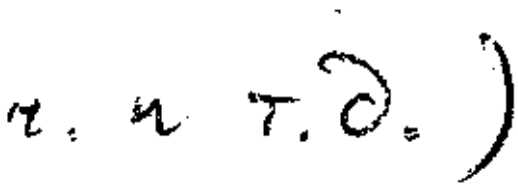


Original source details: GELFAND-P000385

Undeciphered passage. Before $C \circ f$ a formula is crossed repeatedly by horizontal and oblique strokes. Its whole earlier reading has not been recovered.

(Check that this is a module hom.:

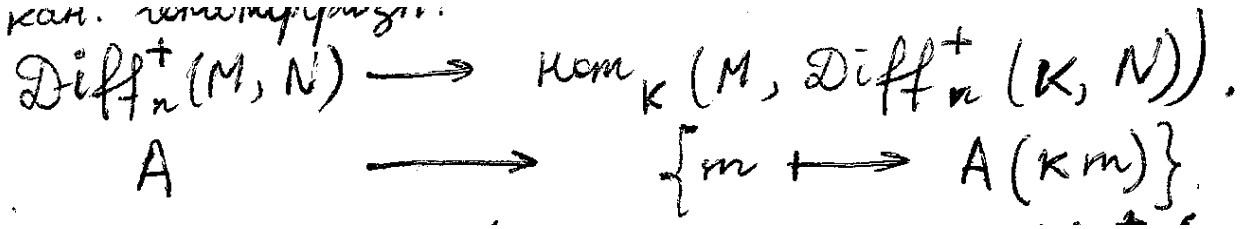
$$kf \longmapsto C \circ kf = C \circ fk = k^+(C \circ f), \quad \text{as req.})$$



Original source details: GELFAND-P000385

Manuscript notation. The calculation ends with the customary Russian abbreviation for “as required”.

Lemma 5. A canon. homomorphism exists:



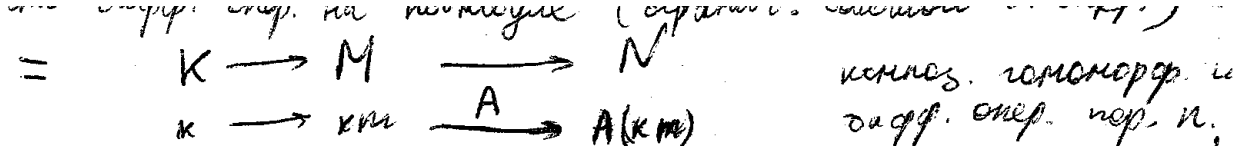
Original source details: GELFAND-P000385

Diagram and typed description.

$$\mathrm{Diff}_n^+(M,N) \longrightarrow \mathrm{Hom}_K(M, \mathrm{Diff}_n^+(K,N))$$

$$A \longrightarrow \{m \longmapsto A(km)\}$$

Proof. 1) Check that $A(km)$ (m is fixed) $\in \mathrm{Diff}_n^+(K,N)$ — — this is a diff. op. on a submodule (restrict. of the larger diff. op.) $=$



Original source details: GELFAND-P000385

Diagram and typed description.

$$K \longrightarrow M \longrightarrow N$$

$$k \longrightarrow km \xrightarrow{A} A(km)$$

canon. hom. and diff. op. of ord. n.

2) Check that $g : m \longmapsto A(km)$ is $\mathrm{Hom}_K(M, \mathrm{Diff}_n^+(K,N))$.



Original source details: GELFAND-P000385

Uncertain reading. The letter before m is densely overwritten, with a loop and lower crossing strokes. Provisionally g; its earlier form is not established.

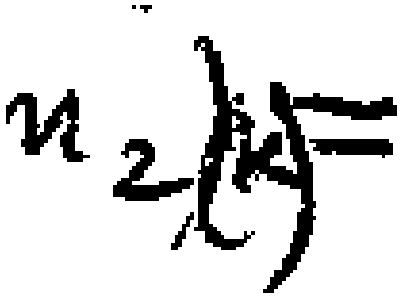
$g \ldots$



Original source details: GELFAND-P000385

Uncertain reading. Before the additivity line is a cancelled earlier prefix with a g-like beginning. Its argument strokes overlap; the complete earlier expression has not been recovered.

$$g(m_1 + m_2)(k) = A(km_1 + km_2) = A(km_1) + A(km_2) = g(m_1)(k) + g(m_2)(k).$$

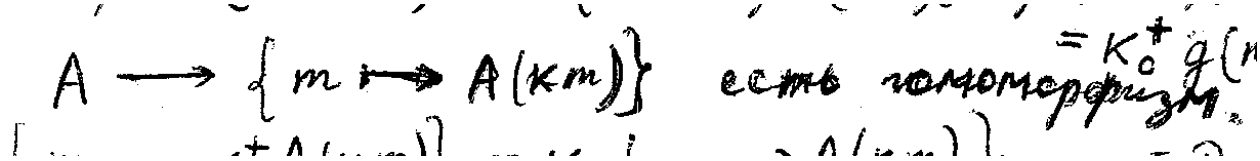


Original source details: GELFAND-P000385

Uncertain reading. After m_2 the right argument boundary and the following (k) are densely reinforced and intersect. Provisionally $g(m_1 + m_2)(k)$; possibly $g(m_1 + m_2, k)$.

$$g(k_0 m)(k) = A(kk_0 m) = A(k_0 km) = (Ak_0)(km)$$
$$= (k_0^+ A)(km) = k_0^+ g(m).$$

3) Check that

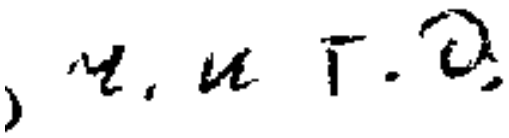


Original source details: GELFAND-P000385

Diagram and typed description.

$A \longrightarrow \{m \longmapsto A(km)\}$ is a homomorphism.

$$k^+ A \longrightarrow \{m \longrightarrow k^+ A(km)\} = k\{m \longrightarrow A(km)\}, \quad \text{as req.}$$



Original source details: GELFAND-P000385

Manuscript notation. The calculation ends with the customary Russian abbreviation for “as required”.

Lemma 6. The constructed maps (the consequence of lemma 4 and lemma 5) are mutually inverse and give a module isom.

$$\mathrm{Diff}_n^+(M, N) \simeq \mathrm{Hom}_K(M, \mathrm{Diff}_n^+(K, N)).$$

Proof. [?]



Original source details: GELFAND-P000385

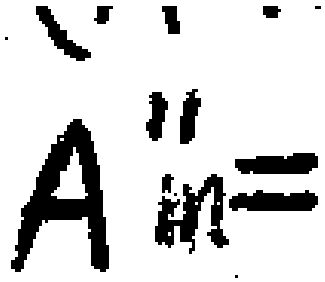
Undeciphered passage. Between “Proof” and “Let” a word is repeatedly cancelled by long horizontal strokes. The whole word has not been recovered.

Let $A \in \mathrm{Diff}_n^+(M,N)$, construct $A' \in \mathrm{Hom}_K(M, \mathrm{Diff}_n^+(K,N))$:

$$(A'm)(k) = A(km).$$

Now from A' construct $A'' \in \mathrm{Diff}_n^+(M,N)$ by the formula

$$A''_m = C \circ A'm = (A'm)(1) = A(m), \quad \text{i.e. } A'' = A$$



Original source details: GELFAND-P000385

Uncertain reading. Below the first A'' is a small reinforced m-like entry. Provisionally a lower m; the earlier strokes are not established. The following $A'm$ is rendered without adding a lower index.

Conversely, let $A' \in \mathrm{Hom}_K(M, \mathrm{Diff}_n^+(K,N))$. Construct $A'' \in \mathrm{Diff}_n^+(M,N)$



Original source details: GELFAND-P000385

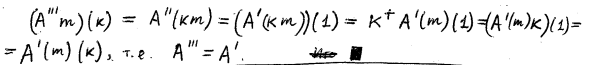
Uncertain reading. In the final $\mathrm{Diff}_n^+(M,N)$ the first argument is densely reinforced. Provisionally M; the complete earlier form underneath has not been recovered.

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[source number: 35]

by the formula $A''m = (A'm)(1)$ and then $A''' \in \mathrm{Hom}_K(M, \mathrm{Diff}_n^+(K,N))$ as

$$(A'''m)(k) = A''(km) = (A'(km))(1) = k^+A'(m)(1) = (A'(m)k)(1) = \\ = A'(m)(k), \quad \text{i.e. } A''' = A'.$$



Original source details: GELFAND-P000386

Manuscript notation. The equals sign is written at the end of the first line and again at the start of the second.

~~[?]~~ ■



Original source details: GELFAND-P000386

Undeciphered passage. After $A''' = A'$ and before the black square, a short word is cancelled by several horizontal strokes. The whole earlier word has not been recovered.

Thus, it is proved that

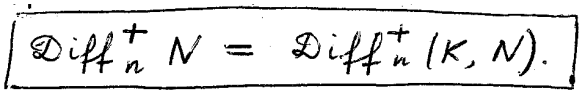


Diagram and typed description.
$\mathrm{Diff}_n^+ N = \mathrm{Diff}_n^+(K, N).$

Original source details: GELFAND-P000386

GELFAND-P000387

15/XI. I. M. Gelfand.

G is a Lie group, X a sp., G acts transitively on X (X is a homogeneous sp.).

Principal homog. sp.: G acts on X without fixed pts. ($xg = y \Rightarrow g$ is unique).

The same in terms of the Lie algebra:

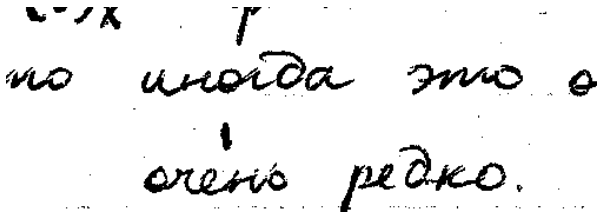
Let a Lie algebra $\mathfrak{g}$ act on X.

$$\xi \in \mathfrak{g} \rightsquigarrow \text{a vector field on } X; \quad a(\xi),$$
$$\xi_1 + \xi_2 \rightsquigarrow a(\xi_1) + a(\xi_2);$$
$$[\xi_1, \xi_2] \longrightarrow [a(\xi_1), a(\xi_2)].$$

Homogeneous sp.: if $\mathfrak{g} \longrightarrow TX_x$ epim. $\forall x$ ($a(\xi)_x$ generate all of X).

(One can prove that sometimes this means ordinary homogeneity)

very rarely.



Manuscript notation. A vertical stroke below "sometimes" leads to the annotation "very rarely".

Original source details: GELFAND-P000387

When can one extend[?] to a group:



Uncertain reading. The final cursive stroke is provisionally read as п in «доп.», probably short for "extend".

Original source details: GELFAND-P000387

when $\mathfrak{g}$ is finite-dimensional.

Principal homog. sp.: map $\mathfrak{g} \longrightarrow TX_x$ isomorphism.

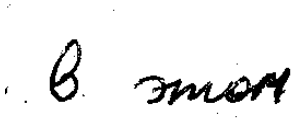
Ordinary case:

fix x_0, one gets $x_0 g = y$. $g \leftrightarrow y$, $X \cong G$ and the transf. is a right translation.

If there is no group, there may be many of them and different ones!

(All of them are candidates for the title of group)

Principal homog. sp. is now understood in this[?] sense.



Uncertain reading. In the word before "sense," the final letters are densely reinforced and intersect earlier strokes. Provisionally "этом"(this); the whole earlier form has not been recovered.

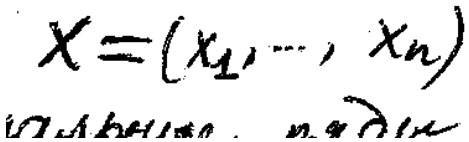
Original source details: GELFAND-P000387

Example.

W_n — formal vect. fields;

$$\sum_{i=1}^{n} a_i(x)\frac{\partial}{\partial x_i}$$

$$X = (x_1, \ldots, x_n)$$



Original source details: GELFAND-P000387

Uncertain reading. The initial cross-shaped sign on the coordinate line is noticeably larger than the components x_i. Provisionally X; possibly a large handwritten x. The case distinction is not finally established.

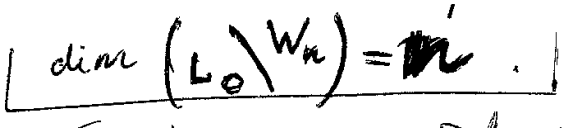
$a_i(x)$ — formal series (power series).

GELFAND-P000388

[source number: 36]

The group does not exist: one cannot move.

In W_n there is a subalgebra $L_0 \subset W_n$, $a_i(0) = 0$ — const. term $= 0$.



Original source details: GELFAND-P000388

Diagram and typed description.

$$\dim(L_0 \backslash W_n) = n.$$



Original source details: GELFAND-P000388

Uncertain reading. After the equals sign, the letter is densely reinforced, especially at left. Provisionally n; the complete earlier strokes have not been recovered.

The algebra L_0 can be brought to a group.



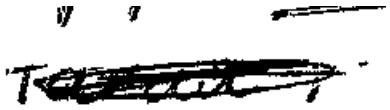
Original source details: GELFAND-P000388

Diagram and typed description. A circled mark: two roughly parallel oblique strokes and a dark spot in the middle. Its number or symbol is not established.



$y = \varphi(x)$ — formal automorphisms, 0 stays in place.

Its Lie alg. ~~then[?]~~ is L_0.



Original source details: GELFAND-P000388

Uncertain reading. Between “Its Lie alg.” and “is L_0” a word is cancelled by several strokes. Provisionally “тогда”(then); the complete earlier reading is not finally established.

Consider W_n, its gr. of automorphisms, its Lie alg. is L_0.

Thm: the Lie alg. of the automorphism gr. of W_n is L_0.

(In the classical case one gets the quotient gr. [?]

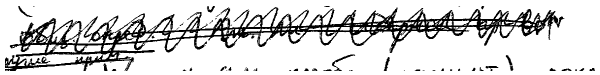


Original source details: GELFAND-P000388

Undeciphered passage. Between “quotient gr.” and “alg.” is a short densely cancelled fragment. Its whole content has not been recovered.

of the alg. by the centre).

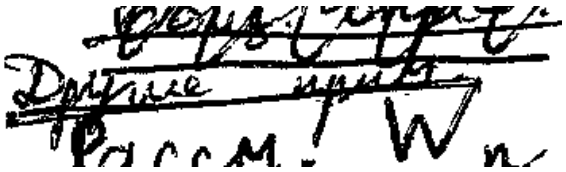
[?]



Original source details: GELFAND-P000388

Undeciphered passage. A long line above the lower cancelled heading is covered by repeated loops and horizontal strokes. The complete earlier line has not been recovered.

~~Other ex.~~[?]



Original source details: GELFAND-P000388

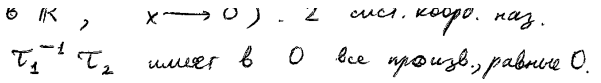
Uncertain reading. After “Other”, the abbreviation is provisionally «прим.» (ex.). The descending p-loop and adjacent strokes are visible. A cancellation line crosses some letters.

Consider W_n and sympl. transf. (Hamiltonian), volume-preserving and so on.

Construction of a principal homog. sp. for W_n.

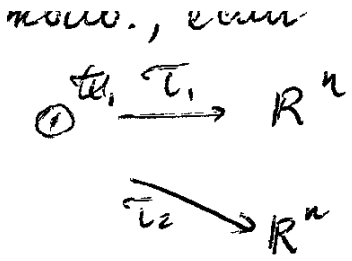
M — a sp. (arbitrary) — a finite-dimensional manifold, $x \in M$.

Consider at pt. x jets of coordinate systems (a coord. syst. is a map of a nbd. U_x into R^n, $x \longrightarrow 0$). 2 coord. systs. are called equiv. if $\tau_1^{-1}\tau_2$ has at 0 all derivs. equal to 0.



Original source details: GELFAND-P000388

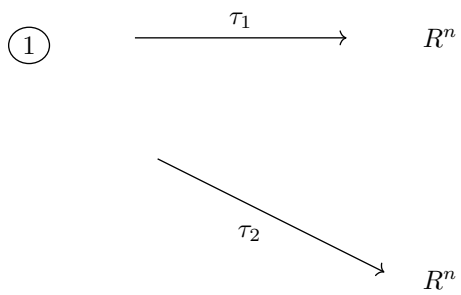
Manuscript notation. The manuscript has $\tau_1^{-1}\tau_2$ and “all derivs. equal to 0”.



Original source details: GELFAND-P000388

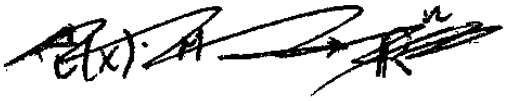
Diagram and typed description. Two separate arrow starts: the upper arrow points right with label τ_1 to R^n; the lower arrow slopes down and right with label τ_2 to a second R^n. Above the upper start is provisionally U_1, with a circled digit 1 below. A cursive stroke crosses the upper letter; no connection between starts is supplied.

U_1[?]



$\xi \in W_n$ acts there.

$\xi\tau(x)$ — another formal coord. syst.



Original source details: GELFAND-P000388

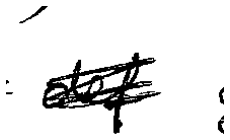
Diagram and typed description. An earlier cancelled scheme: provisionally $\tau(x)$, then A, a rightward arrow and R^n. Long loops and oblique strokes cross the labels and arrow. The exact earlier beginning has not been recovered.



$X = S(M)$ — the sp. of all jets of coord. systems (at all points of M).

Theorem: $S(M)$ — a principal homog. sp. for W_n.

dim M = ~~def[?]~~ defect W_n (by how much it falls short).



Original source details: GELFAND-P000388

Uncertain reading. After dim $M =$ a short earlier beginning is cancelled by horizontal and oblique strokes. Provisionally “деф”(def); the complete earlier form has not been recovered.



Original source details: GELFAND-P000388

Uncertain reading. In the current word “дефект”(defect), the middle letters are densely overwritten. The word itself is legible; all earlier strokes underneath are not established.

It is unclear whether there are other substantive principal homog. spaces.

GELFAND-P000389

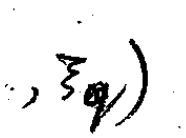
Let G act on X, ω — an invar. diff. form.

It suffices to specify it at the 1st point. Right-inv. forms correspond to forms on G.
Differential = diff. in the cohomology of the Lie algebra $\mathfrak{g}$.

W_n — a Lie alg., $X = S(M)$.

$f(\xi_1, \dots, \xi_q)$ — a multilin. skew-symm. form on $W_n \rightsquigarrow$

$\rightsquigarrow$ an invar. form on $S(M)$.

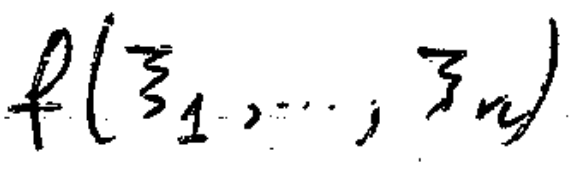


Original source details: GELFAND-P000389

Uncertain reading. The subscript of the last ξ in the first f is densely reinforced and crossed by earlier strokes. Provisionally q; possibly n. The whole earlier subscript form has not been recovered.

$x, D_1, \dots, D_q$ — vectors at the point $x \in X$.

corresponding to them are vectors $\xi_1, \dots, \xi_q \in W_n$. One gets $f(\xi_1, \dots, \xi_n)$ — a form on $S(M)$.



Original source details: GELFAND-P000389

Manuscript notation. The last subscript here is n. The earlier line provisionally reads q.

Right-invar. one can give a definition.

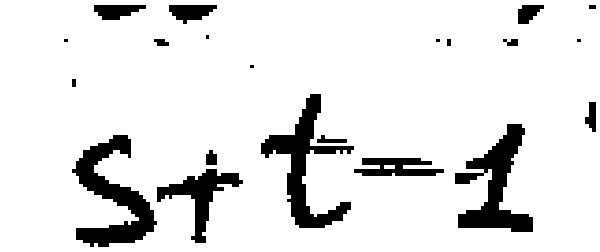


Original source details: GELFAND-P000389

Uncertain reading. Several initial letters in “Right-invar.” are densely overwritten. The current reading is provisionally retained; the whole earlier strokes are not established.

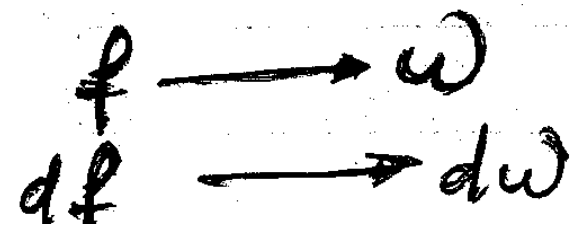
$$(df)(\xi_1,\dots,\xi_{q+1}) = \sum_{s<t}(-1)^{s+t-1}$$

$$f([\xi_s,\xi_t],\xi_1,\dots,\hat{\xi}_s,\dots,\hat{\xi}_t,\dots,\xi_{q+1}).$$



Original source details: GELFAND-P000389

Uncertain reading. The horizontal strokes nearly merge. They are provisionally read as a single retraced minus sign.



Original source details: GELFAND-P000389

Diagram and typed description.

$f \longrightarrow \omega$

$df \longrightarrow d\omega$

($d\omega$ — the ordinary de Rham diff.

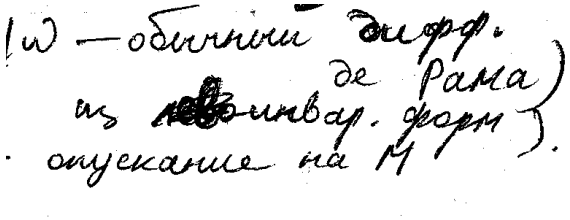
from ~~left[?]~~ invar. forms

Tricks of diff. geometry — lowering to M.



Original source details: GELFAND-P000389

Uncertain reading. Before “invar. forms” is a cancelled joined beginning with a tall loop. Provisionally “лево”(left); the whole earlier entry is not established.



Original source details: GELFAND-P000389

Uncertain reading. At right are two joined curves: the upper is beside the de Rham line; the lower is shorter, beside the invar. forms and lowering lines. Below and right is a separate period. The exact semantic attachment of the curves is not established.

(classes and so on).

see note in DAN, I. M. Gelfand and Kazhdan.

I. Bernstein.



Diagram and typed description. At right in the source raster is a digital yellow comment icon with a black outline, a lower tail and three horizontal bars.

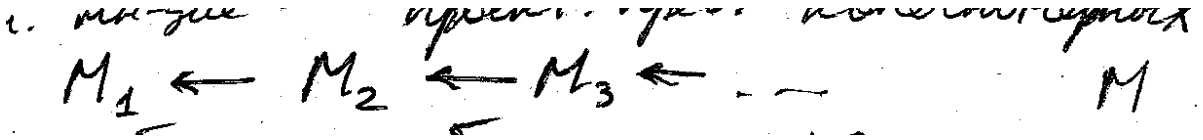


Original source details: GELFAND-P000389

Consider ∞-dimensional manifolds.

Definition:

An ∞-dim. manifold is a proj. limit of finite-dimensional ones

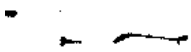


Original source details: GELFAND-P000389

Diagram and typed description.

$M_1 \longleftarrow M_2 \longleftarrow M_3 \longleftarrow \; - \; \sim \qquad\qquad M$

Uncertain reading. After the last arrow are a short point-like stroke and a separate low wavy stroke. The exact continuation mark is not established; it is not normalized to three dots.



Original source details: GELFAND-P000389

All maps are onto maps.

Ring of functions on M: ~~A[?]~~

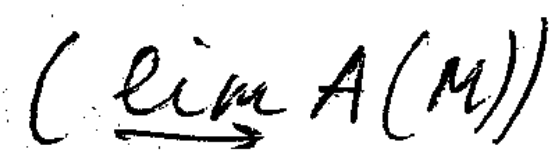
Uncertain reading. After the ring-of-functions heading, a single letter-like sign is cancelled. Provisionally A; the whole earlier form is not established.



Original source details: GELFAND-P000389

$$A(M) = \bigcup A(M_i) \qquad (\varinjlim A(M))$$

Manuscript notation. A right arrow is written below lim. Its argument is M, with no subscript.

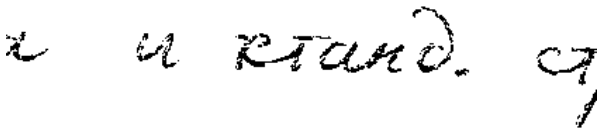


Original source details: GELFAND-P000389

GELFAND-P000390

[source number: 37]

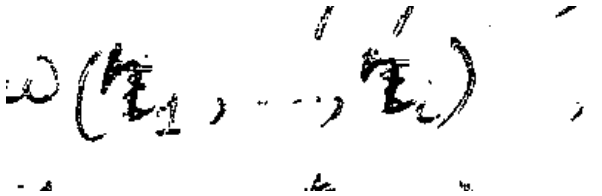
C^∞ — a vector field, C^∞ — a diff. form, and in the std.[?] way one constructs the de Rham compl.



Original source details: GELFAND-P000390

Uncertain reading. The ending of the abbreviation matches the «станд.» (std.) written lower on the page. The initial stroke is irregular, so this reading remains provisional.

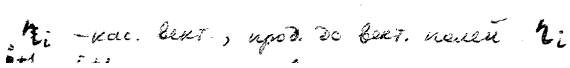
ω — a form; $\omega(\xi_1, \ldots, \xi_i)$,



Original source details: GELFAND-P000390

Uncertain reading. Two argument letters are reinforced over earlier strokes. Provisionally $\xi_1, \ldots, \xi_i$; possibly $\eta_1, \ldots, \eta_i$. The complete earlier outlines have not been recovered.

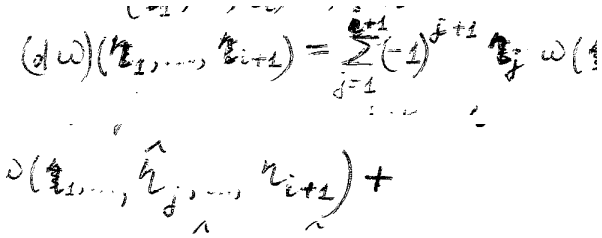
ξ_i — a tangent vect., extended to vector fields η_i.



Original source details: GELFAND-P000390

Uncertain reading. The initial letter with subscript i in the tangent-vector phrase is reinforced. Provisionally ξ_i; possibly η_i. The clean final η_i is rendered separately; the earlier strokes of the initial letter are not established.

$$(d\omega)(\xi_1, \ldots, \xi_{i+1}) = \sum_{j=1}^{q+1} (-1)^{j+1} \xi_j \omega(\xi_1, \ldots, \hat{\eta}_j, \ldots, \eta_{i+1}) + \\ + \sum_{j<k} (-1)^{j+k} \omega([\eta_j, \eta_k], \eta_1, \ldots, \hat{\eta}_j, \ldots, \hat{\eta}_k, \ldots, \eta_{i+1}). \qquad (1)$$



Original source details: GELFAND-P000390

Uncertain reading. In the first row of (1), the two letters at left, the upper summation limit, the letter before ω and the first argument of ω are reinforced. Provisionally $\xi_1, \ldots, \xi_{i+1}$ at left, $q+1$ above, then ξ_j and ξ_1; the corresponding η and $i+1$ are possible. The clean η entries in the tail and second row are retained. The complete earlier strokes have not been recovered.

Using, $S(M)$ we shall construct the Chern classes of M.

$\mathfrak{g}$ — an inf.-dimensional Lie alg. (proj. lim of finite-dimensional ones with the corresp. topol.).

X — a principal homog. sp.

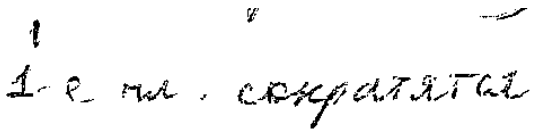
Construct the complex $C(\mathfrak{g})$

$C(\mathfrak{g}) = \{C^i(\mathfrak{g})\}$. — the standard complex of the algebra $\mathfrak{g}$.

$C(\mathfrak{g}) \longrightarrow \Omega(X)$ — the de Rham compl. of X.

A homomorphism (commuting with d) — by formula (1).

The 1st terms will cancel



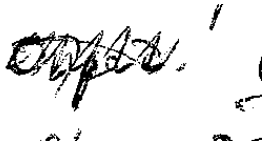
Original source details: GELFAND-P000390

Manuscript notation. The note under the short vertical stroke at right reads “The 1st terms will cancel”.

One gets a map.

$$H^i(\mathfrak{g}) \longrightarrow H^i(X)$$

How does one constr. Chern classes?
~~Def[?]...~~

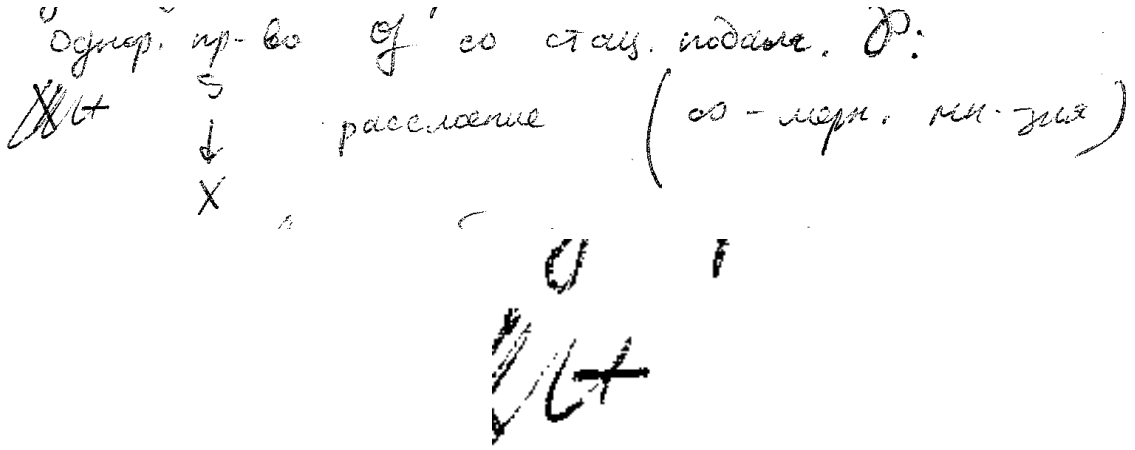


Original source details: GELFAND-P000390

Uncertain reading. Before the new heading is a cancelled earlier beginning. Provisionally “Опр...”(Def...); possibly “Однор...”(Homog...). The ending has not been recovered or completed from the following heading.

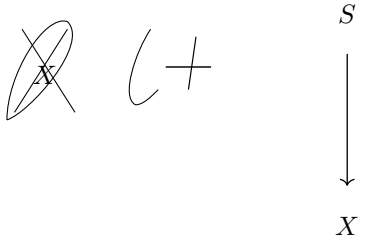
Definition.

$\mathfrak{g} \supset \mathfrak{p}$ — a subalgebra.
A homog. sp. of $\mathfrak{g}$ with isotropy subalg. $\mathfrak{p}$.



Original source details: GELFAND-P000390

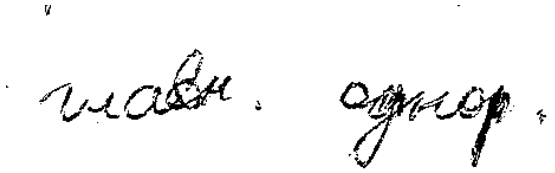
Diagram and typed description. To the left of the current vertical projection from S to X is a cancelled earlier X; beside it are an open hook-like stroke and a separate upper-right crossing resembling a plus. The exact earlier sign is not established. ι^+-, τ^+-or t^+-like forms are possible; the earlier $\mathfrak{g}/\mathfrak{p}$ hypothesis is not confirmed by a clear fraction bar.



a fibration (∞-dim. manifolds)

On tangent spaces the maps are onto

S — a princ. homog. $\mathfrak{g}$-sp. and if $s \in S$,

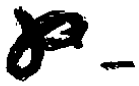


Uncertain reading. In the abbreviations “главн.”(princ.) and “однор.”(homog.), final and middle letters are overwritten and crossed by earlier strokes. The current reading is provisionally retained; the complete earlier outlines are not established.

Original source details: GELFAND-P000390

GELFAND-P000391

$\mathcal{F}_s$ — the fibre of S at pt. s, then $\xi \in \mathfrak{p}$ passes to vector fields, tangent to the fibre, and at the same time $\mathcal{F}_s$ — a principal homogeneous $\mathfrak{p}$-space.



Uncertain reading. The letter before “-space” is densely overwritten. Provisionally $\mathfrak{p}$; possibly $\mathfrak{g}$. The complete original outline is not established.

Original source details: GELFAND-P000391

X — a principal homog. sp.

Chain complex $C(\mathfrak{g})$

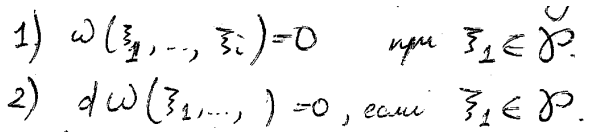
Rel. chain compl. $C(\mathfrak{g}, \mathfrak{p})$.

$$C^i(\mathfrak{g}, \mathfrak{p}) \subset C^i(\mathfrak{g})$$

$\omega(\xi_1, \ldots, \xi_i)$ — a form on $\mathfrak{g}$, belonging to $C^i(\mathfrak{g}, \mathfrak{p})$, if

1) $\omega(\xi_1, \ldots, \xi_i) = 0$ for $\xi_1 \in \mathfrak{p}$.

2) $d\omega(\xi_1, \ldots,) = 0$, if $\xi_1 \in \mathfrak{p}$.



Manuscript notation. In the second condition a comma follows the ellipsis just before the closing parenthesis.

Original source details: GELFAND-P000391

(ω invar. with respect to ~~elem[?]~~ the adjoint repr. of elem. of $\mathfrak{p}$).



Uncertain reading. The cancelled word after “invar. with respect to” is provisionally “элем”(elem.). The full earlier reading is not established.

Original source details: GELFAND-P000391

The forms ω — a complex.

Cohomology — cohom. of the quotient gr. (in the classical case).

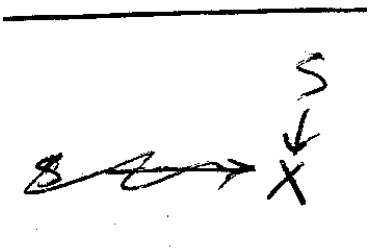
Let there be a homog. $\mathfrak{g}$-space, $\mathfrak{p}$ — an isotropy subalg.

$\exists$ a map

$$C(\mathfrak{g}, \mathfrak{p}) \longrightarrow \Omega(X).$$

Construction:

$\eta \in T_x(X)$,



Original source details: GELFAND-P000391

Diagram and typed description. The current vertical projection is from S down to X. At left is an earlier horizontal right arrow to X with its beginning and small label crossed out. The beginning resembles s; the complete earlier letter and label are not established.



$\eta' \in T_s(S)$, $\eta' \longrightarrow \eta$.

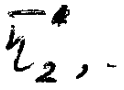


Original source details: GELFAND-P000391

Uncertain reading. The letter before $\in T_s(S)$ is densely overwritten; the separate upper-right stroke is read as a prime. Provisionally η'; possibly ξ'. The original letter is not established.

$\omega \in C(\mathfrak{g}, \mathfrak{p}) \longrightarrow$

$$\overrightarrow{\tilde{\omega}}(\eta_1, \dots, \eta_n) = \omega(\bar{\eta}'_1, \bar{\eta}'_2, \dots, \bar{\eta}'_n)$$



Original source details: GELFAND-P000391

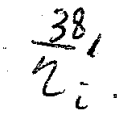
Uncertain reading. Above the second argument $\bar{\eta}_2$ is a dense crossing with a loop-like stroke. The prime in $\bar{\eta}'_2$ is provisional; $\bar{\eta}^*_2$ or a reinforced or overwritten upper stroke is possible. The exact upper mark is not established.

$\bar{\eta}'_i$ — the corresp. elements of the Lie alg. $\mathfrak{g}$.

GELFAND-P000392

[source number: 38]

There is arbitrariness, but $\tilde{\omega}$ does not depend on it — on the choice of $\bar{\eta}'_i$.

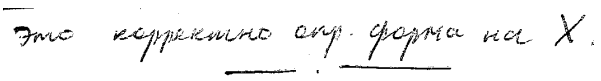


Original source details: GELFAND-P000392

Uncertain reading. The horizontal stroke immediately below page number 38 lies above η'_i. Provisionally $\bar{\eta}'_i$; possibly η'_i with only page number 38 underlined. The attachment of the horizontal stroke is not established.

It does not depend on the choice of point either.

This is a well-def. form on X.



Original source details: GELFAND-P000392

Manuscript notation. The line reads “a well-def. form on X”, in the singular.

M — a manifold, $S(M)$ — a princ. W_n-space (see above).

gl(n) — a subalg. of linear transformations

It acts on $S(M)$ and the action extends to the group

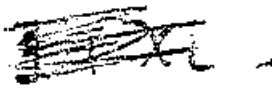
$V_x \longrightarrow V_0 \subset \mathbb{R}^n$, rotates.

Consider $S(M)/GL(n, R)$ corresp. to the pair $(W_n, \text{gl}(n))$.

This is already homotopy equiv. to M.

(in contrast to $S(M)$, which is homotopy equiv. to 1-frames).

~~[...]~~

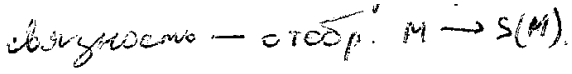


Original source details: GELFAND-P000392

Undeciphered passage. The short cancelled word before “There is a map” has not been recovered.

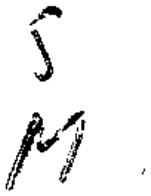
There is a map.

Connection[?] — a map $M \longrightarrow S(M)$.



Original source details: GELFAND-P000392

Uncertain reading. The word provisionally reads “Connection”, as in the lower passage. The written map is $M \longrightarrow S(M)$.



Original source details: GELFAND-P000392

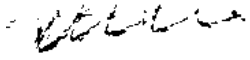
Diagram and typed description. Below the quotient is a separate curved vertical stroke and the label M. In the right margin above the formula are two vertical strokes, one with an upward arrow. Their position is retained; the mark below the quotient is not replaced by an equality assertion.

$$C(W_n, \mathfrak{gl}(n)) \longrightarrow \Omega(S(M)/GL(n,R))$$

M

Description of the cohom. of this alg:

$\psi_1, \psi_2, \ldots, \psi_n$[...ymi?] — generators of degree $2n$.

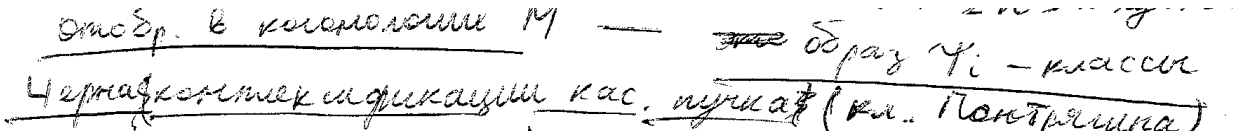


Original source details: GELFAND-P000392

Uncertain reading. After ψ_n are faint letter strokes. The earlier provisional ending “-ыми”(-ymi) is qualified; the whole word is not established.

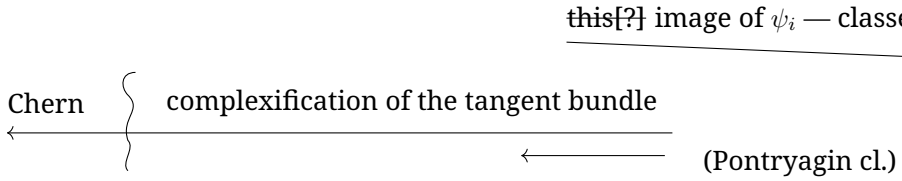
degrees $> 2n \Rightarrow$ one gets 0.

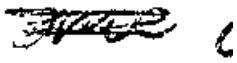
Map to the cohomology of M —



Original source details: GELFAND-P000392

Diagram and typed description. The line reads “Chern [classes] of the complexification of the tangent bundle”. A brace-like mark and arrowed lines link the parts of the entry.





Uncertain reading. Before “image of ψ_i — classes” is a cancelled word, provisionally “это”(this). The full earlier reading is not established.

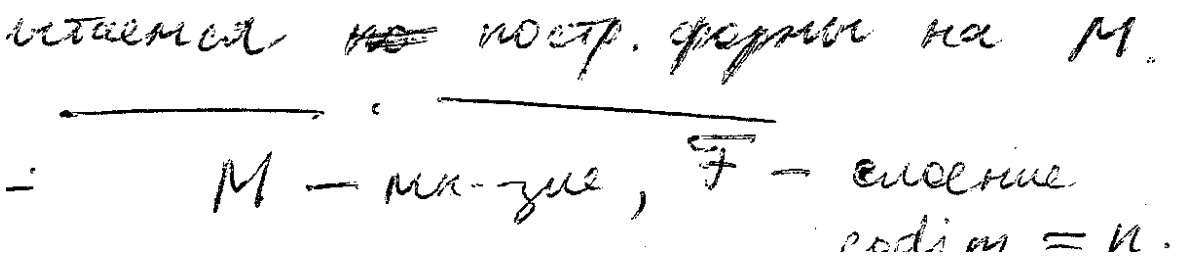
Original source details: GELFAND-P000392

A form on $S(M)/GL(n,R)$ is det. uniquely (invar.), nonuniqueness (choice of connection) appears, when we try ~~not~~[?] to constr. forms on M.



Uncertain reading. Before “constr.” is a short cancelled beginning, provisionally “не”(not). The crossings prevent recovery of the complete original reading.

Original source details: GELFAND-P000392



Original source details: GELFAND-P000392

Diagram and typed description. Above the following heading and line are a separate horizontal arrow with heads at both ends, a small open c-like stroke and a separate long sloping stroke above the foliation sign. Their meanings are not established.



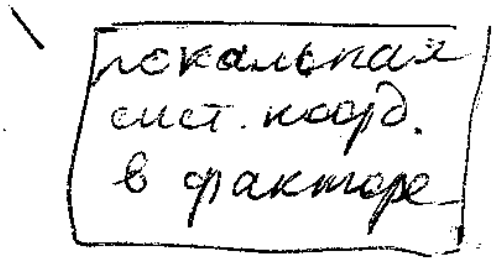
Bott thm. M — a manifold, $\mathcal{F}$ — a foliation, codim $= n$.

GELFAND-P000393

$S_{\mathcal{F}}(M)$. $x \in M$, $U_x \xrightarrow{\text{form.}} U_0 \subset \mathbb{R}^n$, having the properties

1) const. on the leaves; 2) an isom. on a transverse sect. (nondeg.)

(A formal germ of a map).



Original source details: GELFAND-P000393

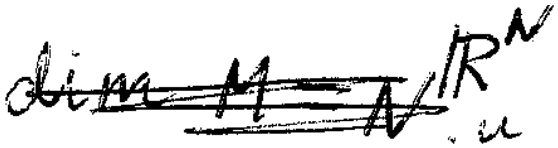
Diagram and typed description. The box reads “local coord. syst. in the quotient”. There is a short oblique stroke at left.

local
coord. syst.
in the quotient.

Consider $S_{\mathcal{F}}(M)/GL(n,R)$.

On $S_{\mathcal{F}}(M)$ no algebra acts.

Consider ~~dim $M \longrightarrow N$~~ $\mathbb{R}^N_n$, in it a standard foliation of codim n.



Original source details: GELFAND-P000393

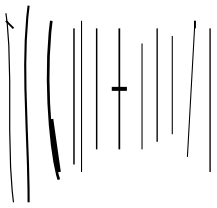
Uncertain reading. After the cancelled dim $M \longrightarrow N$ stands $\mathbb{R}^N$ with a small n lower right. The provisional positional notation is $\mathbb{R}^N_n$; the n may be separate. Its role is not established.

$W_{N,n}$ — the Lie alg. of vector fields preserving the foliation

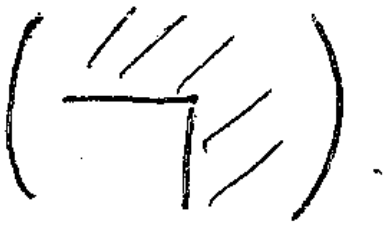


Original source details: GELFAND-P000393

Diagram and typed description. A row of nearly vertical strokes of different lengths and thicknesses, with dark thickenings. Intertwined reinforced strokes are rendered schematically. The exact number of separate lines and the attachment of the thickenings are not established; no points or particular leaves are assigned.

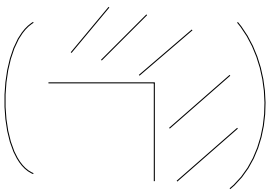


Linear transf.: :



Original source details: GELFAND-P000393

Diagram and typed description.



$S'_{\mathcal{F}}(M)$, $x \in M$, $U_x \overset{\text{form.}}{\longrightarrow} U_0 \subset \mathbb{R}^N$

leaves of the foliation $\longrightarrow$ leaves of the form. foliation.

$S'_{\mathcal{F}}(M)$ — a princ. homog. $W_{N,n}$-space. !

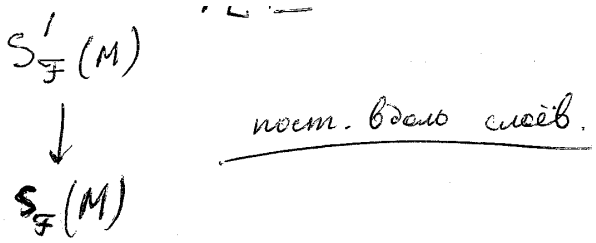
$W_{N,n} \supset L$ — preserving each leaf



Original source details: GELFAND-P000393

Diagram and typed description. Between the quotient and W_n are two reinforced wavy upper strokes and a lower nearly straight stroke. The form is rendered as separate strokes; a standard relation is not imposed.

$$W_{N,n}/L \approx W_n.$$



Original source details: GELFAND-P000393

Diagram and typed description.

$$\begin{array}{c} S'_{\mathcal{F}}(M) \\ \downarrow \\ S_{\mathcal{F}}(M) \end{array}$$

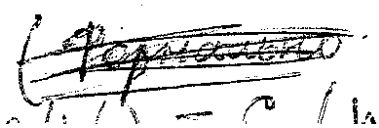
const. along the leaves.

The pair $(S'_{\mathcal{F}}(M), S_{\mathcal{F}}(M))$ — a homogeneous $(W_{N,n}, L)$-space.

GELFAND-P000394

[source number: 39]

~~(Formally[?]~~



Original source details: GELFAND-P000394

Uncertain reading. At the beginning an opening parenthesis and a word are cancelled, provisionally “Формально”(Formally). The complete earlier reading and superimposed strokes are not established.

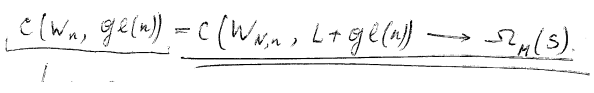
(the quotient group does not act on the quotient space)

$$C(W_n) = C(W_{N,n}, L) \longrightarrow \Omega(S_{\mathcal{F}}(M))$$

$S = S_{\mathcal{F}}(M)/GL(n, R)$ — homotopy equiv. to M.

$(S'_{\mathcal{F}}(M), S)$ — a homog. princ. sp. relative to $(W_{N,n}, L + \mathfrak{gl}(n))$

$$C(W_n, \mathfrak{gl}(n)) = C(W_{N,n}, L + \mathfrak{gl}(n)) \longrightarrow \Omega_M(S).$$



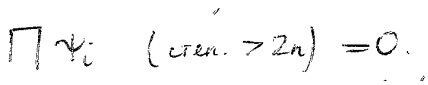
Original source details: GELFAND-P000394

Manuscript notation. The right side of the formula is doubly underlined. A short separate stroke lies below the left side.

these are known (generators) $\psi_1, \ldots, \psi_n$ — Izv. A. N., 1971, Gelfand, Fuks.

$T(M)/T(\mathcal{F})$ — Chern classes — images of the cl. $\psi_1, \dots, \psi_n$.

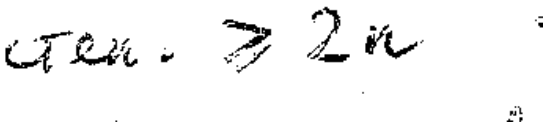
$$\prod \psi_i \quad (\text{deg. } > 2n) = 0.$$



Original source details: GELFAND-P000394

Manuscript notation. The displayed formula has deg. $> 2n$.

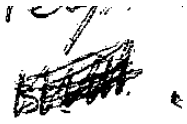
Hence for $T(M)/T(\mathcal{F})$ a product of Chern cl. deg. $\geqslant 2n$ must be 0!! — this is Bott's thm. —



Original source details: GELFAND-P000394

Uncertain reading. Below $>$ is a separate sloping stroke, provisionally rendered as $\geqslant$. The preceding display has $>$ without that stroke.

— an obstruction to being homotopically integrable [...]



Original source details: GELFAND-P000394

Undeciphered passage. The densely cancelled word after "integrable" has not been recovered.

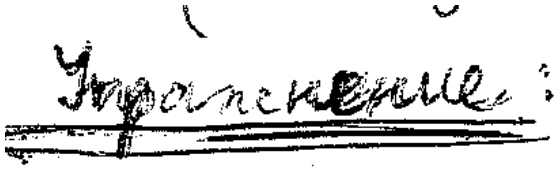
. . .

Arnold.

$$\frac{dw}{dz} = \frac{P(z,w)}{Q(z,w)} \quad \text{on } CP^2 \setminus \text{several singular points.}$$

One gets an invariant of the foliation. (on 3-dimensional complex manifold: point — vector)

Arnold's question — characterize geometrically, prove integrality (rationality) (a generalization of Poincaré rotation numbers).



Original source details: GELFAND-P000394

Uncertain reading. The current "Exercise" is reinforced over earlier strokes. The complete earlier outlines have not been recovered. The two lines beneath the heading with left ends are rendered separately.

Exercise:

define rotation numbers as such an invariant.

GELFAND-P000395

[source number: 43]

14 February.

L. Vaserstein.

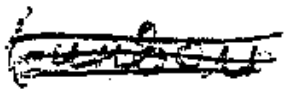
A — an assoc. ring with 1. (not necessarily commut.). V — a module over A. (right).

A quadr. form over a comm. ring — in A^n — is $A^n \longrightarrow A$, written in the form $\sum a_{ij}x_ix_j$. The matrices are det. nonuniquely.

$\exists\, 2^{-1} \Rightarrow$ there is a one-to-one corresp. with symm. matrices.

——— · ———

An involution on A: $A \longrightarrow A$.

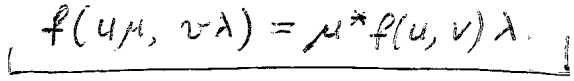


Original source details: GELFAND-P000395

Undeciphered passage. Below the arrow a short word is cancelled by horizontal and slanting strokes. Separate loops are visible; the full reading is not established.

$$(x-y)^* = x^* - y^*; \qquad (xy)^* = y^*x^*; \qquad (x^*)^* = x.$$

A sesquilin. form: $f(u,v) \in A$, $u, v \in V$; additive in all variables;



Original source details: GELFAND-P000395

Diagram and typed description.

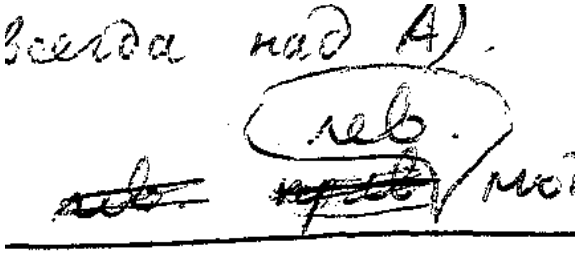
$f(u\mu, v\lambda) = \mu^* f(u,v)\lambda.$

(see Bourbaki). (There are no bilin. forms over a noncomm. ring)

$\mathrm{Hom}(V, V^*)$ — these are $1,5$-lin. forms.

$V^* = \mathrm{Hom}_A(V, A)$ (of two right modules).

(Hom is always over A).



Original source details: GELFAND-P000395

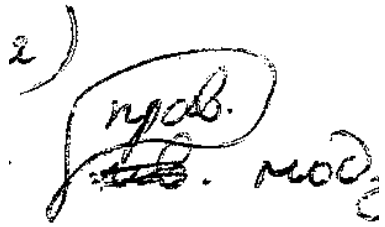
Uncertain reading. Two earlier abbreviations are cancelled. Provisionally “left.; right.”; the complete earlier strokes are not established. Above the second, “left.” is clearly inserted and circled; its pointer crosses the earlier word.

left.

V^* ~~left.[?]~~ ~~right.[?]~~ module.

(generally speaking).

But the involution makes it poss. to consider V^* as



Original source details: GELFAND-P000395

Uncertain reading. Before “module” an earlier abbreviation is cancelled, provisionally “left.”. The complete earlier strokes are not established. Above it “right.” is clearly inserted and circled, with a pointer down and left.

right.

~~left.[?]~~ module.

$$a \longmapsto f(a); \qquad \lambda f(a); \qquad a \longmapsto \lambda f(a).$$

$L \in \mathrm{Hom}(V, V^*)$ (a hom. of right A-modules).

$$f(u, v) = L(u)v.$$

GELFAND-P000396

A quadr. form — not quite that.



Undeciphered passage. At left before the ring line a word is densely cancelled. A large initial stroke and separate loops are visible; the complete word is not established.

Original source details: GELFAND-P000396

A — a ring, $*$ — an invol., $\varepsilon \in$ the centre of A, $\varepsilon \cdot \varepsilon^* = 1$. ($\varepsilon$ — the symmetry of the quadr. form). (most often one encounters $\varepsilon = 1$ or -1).
Λ — a subgroup of the additive gr. of the ring, having the properties:

$$\{x - \varepsilon x^*\}_{x \in A} \subset \Lambda \subset \{x = -\varepsilon x^*\}_{x \in A};$$
$$a^* \lambda a \in \Lambda \quad \text{for } \lambda \in \Lambda \text{ and } a \in A.$$

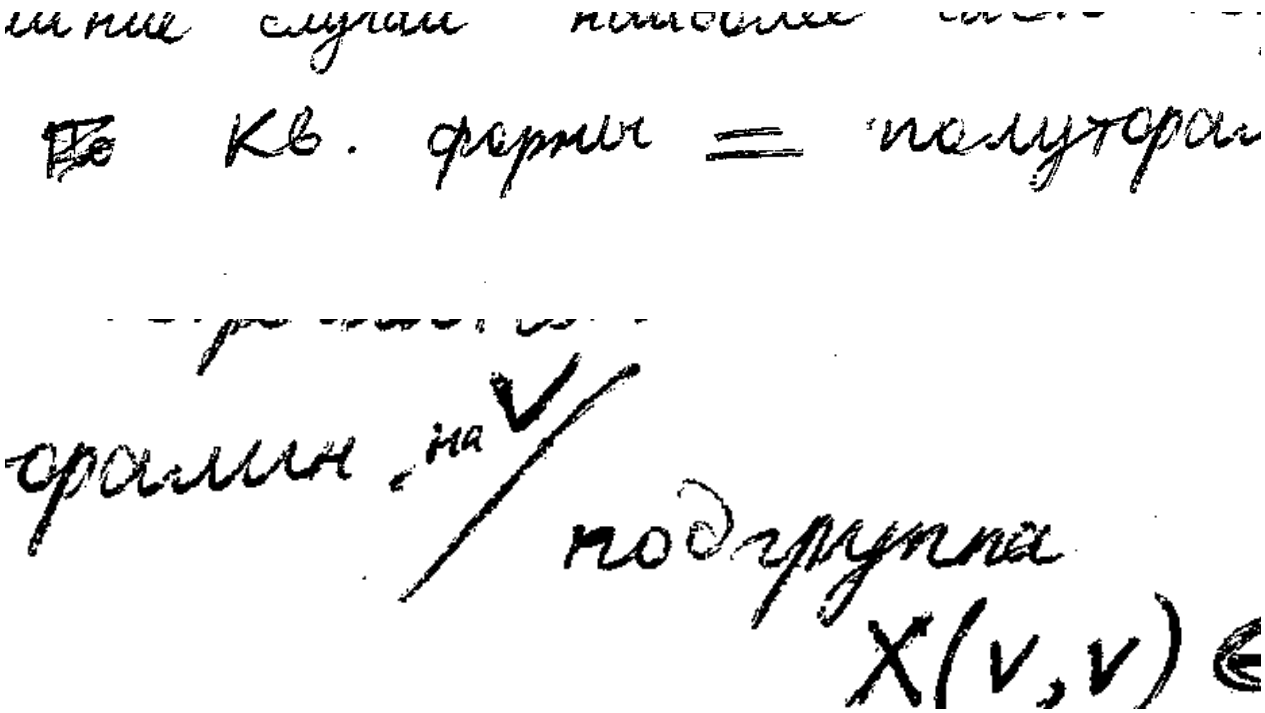


Original source details: GELFAND-P000396

Uncertain reading. At right below “properties” are a separate open bracket-like stroke and a small looped sign. “э” or “д” is possible; no verbal reading is established.



The extreme cases occur most often.



Original source details: GELFAND-P000396

Diagram and typed description. The slanting stroke beside "on V" descends to the left towards the conditions on X.

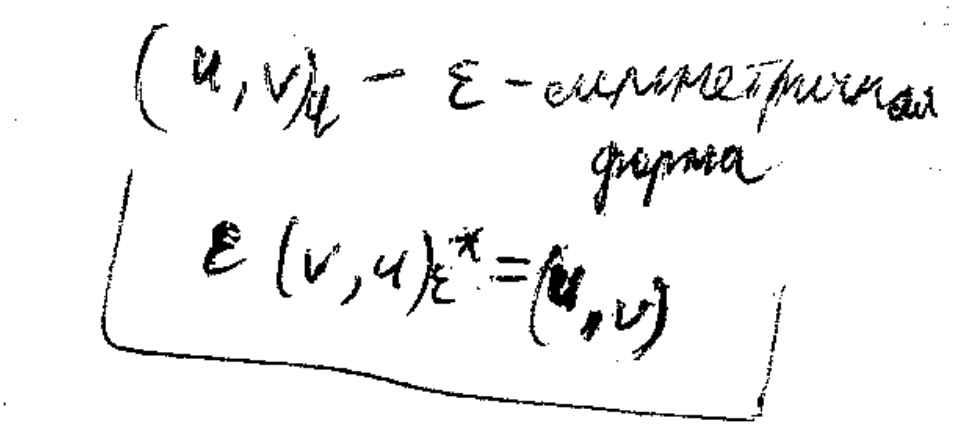
~~I.e.~~ Quadr. forms $\equiv$ sesquilin. on V / subgroup:

$$X(v,v) \in \Lambda; \qquad X = -\varepsilon X^*,$$
$$\text{i.e. } X(u,v) + \varepsilon X(v,u)^* = 0.$$

From a quadr. form q one constr. a scalar product.

$$(u,v)_q = Q(u,v) + \varepsilon Q(v,u)^* \qquad \text{(linearization)}.$$

$Q \in q$; (does not depend on the choice of representative). It is a $1/2$-lin. form.

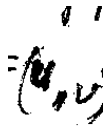


Original source details: GELFAND-P000396

Diagram and typed description. At right are two lines with a frame open at the top; in the lower line the left subscript is ε and the right pair has no subscript.

$(u,v)_q$ — an ε-symmetric form.

$$\varepsilon(v,u)^*_\varepsilon = (u,v)$$



Original source details: GELFAND-P000396

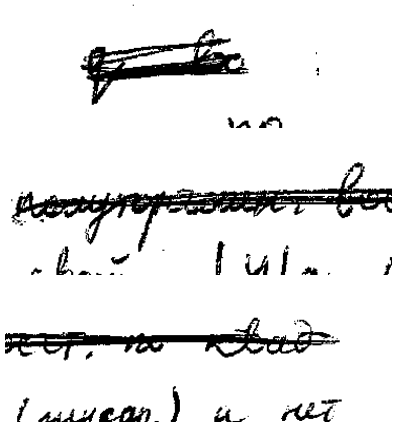
Uncertain reading. The first letter of the right pair in the lower line is reinforced by dense strokes. Provisionally u; the complete earlier strokes are not established.

Length map: $V \longrightarrow A/\Lambda$.

$$|u|_q = Q(u,u) + \Lambda.$$

(also does not depend on the representative)

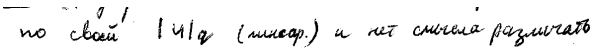
Λ max $\Rightarrow$ ~~[q?] [...] sesquilin. reconstr. from quadr[...]~~



Original source details: GELFAND-P000396

Undeciphered passage. After the arrow a short expression with a q-like sign and a phrase are cancelled. Parts reading “sesquilin.”, “reconstr.” and “from quadr...” are distinguishable; the complete earlier expression and ending have not been recovered.

quadr. is reconstr. from its $|u|_q$ (linear.) and there is no sense in distinguishing.



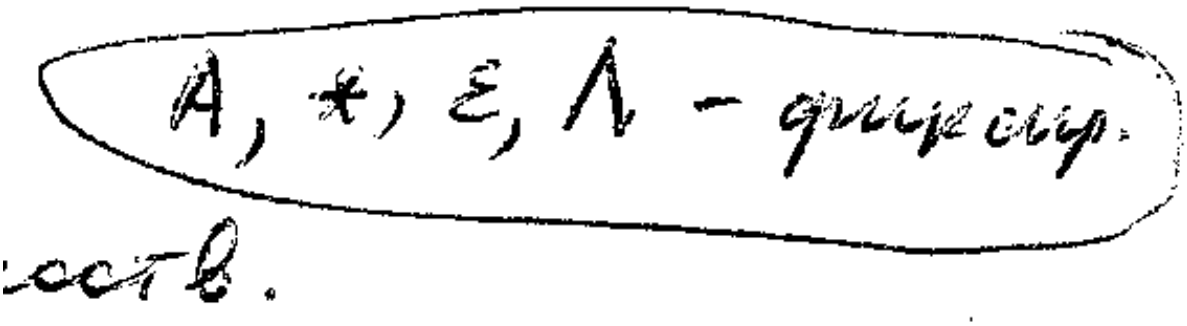
Manuscript notation. The last line has $|u|_q$.

Original source details: GELFAND-P000396

GELFAND-P000397

[source number: 44]

Example: (A Hamiltonian or hyperbolic quadr. form)



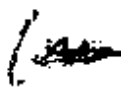
Original source details: GELFAND-P000397

Diagram and typed description. At right the parameters and the word “fixed” are circled.

$A, *, \varepsilon, \Lambda$ — fixed.

U — an A-module (any). $H(U) = U \oplus U^*$ with the quadr. form corresp. to

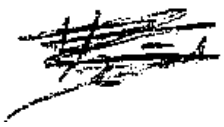
$$Q(u + f, u' + f') = fu'.$$



Undeciphered passage. In parentheses before “1,5 lin.” a short sign is densely cancelled. Its complete initial reading is not established.

Original source details: GELFAND-P000397

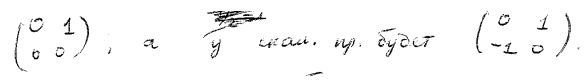
(1,5 lin.).



Undeciphered passage. Below fu' a separate formula with vertical, horizontal and slanting strokes is cancelled. The complete expression is not established.

Original source details: GELFAND-P000397

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \text{and the scalar prod. will have } \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$



Original source details: GELFAND-P000397

Manuscript notation. The lower-left entry of the second matrix is -1.

Witt group: from nonsing. forms: i.e. $(u,v)_q = (L(u)v)$, $L : V \longrightarrow V^*$, ~~corresp. scalar product~~, L is an isomorphism.



Original source details: GELFAND-P000397

Uncertain reading. Immediately before (u,v) a sign with a tall f-like stroke is densely overwritten. Its complete earlier outline and reading are not established; a separate initial f is not asserted.

Witt gr.: $(V,q) \perp (V',q')$ — the orthog. direct sum of two modules with quadr. forms ($A, *, \varepsilon, \Lambda$ fixed).

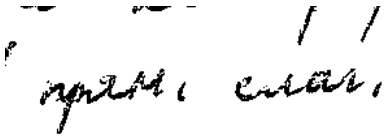
Nonsing. quadr. forms on finitely gener.



Original source details: GELFAND-P000397

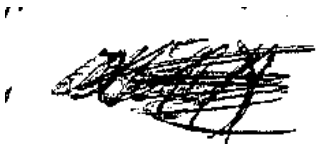
Undeciphered passage. Between “gener.” and “projective” a short fragment is densely cancelled. Its complete reading is not established.

projective modules. (dir. summ.[?],



Original source details: GELFAND-P000397

Uncertain reading. The start of the parentheses is provisionally “прям. слаг.” (dir. summ.); the joined middle letters and the ending of the second abbreviation allow other readings. The abbreviations are not expanded.



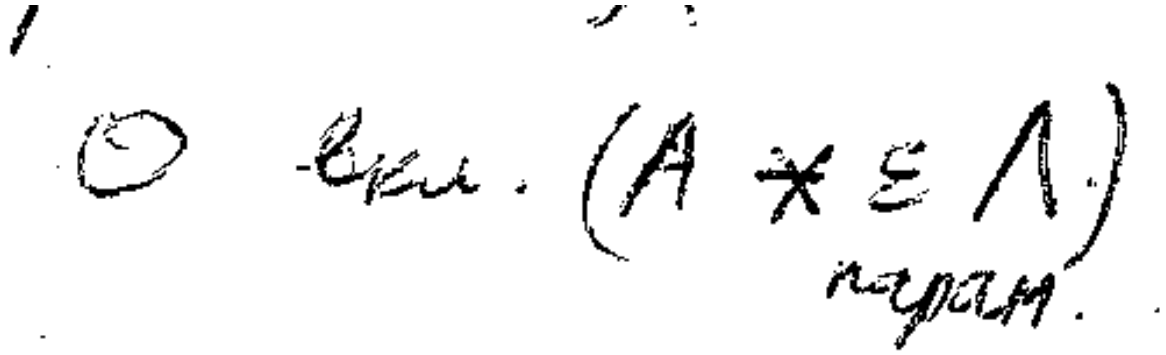
Original source details: GELFAND-P000397

Undeciphered passage. After the first two abbreviations in parentheses a word is densely cancelled. The whole earlier word has not been recovered.

finite-dimensional free ones).

They are a semigroup. From it one constr. a group (Grothendieck’s), this is the Witt gr. Over a field there is a cancellation law (Witt’s thm.).

$K_0O(A)$; O — orthog.



Original source details: GELFAND-P000397

Diagram and typed description. The line has O incl. $(A * \varepsilon\Lambda)$; below the parameters is the label "param.".

O incl. $(A * \varepsilon\Lambda)$
param.

Witt—Grothendieck. Witt, quotient by Hamilton.



Original source details: GELFAND-P000397

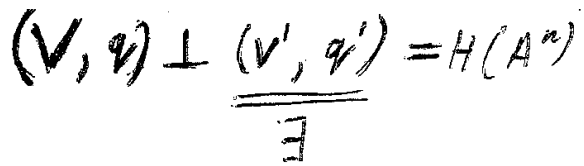
Diagram and typed description. At right of these lines are two separate nearly vertical strokes.



Ex. A — a division ring $\Rightarrow$ the fund. cancellation thm.

Thm. 1. $\forall$ finitely gener. projective module with a quadr. form is a direct summand in a Hamilton. (finite-dim.).

$$(V, q) \perp \underset{\exists}{\underline{\underline{(V', q')}}} = H(A^n).$$



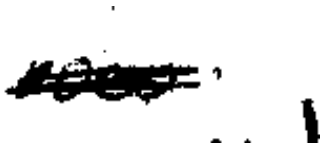
Manuscript notation. An $\exists$ sign sits under the doubly underlined pair (V', q').

Original source details: GELFAND-P000397

Remark: if cancellation by $\forall$ modules is allowed, then the Witt gr. $= 0$. $(a + b + a + b + \ldots)$.

GELFAND-P000398

Examples: 1 A — a field, $*$ the identity, $\varepsilon = 1, -1$ (the theory of symm. and

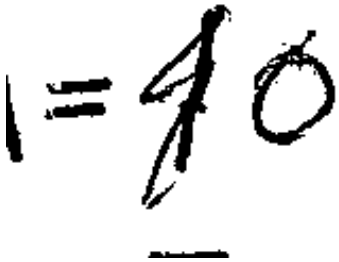


Original source details: GELFAND-P000398

Undeciphered passage. Between “and” and the following abbreviation a short word is densely cancelled. Its complete reading has not been recovered.

sympl. forms.) W.-G. gr. $= Z$.

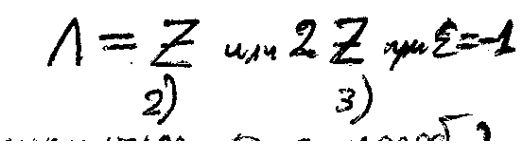
2 $A = Z$, $*$ — the identity; $\varepsilon = +1, -1$; 1) $\Lambda = 0$ for $\varepsilon = 1$;



Original source details: GELFAND-P000398

Uncertain reading. Between the equals sign and 0 is a separate tall corrected or cancelled sign: a left slanting side, a right stem and a cross-stroke. 4, a cancelled one, or an earlier A- or Λ-like letter are possible. Its reading is not established.

$\Lambda =$  0



Original source details: GELFAND-P000398

Diagram and typed description. Below Z is the label “2)”, and below $2Z$ is “3)”.

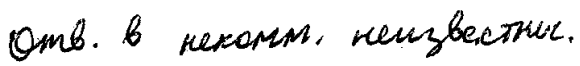
$\Lambda = \underset{2)}{Z}$ or $\underset{3)}{2Z}$ for $\varepsilon = -1$.

1) classif. of quadr. forms: invariants — rank and signature (for nonsing. ones). the cancellation thm. is false.

2) symplectic: they are isom. to Hamilton.

3) — an addit. invariant — the Arf invariant.

3 The group ring of a group. $A = Z[g]$. In the noncomm.[?] case the answers are unknown.



Original source details: GELFAND-P000398

Uncertain reading. The abbreviation is provisionally read as «некомм.» (noncomm.).

$g = Z_p \times Z_q$; K_0 is unknown.



Original source details: GELFAND-P000398

Uncertain reading. The last subscript in $Z_p \times Z_q$ is densely overwritten, with a loop and a lower tail. Provisionally q; the complete earlier strokes are not established.

This is of interest for topology (there Λ is minimal and sometimes maximal).

Taking Matrices over $\underline{\underline{A}}$ does not change $K_0, K_1 \ldots$

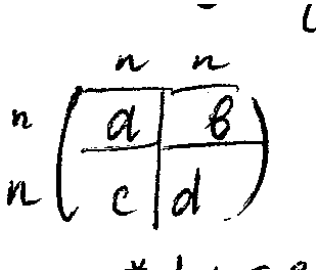
Remark. (there are $80 \cdot 10^6$ nonisom. quadr. forms in 32 variables) with det $= 1$.

$K_1 O(A)$; $O = (A, *, \varepsilon, \Lambda)$. $O(H(A^n))$.



Original source details: GELFAND-P000398

Uncertain reading. In A below the superscript n are two parallel left strokes and a dense apex. The current A is readable; reinforcement and an earlier A- or Λ-like layer cannot be separated confidently. The earlier letter is not established.

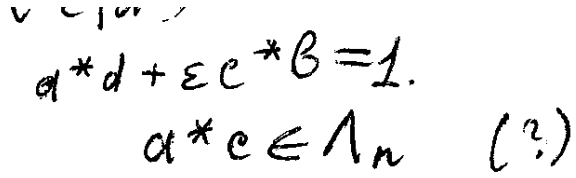


Original source details: GELFAND-P000398

Diagram and typed description. Within parentheses the blocks a, b, c, d are separated by horizontal and vertical lines; n stands above both columns and to the left of both rows.

$$\begin{array}{cc} & \begin{matrix} n & n \end{matrix} \\ \begin{matrix} n \\ n \end{matrix} & \left(\begin{array}{c|c} a & b \\ \hline c & d \end{array}\right) \end{array}$$

$$a^* d + \varepsilon c^* b = 1.$$
$$a^* c \in \Lambda_n \quad (?)$$



Manuscript notation. The second line ends with a separate $(?)$.

Original source details: GELFAND-P000398

(the scalar product $= 1$, and the length 0).

$$O(H(A^n)) \subset O(H(A^{n+m})).$$

~~$O(H(A^\infty))$.~~ $O(A)$ — their union.

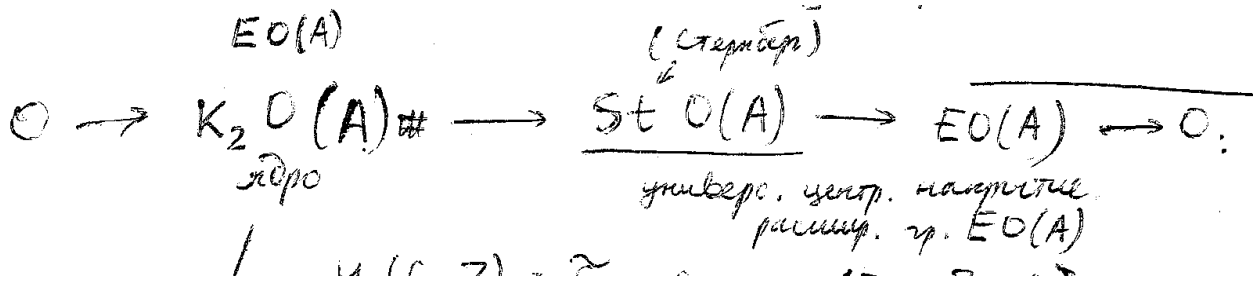
GELFAND-P000399

[source number: 45]

$$O(A)/\underline{[O(A),O(A)]} = K_1O(A).$$

Whitehead lemma $[O(A),O(A)]$ is gener. by elem. matrices

$$\begin{pmatrix}1 & b\\ 0 & 1\end{pmatrix}, \qquad \begin{pmatrix}1 & 0\\ c & 1\end{pmatrix}.$$



Original source details: GELFAND-P000399

Diagram and typed description. At left is $EO(A)$; a slanting arrow from "(Steinberg)" points to the underlined $StO(A)$. Below $K_2O(A)$ is "kernel", and below $StO(A)$ are two lines of caption. The cancelled loops after the kernel are rendered separately.

$$0 \longrightarrow \underset{\text{kernel}}{K_2O(A)} \longrightarrow \underset{\substack{\text{univ. centr. cover}\\ \text{extension of the gr. } EO(A)}}{StO(A)} \longrightarrow EO(A) \longrightarrow 0$$

(Steinberg) ↙ above $StO(A)$; $EO(A)$ above the last arrow.



Undeciphered passage. After $K_2O(A)$ before the arrow two small loop-like signs are crossed by strokes. Their complete original expression has not been recovered.

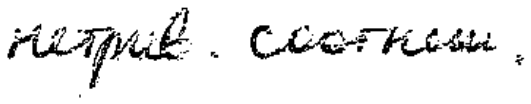
Original source details: GELFAND-P000399

$$H_2(G,Z) \longrightarrow \widetilde{G} \longrightarrow G \qquad ([G,G]=G).$$



Uncertain reading. The first letter with subscript 2 has an M-shaped bend. The reading H_2 is provisional and the manuscript form is shown here.

Original source details: GELFAND-P000399



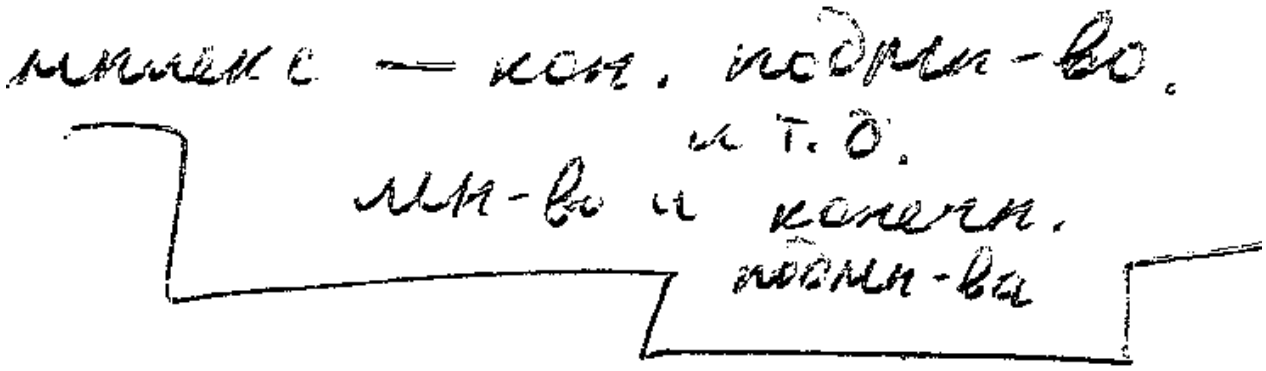
Uncertain reading. The note below the line provisionally reads "nontriv. rels." The attachment of the separate slanting stroke at left is uncertain.

Original source details: GELFAND-P000399

/

nontriv. rels.[?]

Abstr. simplicial scheme. A simplex — a finite subset. a set and finite subsets and so on.



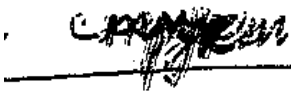
Original source details: GELFAND-P000399

Diagram and typed description. Below the words about a set and finite subsets is a stepped bracket-like line: a short upper horizontal at left, a descent, a long horizontal, a slanting descent, a lower horizontal, a rise at right and a short continuation to the right. There are no labelled vertices or arrows; the role of the steps is not established.



$$K_n O(A) = \pi_{n-1}(O(A)).$$

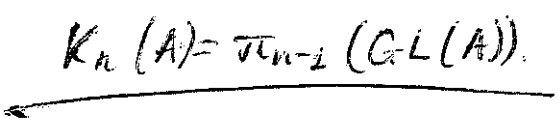
Simpl. str.[?] in $O(A)$.



Original source details: GELFAND-P000399

Uncertain reading. The abbreviation after “Simpl.” is densely overwritten. Provisionally “стр.” (str.); the longer “структ.” (strukt.) is possible. The complete earlier strokes are not established.

On finite $O(H(A^n))$.

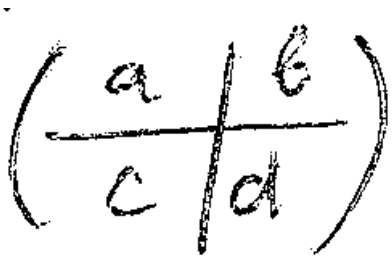


Original source details: GELFAND-P000399

Diagram and typed description. At right the formula is underlined by a long stroke with a left-pointing head at its left end.

$$K_n(A) = \pi_{n-1}(GL(A))$$

Matrices of order $(2n)$.

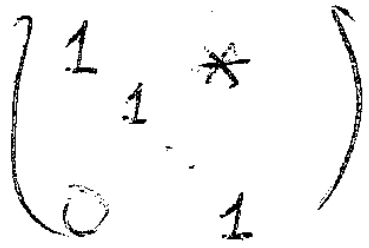


Original source details: GELFAND-P000399

Diagram and typed description. Within parentheses a, b, c, d are separated by horizontal and vertical lines.

$$\left(\begin{array}{c|c} a & b \\ \hline c & d \end{array}\right)$$

Which finite set is a simplex?



Original source details: GELFAND-P000399

Diagram and typed description. Within parentheses are three diagonal ones, $*$ at upper right, 0 at lower left and two small separate strokes between the middle and lower ones.

$$\begin{pmatrix} 1 & & * \\ & 1 & \\ & \cdot & \\ 0 & & 1 \end{pmatrix}$$

Any finite subset in this subgr. is a simplex. and right and left translations are also a simplex. (All simult. reduced to Δ form). K_1 and K_2 defined earlier agree with these.



Original source details: GELFAND-P000399

Undeciphered passage. At left before the lower line a short fragment is cancelled. The large initial stroke resembles “Э”; the whole word is not established.

Comm. relations. — relns. of elem. matrices. Triv. relns. — also conn. with π_2.

GELFAND-P000400

21/II. Kazhdan[?].



Original source details: GELFAND-P000400

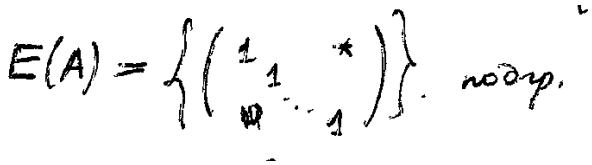
Uncertain reading. The middle strokes of the surname run together. The MCCME catalogue associates the talk on 21 February 1972 with D. A. Kazhdan (record 247).

Let A be a ring. Then $K_0(A)$ is the Groth. gr. of proj. A-modules of fin. rank; $K_1(A) = GL(\infty, A)/E(A)$.



Original source details: GELFAND-P000400

Manuscript notation. The beginning of the line reads “of fin. rank”.



Original source details: GELFAND-P000400

Diagram and typed description. Within braces is a matrix schematic: three diagonal ones, $*$ at upper right, three small diagonal strokes and an overwritten lower-left sign. Its provisional candidate 0 is qualified separately; no finite size is specified.

$$E(A) = \left\{ \begin{pmatrix} 1 & & * \\ & 1 & \\ & \cdot & \\ 0 & & 1 \end{pmatrix} \right\}$$

. subgr.



Original source details: GELFAND-P000400

Uncertain reading. The lower-left sign is densely overwritten: a loop is crossed by vertical strokes. The candidate is 0; the earlier sign is not established.

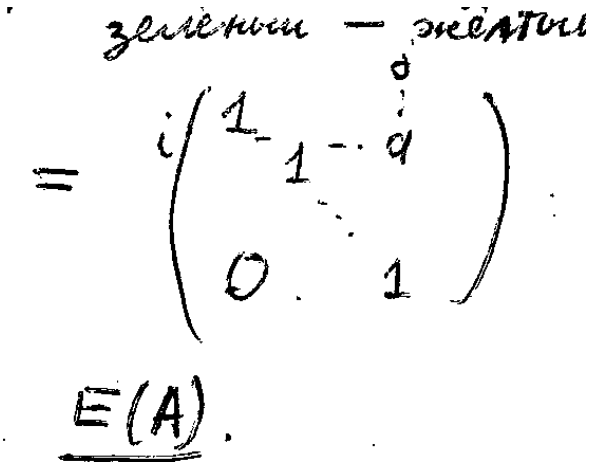


Original source details: GELFAND-P000400

Undeciphered passage. After “subgr.” is a separate short densely cancelled fragment; its initial entry has not been recovered.

$$K_2(A) = \mathrm{Ker}\{St(A) \longrightarrow E(A)\}.$$

(Gr. of Helmholtz—Grothendieck — Helmholtz when changing colours — subtract.) green — yellow = blue.



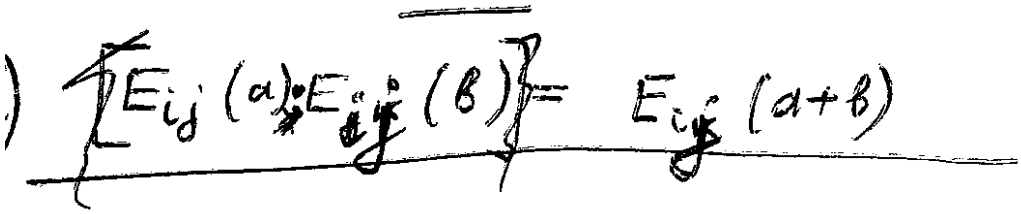
Original source details: GELFAND-P000400

Diagram and typed description. Within parentheses are three visible ones with intervening dots on a descending diagonal, 0 at lower left; a at upper right, short dotted guides to the left of and above a. At left is i, above a a small j-like label. Below $E_{ij}(a)$ is a separate $i \neq j$. The full matrix size is not specified.

$$E_{ij}(a) = \quad i \neq j \qquad \begin{matrix} & & j[?] \\ & & \vdots \\ 1 & & \\ & \ddots & a \\ & 1 & \\ & & \ddots \\ 0 & & 1 \end{matrix}$$

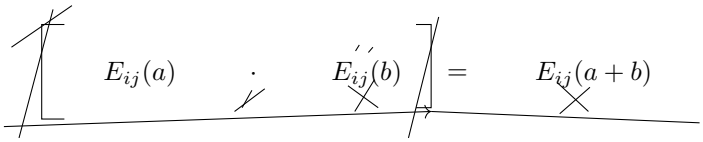
The gr. gener. by them, $\underline{\underline{E(A)}}$.

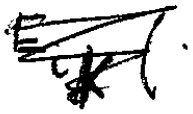
1)

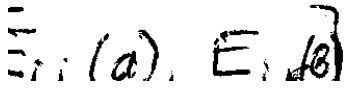


Original source details: GELFAND-P000400

Uncertain reading. The current candidate is $E_{ij}(a) \cdot E_{ij}(b) = E_{ij}(a+b)$. Earlier square brackets are crossed out, and a comma-like stroke near the dot is cancelled. The indices of the second and right-hand E are overwritten; their earlier strokes have not been recovered. Beneath the line is a long rule with a rightward head at the old right bracket and a continuation below the right-hand side.

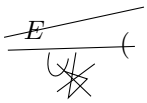




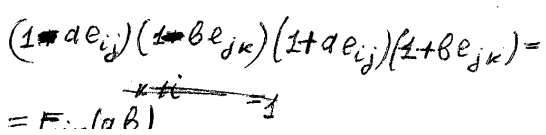


Original source details: GELFAND-P000400

Uncertain reading. Below at left a cancelled unfinished prefix E has overwritten i-/j-/k-like lower strokes and an opening parenthesis. After it is a blank interval. No written argument or closing parenthesis is visible; the exact indices are not established.

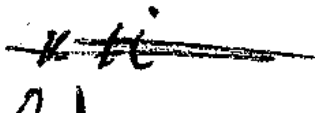


$$(1+ae_{ij})(1+be_{jk})(1+ae_{ij})(1+be_{jk}) = 1.$$



Original source details: GELFAND-P000400

Uncertain reading. In the first two factors the signs after 1 are densely overwritten; the candidates are plus and minus. Plus candidates are provisionally retained in the line; the final two plus signs are separately visible. The result 1 is written lower down at right after the equals sign.

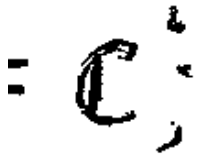


Original source details: GELFAND-P000400

Undeciphered passage. Below the product before the separately written 1 a short entry with u- or v-like and index-like strokes is cancelled; the complete earlier result has not been recovered.

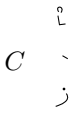
2) $[E_{ij}(a), E_{jk}(b)] = E_{ik}(ab)$. and these are all the relations for now.

In $GL(4,\mathbb{C})$ the gr. of matrices with $\det = 1$ will be gener. by elementary ones. i.e. $K_1(A) = \mathbb{C}$ for $A = \mathbb{C}$;



Original source details: GELFAND-P000400

Uncertain reading. Above and to the right of the final C is a small upper i-like or stroke-like sign, a separate dot below and further below a dot with a tail. The main C is readable; the upper sign is not identified.

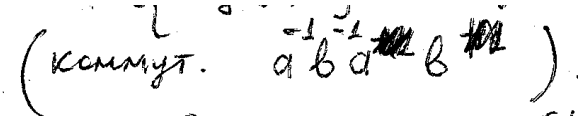


and this holds for any field.

GELFAND-P000401

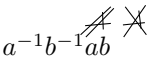
[source number: 47]

Consid. $\{E_{ij}(a)\}$ with the indicated relns. (commut. $a^{-1}b^{-1}ab$).

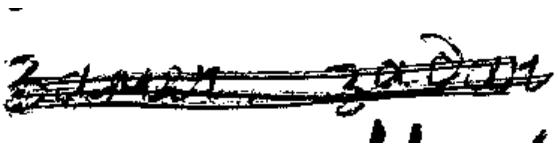


Original source details: GELFAND-P000401

Uncertain reading. Above the final a and b two exponents with minus-like and one-like strokes are cancelled. The current first two exponents $-1, -1$ are clear; the complete earlier final exponents are not established.

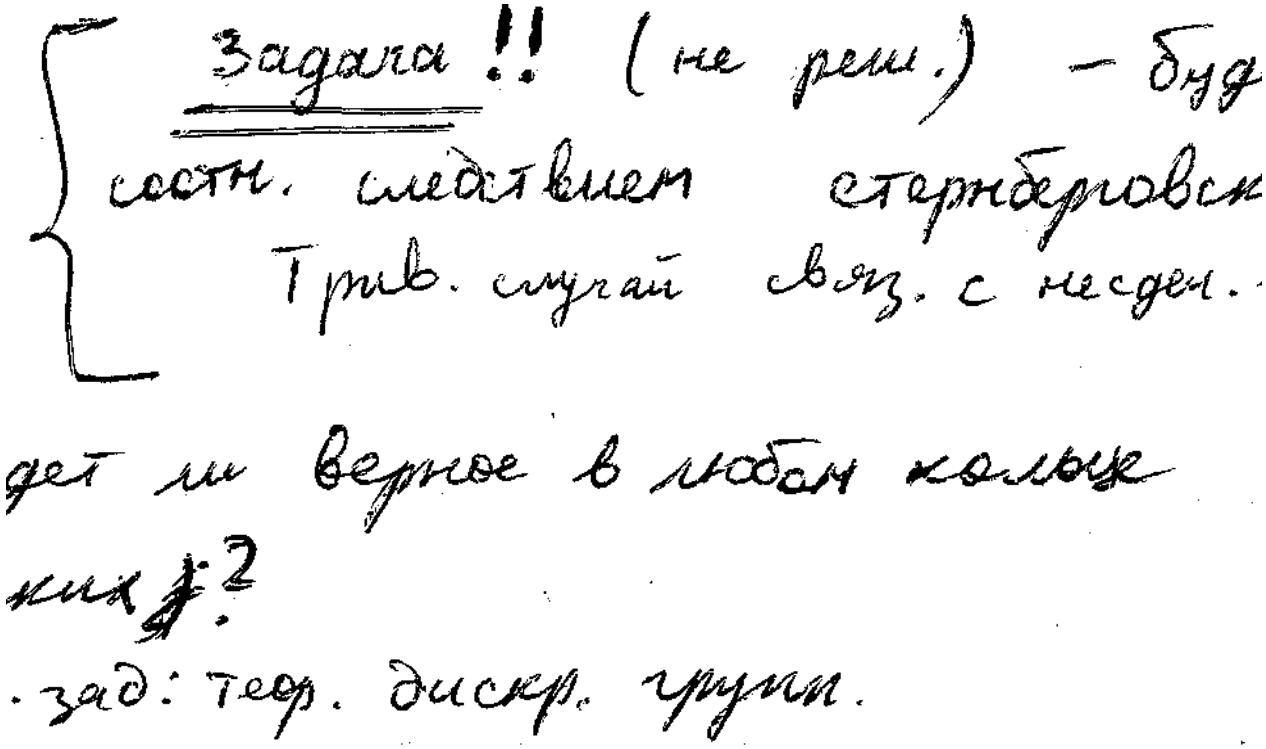


The gr. gener. by them is called $St(A)$ (an abstr. group). (Relns. in the ring, apart from the standard ones, are called $K_2(A)$) Unkn. whether these generate all relns. valid in any ring. (all Steinberg's relns.).



Original source details: GELFAND-P000401

Undeciphered passage. Before the new problem heading a line is cancelled. Parts of "Задача" (Problem) and the final word are distinguishable; the entire earlier formulation has not been recovered.



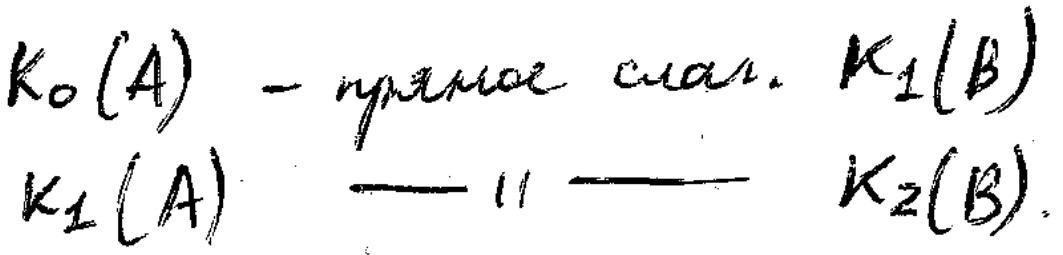
Original source details: GELFAND-P000401

Diagram and typed description. At left a brace joins the new heading, the question and the following “Triv. case”. Two rules are beneath the word “Problem”; the exclamation marks are to its right. A short stroke is cancelled before the question mark.

Problem!! (unsolv.) — will a reln. valid in any ring follow from the Steinberg ones ~~[?]~~?

The triv. case is conn. with an unsolv. problem of the theory of discrete groups.

A, $B = A[t, t^{-1}]$.



Original source details: GELFAND-P000401

Diagram and typed description. In the second line two short nearly vertical ditto strokes are between two horizontal rules. They are reproduced without expanding the repeated words.

$K_0(A)$ — direct summ. $K_1(B)$

$K_1(A)$ — " — $K_2(B)$.

1) $K_0(A) \longrightarrow K_1(B)$. $K_1(B)$ — autom. of a free module. Shall we specify an autom. of a proj. module? $L \in \mathrm{Aut}\, P$; $P + Q = A^n$.



Original source details: GELFAND-P000401

Uncertain reading. At right the formula $L \oplus 1 \in \mathrm{Aut}\, A^n$ is underlined. After 1 is a small dot; two more small marks are below and to the right of A^n. Their role is not established; they are rendered as separate strokes and dots.

$$L \oplus 1 \in \mathrm{Aut}\, A^n$$

This is a well-defined def. in K_1.

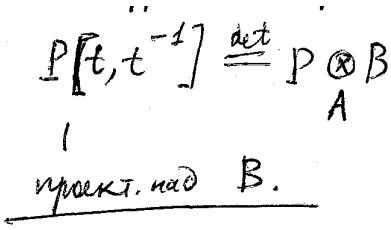
$$P[t, t^{-1}] \overset{\text{def}[?]}{=} P \otimes_A B.$$



Original source details: GELFAND-P000401

Uncertain reading. Above the equals sign is a short underlined Latin label: provisionally def; the first two letters are de, the final tall crossed stroke with a lower hook is t-/f-like. Its exact letter reading is not established.

de

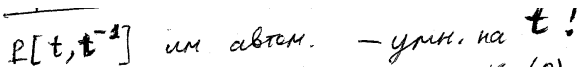


Original source details: GELFAND-P000401

Diagram and typed description. Below the equation at left is a separate short vertical stroke. The words “proj. over B.” are underlined by a long rule with a leftward head.

proj. over B.

$P[t, t^{-1}]$ has an autom. — mult. by t!



Original source details: GELFAND-P000401

Manuscript notation. The line ends “mult. by t” with an exclamation mark.

From it we construct



Original source details: GELFAND-P000401

Undeciphered passage. After “construct” a short word is crossed out. Provisionally “авто” / “авто…” (avto / avto…) can be distinguished; dense correction covers the end. The exact whole word and ending are not established.

an element of $K_1(B)$.

GELFAND-P000402

Reverse projection:

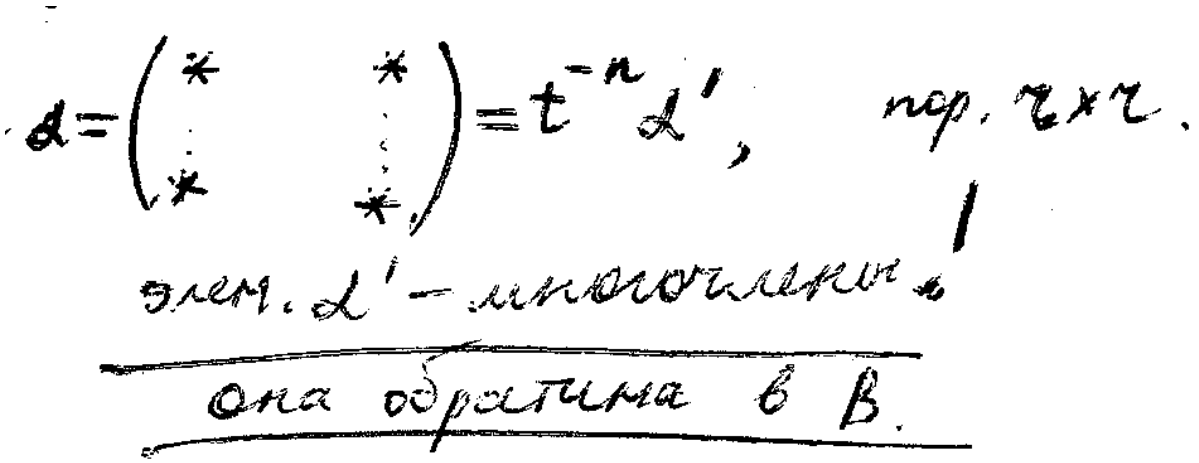


Original source details: GELFAND-P000402

Manuscript notation. The heading “Reverse projection” has a left-pointing stroke beneath it.

$$A^n \longrightarrow (B^n, t) \simeq (B, t^n).$$

An elem. of $K_1(B)$ is a matrix



Original source details: GELFAND-P000402

Diagram and typed description. Within parentheses are four asterisks in two outer columns with dots between top and bottom. Its size is separately stated as $r \times r$; the schematic does not specify the number of intervening rows.

$$L = \begin{pmatrix} * & * \\ \vdots & \vdots \\ * & * \end{pmatrix} = t^{-n} L'$$

of order $r \times r$. The elems. of L' are polynomials! It is invertible in B.

$$L' : A[t]^r \longrightarrow A[t]^r.$$



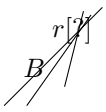
Original source details: GELFAND-P000402

Uncertain reading. Inside the first square brackets the letter is densely overwritten, with B-like loops; provisionally t. The exponent after the bracket is r-/n-like; the line uses the candidate r. The earlier strokes are not established.

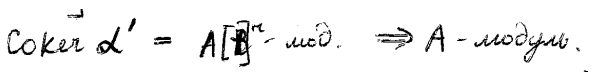


Original source details: GELFAND-P000402

Uncertain reading. Before the current right-hand $A[t]^r$ a B-like expression with a densely cancelled upper exponent is crossed out; the candidates are B^r, B^n. The drawn exponent is provisional.



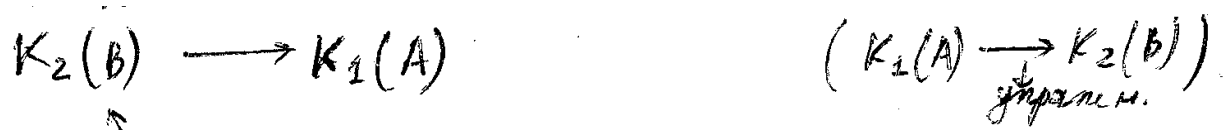
Coker $L' = A[t]^r$-mod. $\Rightarrow$ A-module.



Original source details: GELFAND-P000402

Uncertain reading. In $A[t]^r$ after Coker the letter inside the brackets is overwritten; the exponent is r-/n-like. The line provisionally uses t, r; the equals sign and abbreviation "-mod." are written in the source.

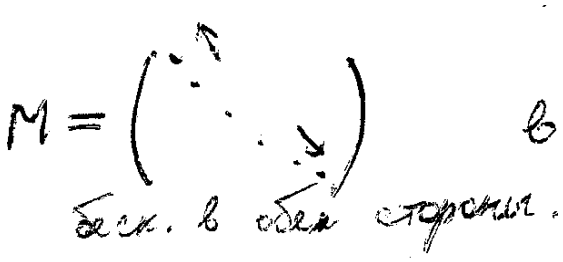
It is fin. gener. and projective over A. (like a Fredholm oper. — finite cokernels).



Original source details: GELFAND-P000402

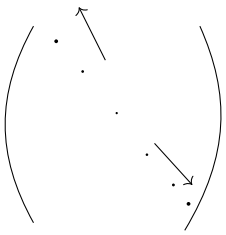
Diagram and typed description. At left is an arrow $K_2(B) \to K_1(A)$. At right in parentheses is the opposite direction $K_1(A) \to K_2(B)$; below its arrow is a short downward arrow to "exercise".

$$K_2(B) \longrightarrow K_1(A) \qquad (K_1(A) \longrightarrow K_2(B)) \downarrow \text{exercise}$$

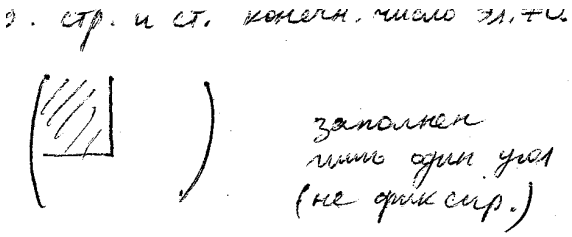


Original source details: GELFAND-P000402

Diagram and typed description. Within parentheses is a descending diagonal row of dots; beside it are separate arrows upwards-left and downwards-right.

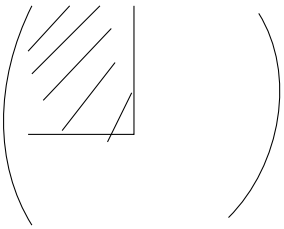
$M =$ 

infinite in both directions. In each row and col. a finite number of elems. $\neq 0$. $M^+ \subset M$, infinite in one dir.

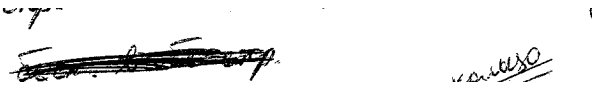


Original source details: GELFAND-P000402

Diagram and typed description. Within parentheses the upper-left corner is filled: a vertical line is to the right of the hatching and a horizontal below it; inside are slanting strokes. The rest is unfilled, and no finite size is specified.

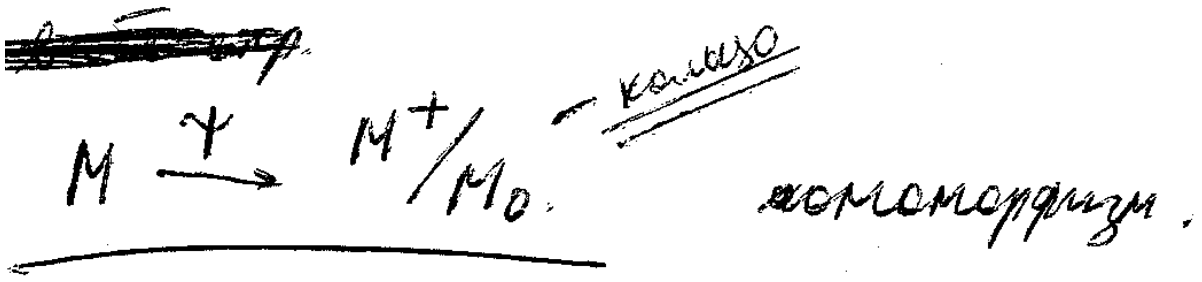


only one corner is filled (not fixed). $M_0 \subset M^+$.



Original source details: GELFAND-P000402

Undeciphered passage. After $M_0 \subset M^+$ a phrase is cancelled. An initial б-like sign and separate letter loops are visible; the whole phrase has not been recovered.



Original source details: GELFAND-P000402

Diagram and typed description. The label Ψ is above the arrow from M to M^+/M_0; "ring" is doubly underlined, and "homomorphism." is lower at right. Beneath is a long underlining rule.

$$M \xrightarrow{\Psi} M^+/M_0$$

ring,

homomorphism.

M_0 — a finite number of elems. different from 0.

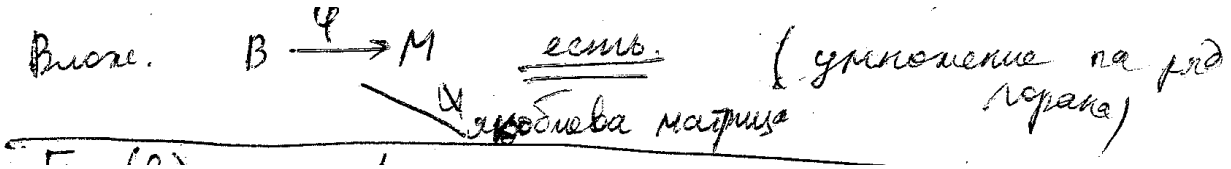


Original source details: GELFAND-P000402

Uncertain reading. At M with lower 0 there is an additional densely corrected upper sign with loops, resembling 0; its precise initial position and reading are not established.

M_0

An embedding.



Original source details: GELFAND-P000402

Uncertain reading. A horizontal arrow labelled φ goes from B to M; at right "exists." is doubly underlined. From M a slanting arrow downwards-right labelled Ψ points to a caption, provisionally "якобиева матрица" (yakobieva matrix, Jacobian matrix); the beginning of the first word is densely overwritten.

$$B \xrightarrow{\varphi} M$$

exists.

Ψ

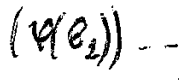
Jacobian[?] matrix

(exercise on a Laurent series)

$$E_{ij}(b_1)\ldots = 1$$

— a reln. of elem. matrices — an element of K_2 — — a certain reln. (not a Steinberg one).

$$E_{ij}(\Psi(\varphi(e_2)))\ldots = 1.$$



Original source details: GELFAND-P000402

Uncertain reading. In the last line inside φ is a looped e-like letter and a small lower sign with a bend and lower rule. The current candidate is e_2; the historical candidates e_i, b_i are not excluded. The preceding argument b_1 has not been copied here.

GELFAND-P000403

[source number: 48]

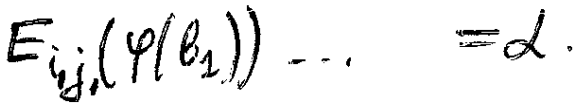
$\ldots M^+/M_0$ “all oper. po vn. nepr.”



Original source details: GELFAND-P000403

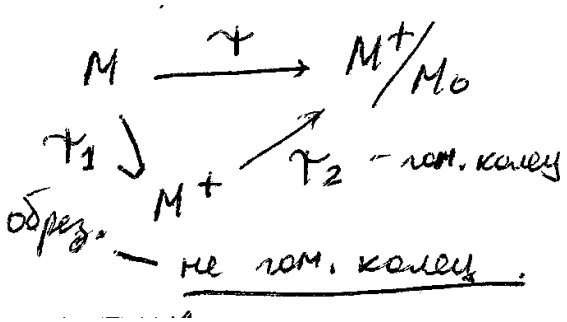
Uncertain reading. Within quotation marks is “все опер. по вн. непр.” (all oper. po vn. nepr.). The abbreviations “вн. непр.” are left unexpanded; their full sense is not established.

$$E_{i_1 j_1}(\varphi(b_1))\ldots = L.$$
$$\Psi(L) = 1.$$



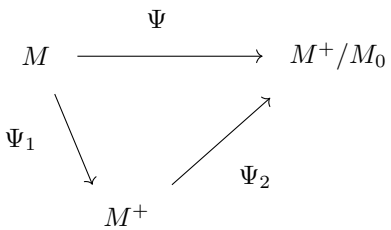
Original source details: GELFAND-P000403

Manuscript notation. Separate small ones follow i and j.

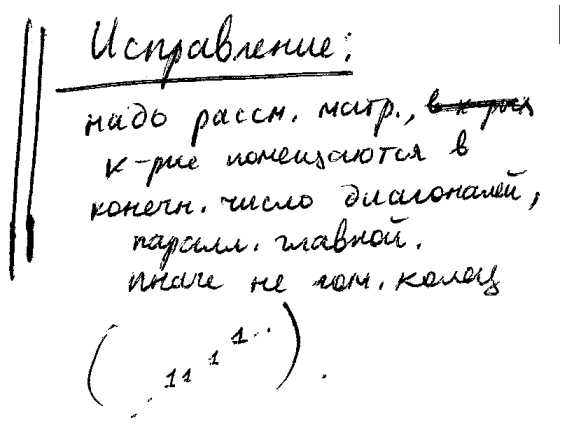


Original source details: GELFAND-P000403

Diagram and typed description. The upper arrow $M \to M^+/M_0$ is labelled Ψ; at left an arrow downwards-right labelled Ψ_1 goes to M^+, and from there an arrow upwards-right labelled Ψ_2.



Ψ_2 — a ring hom.; Ψ_1 — trunc. — not a ring hom..

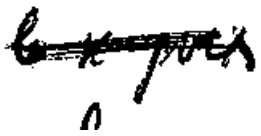


Original source details: GELFAND-P000403

Diagram and typed description. The correction at right is separated by two long nearly vertical rules; its heading is underlined.

Correction:

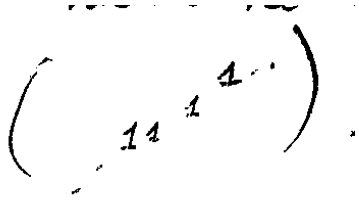
we must consid. matrices,



Original source details: GELFAND-P000403

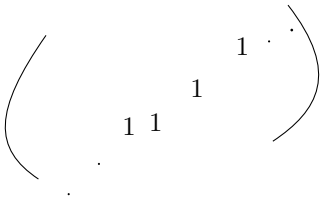
Undeciphered passage. After “matrices” a word with distinguishable в-like and p-like strokes is cancelled; its complete reading has not been recovered.

which fit in a finite number of diagonals parall. to the principal one. Otherwise not a ring hom.

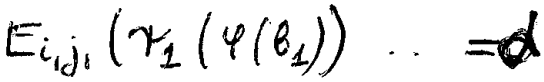


Original source details: GELFAND-P000403

Diagram and typed description. Within parentheses are four one-like signs: two beside each other below, one higher, and one still higher at right; at both ends are separate diagonally positioned dots. Full rows, band boundaries and the remaining entries are not labelled.



Consider $E_{i_1 j_1}(\Psi_1(\varphi(b_1)))\ldots = L$.



Original source details: GELFAND-P000403

Manuscript notation. Separate small ones follow i and j.

$L \neq 1$, since Ψ_1 is not a ring hom.



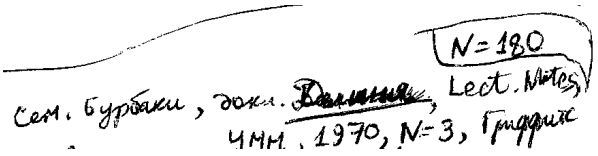
Original source details: GELFAND-P000403

Uncertain reading. The sign after the inequality is densely overwritten: a one-like stem with a lower rule and superimposed loops is visible. The current candidate is 1; the earlier sign is not established.

$L \neq$

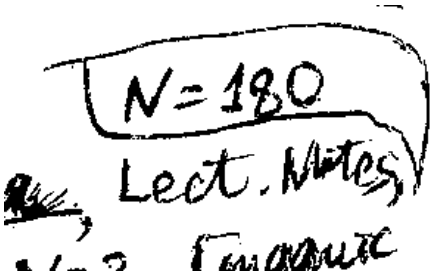
But $\Psi_2(L) = 1$. L is a matrix over M^+, hence it is over M^0 — a finite matrix, def. an elem. of $K_1(A)$.

Bourbaki Sem., report by Delinya[?];



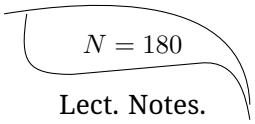
Original source details: GELFAND-P000403

Uncertain reading. The overwritten surname provisionally reads “Deligne”. The reference to Lect. Notes 180 agrees with Pierre Deligne’s “Travaux de Griffiths”, Bourbaki talk 376.



Original source details: GELFAND-P000403

Diagram and typed description. At right $N = 180$ is within a shaped frame, with Lect. Notes. below it. The frame edge bends downwards at right.



UMN, 1970, $N = 3$, Griffiths.

Gabrielov. Work of Griffiths on var. Hodge structure.

X — a nonsingular complex[?] alg. projective variety; dim $X = d$.



Original source details: GELFAND-P000403

Uncertain reading. The abbreviation after “nonsingular” is provisionally “компл.” (complex); its middle joined strokes and ending are not fully separate. The complete word is not written out here.

$$H^n(X, Z), \qquad H^n(X, C) = H^n(X, Z) \otimes_Z C.$$

X is Kähler. On it there is an integral form η of second pr.[?] of type $(1, 1)$.



Original source details: GELFAND-P000403

Uncertain reading. After “second” is a short abbreviation with п-like and p-like strokes, provisionally “пр.” (pr.); a compressed middle о in “пор.” (por., order) is possible. Its exact complete letter reading is not established.

$$H^n(X, C) = \bigoplus_{p+q=n} H^{p,q}.$$

$\eta \in H^{1,1}$, closed and def. a cohom. class. it is given a priori. η is called a polarization.

GELFAND-P000404

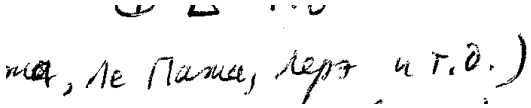
$L : H^n(X, C) \longrightarrow H^{n+2}(X, C)$ — the oper. of mult. by η.

$$\text{Put } H_0^n = \text{Ker}(L^{d-n+1}; H^n \longrightarrow H^{2d-n+2}).$$

primitive cohomology. In cohom. there is a natural scalar product. One can prove that H_0^n is the orthog. complement of the image of L.

$$H^n(X, C) = \bigoplus L^i H_0^{n-2i}.$$

(decomp. of Hodge, Le Pazha, Lert[?], and so on)

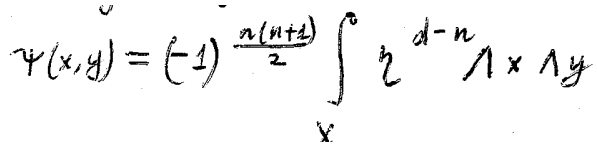


Original source details: GELFAND-P000404

Uncertain reading. After “Ходжа” (Hodge) is written “Ле Пажа” (Le Pazha), then a surname or abbreviation beginning “Лер” and ending in a t-like stroke. Provisionally “Лерт” (Lert);“Лерэ” (Lere) and “Лефт” (Left) are possible. A full name is not expanded here.

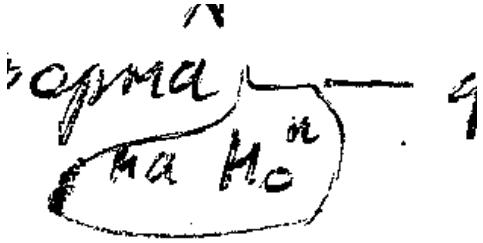
(i.e. everything decomposes into primitives). $x, y \in H_0^n$. Consid.

$$\Psi(x, y) = (-1)^{\frac{n(n+1)}{2}} \int_X \eta^{d-n} \wedge x \wedge y.$$



Original source details: GELFAND-P000404

Uncertain reading. The two tall chevrons between the factors are provisionally read as exterior products.



Original source details: GELFAND-P000404

Diagram and typed description. Below the word “form” is an insertion “on H_0^n” within a curved enclosure connected above to the line.

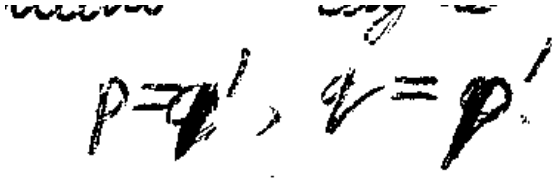
This is a bilin. form — the Hodge form.

on H_0^n

Properties:

1) $\Psi(x, y) = (-1)^n \Psi(y, x)$.

2) $x \in H_0^{p,q}$, $y \in H_0^{p',q'} \Rightarrow \Psi(x,y) \neq 0$ only when $p = q'$, $q = p'$.

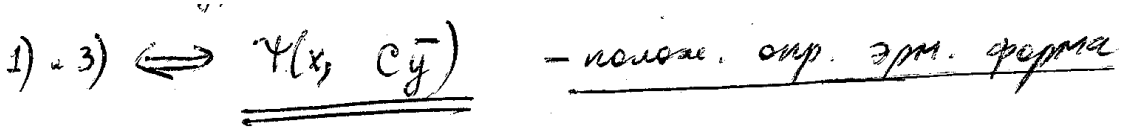


Uncertain reading. After $p =$ the q' is densely overwritten; after $q =$ the letter p' has repeated strokes. The current candidates are $p = q'$, $q = p'$; earlier letters have not been recovered.

Original source details: GELFAND-P000404

3) $0 \neq x \in H_0^{p,q} \Rightarrow i^{p-q}\Psi(x,\overline{x}) > 0$. (in cohom. there is complex conjugation, $\overline{x}$ is the complex conjugate) Let C be the Weil autom. — mult. by i^{p-q} in $H^{p,q}$. (mult. by i in the tangent space). $\Psi(x,y)$ is inv. under the Weil autom.

1) and 3) $\Longleftrightarrow$



Original source details: GELFAND-P000404

Diagram and typed description. The formula with $C\overline{y}$ is doubly underlined; the lower rule has a leftward head. At right "pos. def. Hermitian form" is underlined by one long rule.

$\Psi(x, C\overline{y})$ $\Longleftarrow$ — pos. def. Hermitian form

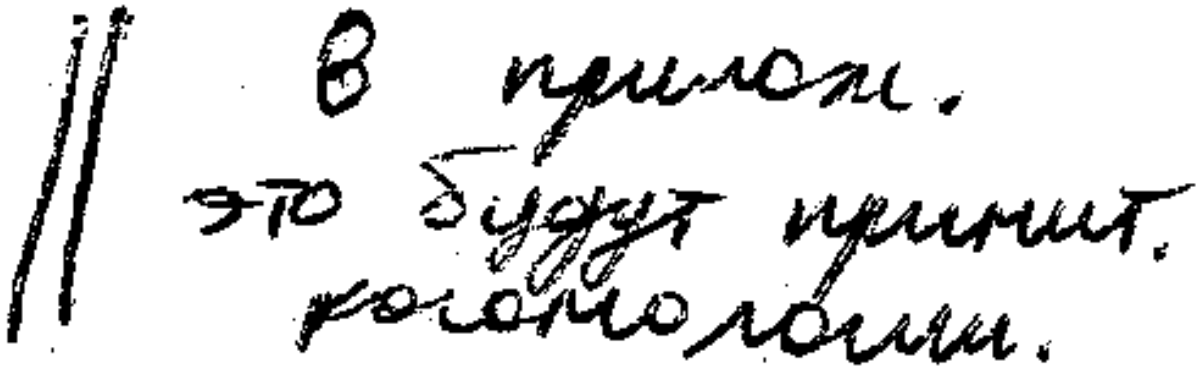
GELFAND-P000405

[source number: 49]

The basic idea of Deligne — one can forget about manif. and retain only the Hodge struct. Cohom. — lin. spaces. A problem of lin. algebra.

$H_{\mathbb{Z}}$ — a finite-type $\mathbb{Z}$-module

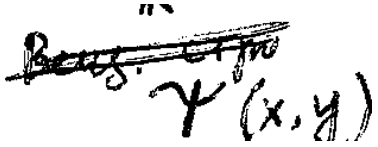
$H_{\mathbb{R}} = \mathbb{R} \otimes H_{\mathbb{Z}}$; $\quad H_{\mathbb{C}} = H_{\mathbb{Z}} \otimes \mathbb{C}$.



Original source details: GELFAND-P000405

Diagram and typed description. At right beside a double vertical rule:

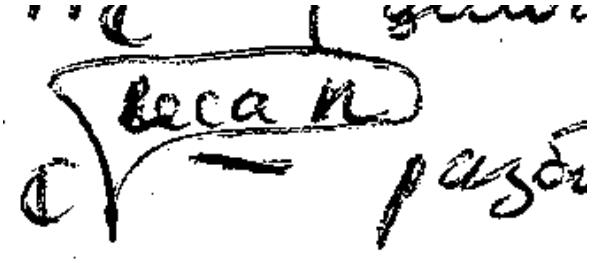
// in applic. these will be primit. cohomology.



Original source details: GELFAND-P000405

Uncertain reading. [Crossed out above the formula:] “Real struct.[?]” (“Вещ. стр.”); the ending “стро” is possible. Strokes obscure the end of the abbreviation.

$\psi(x,y)$ — a bilin. form in $H_{\mathbb{C}}$ (integral — takes integer values on $H_{\mathbb{Z}}$).



Original source details: GELFAND-P000405

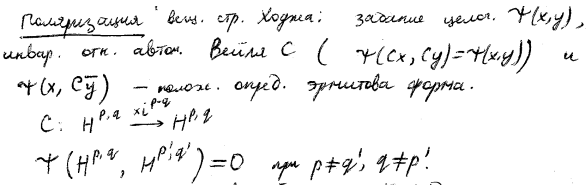
Diagram and typed description. After $H_{\mathbb{C}}$, “of weight n” is inserted within a curved enclosure.

A real Hodge struct. on $H_{\mathbb{C}}$ (of weight n) — a decomposition

$$H_{\mathbb{C}} = \bigoplus_{p+q=n} H^{p,q}, \qquad H^{p,q} = \overline{H^{q,p}}.$$

A polarization of a real Hodge struct.: specifying an integral $\psi(x,y)$, inv. under the Weil autom. C ($\psi(Cx,Cy)=\psi(x,y)$) and $\psi(x,C\bar{y})$ — a pos. def. Hermitian form.

$$C: \quad H^{p,q} \xrightarrow{\times i^{p-q}} H^{p,q},$$
$$\psi(H^{p,q}, H^{p',q'}) = 0 \quad \text{when } p \neq q',\ q \neq p'.$$

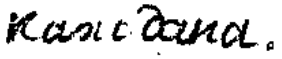


Original source details: GELFAND-P000405

Manuscript notation. The bar in $\psi(x,C\bar{y})$ is over y.

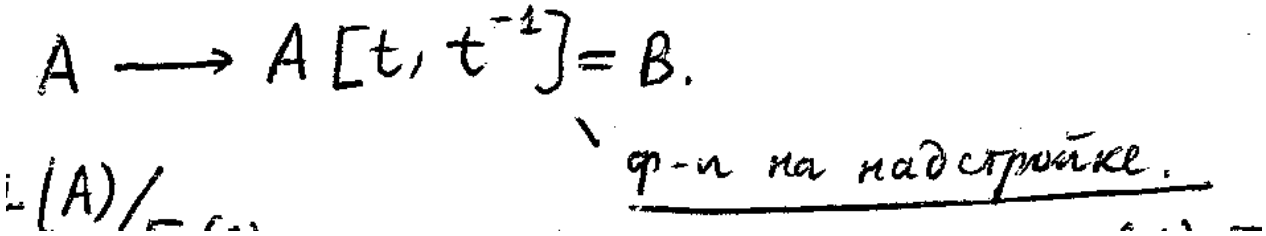
This is called a polarized Hodge structure.

27/II. Kirillov. — repetition of Kazhdan[?].



Original source details: GELFAND-P000405

Uncertain reading. After “repetition” the surname reads “Ка...дана”. “Каждана” (Kazhdan) is provisional because the middle strokes run together.



Original source details: GELFAND-P000405

Diagram and typed description. A short slanted stroke starts under the closing bracket and leads to the underlined caption. Its end is unheaded.

$A \longrightarrow A[t, t^{-1}] = B.$

fns. on the suspension.

$$K_1 = GL(A)/E(A).$$



Original source details: GELFAND-P000405

Diagram and typed description. Two separate horizontal arrows: the upper one points right, the lower one left.

$K_0(A) \;\rightleftarrows\; K_1(B).$

$M \in GL(B)$; $\quad M = t^{-k}$[crossed-out sign]M_1,



Original source details: GELFAND-P000405

Uncertain reading. Between t^{-k} and M_1 a separate sign is densely crossed out. Its earlier shape cannot be separated; candidates are a crossed-out multiplication sign or another separate symbol.

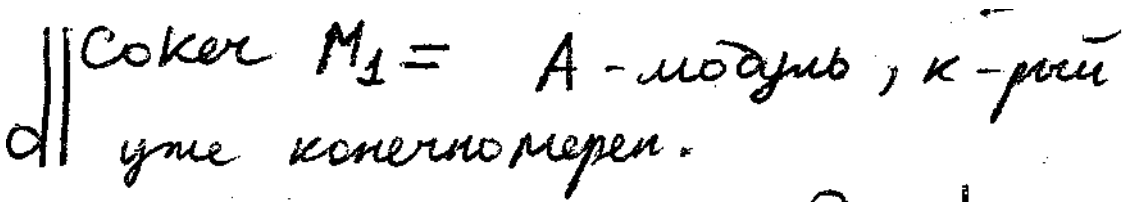
M_1 — a polyn. matrix mstr.[?] $k \times k$.



Original source details: GELFAND-P000405

Uncertain reading. Below “polyn. matrix” the abbreviation is provisionally read “мстр.” (mstr.); “матр.” (matr.) is possible. The middle letter cannot be separated reliably.

$$A[t]^k \xrightarrow{M_1} A[t]^k, \qquad \ker = 0.$$



Original source details: GELFAND-P000405

Diagram and typed description. At right of the line beside a double vertical rule:

// Coker $M_1 = A$-module, which is already finite-dimensional.

(actually one needs the difference between kernel and cokernel) Elem. transformations do not change the cokernel.

GELFAND-P000406

$$M = t^{-k}M_1 = t^{-\ell}M_2.$$



Original source details: GELFAND-P000406

Diagram and typed description. In the margin — a long lower rule and an additional short one at left:

one must subtract the excess.



Original source details: GELFAND-P000406

Diagram and typed description. Crossed out:

~~M~~

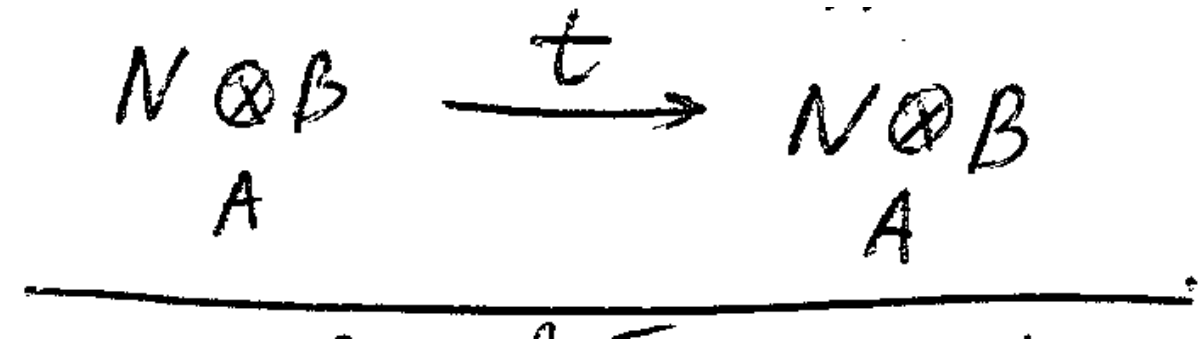
The inverse map:

$$N \longrightarrow N[t, t^{-1}] = N \otimes_A B.$$



Original source details: GELFAND-P000406

Uncertain reading. The sign has two horizontal bars and a separate small downward stroke; provisionally the equality sign is retained. The meaning of the lower mark is not established.



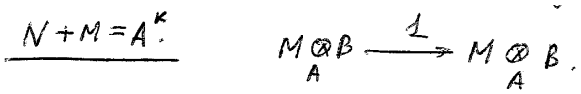
Original source details: GELFAND-P000406

Diagram and typed description. A long horizontal rule is drawn below the next line.

$$N \otimes_A B \xrightarrow{t} N \otimes_A B$$

complete to a free module (otherwise one does not obtain a matrix).

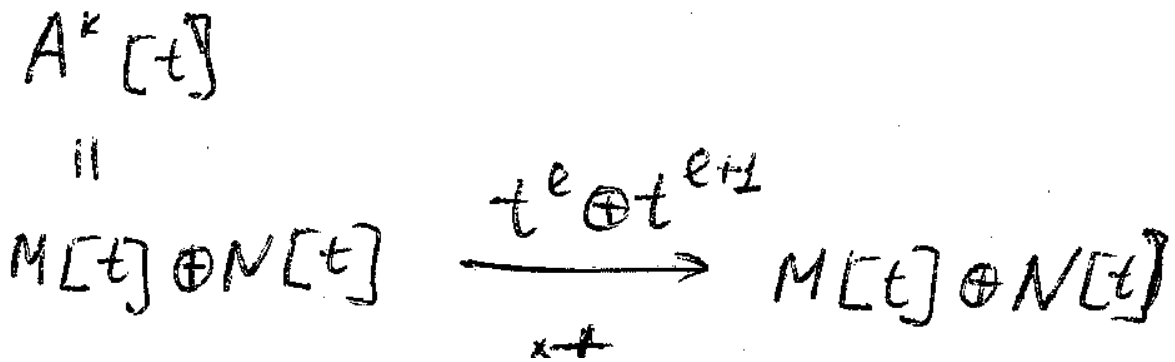
$$\underline{N + M = A^k.} \qquad M \otimes_A B \xrightarrow{1} M \otimes_A B.$$



Manuscript notation. $N + M = A^k$ is written at left. The right-hand arrow is labelled 1.

Original source details: GELFAND-P000406

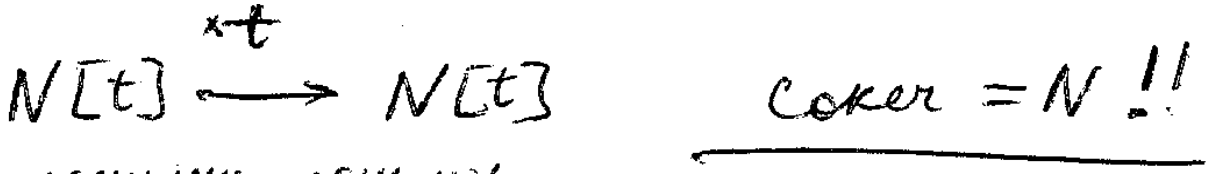
One obtains a map $B^k \longrightarrow B^k$
multiply by t^ℓ to obtain a polynomial one? [At right:] One need not multiply.



Original source details: GELFAND-P000406

Diagram and typed description. $A^k[t]$ is above the left-hand $M[t] \oplus N[t]$; between them are two vertical rules.

$$\begin{array}{ccc} A^k[t] & & \\ \| & & \\ M[t] \oplus N[t] & \xrightarrow{t^\ell \oplus t^{\ell+1}} & M[t] \oplus N[t] \end{array}$$



Original source details: GELFAND-P000406

Diagram and typed description. At right the equality is underlined and has two exclamation marks.

$$N[t] \xrightarrow{\times t} N[t] \qquad \underline{\text{Coker} = N!!}$$

And the ends have met.

$$1 \longrightarrow K_2(B) \longrightarrow St(B) \longrightarrow E(B) \longrightarrow 1.$$

Swan, Tate. Two talks in Nice from [crossed out: an unreadable word] different points of view.



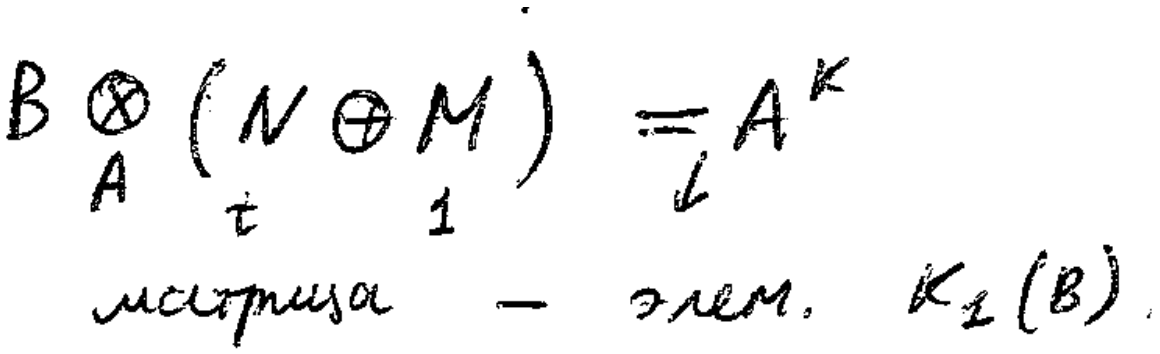
Undeciphered passage. The word after "Two talks in Nice from" is entirely covered by dense crossing-out; its reading is not established.

Original source details: GELFAND-P000406

Gelfand.

$$K_0(A) \longrightarrow K_1(B).$$

N — a proj. module.



Original source details: GELFAND-P000406

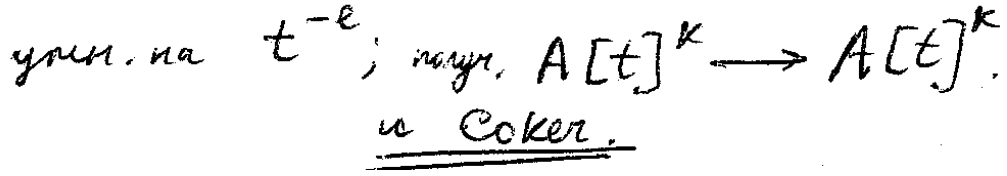
Diagram and typed description. Below N and M are written t and 1; a downward arrow leads from the equality to the caption.

$$B \otimes_A (\underset{t}{N} \oplus \underset{1}{M}) = A^k$$

$$\downarrow$$

a matrix — an elem. of $K_1(B)$.

$B^k \longrightarrow B^k$ multiply by $t^{-\ell}$; one obtains $A[t]^k \longrightarrow A[t]^k$.



Original source details: GELFAND-P000406

Diagram and typed description. The concluding words are doubly underlined.

and Coker.

GELFAND-P000407

[source number: 50]

$$0 \longrightarrow K_1 A \longrightarrow K_1 A[t] \oplus K_1 A[t^{-1}]$$
$$\longrightarrow K_1 A[t, t^{-1}] \longrightarrow K_0 A \,; \longrightarrow 0.$$



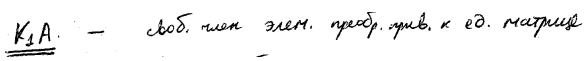
Original source details: GELFAND-P000407

Uncertain reading. Immediately after $K_0 A$ are a detached upper dot and a lower comma-like stroke; provisionally ";" in the line. The exact nature of the mark is not established.

$K_0 A$;

$$K_1A[t] = K_1A \oplus \text{Nil } A$$

$M : A[t]^k \longrightarrow A[t]^k$ an invertible matrix up to elem. transformations. For regular rings Nil $A = 0$. K_1A. — the constant term reduc. to the identity matrix by elem. transf.



Original source details: GELFAND-P000407

Manuscript notation. The abbreviation before “to the identity matrix” reads «прив.».

Now $M = 1 + tN(t)$; and invertible. one can make N independent of t; t^2 is broken up by increasing the order of the matrix (reduction to an equation of the first order).

$$1 + tN \text{ invertible in polynomials} \quad \Longrightarrow \quad N^k = 0.$$

Nil$A$ — nilpotent matrices.



Original source details: GELFAND-P000407

Diagram and typed description. Two horizontal segments with a detached dot between them.

The K-functor in Quillen’s sense.

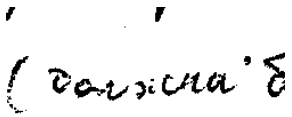
Morava. (Kolumb.[?] Univ.).



Original source details: GELFAND-P000407

Uncertain reading. The abbreviation before “Univ.” is provisionally read «Колумб.» (Kolumb.). The first vowel could be «а».

Quillen’s lemma. (it must be one of the first in topology).



Original source details: GELFAND-P000407

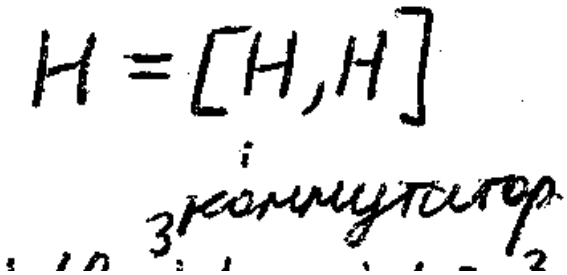
Manuscript notation. The first word in parentheses is «должна» (must).



Original source details: GELFAND-P000407

Manuscript notation. «Быть» (be) follows «должна».

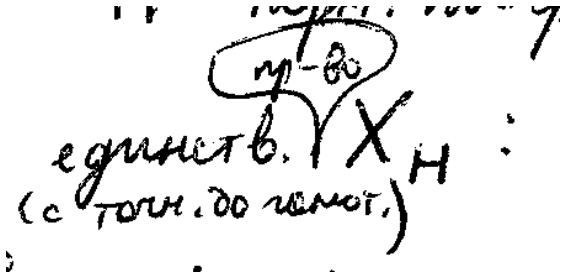
X — a space, H — a normal subgroup of $\pi_1(X)$, with



Original source details: GELFAND-P000407

Diagram and typed description. Below the condition are two small separate vertically placed strokes/dots and a caption; no arrowhead is visible.

$H = [H, H]$
:
commutator



Original source details: GELFAND-P000407

Diagram and typed description. Between "unique" and X_H, "space" is inserted in a curved enclosure with a lower tip.

Then ∃ a unique space X_H:

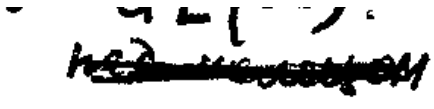
(up to homotopy),

$$X_H = X \cup e^2 \cup e^2 \cup \cdots \cup e^3 \cup \cdots \cup e^3,$$
$$\text{such that} \quad \pi_1(X_H) = \pi_1(X)/H.$$
$$\text{and} \quad H_*(X, \mathbb{Z}) \simeq H_*(X_H; \mathbb{Z}).$$

(the homology ring with integer coeffs.) Corresp. functorially in X. Proof by means of obstruction theory.

GELFAND-P000408

Let $X = BGL(R)$. — the classif. space of $GL(\infty)$. [Below $GL(\infty)$, crossed out: ned...naya.]



Original source details: GELFAND-P000408

Uncertain reading. In the crossed-out word the beginning "нед..." (ned...) and ending "...ная" (...naya) are distinguishable; the middle is covered by a line. The candidate "недискретная" (non-discrete) is not established in full.

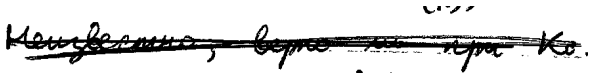
(R — any ring). everything without topology — a discr. group (has nothing in common with Grassm.).

$$E(R) = [E(R), E(R)]$$

the subgroup generated by elem. matrices.

$$\pi_1(BGL(R)) = GL(R).$$
$$\pi_1(X_{E(R)}) = GL(R)/E(R) \text{ — its fund. group } K_1(R).$$

Now $K_i(R) = \pi_i(X_{E(R)})$. [Crossed out: It is unknown whether it is true [...] for K_0.]



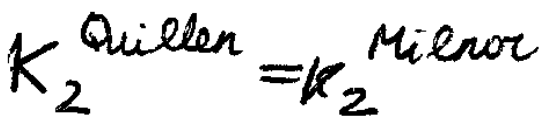
Original source details: GELFAND-P000408

Undeciphered passage. The middle words of the crossed-out line are not recovered; the surviving beginning and end are given in the line.

K_0: the def. must be changed, since $\pi_0(\ ) = \{1\}$.

Quillen: if R is a field, then

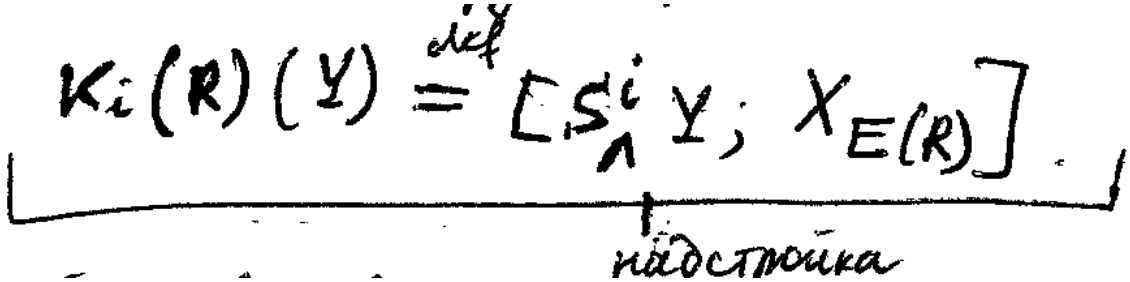
$$K_2^{\text{Quillen}} = k_2^{\text{Milnor}}.$$



Original source details: GELFAND-P000408

Uncertain reading. On the right, the letter before the subscript 2 is distinctly smaller than the left K and has a k-like form. Provisionally typeset as k_2; its case is not inferred from the expected group notation.

There is also Volodin's definition. There exists a theory, applicable to top. groups. Then for R and C one obtains the usual K-functor.



Original source details: GELFAND-P000408

Diagram and typed description. The $\wedge$ sign is lowered below the main line. Below the formula is a lower bracket with a short stroke to the caption "suspension"; the exact scope of the caption is not established.

$$K_i(R)(Y) \stackrel{\text{def}}{=} [S^i \underset{\wedge}{} Y; X_{E(R)}].$$

suspension

Novikov's problem: define the Chern char. in alg. K-theory. Here this reduces to computing $H_*(X_{E(R)}; Q)$.

GELFAND-P000409

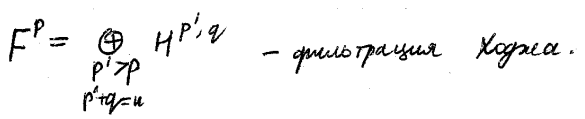
[source number: 51]

A. Gabrielov.

X — an alg. projective manifold.

$$H^n(X,\mathbb{C}) = \bigoplus_{p+q=n} H^{p,q},$$

$$F^p = \bigoplus_{\substack{p'>p \\ p'+q=n}} H^{p',q} \text{ — the Hodge filtration.}$$

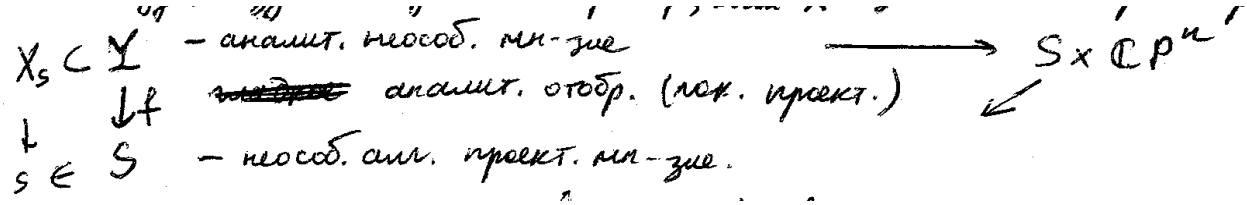


Manuscript notation. Below the direct-sum sign are $p\prime > p$ and $p\prime + q = n$.

Original source details: GELFAND-P000409

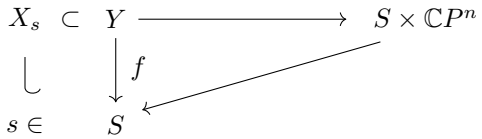
It moves holomorphically! The behaviour of this filtr. will be studied when X depends on a parameter

$X_s \subset Y$ — an analytic nonsing. manifold.



Original source details: GELFAND-P000409

Diagram and typed description. The family diagram and a separate short vertical stroke with a hook below X_s; below it, nearby, $s \in S$.



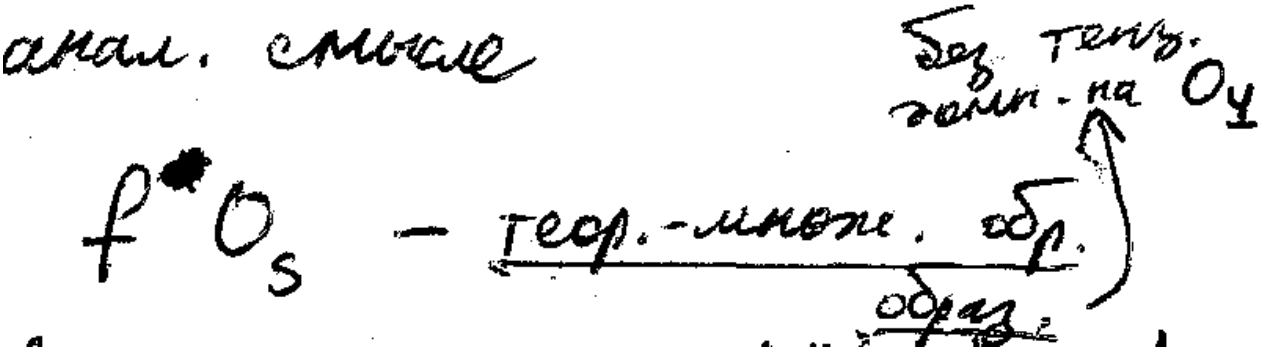
[Before the following words, crossed out:]



Undeciphered passage. The crossed-out word before “analytic map” is not recovered in full. The candidates “гладкое” (smooth) and “полное” (complete) are not fully confirmed.

Original source details: GELFAND-P000409

analytic map (loc. project.). S — a nonsing. alg. project. manifold. smooth — local triviality in the analytic sense. O_S — the sheaf of analytic functions on S;



Original source details: GELFAND-P000409

Diagram and typed description. The underlined definition, “image” inserted below, and a curved arrow to the margin.

Without tensor mult. by O_Y.

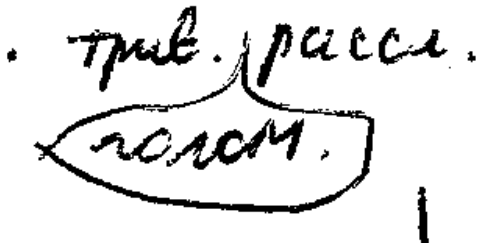
f^*O_S — set-theor. inv. image.



Original source details: GELFAND-P000409

Uncertain reading. In the first f^*O_S the superscript is overwritten into a dark blot. Provisionally $*$; $\circ$ is possible. The later clear f^* and f° do not establish this written sign.

$R^n f_* f^\circ O_S$ —



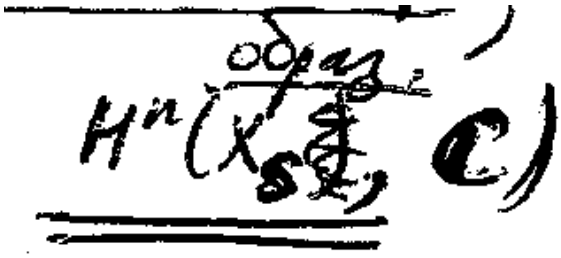
Original source details: GELFAND-P000409

Diagram and typed description. The enclosed insertion “holom.” attached to “bundle”; below to the right is a separate vertical stroke.

loc. triv. bundle

holom.

with fibre



Original source details: GELFAND-P000409

Uncertain reading. After X_s, before the separator, are crossed-out vertical and loop-like strokes. The previous signs are not read; the strokes are shown without asserting a letter Z. Below the left part of the formula is a double underline.

$H^n(X_s; \mathbb{C})$



Original source details: GELFAND-P000409

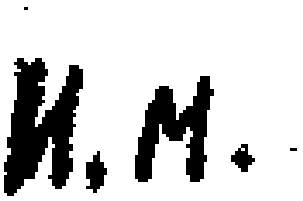
Uncertain reading. Below the formula to the right is a crossed-out fragment ending “...оем” (...oyem); “слоем” (with fibre) is possible, but not established in full. Above it is a short vertical stroke.

~~with fibre[?]~~

in the fibres, a holom. filtration by subbundles. The relative de Rham complex:

$$\Omega^*_{Y/S},$$
$$\Omega^1_{Y/S} = \Omega^1_Y / f^*\Omega^1_S; \qquad \Omega^p_{Y/S} = \bigwedge_p \Omega^1_{Y/S}.$$
$$0 \longrightarrow f^\circ O_S \longrightarrow O_Y \longrightarrow \Omega^1_{Y/S} \longrightarrow \dots.$$

($f^*O_S = O_Y$, $f^\circ O_S$ — constant along the fibres). (a resolution of the sheaf f°). I. M.[?] $H^{p,q}$ forms in $dz, d\bar{z}$ move non-analytically, since $d\bar{z}$ transforms non-analytically.



Uncertain reading. The two letters in the left margin before $H^{p,q}$ provisionally read «И. М.» (I. M.). The first is closely written and also resembles «Н».

Original source details: GELFAND-P000409

$H^{0,q}$ is clear (dz does not enter).

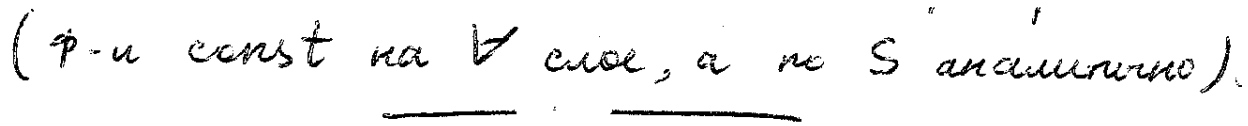


Original source details: GELFAND-P000409

Diagram and typed description. The last line of the leaf, with an unclosed parenthesis: $d\bar{z}$ remains, the number does not increase. (the holom. part increases.

GELFAND-P000410

S — a point, $f^\circ O_S$ — the sheaf of constants, $R^n f_* f^\circ O_S$ — cohom., depending holom. on the parameter.



Original source details: GELFAND-P000410

Diagram and typed description. Two separate horizontal segments below words in parentheses.

(functions const on $\forall$ fibre, but on S analytically).

Consider the Leray spectr. seq.:

$$E_1^{p,q} = R^q f_* \Omega^p_{Y/S} \longrightarrow R^n f_* f^\circ O_S$$

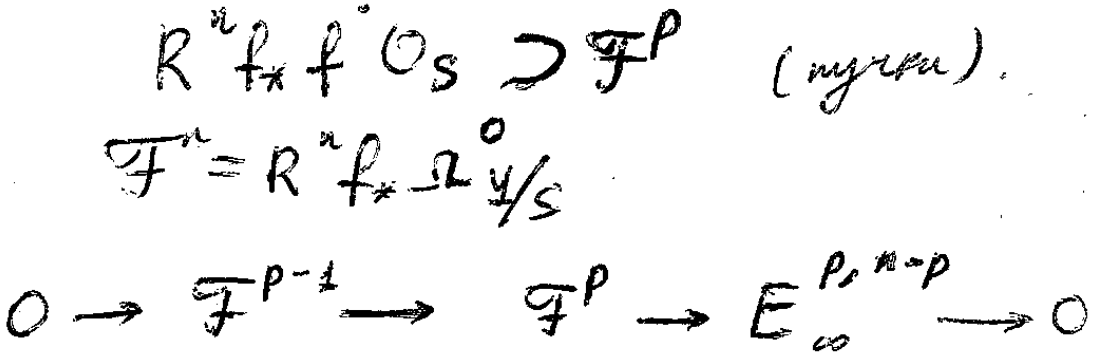
it induces a filtration

Manuscript notation. The subject is «она» (it).

она индуцирует фильтрацию

$R^n f_* f^\circ O_S \supset \mathcal{F}^p$ (пучки).

Original source details: GELFAND-P000410

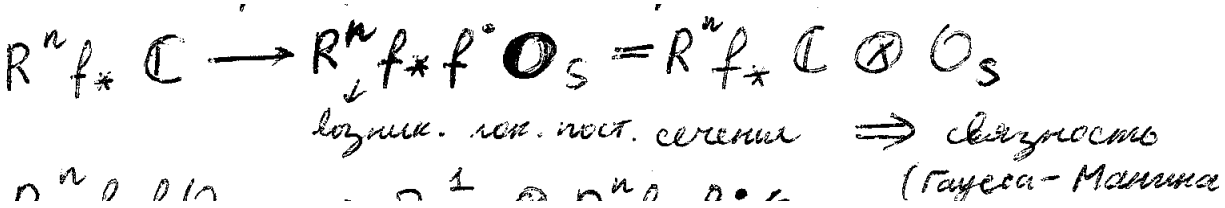


Original source details: GELFAND-P000410

Diagram and typed description. The filtration and exact sequence.

$$R^n f_* f^\circ O_S \supset \mathcal{F}^p \quad \text{(sheaves)}.$$
$$\mathcal{F}^n = R^n f_* \Omega^0_{Y/S}$$
$$0 \longrightarrow \mathcal{F}^{p-1} \longrightarrow \mathcal{F}^p \longrightarrow E_\infty^{p,n-p} \longrightarrow 0$$

The spectr. seq. degenerates (all differentials are zero) In the bundle $R^n f_* f^\circ O_S$ there is a filtration by holom. bundles with fibres F^p.

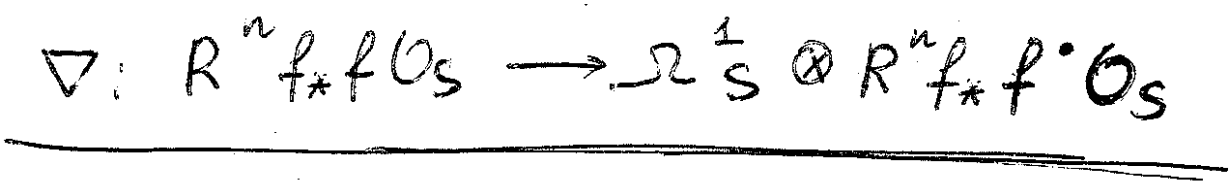


Original source details: GELFAND-P000410

Diagram and typed description. A short arrow from the formula to the lower line about sections.

$$R^n f_* \mathbb{C} \longrightarrow R^n f_* f^\circ O_S = R^n f_* \mathbb{C} \otimes O_S$$
$$\downarrow$$

loc. constant sections arise ⇒ a connection
(Gauss-Manin)



Original source details: GELFAND-P000410

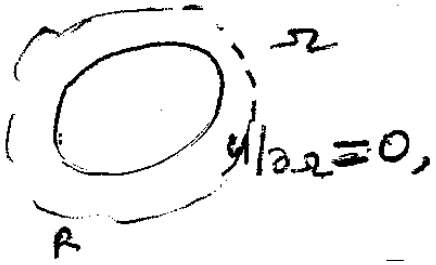
Diagram and typed description. The connection formula and several nearly coincident horizontal strokes below it.

$$\nabla : \quad R^n f_* f O_S \longrightarrow \Omega^1_S \otimes R^n f_* f^{\circ} O_S$$

GELFAND-P000411

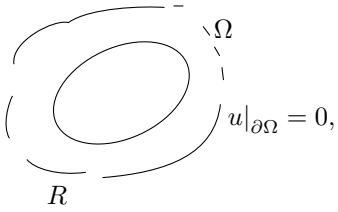
[source number: 52]

6/III. M. Sh. Birman.



Original source details: GELFAND-P000411

Diagram and typed description. A tilted solid oval inside a wider broken contour; Ω at the upper right, R at the lower left, and the condition at the right. The exact assignment of the region Ω to the contours is not established.



$u_{tt} = \Delta u.$



Original source details: GELFAND-P000411

Diagram and typed description. A phrase with a wavy underline.

Suppose Huygens' principle holds.

$\int_\Omega (|\nabla u|^2 + |u_t|^2)\, dx$ — is conserved.



Original source details: GELFAND-P000411

Diagram and typed description. Incoming and outgoing solutions — do not exhaust the subsp. with finite energy.

$K_+ \oplus K_- = \mathcal{H} \ominus K_0$; (consider the initial data of solutions). K_0 — solutions decaying exp. in time, but concentrated near the boundary. f, $\frac{\partial f}{\partial t}\Big|_{t=0} = f_1$. Initial data — pairs (f, f_1) in the energy metric. K_+ — an outg. wave and at $t = 0$ already concentrated outside the sphere of radius R.



Original source details: GELFAND-P000411

Diagram and typed description. A line with two separate underline segments and a dot between them.

see Lax and Phillips.

Words. (key)

1. Characteristic functions. — part 2 of the book by Gohberg and Krein.
2. Scattering,
3. “the second sheet of the resolvent”
4. S-matrix theory.
5. complex poles of



Original source details: GELFAND-P000411

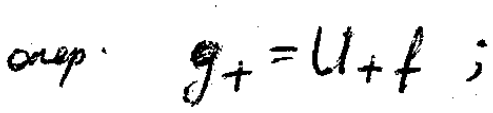
Diagram and typed description. Two separate segments below the words and a dot between them.

self-adjoint oper.

H is close to H_0 if $\forall f\ \exists g_\pm$.

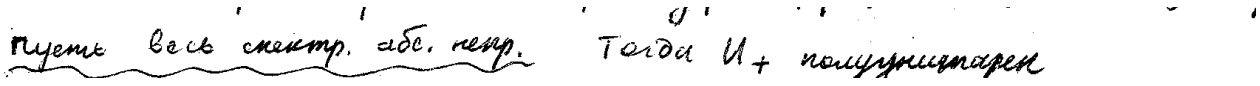
$$\|e^{-iH_0t}f - e^{-iHt}g_\pm\| \longrightarrow 0 \quad \text{as } t \longrightarrow \pm\infty.$$

On the abs. cont. spectrum there is an oper. $g_+ = U_+ f$; U_+ — a partial isometry.



Manuscript notation. Before $g_+ = U_+ f$ the abbreviation is «опер.» (oper.).

Original source details: GELFAND-P000411

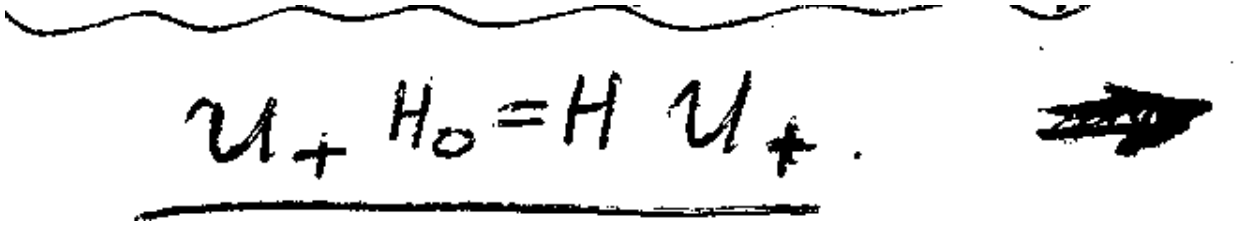


Original source details: GELFAND-P000411

Diagram and typed description. A phrase with a wavy underline.

Suppose the whole spectrum is abs. cont.

Then U_+ is semi-unitary



Original source details: GELFAND-P000411

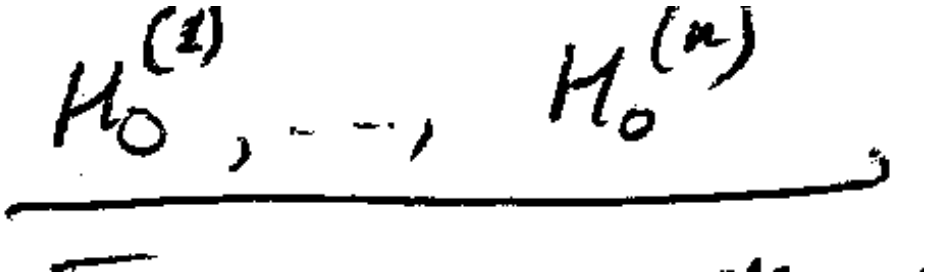
Uncertain reading. After the underlined identity is a separate densely crossed-out sign; its previous form is not established. The visible horizontal and oblique strokes are reproduced without choosing ⇒.

$$U_+H_0 = HU_+.$$

H contains a part unitarily equiv. to H_0.

Problem: when does the range of U_+ exhaust the whole abs. cont. subsp. of H? This is single-channel scattering (in physical language). (i.e. $\exists$ complete wave operators)

multichannel: To describe H one needs



Original source details: GELFAND-P000411

Diagram and typed description. A list of operators with a lower horizontal stroke turned upward at the right.

$$H_0^{(1)}, \ldots, H_0^{(n)}$$

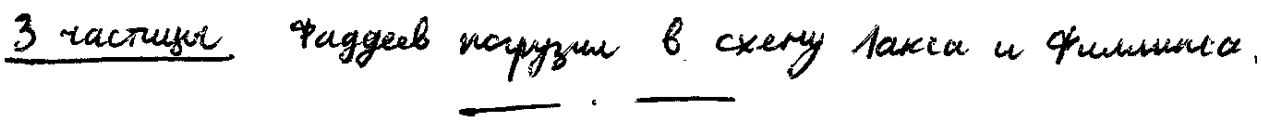
so that the ranges of U^+ cover the whole space and are orthogonal. $H_0^{(i)}$, $i = 1, \ldots, n$ — a collection of operators; H is called admissible if the oper. $U_+^{(i)}$

such that their ranges $U_+^{(i)}$ are pairwise orthogonal. and gen.[?] the abs. cont. spectrum of H.



Original source details: GELFAND-P000412

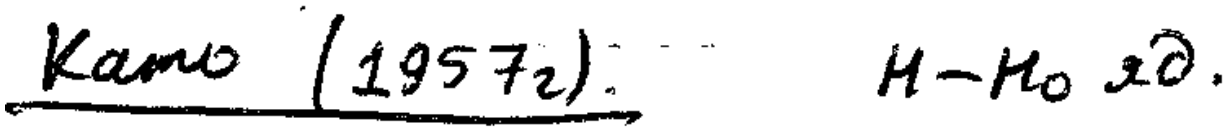
Uncertain reading. The abbreviation after "and" has a dense beginning and a distinguishable ending "...рожд." (...rozhd.). Provisionally "порожд." (gen.); it is not expanded into a full verb.



Original source details: GELFAND-P000412

Diagram and typed description. "3 particles" is underlined; below the middle of the line are two separate segments and a dot.

3 particles Faddeev embedded in the scheme of Lax and Phillips.



Original source details: GELFAND-P000412

Diagram and typed description. Below the name and date is a line ending in a point to the right.

Kato (1957). $\longrightarrow$ $H - H_0$ nuclear.

$\varphi(H) - \varphi(H_0) \in$ nuclear. — Birman, 1962.



Original source details: GELFAND-P000412

Uncertain reading. The second function sign, before (H_0), is heavily overwritten. Provisionally φ; a ψ-like form is not excluded. The clear signs in the later formula do not recover the previous strokes.

$S = U_+^* U_-$, — the scattering oper.

$S(\lambda)$ — the scattering matrix. (corresp. decomp. of H_0).

$S(\lambda) - I_\lambda \in$ nuclear. Birman — Krein

$$\det S(\lambda) = e^{-2\pi i \xi(\lambda)}$$

$$\mathrm{Sp}[\varphi(H) - \varphi(H_0)] = \int \frac{d\varphi(\lambda)}{d\lambda} \xi(\lambda)\, d\lambda.$$

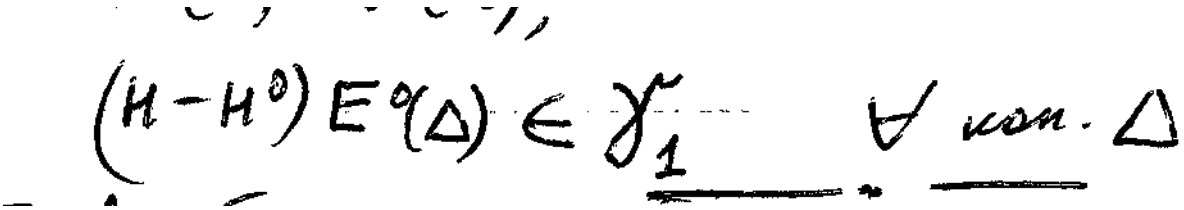
(this defines $\xi(\lambda)$)



Original source details: GELFAND-P000412

Diagram and typed description. A dividing line.

$$D(H) = D(H_0);$$

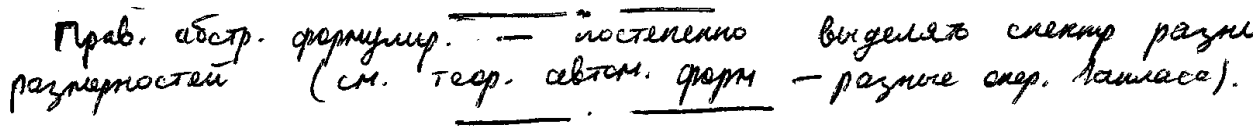


Original source details: GELFAND-P000412

Uncertain reading. A tall loop-like letter with a forked top and subscript 1. Provisionally $\mathcal{J}_1$; the candidate γ_1 remains. The two separate underline segments and dot are retained.

$(H - H^0)E^0(\Delta) \in \mathcal{J}_1 \qquad \forall$ bounded Δ

— sufficient for the existence of wave operators. The proper abstract formulation — gradually separate the spectrum of different dimensions (see



Original source details: GELFAND-P000412

Diagram and typed description. Two underline segments with a dot between them.

theory of coupled forms

— different Laplace operators).

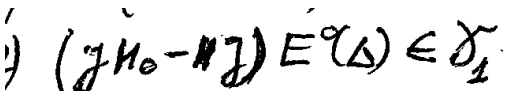
On a more general def. of wave operators.

$$U_\pm = \lim_{t \to \pm\infty} e^{iHt} J e^{-iH_0 t}$$

Corresp. to different spaces and classical scattering.

$$1) \quad D(H) = D(H_0)$$

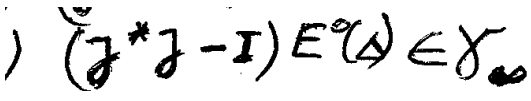
$$2) \quad (JH_0 - HJ)E^0(\Delta) \in \mathcal{J}_1$$



Original source details: GELFAND-P000412

Uncertain reading. In condition 2 is the same loop-like letter with subscript 1. Provisionally $\mathcal{J}_1$; γ_1 is possible. The letter is not established from the expected operator ideal.

$$3) \quad (J^*J - I)E^0(\Delta) \in \mathcal{J}_\infty$$



Original source details: GELFAND-P000412

Uncertain reading. In condition 3 is a loop-like letter with a lower horizontal loop. Provisionally $\mathcal{J}_\infty$; the candidate is γ_∞. The letter and reading of the lower loop remain qualified.

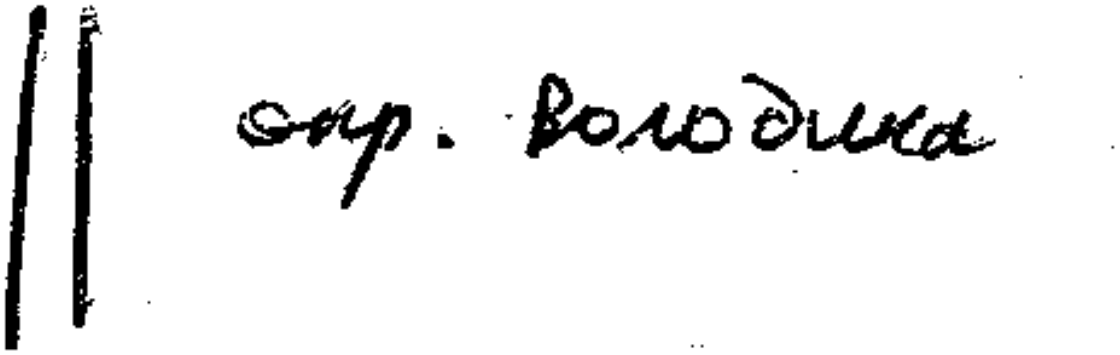
$J = U_+$ — trivially exists. One must ≈ guess U_+.

GELFAND-P000413

[source number: 53]

L. Vaserstein. Different K-theories.

1) A — an assoc. ring $\quad K_n = \pi_{n-1}(GL(A))$.



Original source details: GELFAND-P000413

Diagram and typed description. A double vertical stroke and a caption to the right.

Volodin's def.

def. of an abstr. simpl. scheme in $GL(A)$

K_1 coincides with Whitehead's; K_2 with Milnor's.



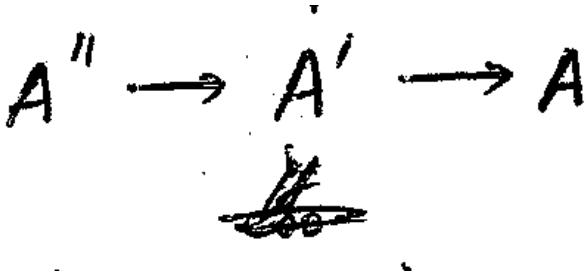
Original source details: GELFAND-P000413

Diagram and typed description. A long dividing line.

2) Derived functors (Swan; Gershtein) of GL.

$K_n^{GS} A$

Consider a free resolution of the ring A:

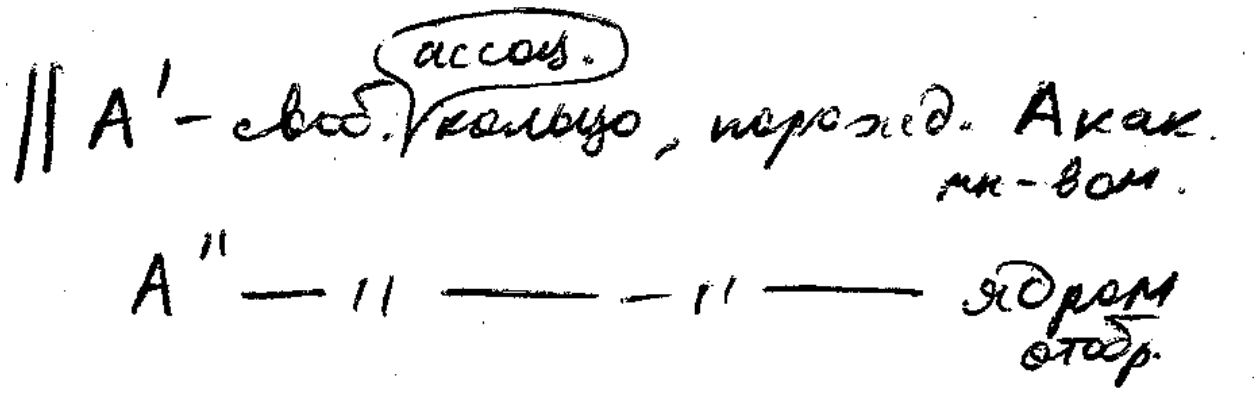


Original source details: GELFAND-P000413

Uncertain reading. Below A' is a crossed-out expression, provisionally K_0. Its meaning is not recovered from context.

$A'' \longrightarrow A' \longrightarrow A$

~~K_0~~



Original source details: GELFAND-P000413

Diagram and typed description. A double vertical stroke; “assoc.” in an enclosure with an insertion tip; ditto strokes in the second line.

assoc.

$||\ A'$ — a free ring, generated by A as a set.

A'' — $||$ — $||$ — the kernel of the map.

$$GL(A'') \longrightarrow GL(A') \longrightarrow GL(A)$$

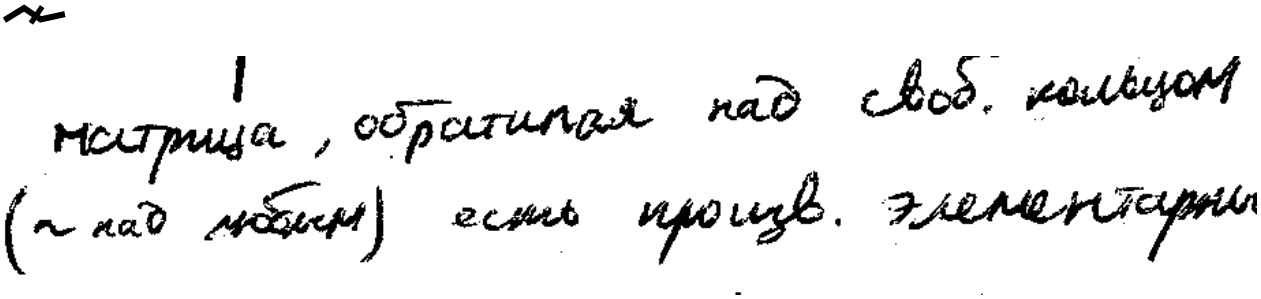
is no longer exact..

K_1 coincides ([crossed-out mark] K_1 of a free ring $= 0$).



Original source details: GELFAND-P000413

Uncertain reading. Before the following K_1 is a separate densely inked mark. The previous sign is not determined.



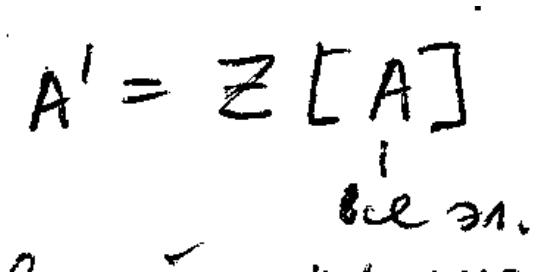
Original source details: GELFAND-P000413

Diagram and typed description. A separate vertical stroke above the marginal explanation.

|

a matrix invertible over a free ring
(∼ over any) is a product of elementary matrices.

(For simplicity the rings are taken without identity).



Original source details: GELFAND-P000413

Diagram and typed description. Below $[A]$ is a short stroke to the lower caption.

$A' = Z[A]$
|
all elem.

— noncommut. polynomials without a constant term.

The derived functor is defined using simplicial constructions.



Original source details: GELFAND-P000413

Diagram and typed description. Two horizontal segments with a detached dot between them.

$$1 \longrightarrow A'' \longrightarrow A' \longrightarrow A$$
$$K_{n+1}(A) = K_n(A'')$$

(an inductive formula): define everything through K_1.

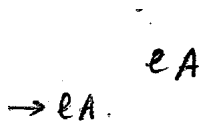
The problem of the Steinberg relations was done by L. V. and follows from the coincidence of these K-functors with Volodin's

A — a ring

MA — matrices over A.

ℓA — in $\forall$ row and column there are only finitely many nonzero elements.

$MA \longrightarrow \ell A$.

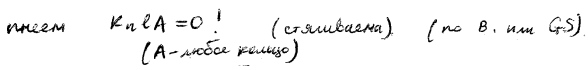


Original source details: GELFAND-P000413

Uncertain reading. In the first ring notation and in the adjacent arrow is a loop-shaped lowercase letter. Provisionally ℓ in both places; the candidate e remains. Similarity of the forms does not establish the choice of letter.

GELFAND-P000414

we have $K_n \ell A = 0$! (contractible) (according to V. or GS).

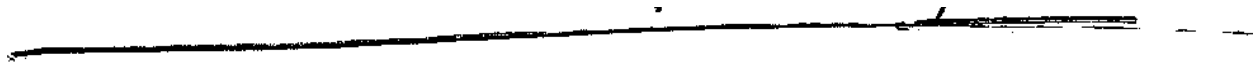


Original source details: GELFAND-P000414

Manuscript notation. The first line reads «имеем» (we have) and «стягиваема» (contractible).

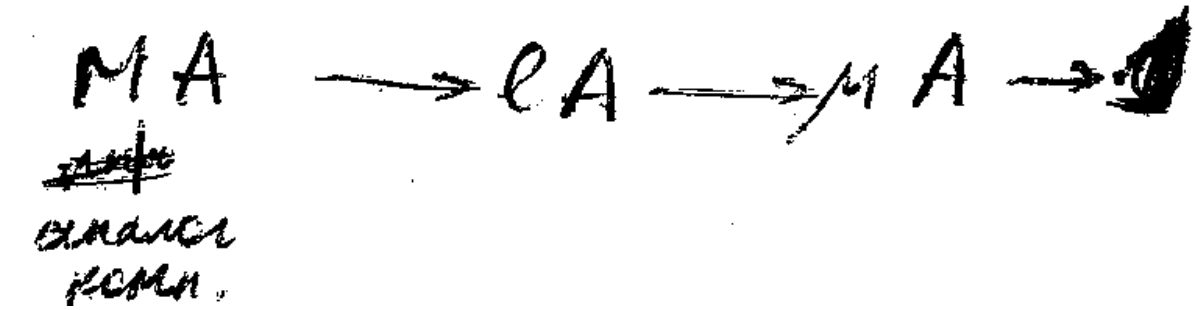
(A — any ring)

Proof $\sim$ Kuiper's theorem.



Original source details: GELFAND-P000414

Diagram and typed description. A horizontal line.



Original source details: GELFAND-P000414

Uncertain reading. At the end of the sequence is an inked sign with a rounded lower loop; 0 is only a candidate. Below MA are crossed-out strokes.

$$MA \longrightarrow \ell A \longrightarrow \mu A \longrightarrow \emptyset$$

analogue of compact.



Original source details: GELFAND-P000414

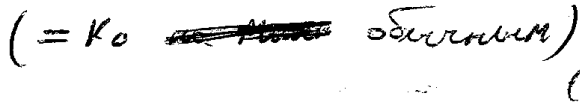
Undeciphered passage. The crossed-out word immediately below MA is not read in full. The candidates "матр." (matr.) and "комп." (compact.) are not asserted.

(matrices with finitely many nonzero elements)

$$\underline{\underline{K_n A = K_n MA.}}$$

$$\underline{\underline{K_n A = K_{n+1} \mu A.}} \quad \text{(from the exact sequence)}$$

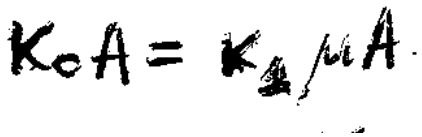
From this one obtains K_0 ($= K_0$ [crossed out: ...] ordinary) — full (a free module $\neq 0$).



Original source details: GELFAND-P000414

Undeciphered passage. The phrase before "ordinary" inside the parenthesis is not recovered in full.

$$K_0 A = K_1 \mu A.$$



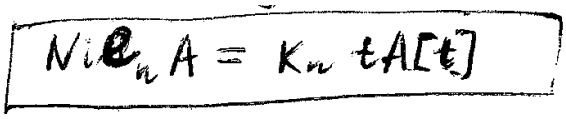
Original source details: GELFAND-P000414

Uncertain reading. The subscript on the second K is reinforced/overwritten. Provisionally 1; the former form is not established from the shift formula.

From this, K_n $(-\infty < n < \infty)$.

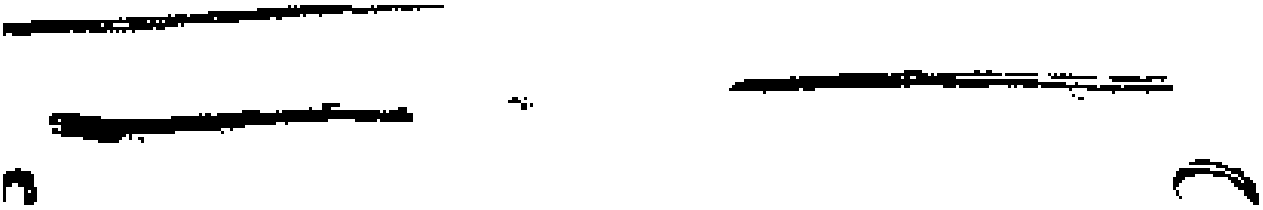
This coincides with Bass's K_n.

$$\boxed{\mathrm{Nil}_n\, A = K_n\, tA[t]}$$



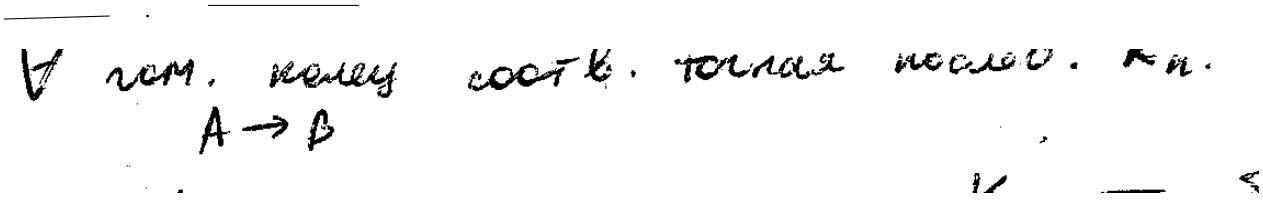
Original source details: GELFAND-P000414

Uncertain reading. The last letter of "Nil" is heavily retraced. "Nil" is retained provisionally.



Original source details: GELFAND-P000414

Diagram and typed description. Two separate horizontal segments, the right one higher, with a dot between them.



Original source details: GELFAND-P000414

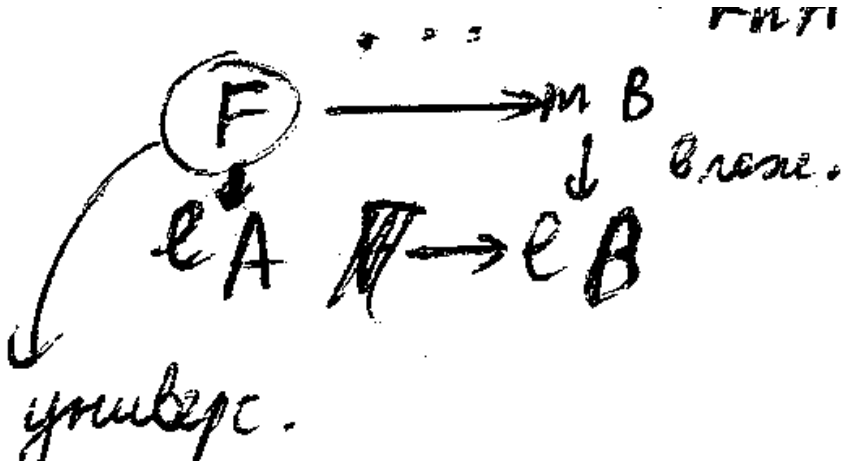
Diagram and typed description. Below "hom." is the inserted line $A \longrightarrow B$.

To $\forall$ ring hom. corresp. an exact seq. of K_n.

$A \longrightarrow B$

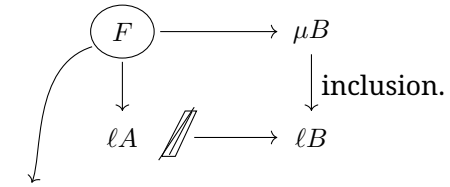
and a new ring, so that relative K_n = absolute of the new ring

$$A \xrightarrow{f} B$$
$$\cdots \longrightarrow K_n A \longrightarrow K_n B \longrightarrow K_n F \longrightarrow K_{n-1} A \longrightarrow \dots$$



Original source details: GELFAND-P000414

Diagram and typed description. A square with circled F, four arrows and an arc to “universal”. Before the lower arrow is a crossed-out mark; the former sign is not recovered. The upper-right letter is μ-/m-like: provisionally μB; mB and MB are not excluded. The notation is not replaced from the expected inclusion.



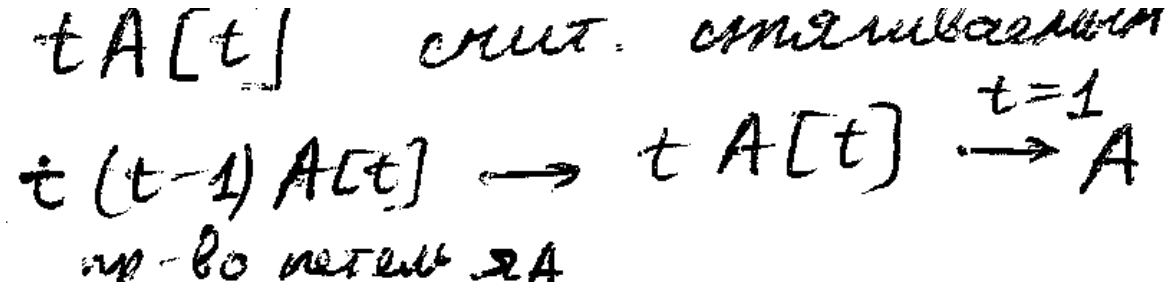
(fibred product; inverse image)

pullback.

In L. V.’s opinion, Quillen’s theory coincides with these.

Karoubi’s theory does not coincide.

$tA[t]$ is considered contractible



Original source details: GELFAND-P000414

Diagram and typed description. A lower caption for $t(t-1)A[t]$ and the label $t = 1$ at the right arrow.

$$t(t-1)A[t] \longrightarrow tA[t] \xrightarrow{t=1} A$$

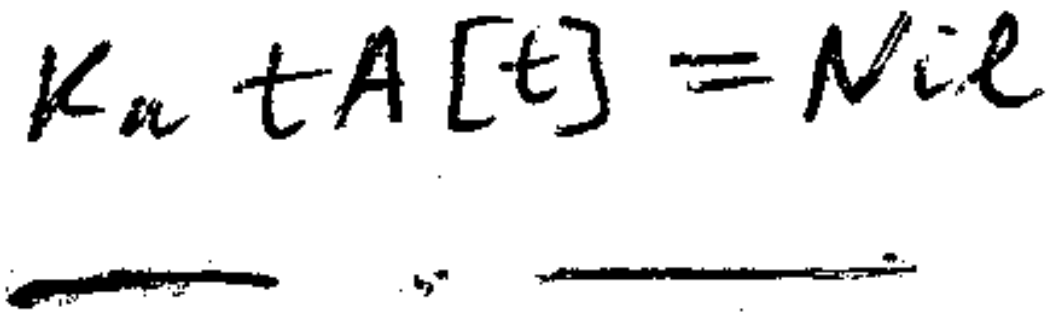
loop space ΩA

GELFAND-P000415

[source number: 54]

$K_{n+1}A = K_n\Omega A$: According to Karoubi.

In the present K-theor.



Original source details: GELFAND-P000415

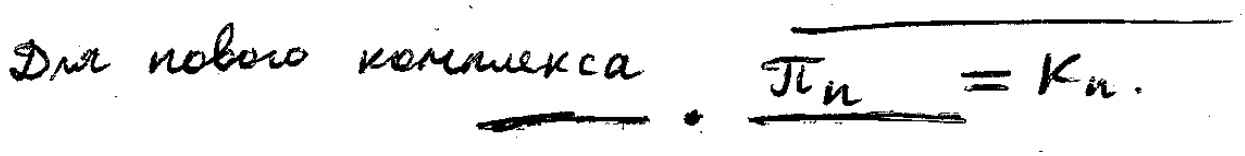
Diagram and typed description. Below the formula are two separate horizontal strokes and a small mark between them.

$K_n\, tA[t] = Nil$

On Quillen's theorem:

Let G be any group, $E = [G, G]$. and $E = [E, E]$. Consider the universal classif. space BG with a cell structure. Publ. Math., 34. $\pi_1 BG = G$, the others 0. $(BG = K(G, 1))$.

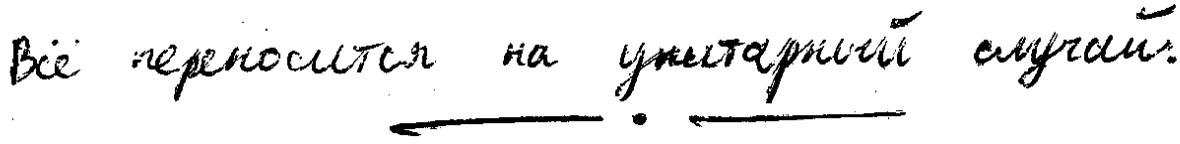
Make minimal cell attachments, turning it into a group (H-space). Homology does not change. E is killed.



Original source details: GELFAND-P000415

Diagram and typed description. Below the end of "For the new complex" is a separate stroke, then a dot and a stroke below π_n.

For the new complex $\pi_n = K_n$.



Original source details: GELFAND-P000415

Diagram and typed description. Below the line are two separate strokes and a dot between them; both strokes point left.

Everything carries over to the unitary case.

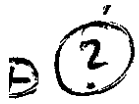
Theor. Valonera/Vagonera[?] on Laurent extensions.



Original source details: GELFAND-P000415

Uncertain reading. The underlined surname is provisionally “Валонера/Вагонера” (Valonera/Vagonera); the middle letters are joined. The full reading is not established.

$$K_2A[t,t^{-1}] = K_2A \oplus K_1A \oplus \textcircled{?}.$$

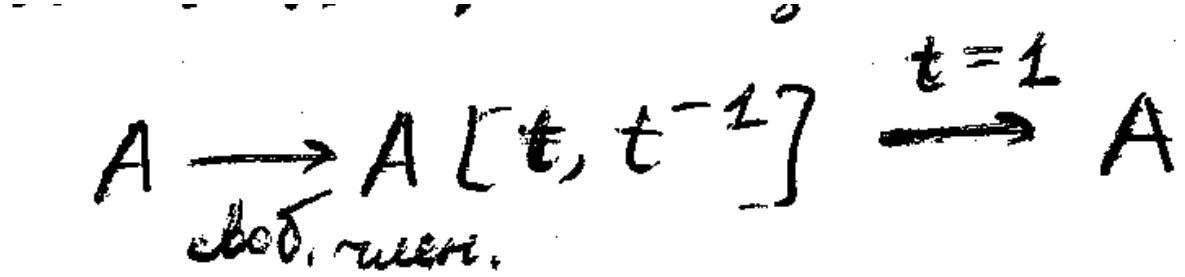


Original source details: GELFAND-P000415

Uncertain reading. The upper curve of the last circled sign is 2-like, with a separate dot below. The visible strokes are reproduced; “?” remains a candidate, and no numeric summand is asserted.

the theor. holds $\forall n$, but from this case all the others follow.

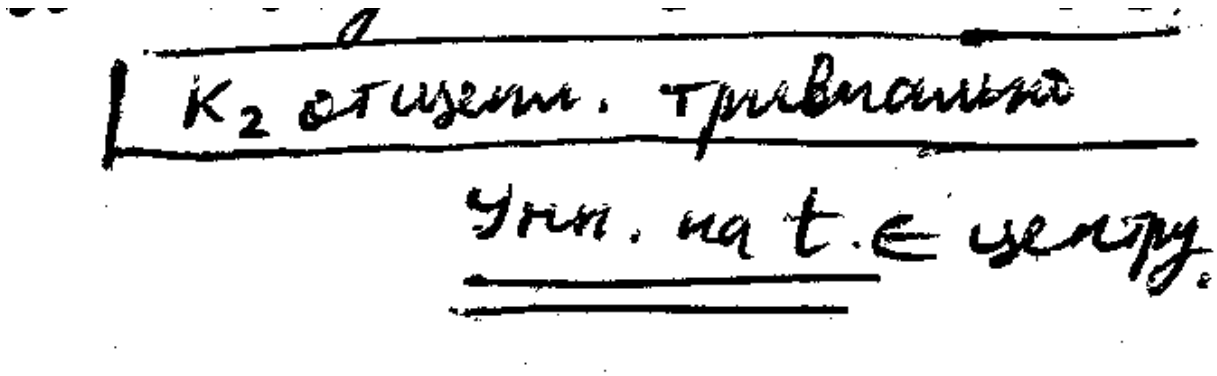
Proof.



Original source details: GELFAND-P000415

Diagram and typed description. The first caption is below its arrow, the second above its arrow.

$$A \xrightarrow{\text{constant term.}} A[t,t^{-1}] \xrightarrow[t=1]{} A$$

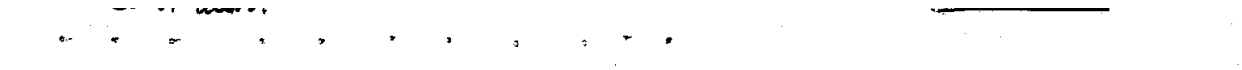


Original source details: GELFAND-P000415

Diagram and typed description. At right is an enclosure with upper, left and lower sides; the right side is open. There are two strokes below the following words.

K_2 relative is trivial

Mult. by $t \in$ the centre.

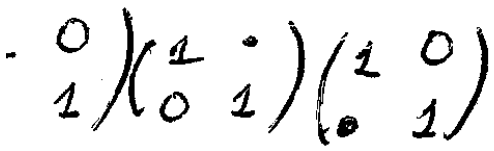


Original source details: GELFAND-P000415

Undeciphered passage. Between the diagram and the matrices is a row of eleven small separate dots/short strokes. Their purpose and any earlier inscription are not established.

$$\begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} = \begin{pmatrix} 1 & a-1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & \cdot \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \circ & 1 \end{pmatrix}.$$

(Whitehead's formula). (the dots are recovered uniquely).



Original source details: GELFAND-P000415

Uncertain reading. The upper-right entry of the third matrix is a dot. The lower-left entry of the fourth is a thick 0-like dot with a small opening; its contour is reproduced, but its value is unwritten.

g is a matrix over A. (defines an elem. of K_1)

$$\begin{pmatrix} gt & 0 \\ 0 & g^{-1}t^{-1} \end{pmatrix} \qquad \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix} \qquad \begin{pmatrix} g & 0 \\ 0 & g^{-1} \end{pmatrix}.$$



Original source details: GELFAND-P000415

Uncertain reading. The upper-left entry of the middle matrix is overwritten; provisionally t, with the earlier strokes unrecovered. There are no equality signs between the three matrices.



Now decompose them into elem.



Original source details: GELFAND-P000415

Uncertain reading. After the abbreviation "элем." (elem.) is a faint slanted chain of broken strokes with two darker spots. Its connection to the inscription and the punctuation are not established.



one obtains relations —

— an elem. of K_2

GELFAND-P000416

The converse: An elem. of $K_2A[t,t^{-1}]$.



Original source details: GELFAND-P000416

Diagram and typed description. The underlined “Laurent series” and a horizontal arrow with a wavy beginning.

Laurent series. ⇝ Mult. by it — an infinite matrix

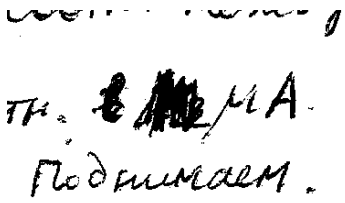


Original source details: GELFAND-P000416

Undeciphered passage. Before “One obtains relations” is a separate densely crossed-out formula. K-like, a lower 2-like and A-like fragments are distinguishable; the full inscription is unrecovered.



One obtains relations between infinite matrices. Cut off ⇒ relations $\ell A/\mu A$. (mod. finite matrices).



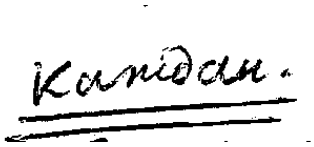
Original source details: GELFAND-P000416

Uncertain reading. The numerator is overwritten: provisionally ℓA, with MA possible beneath the dense earlier sign; the denominator is μA. The earlier strokes are unrecovered.

~~ℓA~~$/\mu A$

We need $\ell A/\mu A$. Lift.

Kazhdan/Kazdan[?].



Original source details: GELFAND-P000416

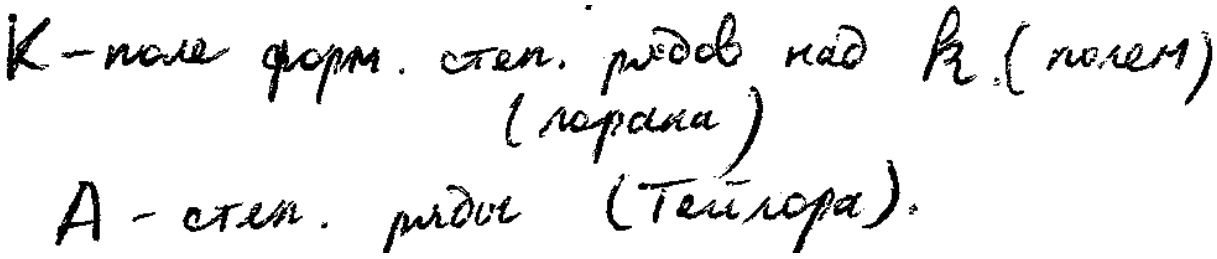
Uncertain reading. The doubly underlined heading is provisionally “Каждан/Каздан” (Kazhdan/Kazdan); the join after “Ка” is ambiguous.



Original source details: GELFAND-P000416

Uncertain reading. The crossed-out heading is provisionally “Let there be[?] 2 formal power series”; “Let 2 formal power series be given” is also possible. The second word merges with the crossing stroke.

~~Let there be[?] 2 formal power series~~



Original source details: GELFAND-P000416

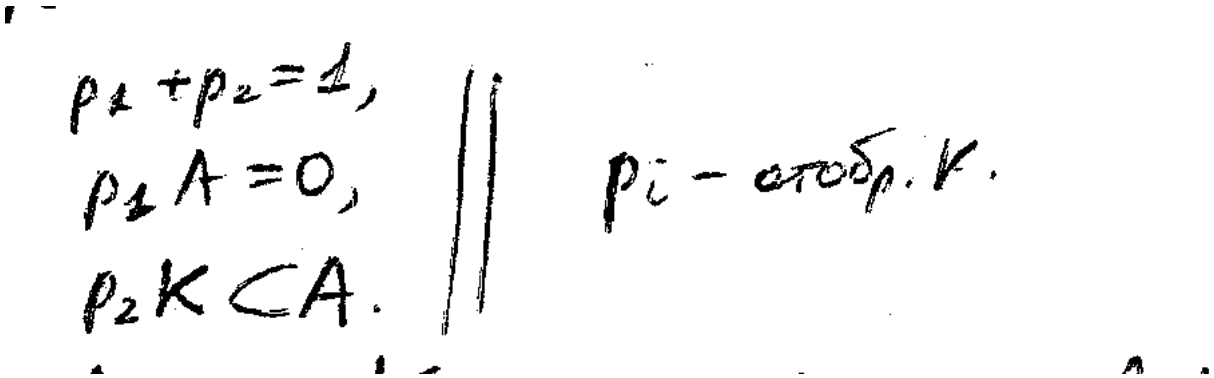
Diagram and typed description. A parenthetical caption is inserted below “power series”.

K is the field of formal power series over k (a field).

(Laurent)

A is power series (Taylor). $k \subset K \supset A$.

Let



Original source details: GELFAND-P000416

Diagram and typed description. To the right of the three conditions are two separate vertical rules and a caption.

$$\begin{aligned} p_1 + p_2 &= 1, \\ p_1 A &= 0, \\ p_2 K &\subset A. \end{aligned}$$

p_i — maps of K.

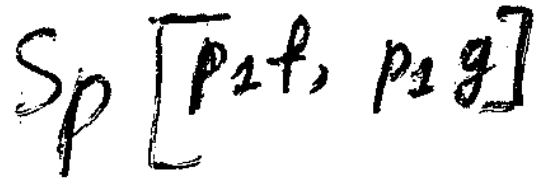
Let $f, g \in K$. The residue of $\underline{f\,dg}$ — ?



Original source details: GELFAND-P000416

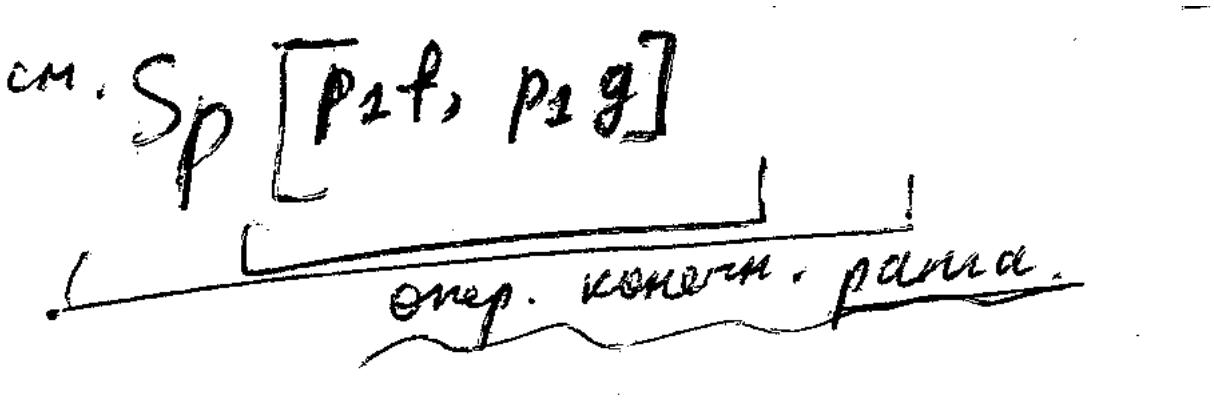
Uncertain reading. The parenthetical note is provisionally “(без парам.!)” (without param.!); the abbreviation is unexpanded. The readings “без лоран.!” (without Laurent.!) and “без полюса!” (without a pole!) are weaker candidates; the first sign of the abbreviation is closer to “п”.

Consider $Sp[p_1 f, p_1 g]$.



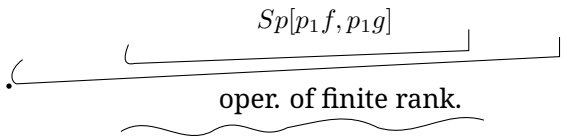
Original source details: GELFAND-P000416

Uncertain reading. In both arguments after p are small lower signs, provisionally $1,1$; $2,1$ and $1,2$ are possible. The first sign especially merges with p.



Original source details: GELFAND-P000416

Diagram and typed description. Below the commutator are two long lower contours with short raised ends. There is a separate dot at the left end of the outer contour; below is a caption with a wavy underline.



GELFAND-P000417

[source number: 55]

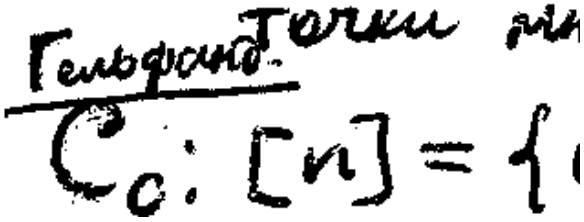
13/III Prof. Morava. Work by G. Segal and K°[?].

раб. G. Segal и K°.

Uncertain reading. A small circle is written above K. It could be the abbreviation for “and Co.” or the notation K^0.

Original source details: GELFAND-P000417

C is a categ., BC the simpl. set corresponding to this category. the points of the set are the objects of this category.



Original source details: GELFAND-P000417

Diagram and typed description. At the left above the beginning of the C_0 line is an underlined surname.

Gelfand.

C_0: $[n] = \{0, 1, \ldots, n\}$. — objects of an auxiliary categ.

Morphisms: $[k] \xrightarrow[\varphi(i)]{} [n]$, $\varphi(1) \leq \ldots \leq \varphi(k)$.

$[k] \xrightarrow[\varphi(i)]{} [n]$ $\quad \varphi(1) \leq \ldots \leq \varphi(k)$.

Uncertain reading. In the first argument of φ is a thick 1; an earlier overwritten 0 is possible underneath.

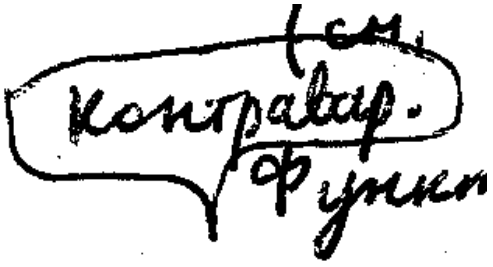
Original source details: GELFAND-P000417

(see Gabriel, Tsisman[?])

(см. главу Габриэль, Цисман)

Uncertain reading. The second surname is provisionally “Цисман” (Tsisman); the joined middle letters are ambiguous, and “Зисман” (Zisman) is possible.

Original source details: GELFAND-P000417



Original source details: GELFAND-P000417

Diagram and typed description. Above the beginning of the next line, “Contravar.” is outlined; the lower contour has a narrow projecting pointer.



A functor A from C_0 to the categ. K is called a simplicial object in the cat. K. A_k is the image of the object $[k]$. $A_0, A_1, \ldots, A_k, \ldots$ and a system of morphisms. If K is the cat. of sets, then this is



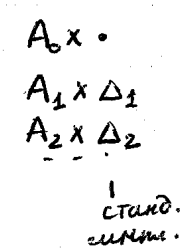
Original source details: GELFAND-P000417

Undeciphered passage. An earlier short word is densely crossed out; its original letters are not fully recovered. A lower loop is visible.



a simplicial set.

The system A_0 is a set of points. The system A_1 — $-\|-$ of segments. and so on.



Original source details: GELFAND-P000417

Diagram and typed description. Three product lines; below are two short strokes and a dot. Under the last triangle are a short vertical stroke and a caption.

$A_0 \times \cdot$

$A_1 \times \Delta_1$

$A_2 \times \Delta_2$

stand.
simpl.

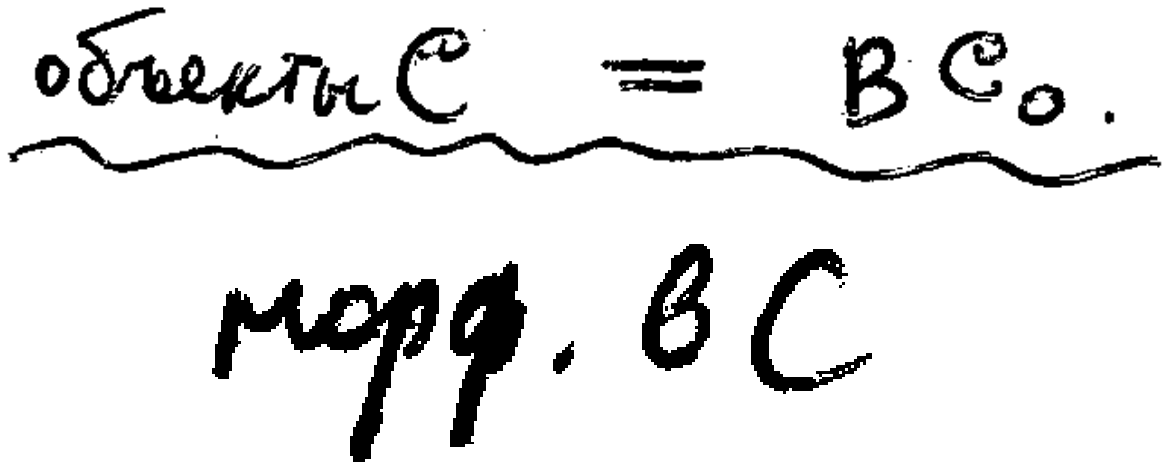
and glue in the natural way. Obtain an actual complex.



Original source details: GELFAND-P000417

Diagram and typed description. Under the line is a long leftward arrow; to the right of its left part is a gap with a separate dot, then the stroke continues.

From a categ. — a simpl.

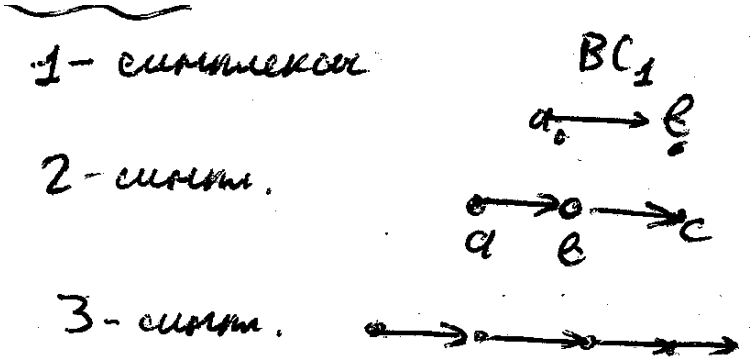


Original source details: GELFAND-P000417

Diagram and typed description. There is a wavy stroke beneath the equality. The next line is "morphisms in C".

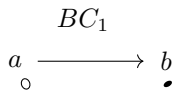
objects $C = BC_0$.

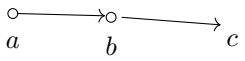
morphisms in C

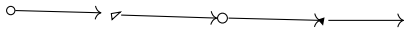


Original source details: GELFAND-P000417

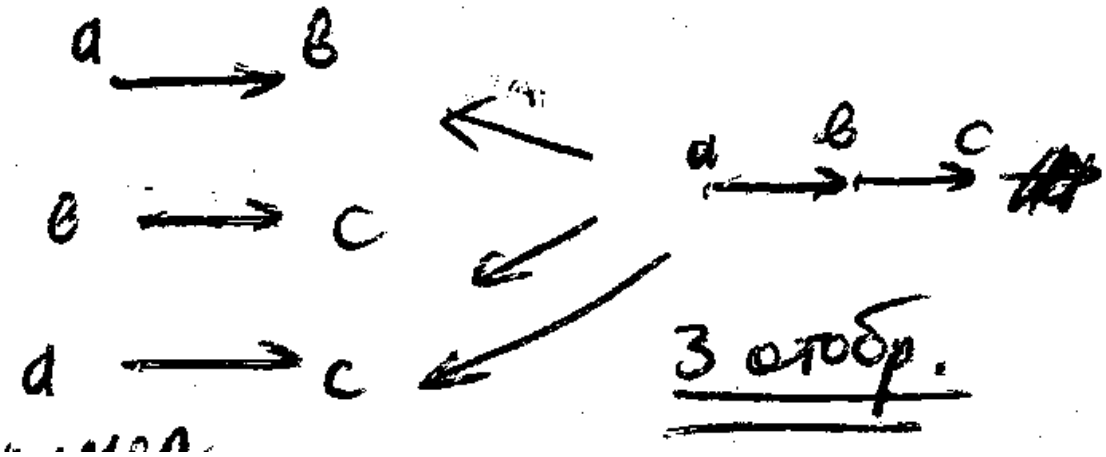
Diagram and typed description. Three lines with captions at the left. In the first, BC_1 is above the arrow; the right letter is provisionally b, possibly an e-like letter. Below a is a small closed contour, and below the right letter a separate thick stroke; the meanings of the lower signs are undetermined. In the second are two round marks and an arrow to c; the middle letter is also provisionally b, possibly e-like. In the third are four marked positions, the third and fourth merging with arrowheads; after the fourth is a further rightward arrow, with no added endpoint.

1-simplices 

2-simpl. 

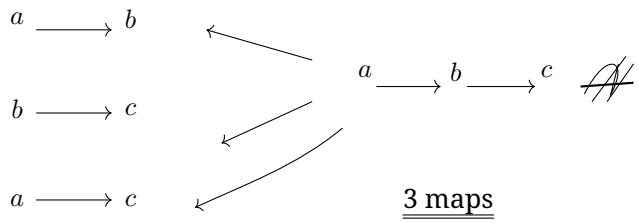
3-simpl. 

GELFAND-P000418



Original source details: GELFAND-P000418

Diagram and typed description. At the right is the chain $a \to b \to c$, with a separate densely crossed-out a-like sign after c. Three oblique arrows point left to the lines $a \to b$, $b \to c$, $a \to c$. Beneath the right chain, "3 maps" is doubly underlined. The earlier overlapping strokes of the last sign are not fully recovered.



Example.

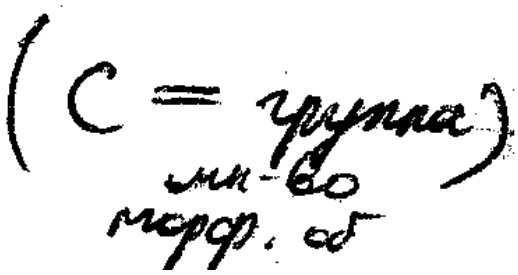
If C has one object, ~~[?]~~ $\forall$ morphism — isomorphism.



Original source details: GELFAND-P000418

Undeciphered passage. After "object" an earlier short inscription is crossed out; its letters are unrecovered. Below it an outlined e is inserted.





Original source details: GELFAND-P000418

Uncertain reading. Below the parenthetical inscription "$C =$ group" is the caption "set of morph. ob[?]". The final joined sign is provisionally read "об" (ob); the abbreviation is unexpanded and the reading is undetermined.

($C =$ group
set of
morph. ob[?])



Original source details: GELFAND-P000418

Diagram and typed description. After $BC_1 = G$, $Mor(e,e)$ is written separately. Below are three short faint marks and a separate dot.

$BC_1 = G$ $\qquad\qquad$ $Mor(e,e)$

$$BC_n = G \times G \times \ldots \times G.$$

def. $\Rightarrow$ Milnor's def. of the classifying space of the group G.

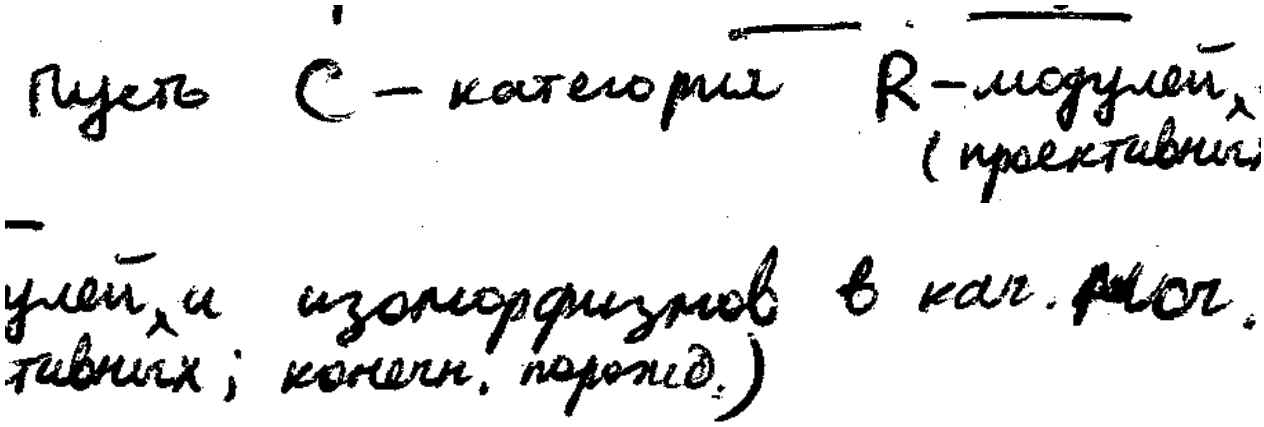


Original source details: GELFAND-P000418

Undeciphered passage. Before "of the group", an earlier short word is densely crossed out and not fully recovered.



since $BC_n \times \Delta_n$ and gluing. G a top. gr. $\Rightarrow$ one may equip with a topology.



Original source details: GELFAND-P000418

Diagram and typed description. Below the R-modules is a parenthetical insertion; between "modules" and "and" is a small angular stroke.

Let C be the category of R-modules and isomorphisms in the cat. Mod.

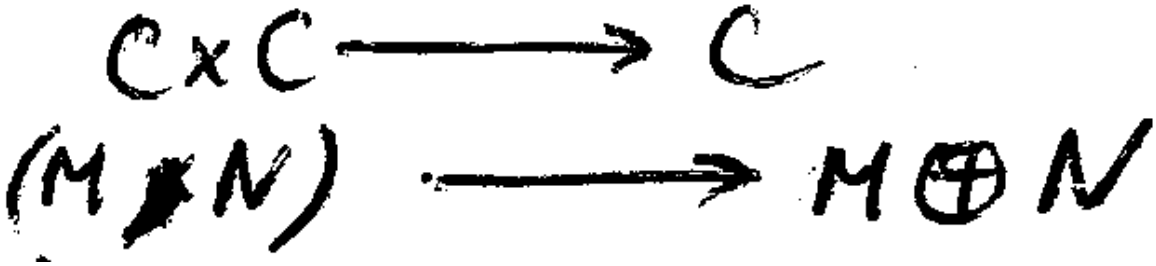
(projective; finitely generated.)



Original source details: GELFAND-P000418

Uncertain reading. The category name is provisionally "Мод." (Mod.); the initial letter merges with dense earlier strokes, and the earlier inscription is unrecovered.

BC.



Original source details: GELFAND-P000418

Diagram and typed description. Two lines of arrows; in the lower one the sign between M and N is overwritten on earlier strokes.

$$C \times C \longrightarrow C$$

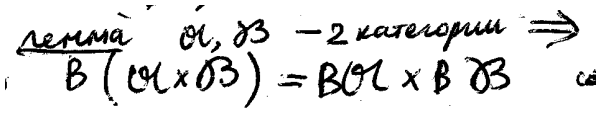
$$(M \;?\; N) \longrightarrow M \oplus N$$



Original source details: GELFAND-P000418

Uncertain reading. In the lower left argument the sign is provisionally $\times$; one diagonal stroke is very thick, and the earlier sign is unrecovered.

Lemma. $\mathcal{A}, \mathcal{B}$ are 2 categories $\Rightarrow B(\mathcal{A} \times \mathcal{B}) = B\mathcal{A} \times B\mathcal{B}$.



Manuscript notation. The category symbols are script capitals $\mathcal{A}$ and $\mathcal{B}$.

Original source details: GELFAND-P000418



Original source details: GELFAND-P000418

Diagram and typed description. In the left margin, “Proof” is crossed out.

~~Proof~~



Original source details: GELFAND-P000418

Undeciphered passage. At the right after the product formula an earlier phrase is densely crossed out and not fully recovered. A separate lower loop is visible beneath the stroke.



$BC \times BC \longrightarrow BC$, i.e. BC is an H-space.



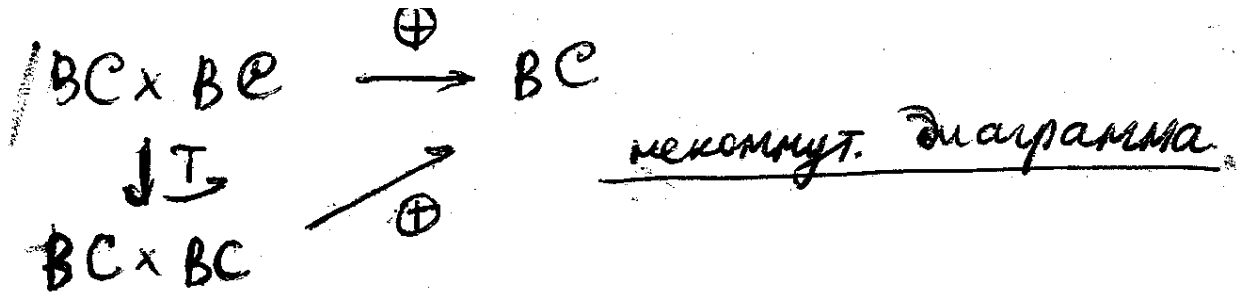
Original source details: GELFAND-P000418

Undeciphered passage. After the doubly underlined heading a short inscription is densely crossed out; its letters are unrecovered. At the left is a separate angular mark.



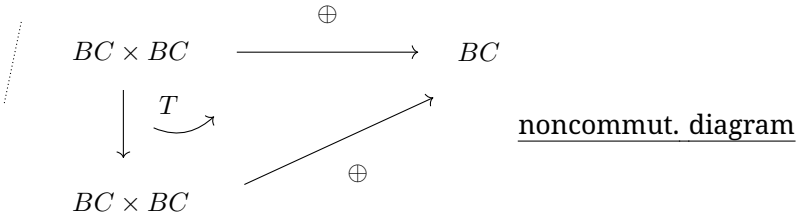
Remark:

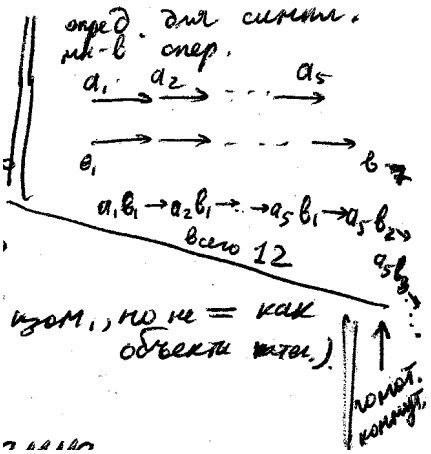
$M \oplus N \neq N \oplus M$ (isom., but not $=$ as objects of the categ.)



Original source details: GELFAND-P000418

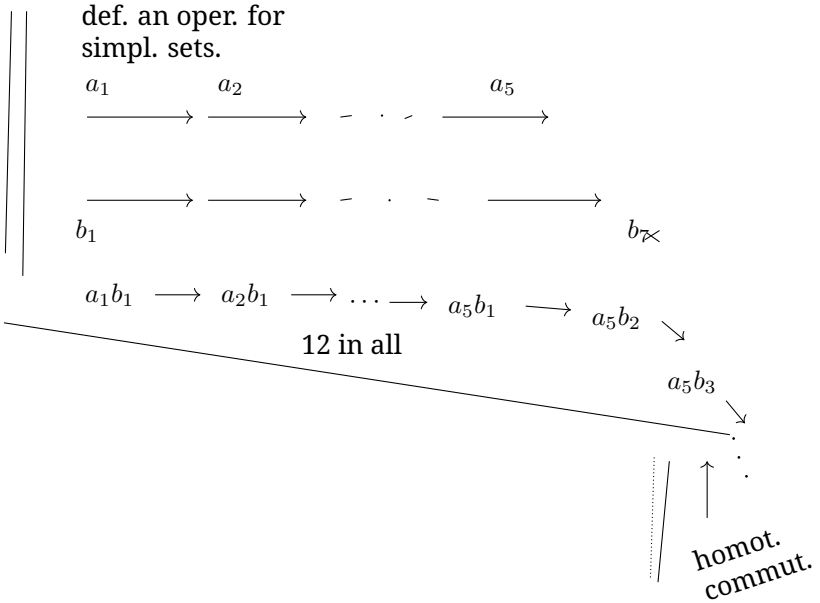
Diagram and typed description. The top and oblique arrows of the triangular diagram are labelled $\oplus$; the vertical arrow points downward and is labelled T. Below T is a separate short curved rightward arrow; at the left is a faint broken stroke.





Original source details: GELFAND-P000418

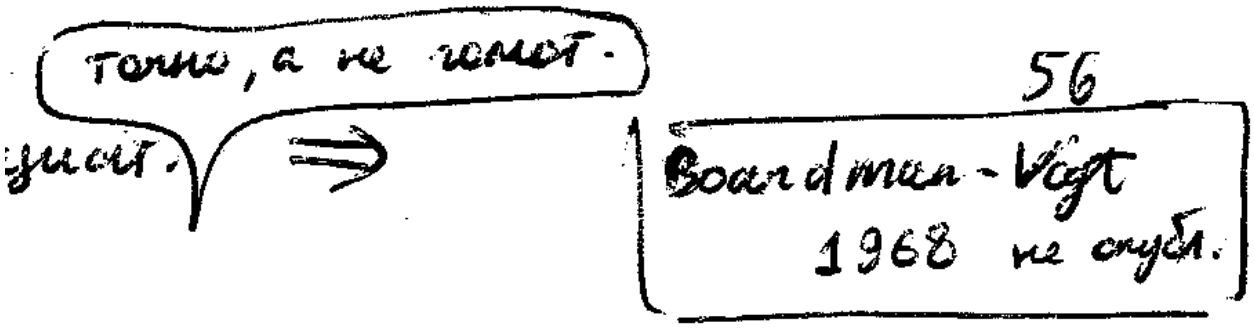
Diagram and typed description. The right-hand insertion has two left vertical rules and an oblique lower boundary. Above the arrows of the upper line are the labels a_1, a_2, a_5; below the outer arrows of the second, b_1, b_7, with the final lower index overwritten and provisionally read 7. Next is written $a_1b_1 \to a_2b_1 \to \cdots \to a_5b_1 \to a_5b_2$, with a_5b_3 below, an arrow and dots. The caption is “12 in all”. At the lower right margin are two further rules, an upward arrow and “homot. commut.”.



GELFAND-P000419

[source number: 56]

Remark. If $X \times X \longrightarrow X$ is commut. and associat.



Original source details: GELFAND-P000419

Diagram and typed description. Above the line, “exactly, and not up to homot.” is outlined with a lower projecting pointer; to the right are a separate frame and the sign ⇒.

exactly, and not up to homot.

⇒

Boardman–Vogt
1968 unpubl.



Original source details: GELFAND-P000419

Diagram and typed description. The sign between X and the product has a wavy upper stroke and two lower horizontal strokes.

$\Rightarrow X \cong \prod_n K(\pi_n, n)$

Theorem. Let $X \times X \longrightarrow X$



Original source details: GELFAND-P000419

Uncertain reading. The first word is provisionally “гомот.” (homot.); the initial letters are overwritten and the earlier strokes cannot be separated.

~~го~~mot.[?]

commut. and assoc. in a canonical way. Then ∃ a classif. space;



Original source details: GELFAND-P000419

Uncertain reading. The overwritten word is provisionally “оно” (it); “это” (this) is possible. The initial strokes merge.

~~Ait~~[?]

is also an H-space with a canonically homot. commut. and assoc. law. BX is this classif. space. $B(BX) \underset{def}{=} B^2X$.



Original source details: GELFAND-P000419

Diagram and typed description. At the left is a vertical sequence with two groups of continuation dots.

C

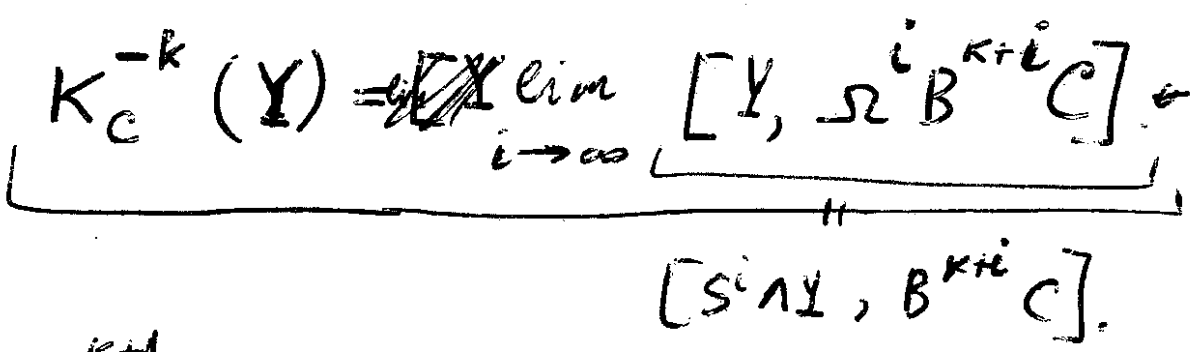
BC

B^2C

$\vdots$

B^kC

$\vdots$



Original source details: GELFAND-P000419

Diagram and typed description. The definition has two lower contours. Two short vertical strokes cross the long contour; beneath is a separate bracketed expression. The cancelled intermediate inscription is provisionally $[Y]$; the small right-hand sign is rendered by separate strokes.

$$K_C^{-k}(Y) = \cancel{[Y]} \lim_{i\to\infty} [Y, \Omega^i B^{k+i} C]$$

$$[S^i \wedge Y, B^{k+i} C]$$



Original source details: GELFAND-P000419

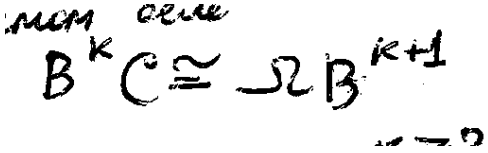
Uncertain reading. Between the equality and lim is an earlier crossed-out bracketed inscription; square brackets and a Y-like letter are visible, provisionally $[Y]$. A fuller inscription is undetermined.



Original source details: GELFAND-P000419

Uncertain reading. After the right bracket is a small sign with a loop, a long rightward stroke and a separate upper dot; below is a further separate mark. An i-like inscription is possible, but its meaning and index position are undetermined.

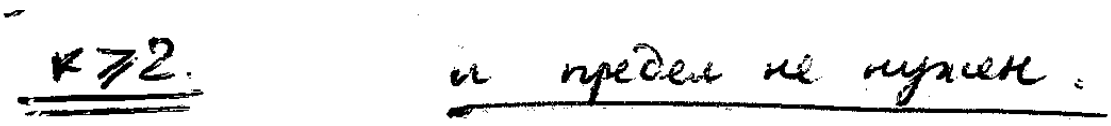
In fact



Original source details: GELFAND-P000419

Uncertain reading. At the right after the exponent $k+1$, the letter C is unwritten. The relation sign has a wavy stroke above two horizontal strokes.

$$B^k C \cong \Omega B^{k+1}$$



Original source details: GELFAND-P000419

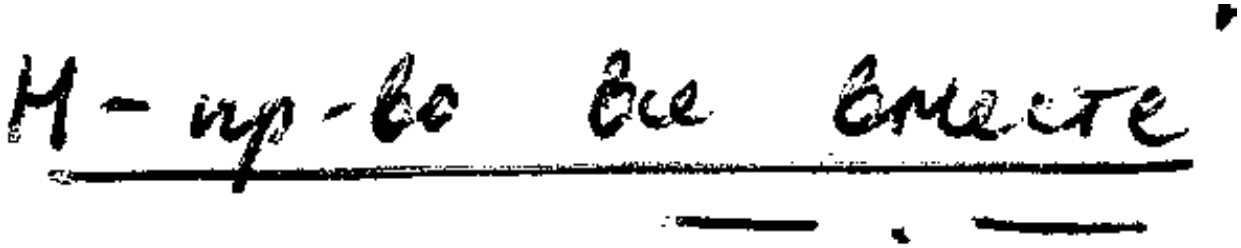
Diagram and typed description. The condition at the left is doubly underlined; at the right the entire phrase, including the initial “and”, is underlined.

$\underline{\underline{k \geq 2.}}$ $\underline{\text{and the limit is unnecessary.}}$

Example 1. $Y = \{*\}$, C the cat. of projective R-mod. $\Rightarrow$ the K-functor according to Quillen.
Example 2. C is the categ. of finite sets.

$$BC = \coprod_n B\Sigma_n$$

(disj. union of all classif. spaces of the symm. groups Σ_n).

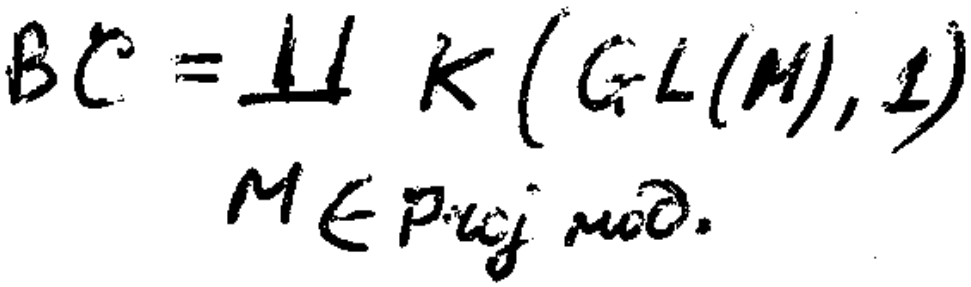


Original source details: GELFAND-P000419

Diagram and typed description. The whole line is underlined; below are two additional short strokes with a dot between them.

An H-space all together

For projective modules also



Original source details: GELFAND-P000419

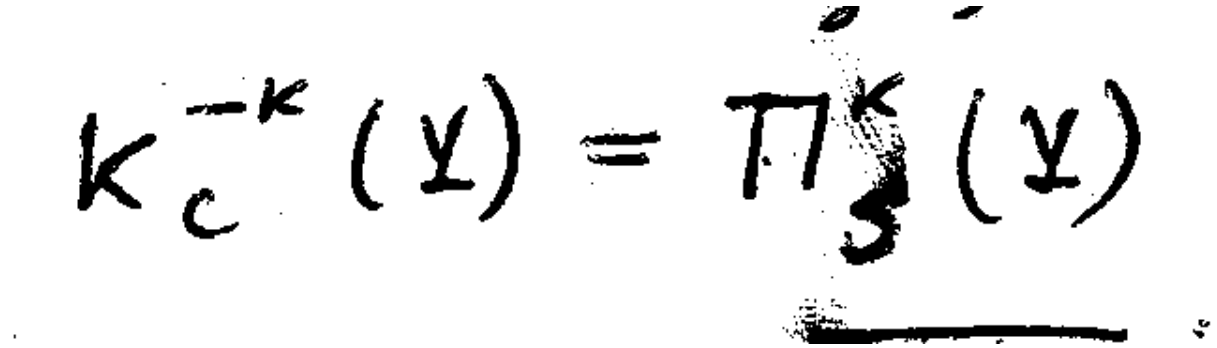
Diagram and typed description. The index $M \in$ Proj mod. is written on a separate line under the union sign.

$BC = \coprod K(GL(M), 1)$

$M \in$ Proj mod.

GELFAND-P000420

For the cat. of finite sets it is known that

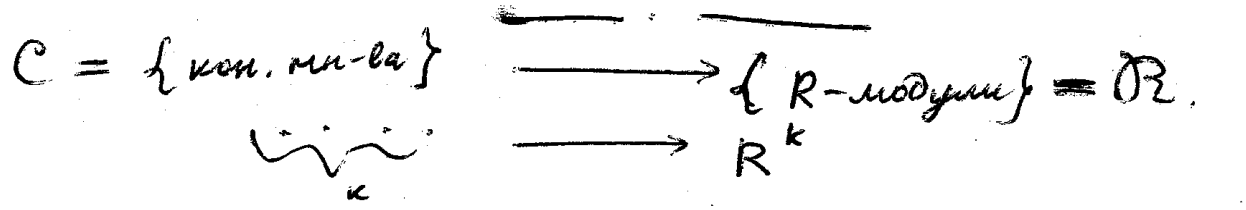


Original source details: GELFAND-P000420

Uncertain reading. The lower index at Π is overwritten and 3-like; provisionally s, possibly 3. The upper k is read separately.

$K_C^{-k}(Y) = \Pi_s^k(Y)$

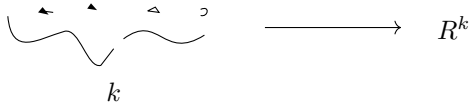
(stable cohomotopy groups.)

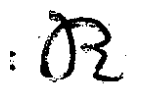


Original source details: GELFAND-P000420

Diagram and typed description. Two lines of arrows. Above the upper arrow are two separate horizontal strokes with a small mark between them. Beneath the left category are four separate small marks, a brace with a gap in its right half and the caption k; the lower arrow leads to R^k.

$C = \{\text{finite sets}\} \longrightarrow \{R\text{-modules}\} = \mathcal{R}$





Original source details: GELFAND-P000420

Uncertain reading. After the equality is a florid R-like letter with large left and upper loops; provisionally $\mathcal{R}$, with the precise font and overlapping strokes undetermined.

$$\Pi_s^k(Y) \longrightarrow K(R)^k(Y).$$

Y a point $\Rightarrow$ or stable homot. groups of spheres. One obtains their ~~[?]~~ homom. into the K-functor of any ring.

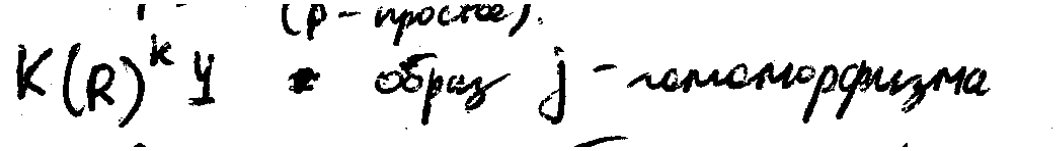


Original source details: GELFAND-P000420

Undeciphered passage. Before “homom.” an earlier separate word is densely crossed out and not fully recovered.



$R = F_p$, Y a point (p prime).



Original source details: GELFAND-P000420

Uncertain reading. Between $K(R)^kY$ and “image” is a small dense sign with two widened parts and a connecting stroke; its exact reading is undetermined.

$K(R)^kY$  image of the j-homomorphism

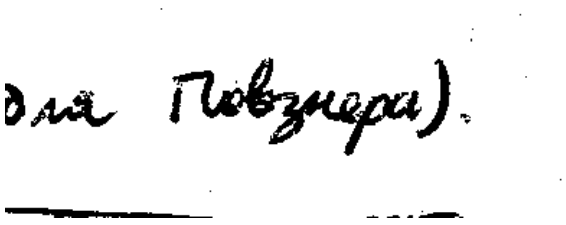


Original source details: GELFAND-P000420

Uncertain reading. Before “-homomorphism” is a letter with a lower loop and a separate upper dot, provisionally j; the case j/J is undetermined.

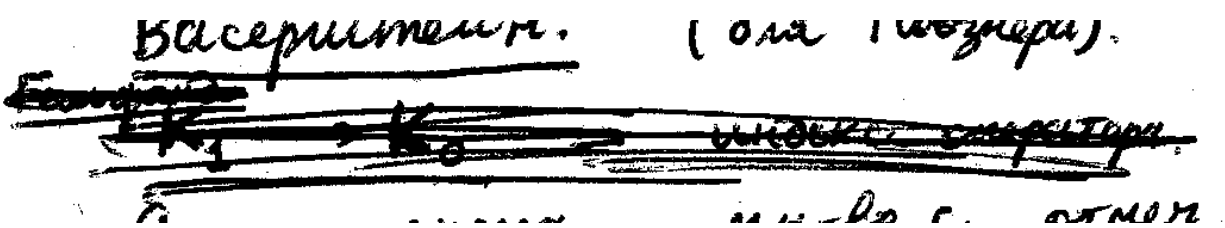
Next time — cobordisms. (in the same exposition).

Vasershtein. (for Povzner[?]).



Original source details: GELFAND-P000420

Uncertain reading. The first join after “П” (P) merges; “Повзнера” (Povzner) and “Певзнера” (Pevzner) remain candidates.



Original source details: GELFAND-P000420

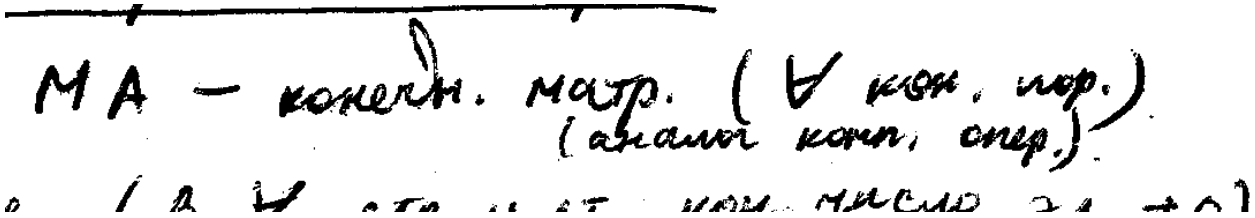
Undeciphered passage. An earlier beginning and a line with K_1, K_0 and the words “new operation” are crossed out. The initial E-like fragment is provisionally rendered “E…”; the whole earlier assertion is unrecovered.

~~E…~~

~~K_1 — K_0 … new operation~~

Simpl. scheme — a set with marked simplices (finite subsets) (together with $\forall$ set all its subsets are included). A topol. space is assigned to it. (see Godeman) homot. groups can be defined combinatorially.

A any ring; (noncommut.)



Original source details: GELFAND-P000420

Diagram and typed description. Above “finite” is a short curved stroke; the parenthetical caption is placed beneath the line.

MA — finite matrices. ($\forall$ finite order.)

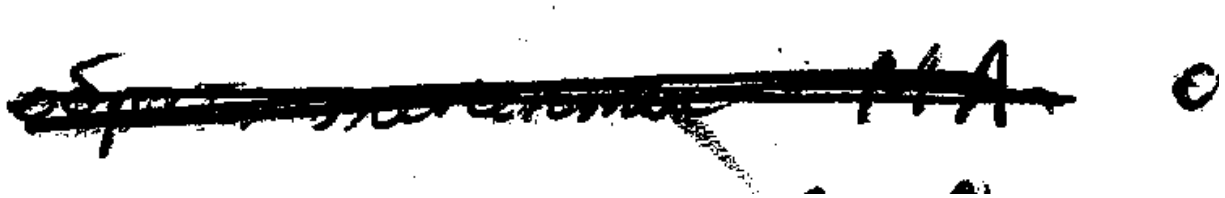
(analogue of compact oper.)

ℓA — row-finite (in $\forall$ row and col. a finite number of elements $\neq 0$) (analogue of bounded oper.)

$\mu A = \ell A / MA$ — the ring of Fredholm oper.

$K_i A$.

$GL(A)$ —



Original source details: GELFAND-P000420

Undeciphered passage. The earlier definition is crossed out; the beginning “invert…” and the final MA are visible, while the middle phrase is unrecovered. A faint oblique stroke descends from the middle part.

~~invert … MA~~



Original source details: GELFAND-P000420

Diagram and typed description. At the right is a new inscription with $1 + MA$ doubly underlined.

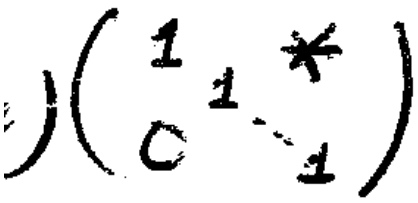
invertible elem. of the form $\underline{\underline{1 + MA}}$

A simplex — a finite subset in upper triang. matrices λ and

GELFAND-P000421

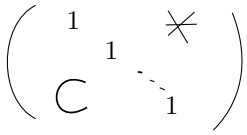
[source number: 57]

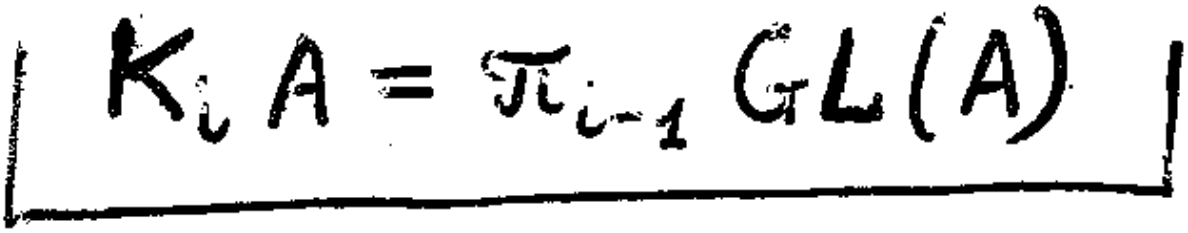
any conjugate of it (suffices by means of permutations) $a_1, \ldots, a_n$ — simpl., if $\exists f, g$ — invertible: fa_ig — upper triang. (with 1 on the diagonal).



Original source details: GELFAND-P000421

Diagram and typed description. A small matrix schematic: three 1-like diagonal entries, an upper-right star, short diagonal strokes and a lower-left $C/0$-like sign open on the right. The full size and lower-left sign are undetermined.

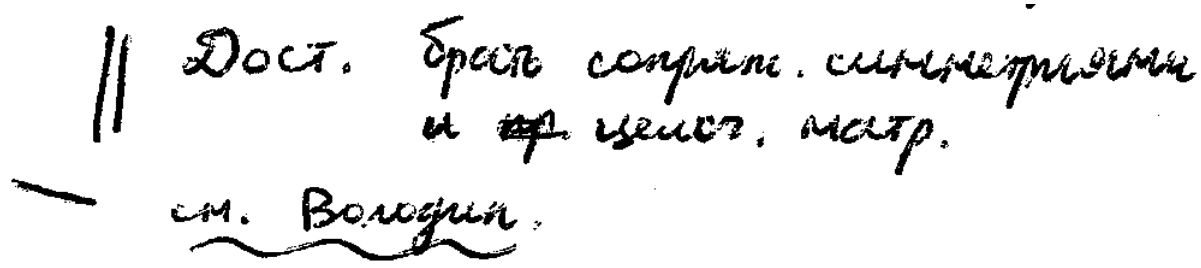




Original source details: GELFAND-P000421

Diagram and typed description. The formula is inside a three-sided enclosure with no top side.

$$K_iA = \pi_{i-1}GL(A)$$

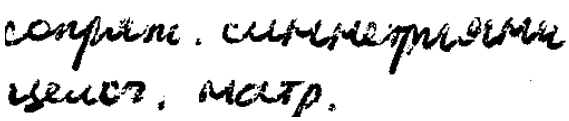


Original source details: GELFAND-P000421

Diagram and typed description. At the right are two vertical rules, two lines of a marginal note, a separate short oblique stroke lower down at the left and a wavy underline beneath “see Volodin.”.

Suffices to take conj. symmetric[?]
and integral matrices.

see Volodin.



Original source details: GELFAND-P000421

Uncertain reading. The adjective before “and” is provisionally “symmetric”; “singular” remains a candidate, and the joined middle is not confidently read.



Original source details: GELFAND-P000421

Undeciphered passage. Before “integral matrices.” a short word is densely crossed out and not fully recovered.

Exercise:



Original source details: GELFAND-P000421

Uncertain reading. find the arrows. The second word is provisional: its middle letters merge.

Theorem.

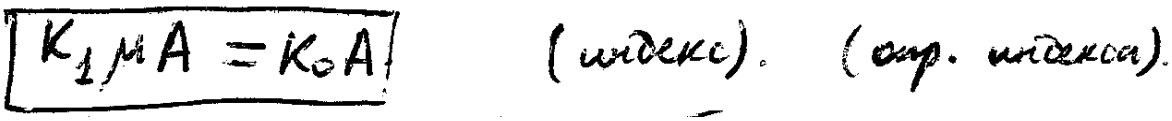


Original source details: GELFAND-P000421

Diagram and typed description. The formula and final period are inside the frame.

$$K_i A = K_{i+1} \mu A.$$

In particular,



Original source details: GELFAND-P000421

Diagram and typed description. The formula is framed; the explanations are outside at the right, with separate periods.

$$K_1 \mu A = K_0 A$$

(index). (def. of index).

The theor. generalizes to a Banach ring $\Rightarrow$ the usual theor. holds.

Question for Quillen's K-functor. And th



Undeciphered passage. A short unfinished beginning after the question; the continuation is unwritten.

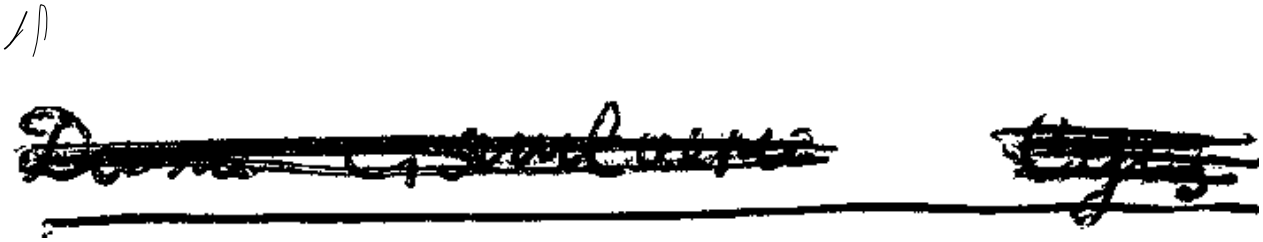
Original source details: GELFAND-P000421

This follows from contractib. of ℓA. Sufficient is $K_i(\ell A) = 0$. Contractibility of $\ell A \Leftrightarrow$ contractib. of the complex $GL(\ell A)$. (even in the strong sense)



Original source details: GELFAND-P000421

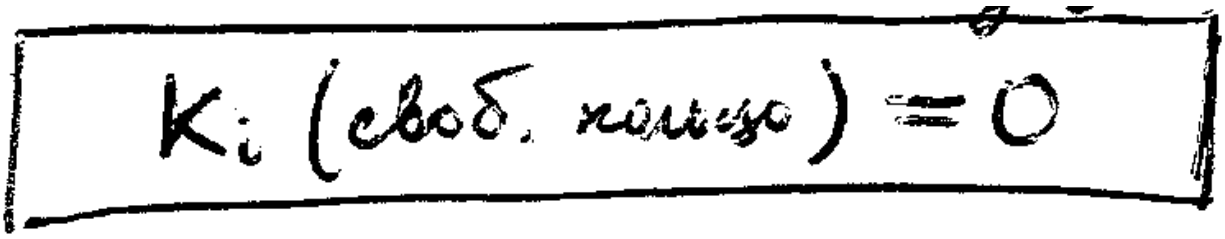
Uncertain reading. On the left margin is a separate slanting $1/i$-like mark and a tall curved stroke; the reading and function are undetermined.



Original source details: GELFAND-P000421

Undeciphered passage. "Proof of contraction" and a following separate inscription are crossed out; the last inscription is not fully recovered.

~~Proof of contraction~~



Original source details: GELFAND-P000421

Diagram and typed description. The formula is inside a closed slightly sloping frame.

$$K_i(\text{free ring}) = 0$$

—

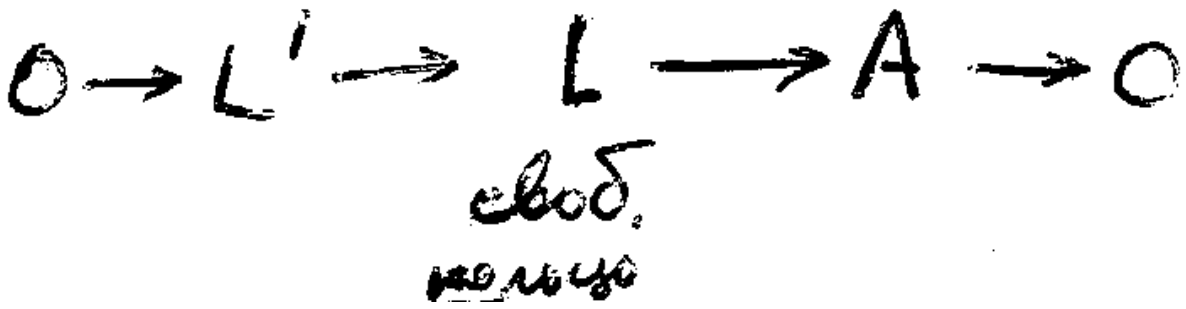


Original source details: GELFAND-P000421

Undeciphered passage. Before “but contractib.” is a separate short crossed inscription, not fully recovered.



but there is no contractib. of a free ring.



Original source details: GELFAND-P000421

Diagram and typed description. The arrow chain has a two-line caption below L. Its last round sign is open on the right.

$$0 \longrightarrow L' \longrightarrow L \longrightarrow A \longrightarrow C$$

free
ring.



Uncertain reading. The final sign after the arrow from A has an open right edge; $0/C$ remain candidates. Its contour is rendered without selecting a value.

Original source details: GELFAND-P000421

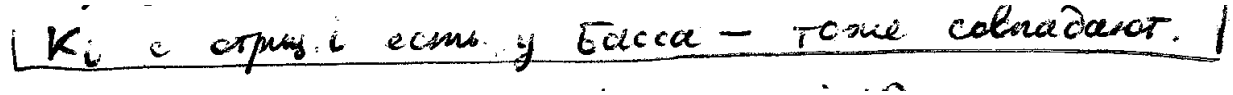


Original source details: GELFAND-P000421

Diagram and typed description. The formula is framed; the parenthetical explanation is at the right.

$K_i A = K_{i-1} L'$ (shift in the other direction)

There is a long exact seq. (everything follows from it) Hence follows agreement with the K-functors of Swan and Gershtein. (K_0, K_1 coincide, and further inductively).



Original source details: GELFAND-P000421

Diagram and typed description. The whole line is enclosed by a bottom rule with short vertical ends; there is no top side.

K_i with negative i occurs in Bass — also coincide.





Original source details: GELFAND-P000421

Undeciphered passage. A separate densely crossed beginning of the final line, not fully recovered.



For regular rings all $K_i = 0$ with $i < 0$.

GELFAND-P000422

Karoubi's theor.: $i > 0$ differs from Bass

No zero divisors — Karoubi also agrees

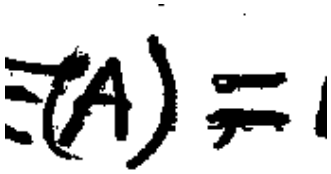
Quillen's theor.:



Original source details: GELFAND-P000422

Undeciphered passage. After the heading and before G is a separate densely crossed earlier fragment, not fully recovered.

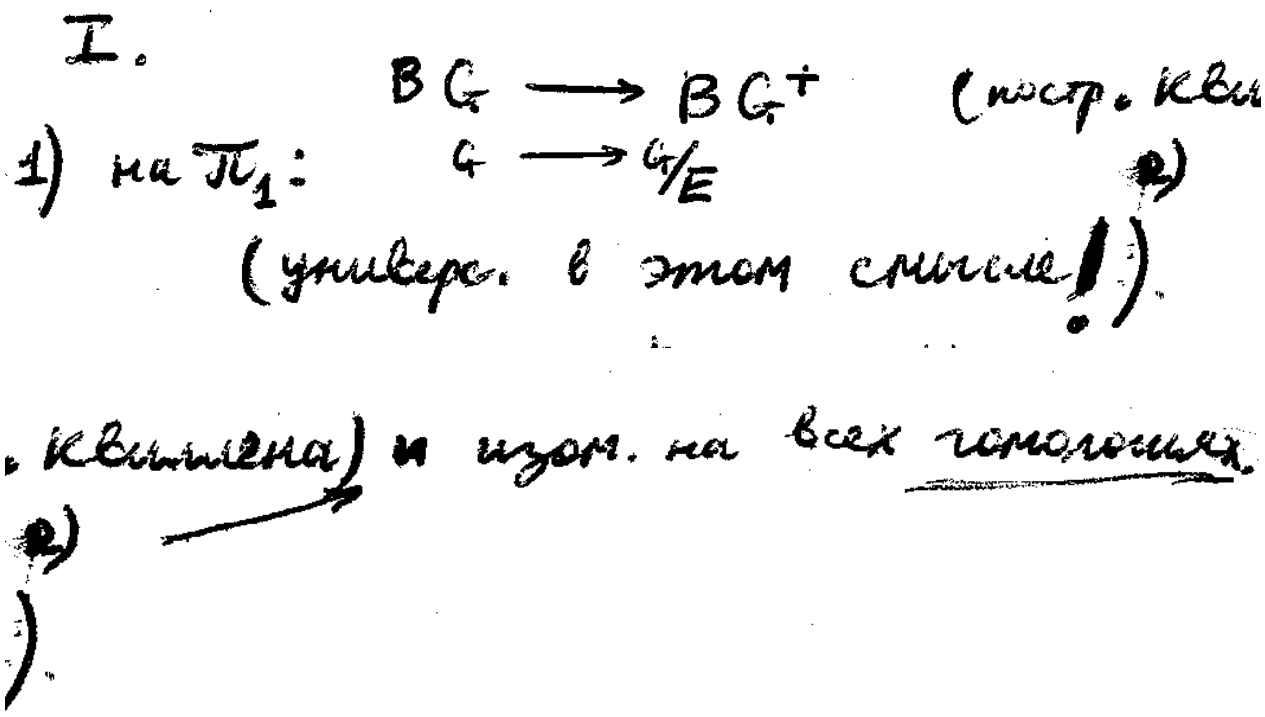




Original source details: GELFAND-P000422

Uncertain reading. After $E(A)$ are two horizontal strokes, with an oblique stroke descending from the lower one. An overwritten equality and a remnant of an earlier $\neq$ are possible; the sign is rendered as strokes without a choice.

$G = GL(A), \quad E = E(A) \quad [E, E] = [G, G].$



Original source details: GELFAND-P000422

Diagram and typed description. I and two numbered marks. The upper arrow leads from BG to BG^+; beneath it is the map on π_1. From "2)" a separate arrow points to the words about homology isomorphism; the last word is underlined.

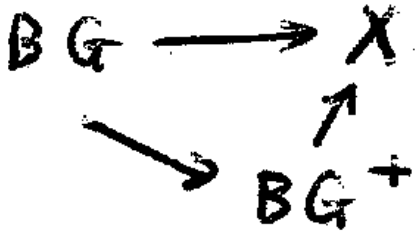
I.

$BG \longrightarrow BG^+$ (Quillen's constr.) and an isom. on all homology.

1) on π_1: $G \longrightarrow G/E$ 2)

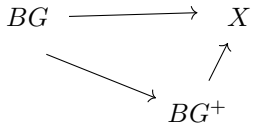
(universal in this sense!).

II. ∃ such a BG^+, which is an H-space and universal, i.e. for ∀ other map into an H-space



Original source details: GELFAND-P000422

Diagram and typed description. Three arrows: $BG \to X$, $BG \to BG^+$ and from the lower BG^+ up to X.



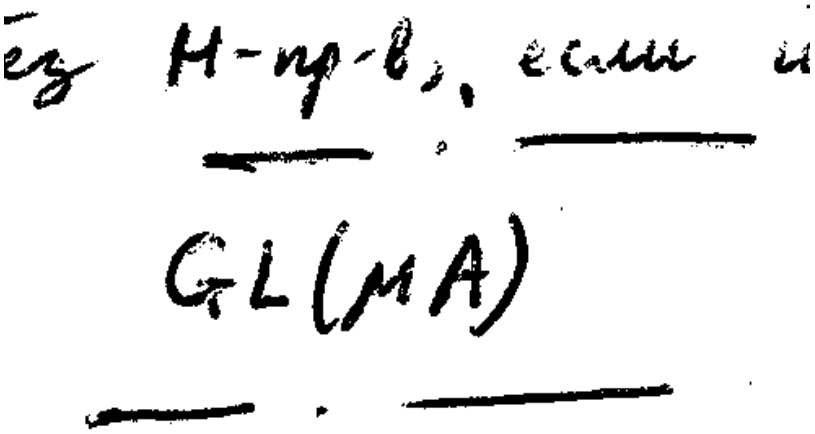
(Morava: and even without H-spaces, if it induces an isom. [?] of homology)



Original source details: GELFAND-P000422

Undeciphered passage. Before “of homology” a short earlier word is crossed out and not fully recovered.





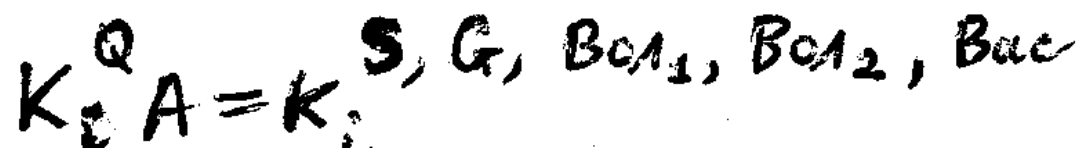
Original source details: GELFAND-P000422

Diagram and typed description. Two rows of separate horizontal strokes each have a small central mark. The upper row is below “H-spaces, if” and the lower beneath “Consider now”. The upper left stroke is pointed to the left.

Consider now $GL(\mu A)$

Quillen's constr.: $K_i^Q A = \pi_i BG^+$, $i \geq 1$.

Theorem: $K_i A = K_i^Q A$.

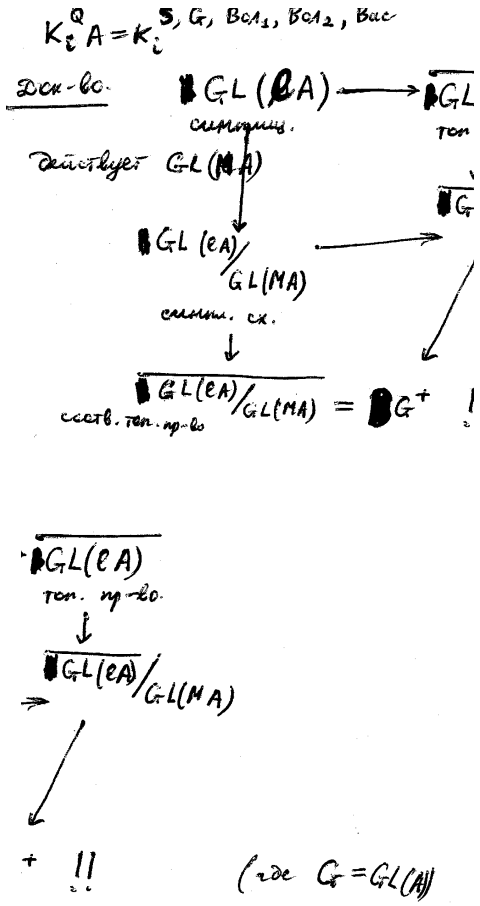


Original source details: GELFAND-P000422

Diagram and typed description. After the equality is just one K_i with the upper list S, G, "Vol" with lower $1, 2$ and "Vas"; the S is heavily inked. No additional A is written on the right.

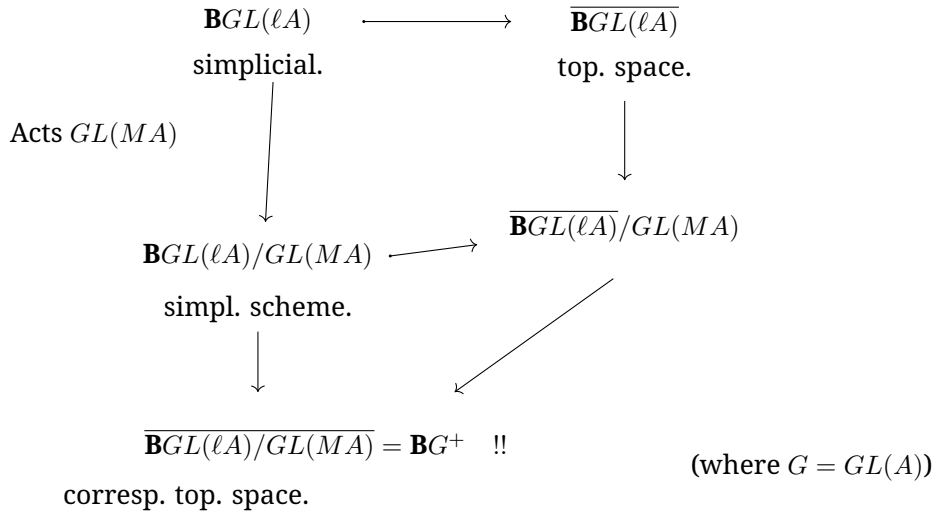
$K_i^Q A = K_i^{\mathbf{S},G,\mathrm{Vol}_1,\mathrm{Vol}_2,\mathrm{Vas}}$

Proof.



Original source details: GELFAND-P000422

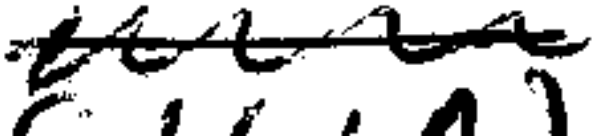
Diagram and typed description. The five nodes with BGL and the lower BG^{+} are provisional: each initial B is heavily blackened and the upper-left ℓ is overwritten; earlier signs are unrecovered. All six arrows, the $GL(MA)$ action, captions and two exclamation marks are retained. At the upper right the bar covers $BGL(\ell A)$, at the middle right only the numerator, and at the bottom the whole quotient.



GELFAND-P000423

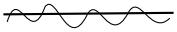
[source number: 58]

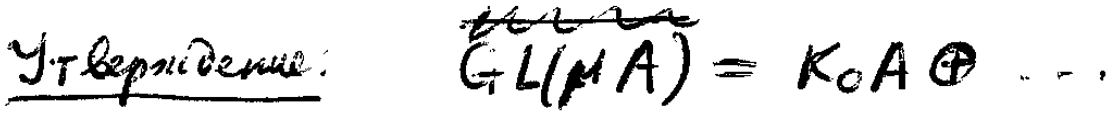
Assertion:



Original source details: GELFAND-P000423

Undeciphered passage. Above $GL(\mu A)$ is an earlier short crossed inscription; separate loops are visible, but it is not fully recovered.

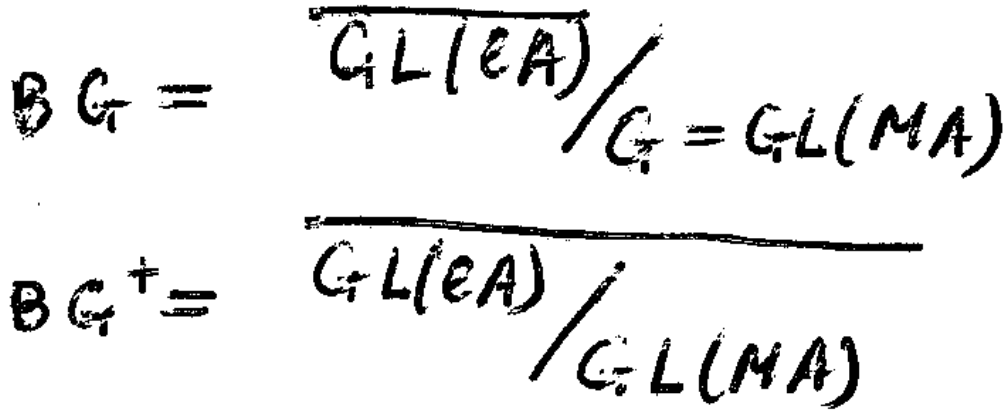




Original source details: GELFAND-P000423

Diagram and typed description. After $\oplus$ are three separate marks: a dot, a short horizontal stroke, a dot.

$GL(\mu A) = K_0 A \oplus \quad \cdot \; - \; \cdot$

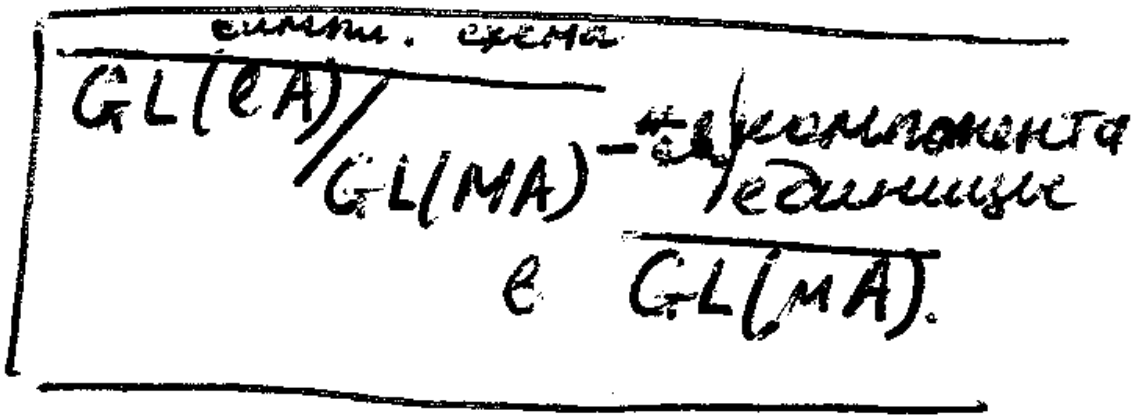


Original source details: GELFAND-P000423

Diagram and typed description. Two inscriptions with oblique quotient strokes. In the first, the upper bar covers only $GL(\ell A)$, and $G = GL(MA)$ is written beneath. In the second, the bar covers the entire quotient.

$$BG = \overline{GL(\ell A)} \,/\, G = GL(MA)$$

$$BG^{+} = \overline{GL(\ell A) \,/\, GL(MA)}$$

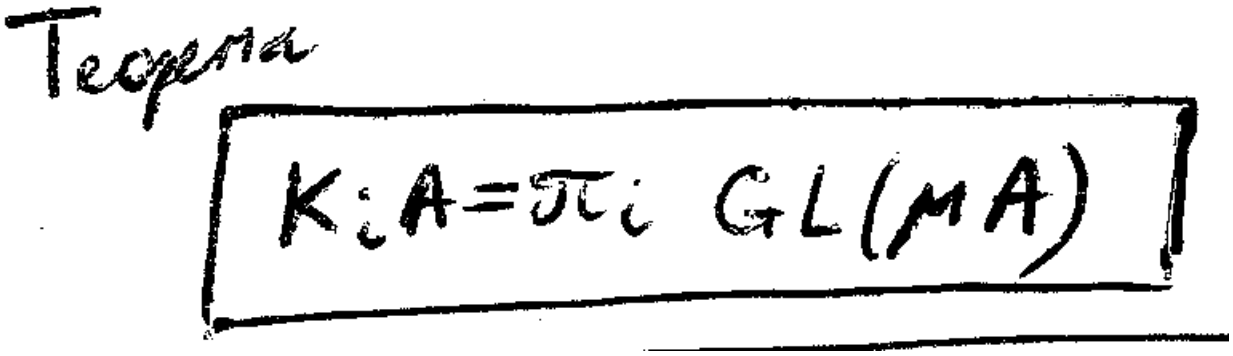


Original source details: GELFAND-P000423

Uncertain reading. At the right is a frame with no right side. The small overwritten mark before "component" has two horizontal and crossing strokes with a separate 1-like inscription; its meaning is undetermined.

simpl. scheme

$GL(\ell A)/GL(MA)$ – component of the identity in $\overline{GL(\mu A)}$.

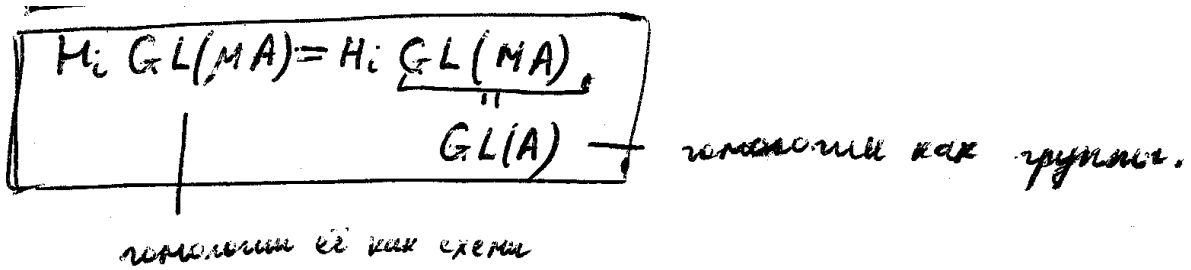


Original source details: GELFAND-P000423

Diagram and typed description. The heading is outside the sloping closed frame; the formula is inside.

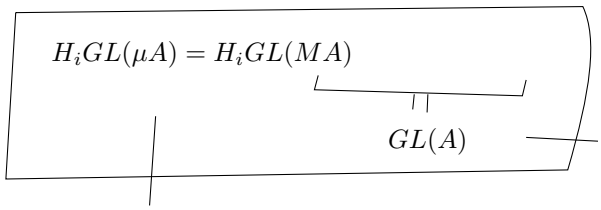
Theorem

$$K_iA = \pi_i GL(\mu A)$$



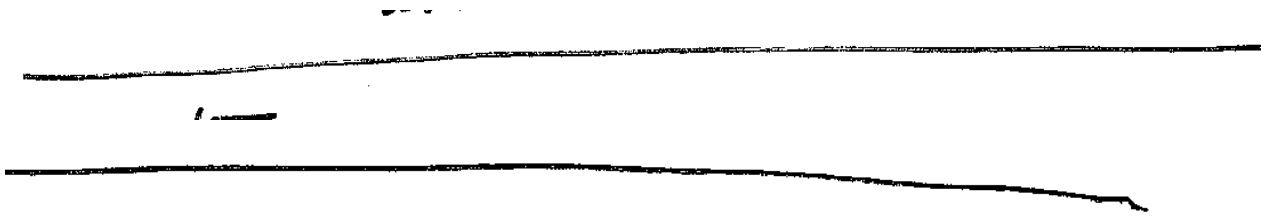
Original source details: GELFAND-P000423

Uncertain reading. Inside the frame is the upper equality; beneath $GL(MA)$ are a bracket and two short vertical strokes, with $GL(A)$ below. The link's meaning and scope of H_i are undetermined. Separate captions at the right and below have leader strokes.



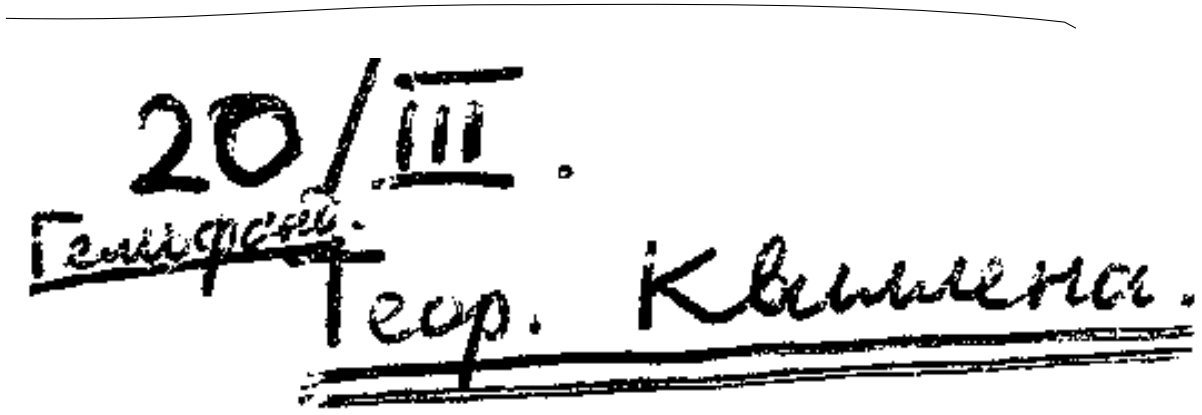
homology as a group.

its homology as a scheme



Original source details: GELFAND-P000423

Diagram and typed description. A long slightly curved transverse rule.



Original source details: GELFAND-P000423

Diagram and typed description. The date has rules above and below III; at the left is an underlined name, and below it a doubly underlined heading.

20/III.

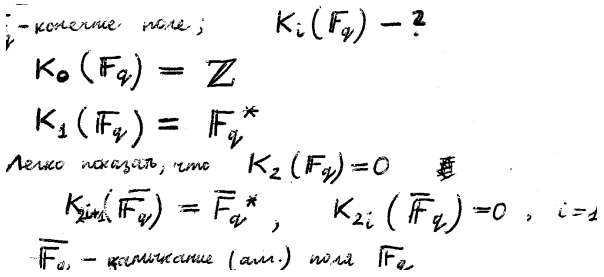
Gelfand.

Quillen's theor.

(proved nonelementarily) $\mathbb{F}_q$ — a finite field; $K_i(\mathbb{F}_q)$ — ?

$$K_0(\mathbb{F}_q) = \mathbb{Z},$$
$$K_1(\mathbb{F}_q) = \mathbb{F}_q^*.$$

It is easy to show that $K_2(\mathbb{F}_q) = 0$.



Original source details: GELFAND-P000423

Manuscript notation. The opening formulas have double-stem F. The first algebraic-closure equation has a single stem on its right-hand side.

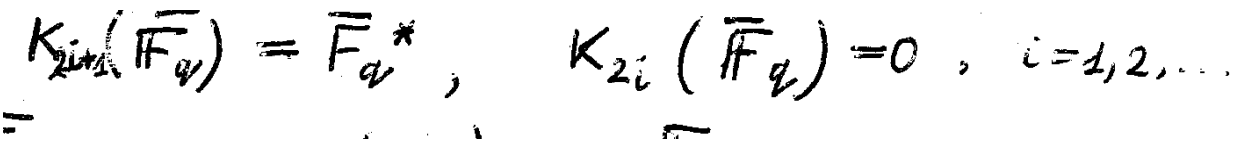


Original source details: GELFAND-P000423

Undeciphered passage. After the equality to zero is a separate small crossed inscription, not fully recovered.







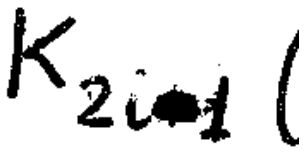
Original source details: GELFAND-P000423

Uncertain reading. The first index is provisionally $2i+1$: $2i$, a dense $+$ and 1 are visible, but an earlier overwritten layer is unrecovered. The range $i = 1, 2, \ldots$ is retained.

$$K_{2i+1}(\overline{\mathbb{F}}_q) = \overline{F}_q^*, \quad K_{2i}(\overline{\mathbb{F}}_q) = 0, \quad i = 1, 2, \ldots.$$

$\overline{\mathbb{F}}_q$ — the (alg.) closure of the field $\mathbb{F}_q$. (like the periodicity theor.)

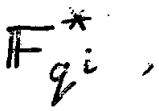
$$K_{2i}(\mathbb{F}_q) = 0,$$



Original source details: GELFAND-P000423

Uncertain reading. Between $2i$ and 1 is a dense overwritten mark, rendered by its contour; $+$ and $-$ remain candidates.

$K_{2i \bullet 1}(\mathbb{F}_q) = \mathbb{F}_q^{*}{}_{i}$, $i = 1, 2, \ldots$.

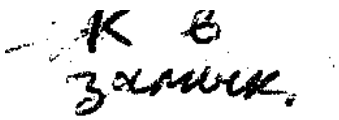


Uncertain reading. At the right of F is a lower q, a small i above and to its right, and an upper star. A candidate is $F^{*}_{q^i}$; the exact two-level placement remains uncertain.

Original source details: GELFAND-P000423

GELFAND-P000424

As i increases, the K-functor approaches elementary K in the clos.



Uncertain reading. The final abbreviation is provisionally read “замык.” (clos.); the short terminal stroke remains uncertain.

Original source details: GELFAND-P000424

Uses the Frobenius automorphism.

Kazhdan: Proof.

Consider

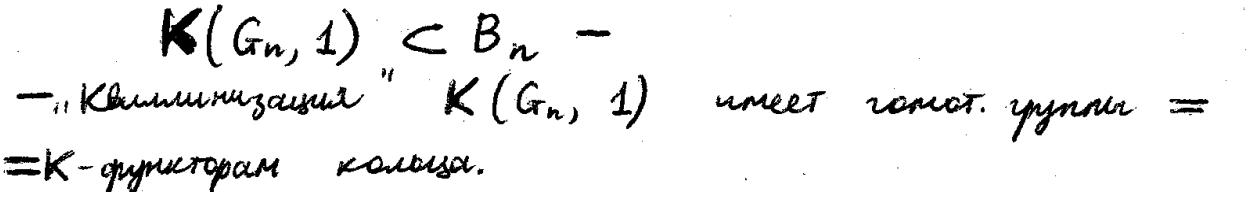


Original source details: GELFAND-P000424

Uncertain reading. The first lower index at G is overwritten; provisionally n, with the earlier sign unrecovered.

$GL(n, F_q) = G_n$

and its classifying space $\underline{K(G_n,1)}$. $GL(n,F_q) \supset SL(n,F_q) = G_n^0$.



Original source details: GELFAND-P000424

Diagram and typed description. After the inclusion is a separate dash; the next line begins with a dash and ends with an equality, with another equality at the start of the bottom line.

$$K(G_n,1) \subset B_n —$$

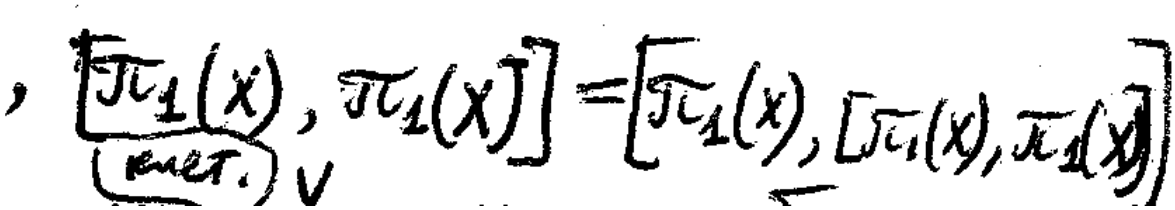
— «Quillenization» $K(G_n,1)$ has homot. groups $=$

$=$ the K-functors of the ring.

Aim: make the fund. group commutative, without changing homology groups. We fill in the commutator subgroup and the resulting 2-dimensional cycles. One can restrict to attaching 3-cycles.

Fuchs.

Existence theor. X — a cell complex,

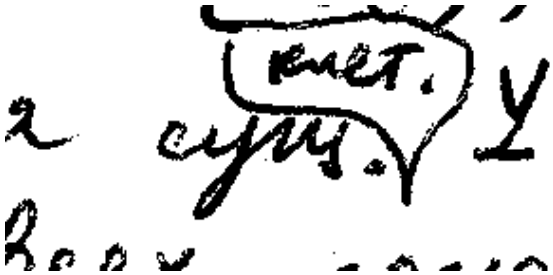


Original source details: GELFAND-P000424

Diagram and typed description. On the right, the outer first argument is $\pi_1(X)$ and the second argument is the inner commutator; all brackets are retained.

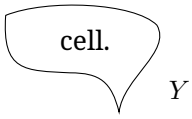
$$[\pi_1(X),\pi_1(X)] = [\pi_1(X),[\pi_1(X),\pi_1(X)]]$$

This condition is necessary and sufficient for the existence of



Original source details: GELFAND-P000424

Diagram and typed description. Above Y is the inserted “cell.” in a curved contour with a pointed lower end.



and a map $f : X \longrightarrow Y$, an isom. of all homology and

всех гомологий и
тивно. 1) $f_* : H_*(X, \mathbb{Z}) \xrightarrow{\sim} H_*(Y, \mathbb{Z})$

Original source details: GELFAND-P000424

Diagram and typed description. At the right is condition 1), a rightward arrow with a wavy stroke above it. At the left is an angular bend and below a long rule; the upper and right sides are open.

1) $f_* : H_*(X, \mathbb{Z}) \xrightarrow{\sim} H_*(Y, \mathbb{Z})$

in Y



Original source details: GELFAND-P000424

Undeciphered passage. Before "2)" a short earlier inscription is crossed out and not fully recovered.

2) $\pi_1(Y)$ is commutative. Hence $\pi_1(Y)$ is computed, namely

Отсюда $\pi_1(Y)$ вычисляется, а именно
$\pi_1(Y) = \pi_1(X) / [\pi_1(X), \pi_1(X)]$.

Original source details: GELFAND-P000424

Diagram and typed description. The formula has an oblique quotient stroke: the numerator is higher at the left, and the commutator lower at the right.

$$\pi_1(Y) = \pi_1(X) \,/\, [\pi_1(X), \pi_1(X)].$$

This is an existence theorem. Such a Y is not unique. (E.g., when $\pi_1(X)$ is commutative).

2a) let Y be simple (the fund. group acts on $\pi_n(Y)$ trivially)

Such a Y is unique ($\Rightarrow$ commutative fund. group, since π_1 acts on itself by conj.) — Whitehead's theor..

It is known that H-spaces are simple.

GELFAND-P000425

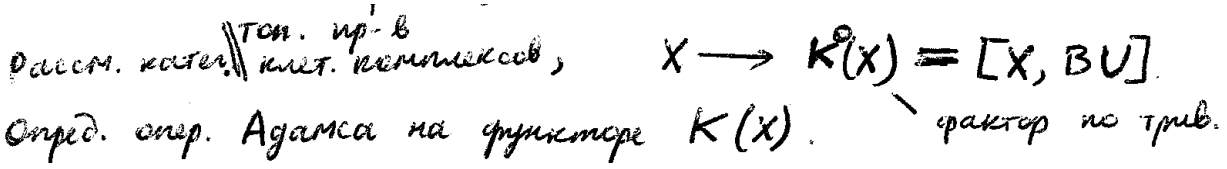
[source number: 59]

Kazhdan — B_n — H-space. (but this is not for every G, only for this one).

$$K_i(G_n) = \pi_i(B_n).$$

Quillen found a topological realization for B_n.

Consider the category of topological spaces of cell complexes, $X \longrightarrow K^0(X) = [X, BU]$. quotient by trivial



Original source details: GELFAND-P000425

Diagram and typed description. Two slanting strokes before the inserted lines; a slanting stroke leads to the annotation under the formula.

Define the Adams operations on the functor $K(X)$.

On U_n consider the function: $u \longmapsto$ eigenvalues $\lambda_1, \ldots, \lambda_n$.

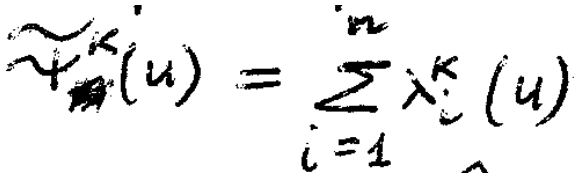


Original source details: GELFAND-P000425

Uncertain reading. The short sign between the arrow and “eigenvalues” is crossed out; the underlying letter is illegible.

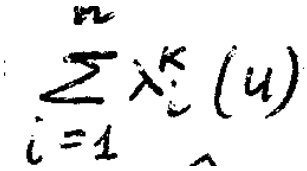
$$\widetilde{\psi}^k(u) = \sum_{i=1}^{n} \lambda_i^k(u)$$

— the character of a virtual representation (difference of characters).



Original source details: GELFAND-P000425

Uncertain reading. Crossed-out subscript beneath the tilded ψ^k; the original index is illegible.



Original source details: GELFAND-P000425

Uncertain reading. The subscript of λ is a dot above a short curved stroke: i or j. The sum limit $i = 1$ is legible.

This representation $\widehat{\psi}^k$: Sp $\widehat{\psi}^k = \widetilde{\psi}^k(u)$.

Construct, for $\forall$ bundle, another with a U action, using the representation $\widehat{\psi}^k$. This is an operation in the K-functor: $\psi^k(\alpha^\dagger)$.

$$\exists \text{ a map } \psi^k : BU \longrightarrow BU,$$

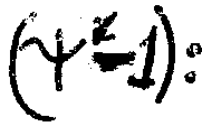
inducing the Adams operation in K. $\alpha^\dagger \in [X, BU]$,



Original source details: GELFAND-P000425

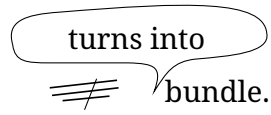
Undeciphered passage. The crossed-out writing after $[X, BU]$ has a ψ-like beginning and a rounded stroke near the end; the remainder is illegible.

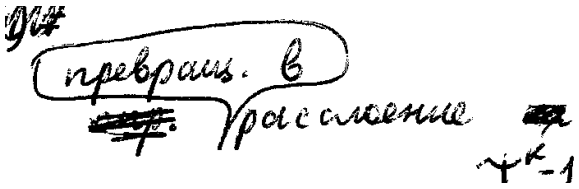
$$(\psi^k - 1) : BU \longrightarrow BU.$$



Original source details: GELFAND-P000425

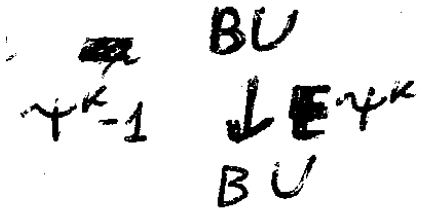
Uncertain reading. The horizontal stroke before 1 has been retraced; the lower stroke merges into it. Reading: $-$ or $=$.





Original source details: GELFAND-P000425

Undeciphered passage. Under the insertion "turns into", the crossed-out fragment before "bundle" is illegible.



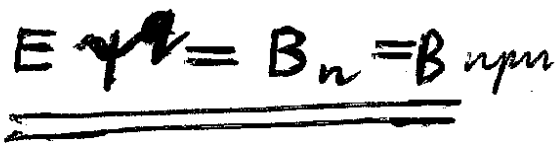
Original source details: GELFAND-P000425

Diagram and typed description. Two BU labels and a down arrow. At left $\psi^k - 1$; at right $E\psi^k$ with no separate arrow. A crossed-out mark is at the left of the upper BU.

$$\begin{array}{ccc} & BU & \\ \psi^k - 1 & \downarrow & E\psi^k \\ & BU & \end{array}$$

$E\psi^k$ — fiber of the map $\psi^k - 1$.

Claim: $E\psi^q = B_n = B$ for large n.



Original source details: GELFAND-P000425

Uncertain reading. The exponent on $E\psi$ has been retraced: k or q. The equality is underlined twice.

Frobenius automorphism $\sim$ Adams operation in topology.

Another def.



Original source details: GELFAND-P000425

Uncertain reading. A looped mark above the line: “e” (the end of the Russian abbreviation) or a separate k.

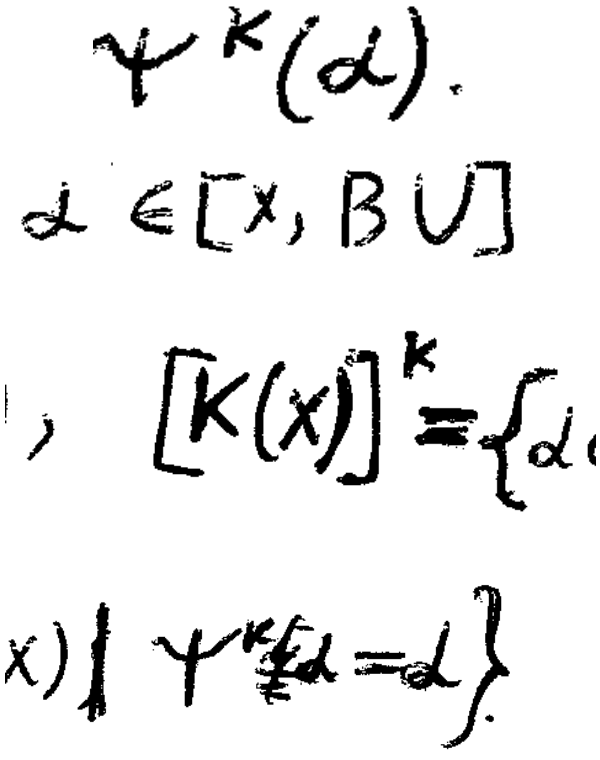
Let $[K(X)]^k \subset K^0(X)$,

$$[K(X)]^k = \{\alpha^\dagger \in K^0(X) \mid \psi^k \alpha^\dagger = \alpha^\dagger\}.$$



Original source details: GELFAND-P000425

Uncertain reading. Crossed-out sign after ψ^k; the original reading is undetermined.



Original source details: GELFAND-P000425

Uncertain reading. † The recurring letter has a low loop and a rising right stroke. It is provisionally read as α; d is also possible.

it is representable and

$$[K(X)]^k = [X, E\psi^k]$$

easy to check.

GELFAND-P000426

$K(G,1) \subset B$. Identify B with $E\psi^q$.





Uncertain reading. Crossed-out traces of $[K(X)]$, a superscript (k or q), and a separate illegible sign to the right.

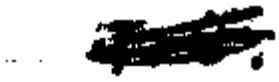
Original source details: GELFAND-P000426

Consider $GL(n, F_q)$ and fix a homomorphism $\mu : \overline{F}_q^* \longrightarrow \mathbb{C}^*$.

On $GL(n, F_q)$ construct a function — the character of a representation.

$\varphi_\mu(g) = \sum \mu(\lambda_i)$, where λ_i — eigenvalues of the matrix g.

We must construct a map $K(G_n, 1) \longrightarrow E\psi^q$ or an element of $[K(G_n, 1), E\psi^q]$, i.e. an element of the K-functor of $K(G_n, 1)$, invariant under ψ^q, =

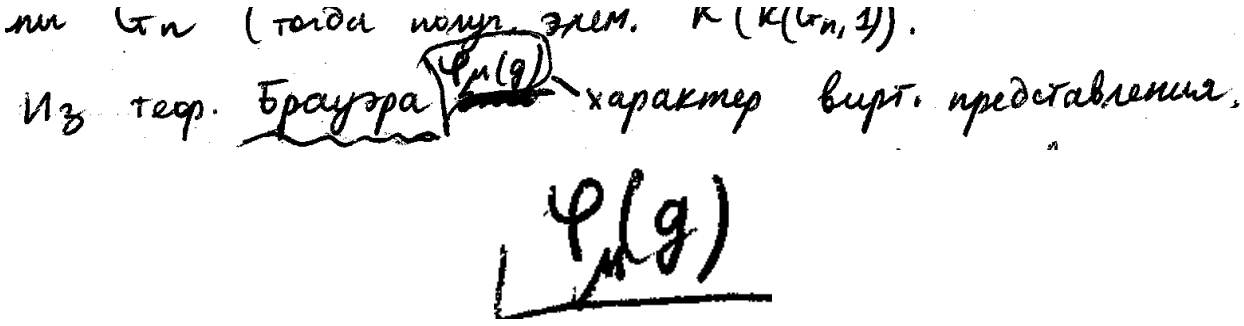


Undeciphered passage. After the invariance clause, the short crossed-out fragment is illegible.

Original source details: GELFAND-P000426

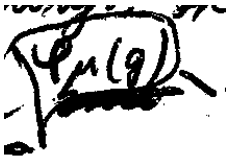
It suffices to give a representation of the group G_n (then we obtain an element of $K(K(G_n, 1))$.

From Brauer's thm. $\varphi_\mu^\dagger(g)$ character of a virtual representation,



Original source details: GELFAND-P000426

Uncertain reading. † In the outlined insertion, the descending stem and the loop favor $\varphi_\mu(g)$. The outline overlaps the letter; $\psi_\mu(g)$ remains possible. A wavy line underlines "Brauer".



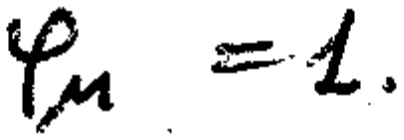
Undeciphered passage. The heavily crossed-out inscription below the insertion is illegible.

Original source details: GELFAND-P000426

The set λ_i^q — eigenvalues too — the same ones $\Rightarrow$ invariance.

On $E\psi^q$ there is an H-space structure $\Rightarrow$ the map extends to a map into B_n (onto the filled-in disks).

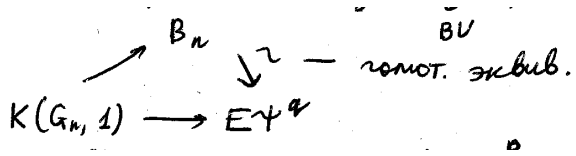
Another way: SL is generated by unipotent matrices, and these are the ones to kill $\Rightarrow$ and on them $\varphi_\mu = 1$.



Original source details: GELFAND-P000426

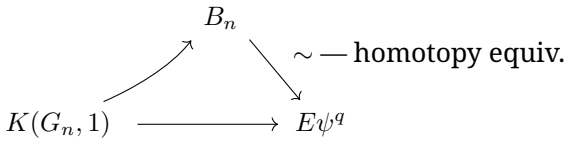
Diagram and typed description. The source reads $\varphi_\mu = 1$.

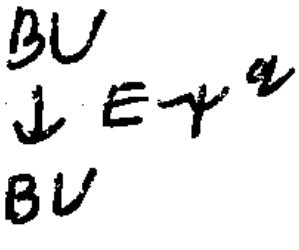
Homotopy groups of $E\psi^q$.



Original source details: GELFAND-P000426

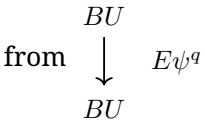
Diagram and typed description. Three arrows; the sign $\sim$ and its caption are by the descending arrow.





Original source details: GELFAND-P000426

Diagram and typed description. Two BU labels, a down arrow, and $E\psi^q$ at right without a separate connection.



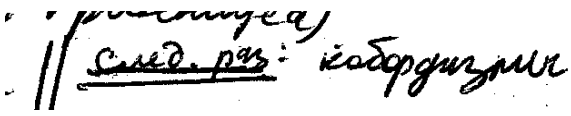
For $E\psi^q$ the homology is computed, and for B_n — ?

The homology groups of $K(G_n, 1)$ are computed. (Quillen too). using étale topology (see Annals).

$$\varphi : G \longrightarrow G$$
$$\varphi(g) = g^{-1} g^\sigma$$

(σ — Frobenius automorphism).

Now write the spectral sequence of the fibration and compute. // Next time: cobordisms.



Original source details: GELFAND-P000426

Diagram and typed description. Two near-vertical slanting strokes separate the marginal note; its opening is underlined twice.

GELFAND-P000427

[source number: 60]

M. Sh. Birman. Kato's theorem.

$H - H_0 \overset{=}{} V \in \mathfrak{S}_1$ (nuclear operator).



Original source details: GELFAND-P000427

Uncertain reading. Between $H - H_0$ and V are several overlapping horizontal strokes: $=$ or $\equiv$.



Original source details: GELFAND-P000427

Uncertain reading. An isolated mark below the page number 60 has two loops and a slanted cross stroke. Possibly f. Its connection to the formulas is undetermined.

Prove that the absolutely continuous parts of H and H_0 are unitarily equivalent. H, H_0 — self-adjoint.

Question (I. M. Gelfand): how can one determine from the collection of spectral types whether they can be realized so that the difference is a nuclear operator?

Spectrum in the quotient ring modulo the nuclear operators — ?

2 facts used:

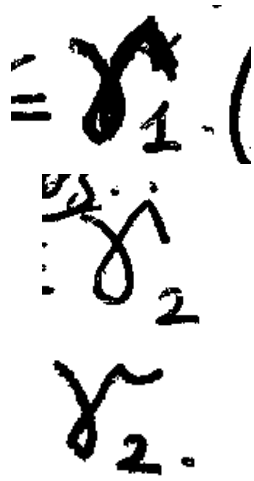
1) $D_1, D_2 \in \gamma_2$; E_λ — resolution of the identity $\Rightarrow$

$$\Rightarrow \quad \frac{d}{d\lambda} D_1 E_\lambda D_2$$

exists almost everywhere and is nuclear ($\sim$ rigging theorem).

(see Gelfand, diff. formula, strongly bounded var.)

2) $D_1 R_z D_2 \longrightarrow D_1 R_{\lambda+i0} D_2$ almost everywhere in γ_2.



Uncertain reading. Each of the three marks has a loop and two upper branches. The letter may be γ or $\mathcal{J}$. The subscripts $1, 2, 2$ are legible.

Original source details: GELFAND-P000427

Operator implementing the unitary equivalence:

$$\int_{-\infty}^{\infty} \frac{d}{d\lambda}(E_\lambda(I + VR_z^0)f, g)\Big|_{z=\lambda\pm i0} d\lambda = (W_\pm f, g).$$

(V is split into 2 Hilbert–Schmidt operators.)

Next time.

27/III Kazhdan.

Artin–Mazur theorem.

If X — set over a field k. any!! From it one can construct a complex C_X

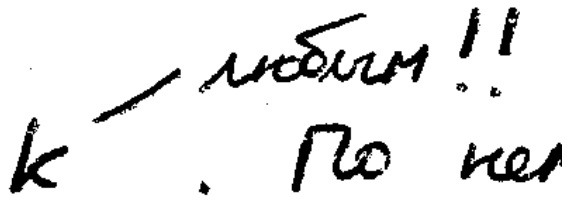


Diagram and typed description. “Any!!” is written above k, with a slanted leader.

Original source details: GELFAND-P000427

If $k = \mathbb{C} \Rightarrow C_X = \widehat{X}_{\mathbb{C}}$ — completion

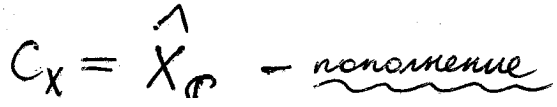
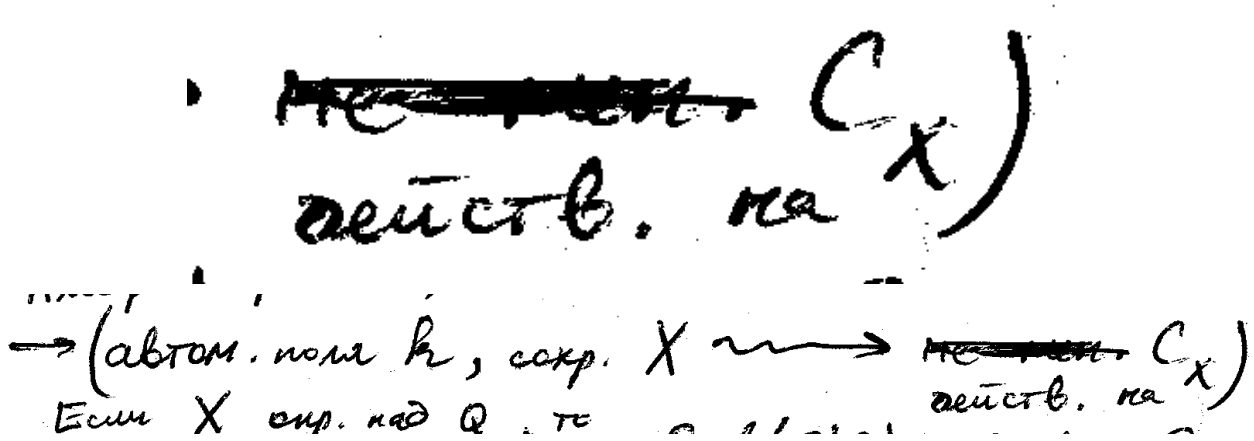


Diagram and typed description. A wavy line underlines “completion”.

Original source details: GELFAND-P000427

Algebraic construction. $\longrightarrow$

$\longrightarrow$ (autom. of the field k, preserving X ~~act on~~ C_X)



Original source details: GELFAND-P000427

Undeciphered passage. Before C_X there is a crossed-out inscription whose beginning resembles “не” (“not”). The remainder is illegible. “Act on” is inserted below.

If X is defined over Q, then Gal$(\mathbb{C} \mid Q)$ acts on C_X.

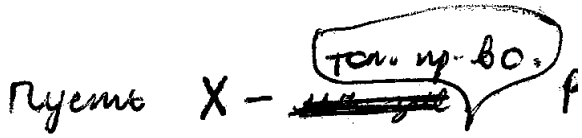


Original source details: GELFAND-P000427

Uncertain reading. Before “then”, below the line, a short mark has a faint slanted continuation. It is unclear whether the continuation is part of the comma.

GELFAND-P000428

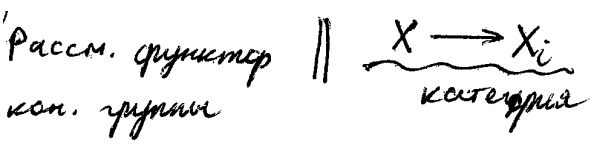
Let X — ~~—~~ top. space.



Original source details: GELFAND-P000428

Undeciphered passage. Horizontal strokes conceal the crossed-out abbreviation. Traces resembling “мн-” or “мн-ие” remain. “Top. space.” was added in the outline above it.

Consider the functor $\| \; \underbrace{X \longrightarrow X_i}_{\text{category}}$



Original source details: GELFAND-P000428

Diagram and typed description. Two vertical strokes precede the arrow diagram. A wavy stroke and a caption are below it.

the images of all $\pi_n(X)$ — finite groups

$\forall \underline{Y}$ consider the set $\underline{Y} \longrightarrow X_i \qquad \prod_i \mathrm{Hom}(\underline{Y}, X_i)$

and compatible $\underline{h_{\widehat{X}}(Y)}$ — a functor.

It is representable and $\widehat{X}$ — the representing object.

———— · ————

I. M. Gelfand.

Standard language.

1°. Category and functor (Grothendieck, "On some questions of hom. alg." — chapter 1).

2°. Topos (topó). + theorem on descent



Uncertain reading. "Descent" is provisional. The middle strokes of the Russian word merge.

Original source details: GELFAND-P000428

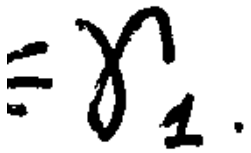
(the definition is in Manin's preface to the article by Mumford)

M. Artin — Grothendieck,

———— · ————

M. Sh. Birman. Theorem of Kato.

$\mathfrak{H}$ — Hilbert space. H, H^0 — operators, $H - H^0 \in \gamma_1$.



Uncertain reading. The ideal sign may be γ_1 or $\mathfrak{S}_1$. The subscript 1 is legible.

Original source details: GELFAND-P000428

$$e^{iHt}e^{-iH^0t}P^0$$

P^0 — projection onto the absolutely continuous subspace of H^0

$$\underset{t\to\pm\infty}{s\text{-}\lim} e^{iHt}e^{-iH^0t}P^0 = W_\pm$$

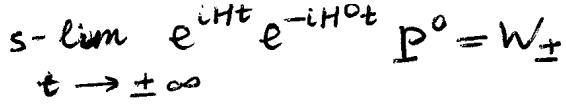


Diagram and typed description. The t under the limit is heavily retraced, followed by an arrow to $\pm\infty$.

Original source details: GELFAND-P000428

Then both W_+ and W_- give an isomorphism of the absolutely continuous spectra of H and H^0 (symmetry of the assumptions!)

$W_\pm H^0 = H W_\pm$ $\quad$ $P^0 \mathcal{H}$ — domain of isometry of $W_\pm$ (on the complement it is 0),

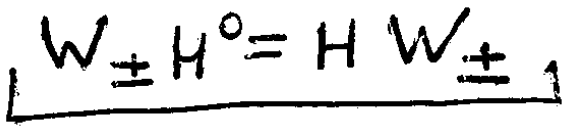


Diagram and typed description. A horizontal bracket with two short upstrokes is below the equality.

Original source details: GELFAND-P000428

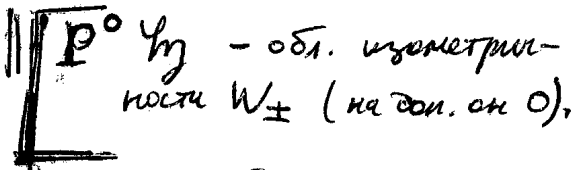


Diagram and typed description. At right, two vertical strokes and a bracket open to the right surround the isometry note.

Original source details: GELFAND-P000428

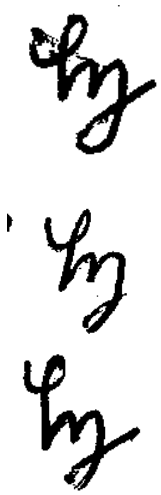
Range: does it coincide with the subspace

$P^{\mathcal{H}}$?



Diagram and typed description. There is a bracket below the subspace symbol, with the question mark outside it.

Original source details: GELFAND-P000428



Uncertain reading. All three occurrences show a looped sign with a long lower tail. $\mathcal{H}$, $\mathcal{H}$, and Ψ are possible readings. The shape is represented by an outline.

Original source details: GELFAND-P000428

GELFAND-P000429

[source number: 61]

Prop. The range of W_+ coincides with $P^{\mathfrak{y}} \Leftrightarrow \exists$ wave oper. W_+ with the roles of H and H_0 interchanged.



Uncertain reading. The ornamental glyph after P has several loops, including a closed lower loop. $\mathfrak{H}$, $\mathcal{H}$, and Ψ are possible readings.

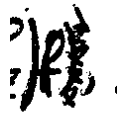
Original source details: GELFAND-P000429

Hence existence $\Rightarrow$ completeness, from the symmetric relation.

$$W_\pm(H_3, H_2)W_\pm(H_2, H_1) = W_\pm(H_3, H_1).$$

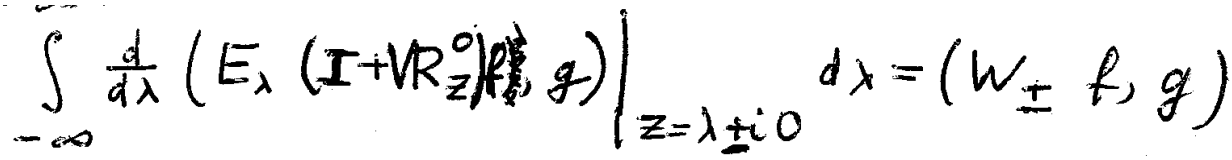
Proof of thm.

$$\int_{-\infty} \frac{d}{d\lambda}(E_\lambda(I + VR_z^0)f, g)\Big|_{z=\lambda\pm i0} d\lambda = (W_\pm f, g)$$



Uncertain reading. A narrow vertical crossing-out stands after f and before the comma. The underlying sign is illegible.

Original source details: GELFAND-P000429



Original source details: GELFAND-P000429

Diagram and typed description. The integral has $-\infty$ below it. No upper limit is visible.

This is the definition of the bilinear form of this operator;

One must be able to: $\frac{dE_\lambda g}{d\lambda}$, and this is possible with $\frac{dDE_\lambda g}{d\lambda}$, where D — a Hilbert–Schmidt oper.

The derivative [constitutes?] directly the absolutely continuous part of the spectrum.



Uncertain reading. The word begins with two open rounded letters. A provisional literal reading is “сставляет”. Alternatives are “составляет” (constitutes) and “оставляет” (leaves).

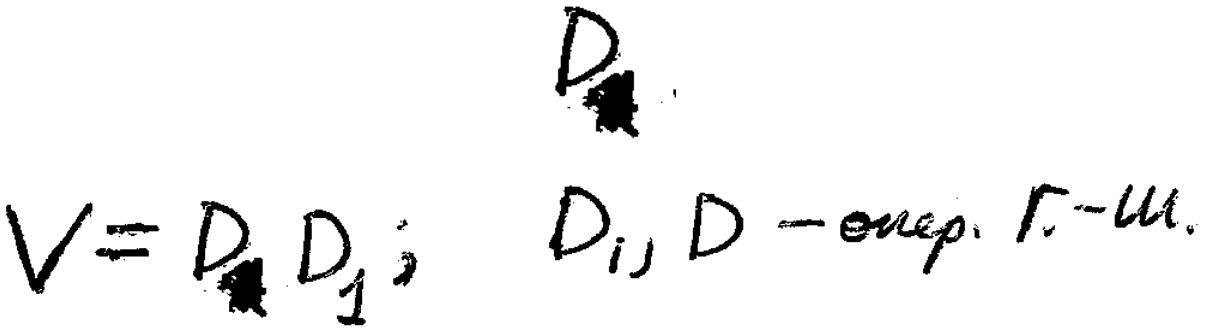
Original source details: GELFAND-P000429

Factor V into 2 Hilbert–Schmidt oper.

Almost everywhere there exists the derivative of $(E_\lambda f, g)$

$$\frac{d(E_\lambda f, g)}{d\lambda} + \left(D_1 R^0_{\lambda + i0} f, \frac{dD^* E_\lambda g}{d\lambda} \right)$$

$V = D_\bullet D_1; \qquad D_i, D$ — Hilbert–Schmidt oper.



Original source details: GELFAND-P000429

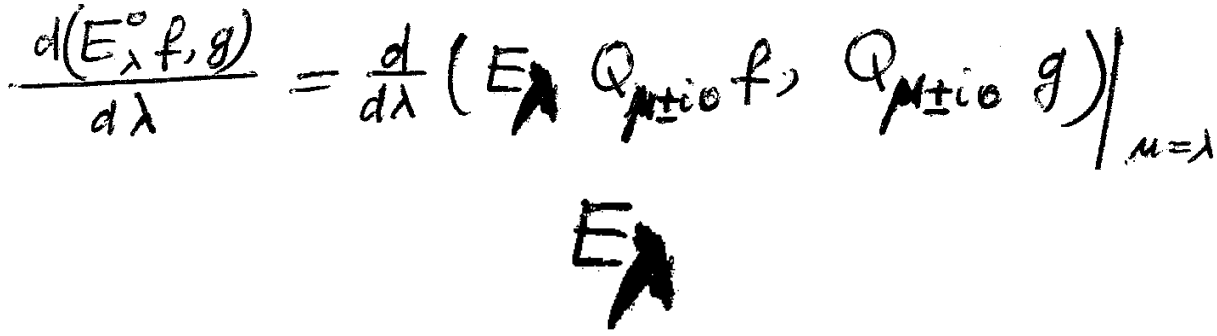
Uncertain reading. The first subscript in the product is heavily overwritten. 2 is possible. The second factor's subscript 1 is legible.

The oper. $W_\pm$ is independent of the decomposition of V as a product.

$$R^0_z - R^0_{\bar z} = Q^*_z (R_z - R_{\bar z}) Q_z; \qquad Q_z = I + V R^0_z;$$

Further,

$$\frac{d(E^0_\lambda f, g)}{d\lambda} = \frac{d}{d\lambda} \left(E_\bullet Q_{\mu \pm i0} f, Q_{\mu \pm i0} g \right) \Big|_{\mu = \lambda}$$



Original source details: GELFAND-P000429

Uncertain reading. The overwritten subscript on E at right resembles λ; μ is possible. Both Q subscripts read $\mu \pm i0$.

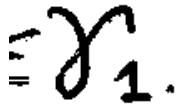
GELFAND-P000430

All D — the required riggings,
see the article in “Izvestiia”, 1967.

$$\int_{-\infty}^{\sigma} \frac{d}{d\lambda}\left(E_\lambda(I+VR_z^0)f,g\right)\Big|_{z=\lambda\pm i0} d\lambda = (E_\sigma W_\pm f, g)$$

More gen. variant:

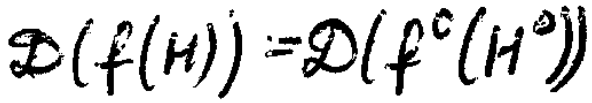
$$E(\Delta)(H-H^0)E^0(\Delta) \in \gamma_1.$$



Uncertain reading. A looped sign with subscript 1. Possible readings are γ_1 and $\mathfrak{S}_1$.

Original source details: GELFAND-P000430

$\mathcal{D}(f(H)) := \mathcal{D}(f^c(H^0))$, f, f^0 grow at infinity.







Uncertain reading. The second f has an open, c-shaped superscript. The later f has an oval superscript, set as 0. At the equality, a high dot and a low fleck sit at the left of the bars, resembling $:=$.

Original source details: GELFAND-P000430

3/IV-72 Minimum.

1. Grothendieck topol. and derived functors.
2. Descent.
3. Simplic.

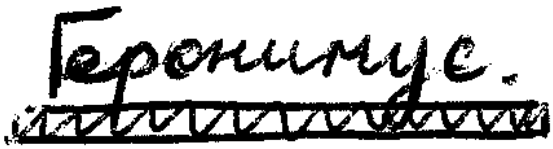


Uncertain reading. “Simplic.” is a provisional reading of “Симплиц.” The middle strokes run together.

Original source details: GELFAND-P000430

Exam: work through Artin's article on the relation between formal and algebraic objects.

Geronimus.

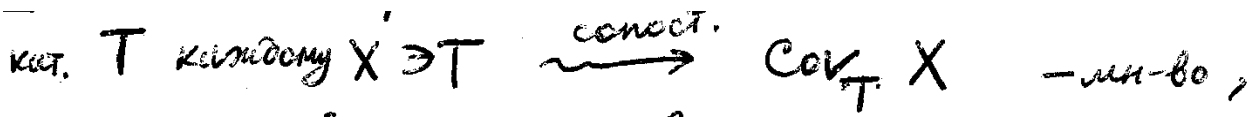


Original source details: GELFAND-P000430

Diagram and typed description. A narrow, diagonally hatched band underlines the name.

Def. Grothendieck topology. —

cat. T to each $X \ni T \overset{\text{assigns}}{\rightsquigarrow} \mathrm{Cov}_T X$ — a set,



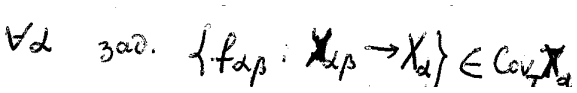
Original source details: GELFAND-P000430

Diagram and typed description. The labelled arrow begins with a wavy stroke. It follows $X \ni T$ as written.

its elements $\{f_\alpha : X_\alpha \longrightarrow X\}$. (a family of morphisms). ($\alpha \in A$ — a set).

1) $f : Y \longrightarrow X$ an isom. in cat. T. Then $\{f : Y \longrightarrow X\}$ is a covering.

2) $\{f_\alpha : X_\alpha \longrightarrow X\} \in \mathrm{Cov}_T X$ and $\forall \alpha$ given $\{f_{\alpha\beta} : \boldsymbol{X}_{\alpha\beta} \longrightarrow X_\alpha\} \in \mathrm{Cov}_T \boldsymbol{X}_\alpha$



Original source details: GELFAND-P000430

Diagram and typed description. Two X glyphs are heavily retraced, in the last family and after Cov_T.

Then

$$\{f_\alpha \circ f_{\alpha\beta} : X_{\alpha\beta} \longrightarrow X\} \in \mathrm{Cov}_T X$$

Mumford, Matematika 13:2

3) $\{f_\alpha : X_\alpha \longrightarrow X\} \in \mathrm{Cov}_T X$, $f : U \longrightarrow X$ — arbitrary morphism.

GELFAND-P000431

[source number: 62]

Then



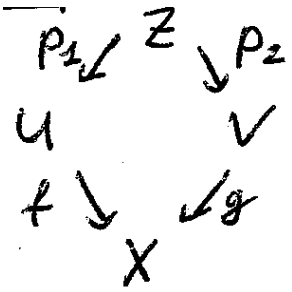
Original source details: GELFAND-P000431

Uncertain reading. Diagonal strokes cover a single sign. φ and 0 are possible readings.

3.1 $X_\alpha \underset{X}{\times} U$ exists (fibre prod.)

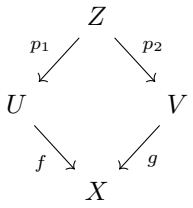
3.2 $\{p_2 : X_\alpha \underset{X}{\times} U \longrightarrow U\} \in \mathrm{Cov}_T\, U$.

Def. of fibre prod.:



Original source details: GELFAND-P000431

Diagram and typed description. The diamond has arrows p_1, p_2, f, g. The note on Z is to its right.



Z univ. — fibre prod.

T — the starting category.

Let A be a fixed abelian category (with infinite direct sums)

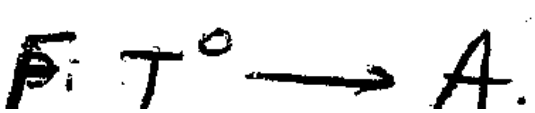
Presheaf: ~~a~~ contravar. functor from T to A:

$$\boldsymbol{F} : T^0 \longrightarrow A.$$



Original source details: GELFAND-P000431

Undeciphered passage. Heavy crossing-out covers a short fragment before “contravar.”



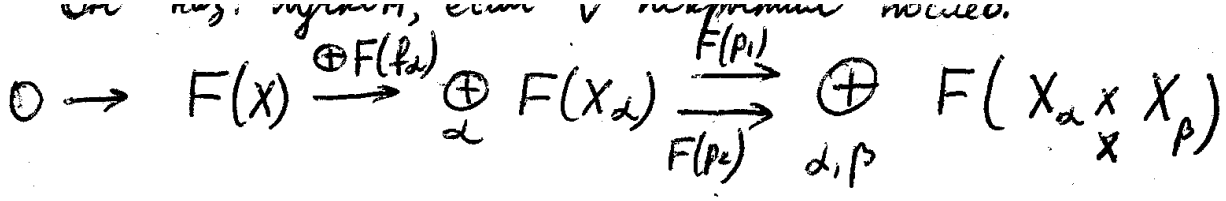
Original source details: GELFAND-P000431

Diagram and typed description. The strokes of F are reinforced. The small oval above T is set as 0.

P_T — the category of presheaves.

It is called a sheaf if $\forall$ coverings the seq.

$$0 \longrightarrow F(X) \xrightarrow{\bigoplus F(f_\alpha)} \bigoplus_\alpha F(X_\alpha) \underset{F(p_2)}{\overset{F(p_1)}{\rightrightarrows}} \bigoplus_{\alpha,\beta} F(X_\alpha \underset{X}{\times} X_\beta)$$



Original source details: GELFAND-P000431

Diagram and typed description. Two direct sums are written. The first labelled arrow reads $\bigoplus F(f_\alpha)$; the parallel arrows read $F(p_1)$ and $F(p_2)$.

S_T — the full subcat. of sheaves in the cat. of presheaves

Morphism of functors

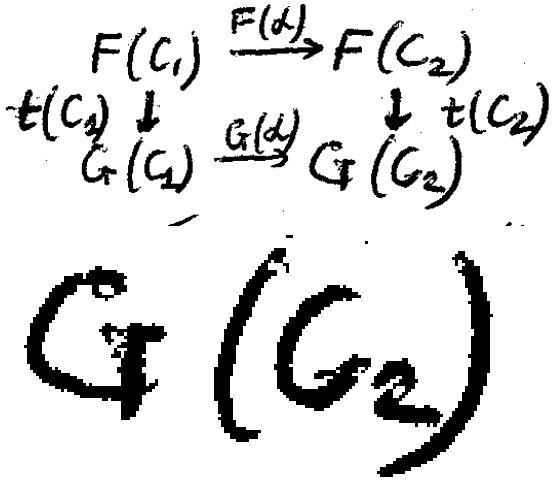
$$F : \mathcal{C} \longrightarrow \mathcal{D} \qquad F \xrightarrow{t} G$$
$$G : C \longrightarrow \mathcal{D}$$



Diagram and typed description. Two rows point to $\mathcal{D}$. At right, an arrow from F to G carries the label t.

Original source details: GELFAND-P000431

$\alpha : C_1 \longrightarrow C_2$



Uncertain reading. The inner glyph at lower right has an inward stroke resembling G. The reading $G(G_2)$ is provisional. $G(C_2)$ is also possible.

$$\begin{array}{ccc} F(C_1) & \xrightarrow{F(\alpha)} & F(C_2) \\ {\scriptstyle t(C_1)}\downarrow & & \downarrow{\scriptstyle t(C_2)} \\ G(C_1) & \xrightarrow{G(\alpha)} & G(G_2) \end{array}$$

Original source details: GELFAND-P000431

This is a deep concept. Nontriviality: there are many different α, and there are exactly 2 of the $t(C_i)$.

GELFAND-P000432

Hence the functors themselves form a “category”.

(up to the fact that morphisms may fail to form a set)

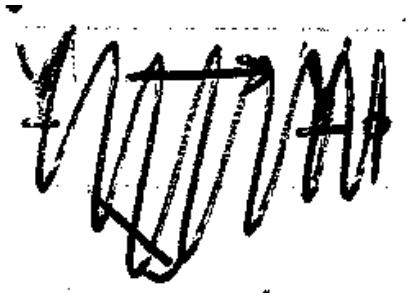
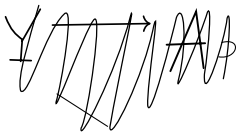
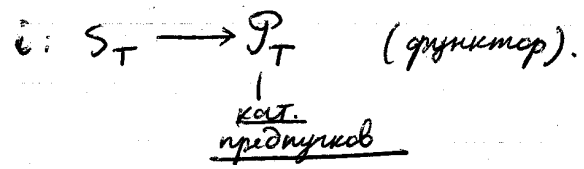


Diagram and typed description. The crossing-out leaves a Y-like left label, a rightward arrow, and an A-like right label visible. A further looped stroke is at the right edge.



Original source details: GELFAND-P000432

Theorem: the category S_T is abelian.



Original source details: GELFAND-P000432

Diagram and typed description. A vertical line without an arrowhead and two underlined rows are below $\mathcal{P}_T$.

$i : S_T \longrightarrow \mathcal{P}_T$ (functor).

|

cat. of presheaves

Idea of proof:

$\exists$ a functor $\# : \mathcal{P}_T \longrightarrow S_T$, adj. to i.

(associated sheaf).

Functors:

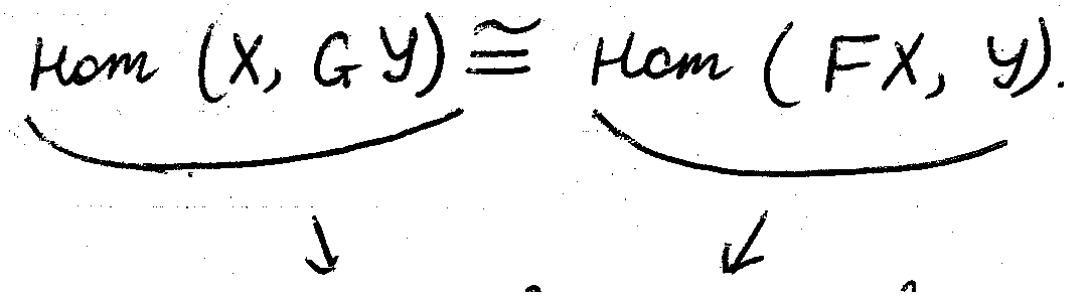
$$F : T_1 \longrightarrow T_2$$
$$G : T_2 \longrightarrow T_1$$

are called adjoint if $\forall X \in T_1$ and $\forall Y \in T_2$ a nat. isomorphism is given

$\mathrm{Hom}(X, GY) \cong \mathrm{Hom}(FX, Y)$.

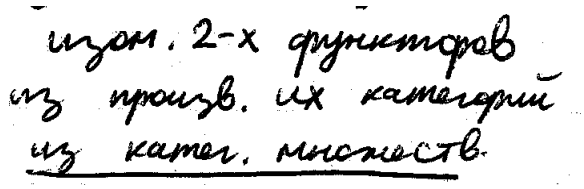
isom. of 2 functors
from the product of their categories
from the cat. of sets.

Def. using a
univ. morphism from a
presheaf to a sheaf
also works



Original source details: GELFAND-P000432

Diagram and typed description. An arc and an arrow run from below each Hom expression toward the shared caption.



Original source details: GELFAND-P000432

Diagram and typed description. Both of the last two lines begin with “from” in the source. The final line is underlined.

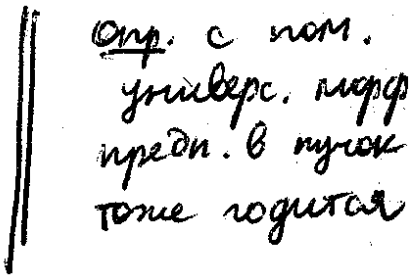
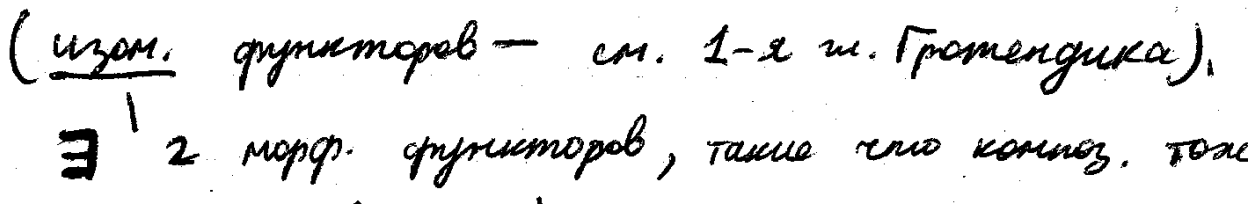


Diagram and typed description. The note is at the right of two vertical strokes.

Original source details: GELFAND-P000432

(isom. of functors — see ch. 1 of Grothendieck).

$\exists$ 2 morphisms of functors whose compositions are identities.



Original source details: GELFAND-P000432

Diagram and typed description. The abbreviation “isom.” is underlined and a separate stroke leads downward to the next line.

(for the proof see Artin).

GELFAND-P000433

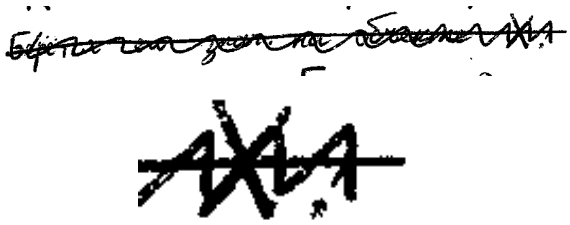
[source number: 63]

Definition. Let T, $X \in T$, $F \in S_T$

$$H^q(T, X, F) \stackrel{\text{def}}{=} R^q i(F)(X).$$

$R^q i$ — a new functor. (also from S_T to $\mathcal{P}_T$).

~~Its value is taken on the object X~~



Uncertain reading. The crossed-out sentence is provisional. There are further slanting strokes and a low mark after X, possibly “, A”.

Original source details: GELFAND-P000433

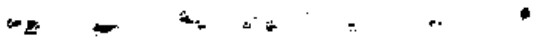
Its value on F — a presheaf (i.e. also a functor)

A presheaf on X — ~~is~~



Uncertain reading. The crossed-out Russian “это” is a provisional reading.

Original source details: GELFAND-P000433



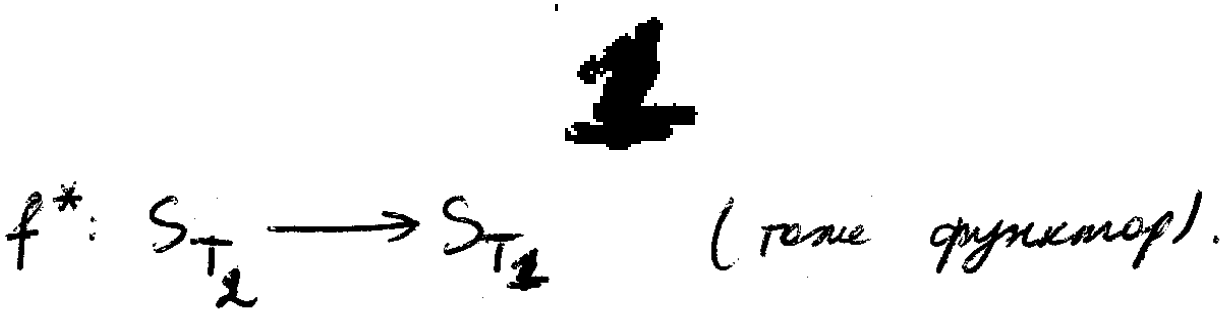
Original source details: GELFAND-P000433

Diagram and typed description. A row of dots and small strokes follows the crossing-out.

Definition.

$f : T_1 \longrightarrow T_2$ morph. of topol. $\underset{\text{def}}{\Leftrightarrow}$ covering $\longrightarrow$ cov.

$f^* : S_{T_2} \longrightarrow S_{T_1}$ (also a functor).



Original source details: GELFAND-P000433

Uncertain reading. The subscript on the right-hand T is heavily retraced. 1 and traces of 2 are possible. The shape is represented by an outline.

$$f^*F(X) = F(f(X)), \quad \text{where } X \in T_1,\ F \in S_{T_2}.$$

Easy to check that this is a sheaf.

Leray spectr. seq.

$f : T_1 \longrightarrow T_2$; $X \in T_1$; $F \in S_{T_2}$.

Then a spectr. seq. exists with second term

$$E_2^{p,q} = H^p(T_1, X, R^q f^*(F)) \Rightarrow H^*(T_2, f(X), F)$$

Let A, B, C be ab. categories

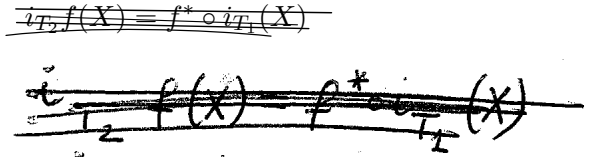
$$A \xrightarrow{F} B \xrightarrow{G} C.$$

Then (under certain cond.) there exists a spectr. seq.

$$R^pG \circ R^qF(X) \Rightarrow R^*(G \circ F)(X)$$

Leray's thm. is a special case of this situation: $\Rightarrow$

GELFAND-P000434



Original source details: GELFAND-P000434

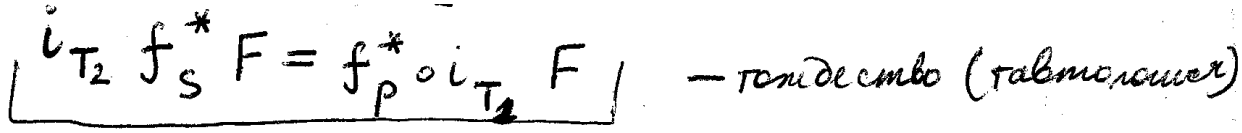
Uncertain reading. The crossed-out formula is provisional. Horizontal strokes obscure the middle relation.

$i_{T_2} f_s^* F = f_p^* \circ i_{T_1} F$ — an identity (tautology)



Uncertain reading. The right-hand index on T is heavily retraced where it meets the bracket. 1 and 2 are possible. The left-hand T_2 is clear.

Original source details: GELFAND-P000434

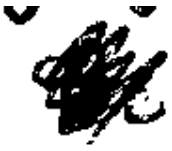


Original source details: GELFAND-P000434

Diagram and typed description. The horizontal line under the equality has two raised ends. The caption is at right.

f_p^* — the same morph. on presheaves

Leray's thm. is a special case because f_p^* is exact in the cat. of presheaves.



Uncertain reading. A separate entry is heavily crossed out. φ is possible.

Original source details: GELFAND-P000434

Classifying topology.

Definition.

Let T be a Grothendieck topol.,

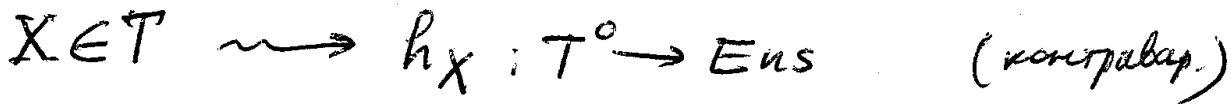


Uncertain reading. "Topol." is the provisional reading of a retraced Russian abbreviation. A wavy line runs below the phrase.

Original source details: GELFAND-P000434

G is a group in cat. T.

$$X \in T \rightsquigarrow h_X : T^0 \longrightarrow \text{Ens} \qquad \text{(contravar.)}$$



Original source details: GELFAND-P000434

Diagram and typed description. The first arrow starts with a wave. The oval above T is set as 0.

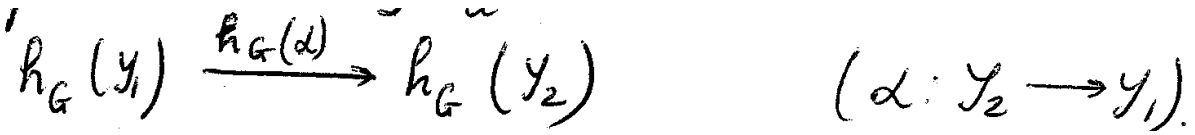
$$h_X(Y) = \mathrm{Hom}(Y, X).$$

Together with G there must be given a group structure on $h_G(Y)$,

functorial in Y and

$$h_G(Y_1) \xrightarrow{h_G(\alpha)} h_G(Y_2) \qquad (\alpha : Y_2 \longrightarrow Y_1).$$

is a group hom.

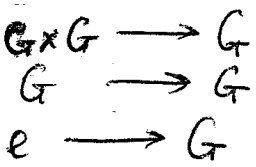


Original source details: GELFAND-P000434

Diagram and typed description. The arrow from $h_G(Y_1)$ to $h_G(Y_2)$ is labelled $h_G(\alpha)$. At right is $\alpha : Y_2 \longrightarrow Y_1$.

Another def.: a collection of 3 morphisms

$$G \times G \longrightarrow G$$
$$G \longrightarrow G$$
$$e \longrightarrow G$$



Original source details: GELFAND-P000434

Diagram and typed description. Three separate unlabelled arrows point rightward, all to G.

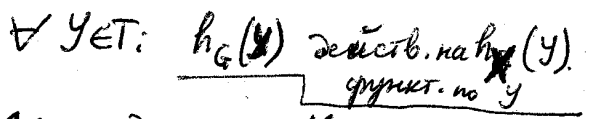
(e — terminal object in the cat.)

related by several commut. diagr.

G acts on X if $\forall Y \in T$: $h_G(Y)$ acts on $h_X(Y)$.

X — G-object. on the left

funct. in Y



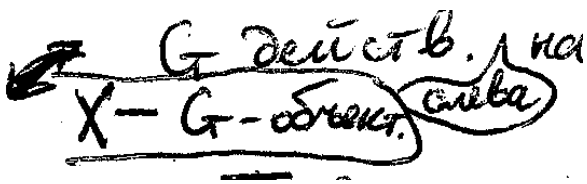
Original source details: GELFAND-P000434

Uncertain reading. Dense strokes obscure the argument of the first h and the subscript of the second. Y and X, respectively, are possible.



Original source details: GELFAND-P000434

Diagram and typed description. A stepped line runs below the formula, with the caption on the lower step.



Original source details: GELFAND-P000434

Diagram and typed description. The outline at left has a point at its lower-left end. “On the left” is in a separate outline.

TG — a new Grothendieck topol. Category —

GELFAND-P000435

[source number: 64]

— G-sets (G-objects).

$X \in T$ and on it an action of G.

Objects [...] of cat. TG — G-objects.



Undeciphered passage. The short crossed-out entry between “Objects” and “cat.” is unreadable.

Original source details: GELFAND-P000435

Morph. — commute with the action (G-morphisms).

Coverings — those cov. $\underbrace{\{f_\alpha : X_\alpha \to X\}}_{\text{in the orig. cat.}}$, which consist of G-morph.

This is the classifying topology.

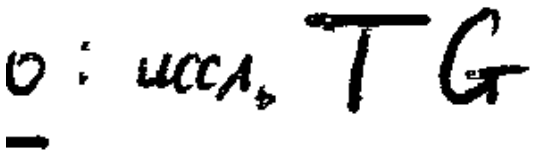
[...] Classif. space: final object, if it exists.



Undeciphered passage. The first sign in the cancelled entry before “Classif. space” resembles M. Two long strokes extend below it. The full reading is unresolved.

Original source details: GELFAND-P000435

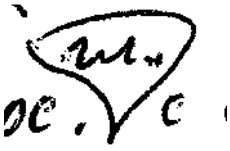
Interesting: иссл.[?] $TG \longrightarrow T$ (forgetf.).



Uncertain reading. The two curves before the last ascending stroke favour «иссл.». «исл.» and «исх.» remain possible.

Original source details: GELFAND-P000435

This is an analogue of a principal bundle (univ. гл.[?] with base the classif. space).



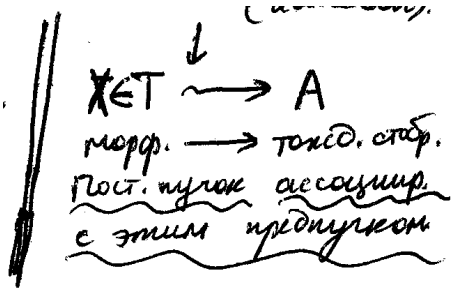
Uncertain reading. The circled insertion provisionally reads «гл.» (principal); «им.» is possible.

Original source details: GELFAND-P000435

Theorem. A const. sheaf is def. in the obvious way by a group A (abelian).

Aim: cohom. of TG from the cohom. of T.

For Quillen's theor. a const. sheaf is needed.



Original source details: GELFAND-P000435

Diagram and typed description. Two rules separate the note on the right. An arrow from the line "(abelian)" points down to the first row; its rightward arrow starts with a wavy stroke. Both closing lines have wavy underlines.

$X \in T \rightsquigarrow A$

morph. $\longrightarrow$ ident. map

Const. sheaf assoc.
with this presheaf

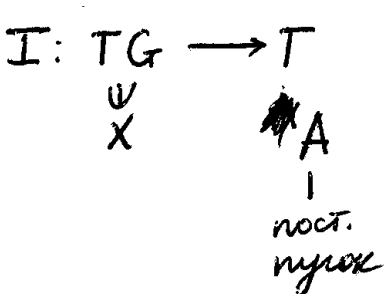
$I : TG \longrightarrow T$.

[...] The Leray spectr. seq., with some addit. restrictions, is arranged thus:



Original source details: GELFAND-P000435

Undeciphered passage. The cancelled entry before "Spectr." has a C-shaped initial curve and loops below the line. The full reading is unresolved.



Original source details: GELFAND-P000435

Diagram and typed description. Below TG is a U-shaped stroke with a middle bar, above X. A blot lies to the left of A; a short rule extends down to the two-line label. The direction of the U-shaped mark is unresolved.

$I : \quad TG \longrightarrow T$

X $\quad$ A

const.
sheaf



Original source details: GELFAND-P000435

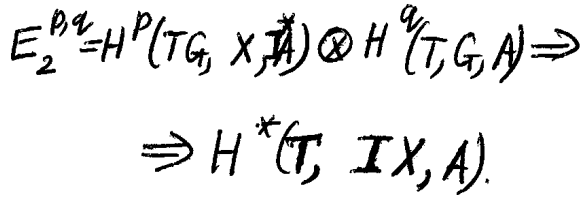
Undeciphered passage. Dense ink covers the entry immediately to the left of the lower A.

$$E_2^{p,q} = H^p(TG, X, A^*) \otimes H^q(T, G, A) \Rightarrow$$
$$\Rightarrow H^*(T, IX, A).$$



Original source details: GELFAND-P000435

Uncertain reading. The coefficient is provisionally typeset as A^*. An overlapping vertical stroke with crossbars is visible below the star; the order of the layers is unresolved.



Original source details: GELFAND-P000435

Diagram and typed description. The sign $\Rightarrow$ closes the first line and is repeated at the start of the next.

X — point.

$$H^p(BG) \otimes H^q(G) \Rightarrow H^*(E).$$

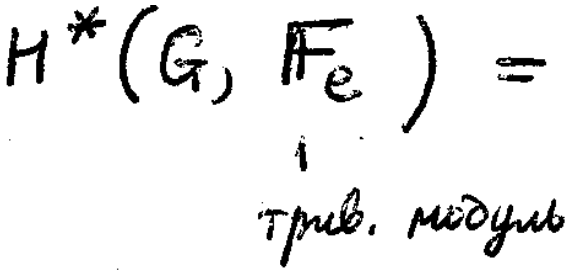
GELFAND-P000436

Quillen's theor.

$G = GL(n, F_q)$

Then

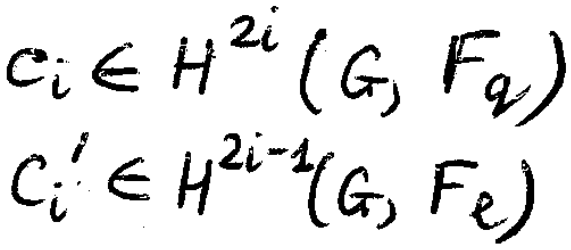
$$H^*(G, \underset{\substack{\uparrow \\ \text{triv. module}}}{F_\ell}) = S[c_1, \ldots, c_n] \otimes \Lambda[c_1', \ldots, c_n']$$



Original source details: GELFAND-P000436

Diagram and typed description. Below the first F_ℓ is a short stroke with an upward point and the label "triv. module".

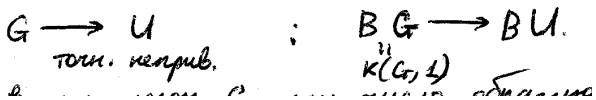
$$c_i \in H^{2i}(G, F_q)$$
$$c_i' \in H^{2i-1}(G, F_\ell)$$



Original source details: GELFAND-P000436

Diagram and typed description. The subscript of F reads q on the first line and ℓ on the second.

Where does this come from?



Original source details: GELFAND-P000436

Diagram and typed description. Below the first arrow is "faithf. irred.". Two short strokes above $K(G,1)$ are below BG.

If $G \longrightarrow U$ (faithf. irred.) ; $BG \longrightarrow BU$. (BG = $K(G,1)$)

hence in the alg. cohom. of G a fin. number of generators

(this is kn.). (from the spectr. seq.)

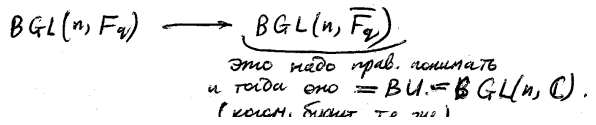
(we have a finite number of gen.)

We need to know this map explicitly.

There is none among Lie groups!

Quillen's rem.: take matr. over the closure of the field F_q

$GL(n, F_q)$.



Original source details: GELFAND-P000436

Diagram and typed description. An arc under the right-hand term connects it to the three lines below.

$BGL(n, F_q) \longrightarrow BGL(n, \overline{F}_q)$

This must be understood prop. and then it $= BU. = BGL(n, \mathbb{C})$. (the cohom. will be the same).



Original source details: GELFAND-P000436

Uncertain reading. The separate low mark after BU and before the reinforced equals sign is set as a dot. Its function is unresolved.

Theor. of Ma[n/m]ford — coincid. of cohom.



Original source details: GELFAND-P000436

Uncertain reading. The connected middle strokes allow «Манфорда» (Manford) and «Мамфорда» (Mamford).

Grothend. theor.

Rem. of I. Bersht./Bernsht.[?]: repres. of $\overline{F}_q$ and $\mathbb{C}$ are descr. in terms of high. weights.



Original source details: GELFAND-P000436

Uncertain reading. After «И.» the joined strokes allow «Бершт.» (Bersht.) and «Берншт.» (Bernsht.).

GELFAND-P000437

[source number: 65]

1[0/8]/IV Geronimus.



Original source details: GELFAND-P000437

Uncertain reading. The second date digit is overwritten. Both 10/IV and 18/IV remain possible. There are rules below and to the right, with rules above and below IV.

Derived category.

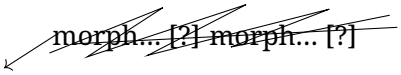
Let A — an abelian category.

$K(A)$ — category of complexes with val. in A;



Original source details: GELFAND-P000437

Undeciphered passage. Two fragments resembling «морф» (morph.) survive the crossing-out. The slanted arrow points down and left.



$$C = \{0 \longrightarrow C_0 \xrightarrow{d_0} C_1 \xrightarrow{d_1} C_2 \xrightarrow{d_2} \dots\}$$

$$H^q(C) = \operatorname{Ker} d_q / \operatorname{Im} d_{q-1}.$$

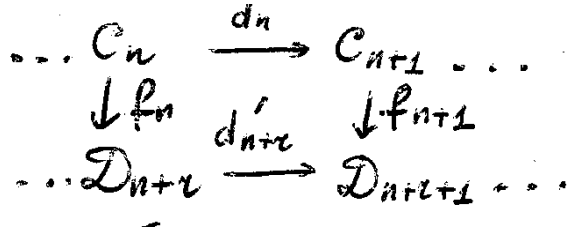


Original source details: GELFAND-P000437

Uncertain reading. Both subscripts of d are darkened. In the denominator the sign before -1 is provisionally q.

$$D = \{0 \longrightarrow D_0 \xrightarrow{d'_0} D_1 \longrightarrow \dots\}$$

map of degree τ:



Original source details: GELFAND-P000437

Diagram and typed description. Both vertical arrows are labelled on the right. Each horizontal arrow is labelled above.

$$\begin{array}{ccccc} \cdots & C_n & \xrightarrow{d_n} & C_{n+1} & \cdots \\ & \downarrow f_n & & \downarrow f_{n+1} & \\ \cdots & D_{n+\tau} & \xrightarrow{d'_{n+\tau}} & D_{n+\tau+1} & \cdots \end{array}$$

Let f and g — 2 maps of degree τ.

They are called homotopic, if $\exists$ a map α of deg. $\tau - 1$:

$$\forall n: \quad f_n - g_n = d_n\alpha_n + \alpha_n d'_n \qquad (f - g = d\alpha + \alpha d').$$



Original source details: GELFAND-P000437

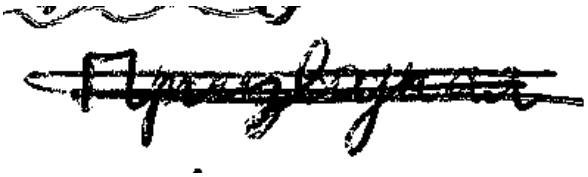
Diagram and typed description. The second α_n is more heavily retraced. Both subscripts on d and both on α read n.

Homot. maps induce the same cohom. morph..

Morphisms in $K(A)$, — morphisms of degrees τ (all) —

— direct sum — quotient by morphisms, homotopic to zero.

~~Derived~~ Category of complexes — is not abelian!



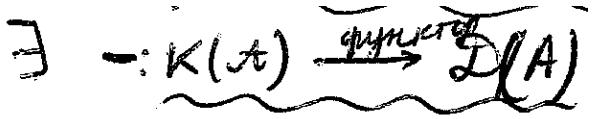
Original source details: GELFAND-P000437

Diagram and typed description. "Derived" is crossed out before "Category".

(because of homot. equival.)

Deriv. categ. for an ab. categ. A, — is the cat. $D(A)$, and

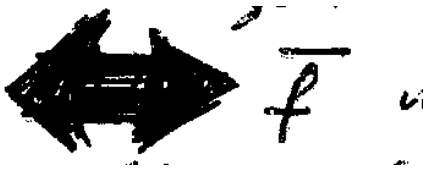
$\exists$ —: $K(A) \xrightarrow{\text{functor}} D(A)$, and $\forall$ morph. of compl. inducing a cohomology isom. is assigned an isom. in cat. $D(A)$.



Original source details: GELFAND-P000437

Diagram and typed description. "Functor" is written above the arrow. A wavy line runs below $K(A)$, the arrow and $D(A)$.

f induces a cohom. isom. $\iff$ $\bar{f}$ isom. in $D(A)$



Original source details: GELFAND-P000437

Diagram and typed description. The two-headed arrow is heavily retraced. A bar is above f.

and $D(A)$ is univ. in this sense.

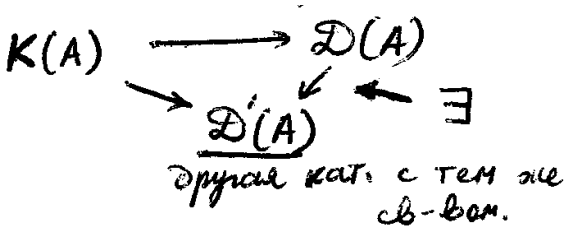
Construction:

consider morphisms the same for morph. inducing cohom.

$D(A)$ is def. by 2 prop.

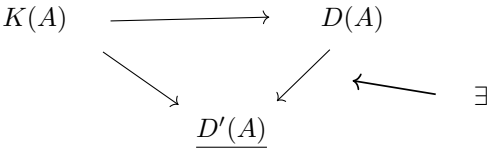
1) f^* isom. $\Rightarrow \bar{f}$ isom.

2) Let



Original source details: GELFAND-P000438

Diagram and typed description. A rule and caption sit below $D'(A)$. To the right of the diagonal arrow, a separate pointer from ∃ points left.



another cat. with the same prop.

$D(A)$ — deriv. category for A

one can pr. (this is a theor.) that then in $D(A)$ f^* isom. $\Leftrightarrow \bar{f}$ isom.

This is not the same as the categ. C with morph. $f \sim g$, if $f^* = g^*$.

In $D(A)$ there are more morphisms.

To maps inducing cohom. isom., inverses are added formally.

It suffices to have a map $C_1 \longrightarrow C_2$, inducing a cohom. isom. — then it is an isom.

Construction. Objects — complexes

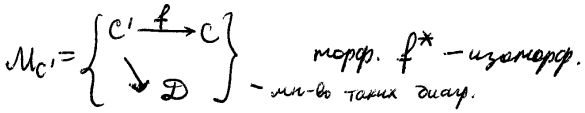
Let C, D — complexes.

~~For...~~ [?] To an object C' associate a diagr.



Original source details: GELFAND-P000438

Undeciphered passage. An opening resembling «Для» (For) and a tall loop survive before “To an object”. Crossing strokes cover the right end.



Original source details: GELFAND-P000438

Diagram and typed description. Two arrows start at C' inside the braces. Only the upper one is labelled f.

$$\mathcal{M}_{C'} = \left\{ \begin{array}{ccc} C' & \xrightarrow{f} & C \\ & \searrow & \\ & & D \end{array} \right\}$$

morph. f^* — isomorph. — set of such diagr.

With resp. to the map C' a proj. system, indexed by objects of the category $C^{[?]}$; ...



Original source details: GELFAND-P000438

Uncertain reading. The joined strokes above C resemble an H or M. A separate prime is unresolved.



Original source details: GELFAND-P000438

Undeciphered passage. Horizontal strokes cover the entry after the category name. Separate lower loops survive.

(contravar. functor of C').

Proj. lim.it[?] of this syst. — set of maps[?] from C to D.



Original source details: GELFAND-P000438

Uncertain reading. A gap and a small dot occur before the letter д. «предел» (limit) and a contracted reading remain possible.



Original source details: GELFAND-P000438

Uncertain reading. An interrupted round stroke precedes б. «отобр.» is provisional, with «отбр.» also possible. Both are abbreviations for maps.

$$f : A \longrightarrow A'$$

Deriv. functor:

$$D(f) : D(A) \longrightarrow D(A').$$



Original source details: GELFAND-P000438

Diagram and typed description. Two separate lengthwise traces run near the right edge.

GELFAND-P000439

[source number: 66]

Ka[z/zh]dan.



Uncertain reading. The joined strokes after “Ka” allow «Каздан» (Kazdan) and «Каждан» (Kazhdan). The surname is doubly underlined.

Original source details: GELFAND-P000439

Let A — an ab. categ. with a suff. number of inj. obj.

Consider the full subcategory of inj. objects. Inj(A)

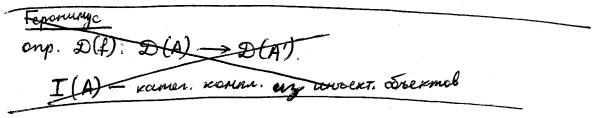
Consider there the categ. of complexes.

Categ. of compl. in Inj$(A) \cong$ the deriv. category of A.



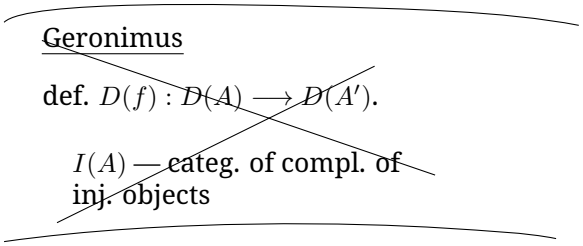
Original source details: GELFAND-P000439

Diagram and typed description. The sign between Inj(A) and “deriv.” has a wave over two strokes.



Original source details: GELFAND-P000439

Diagram and typed description. Two long rules bound the three lines above and below. Two oblique strokes cross out the block.



Morava.

Cobordism.

M, N — manif.[?], compact[?], C^∞, $\partial M = \partial N = \varnothing$.



Uncertain reading. The abbreviation has a loop with a descender after the hyphen. «мн-зия» is provisional, with «мн-гия» also possible.

Original source details: GELFAND-P000439



Uncertain reading. The final rounded stroke favours «компактные»; «компактны» is still possible. Both readings mean “compact”.

Original source details: GELFAND-P000439

$$\dim M = \dim N = n.$$

$$M \sim N \quad \text{cobordant} \quad \Longleftrightarrow \quad \exists\, \text{manif. } W, \quad \partial W = M \cup N$$

(W comp., C^∞).

$$N_n = \{\text{set of this cl. equiv.}\}. \quad = \{\text{cobord. cl.}\}.$$

1) $N_n = \{\text{semigroup}\}$

$$[M] + [N] = [M \cup N] \qquad \text{(disjoint)}$$
$$[M] + [\varnothing] = [M].$$

In | fact



Diagram and typed description. A tall slanting rule separates “In” from “fact”.

Original source details: GELFAND-P000439

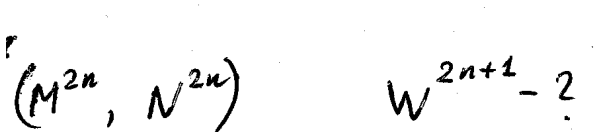
$N_n =$ group, because $[M] + [M] = [\varnothing]$

$$\partial(M \times [0,1]) = (M \times 0) \cup (M \times 1).$$

Spec. case: 1) M oriented

2) M complex

$$(M^{2n}, N^{2n}) \qquad W^{2n+1} - ?$$



Uncertain reading. After the dash, the upper part of the sign resembles a 2. There is a separate dot below. “?” is a possible reading.

Original source details: GELFAND-P000439

GELFAND-P000440

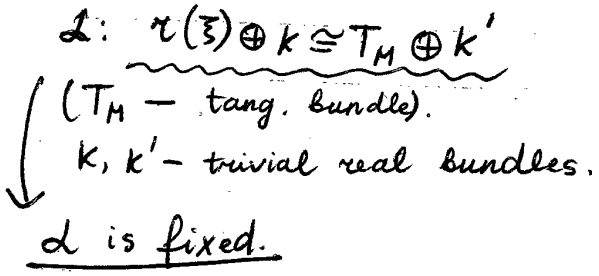
Weak compl. manif.

$(M, \xi) = \{M$ — manif.; ξ — compl. vect. bundle over $M\}$.



Undeciphered passage. Below the pair is a short stroke and a crossed-out sign with a loop. Its lettering is unresolved.

Original source details: GELFAND-P000440



Original source details: GELFAND-P000440

(T_M — tang. bundle).

k, k' — trivial real bundles.

α is fixed.

Diagram and typed description. A wavy line is below the isomorphism. At left a long curved arrow points down to the underlined condition.

$\alpha : \tau(\xi) \oplus k \cong T_M \oplus k'$

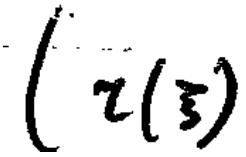
(T_M — tang. bundle).

k, k' — trivial real bundles.

α is fixed.

($\tau(\xi)$ — real ... sous-jacent bundle).





Original source details: GELFAND-P000440

Uncertain reading. Both signs before ξ have a cross stroke and a curved stem. They are provisionally set as τ; r is possible.



Original source details: GELFAND-P000440

Undeciphered passage. A single sign after "real" is crossed out. An oblique stroke extends below it.

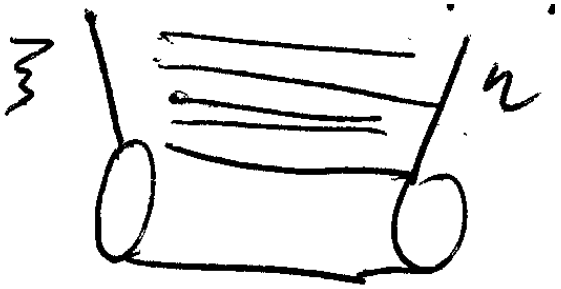
$[M, \xi, \alpha] \longrightarrow M$ ...

class of equiv. three objects



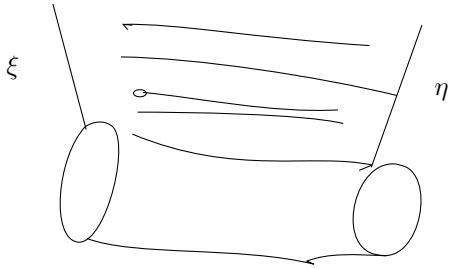
Original source details: GELFAND-P000440

Undeciphered passage. A short crossed-out entry remains to the right of M. Parts of loops show through the strokes.



Original source details: GELFAND-P000440

Diagram and typed description. The lower line connects two ovals. The slanted uprights are labelled ξ and η. Two of the five inner strokes reach the right upright. The third has a small loop at left. The role of the angular ends is unresolved.



S^1 — weak complex manif.

$T_{S^1} =$ triv.

Every W. c. m. is cobord. to compl. man.).

Ω_n^U — group | cobord. n-dim man. (w-compl.).

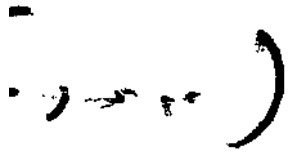
Film and conn. bundle. $\dots$ Σ.



Original source details: GELFAND-P000440

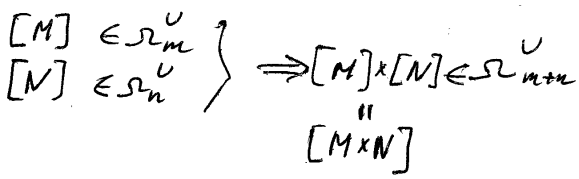
Undeciphered passage. Overlapping strokes cover a loop and diagonals before Σ. The earlier sign is unreadable.

$(W, \Sigma, \dots)$ — also W. c. m.



Original source details: GELFAND-P000440

Uncertain reading. Three separate groups of strokes follow the comma after Σ. An ellipsis remains a tentative reading.



Original source details: GELFAND-P000440

Diagram and typed description. A brace gathers two lines. Two short vertical strokes sit between the product of classes and the expression below.

$$\left.\begin{array}{l}[M] \in \Omega_m^U \\ [N] \in \Omega_n^U\end{array}\right\} \Rightarrow [M] \times [N] \in \Omega_{m+n}^U$$
$$||$$
$$[M \times N]$$

GELFAND-P000441

[source number: 67]

$$\Omega_*^U = \bigoplus_{n \geq 0} \Omega_n^U$$

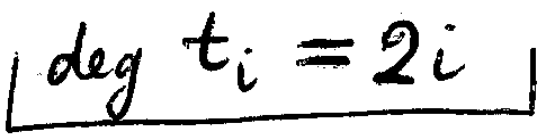
This | is a graded ring.



Original source details: GELFAND-P000441

Diagram and typed description. A short vertical stroke stands between “This” and “is”. The phrase has a wavy underline.

Th. $\Omega_*^U \cong \mathbb{Z}[t_0, t_1, \dots, t_n, \dots]$.



Original source details: GELFAND-P000441

Diagram and typed description. The enclosure is open at the top.

$\deg t_i = 2i$

(Ord$[M] \neq 2$ — compl. structure!).

Ω_n^U is group

Proof: $-[M] = [M \overset{\text{opposite}}{\text{orientation}}]$.

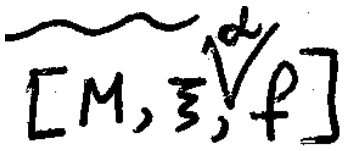
for orient cobord.

Compl. case — something similar.

X — topological space.

$X \mapsto \Omega_n^U(X)$ — group (functor of X).

$[M, \xi, \sqrt[\alpha]{} f]$ ~~[bord...?]~~



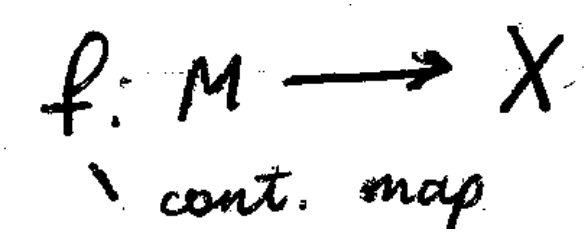
Original source details: GELFAND-P000441

Uncertain reading. The letter above the deep insertion mark reads α. The left end of the mark hooks back. A comma sits below it.



Original source details: GELFAND-P000441

Undeciphered passage. Strokes resembling “bord” or “cobord” remain under the cancellation. The full reading is unresolved.

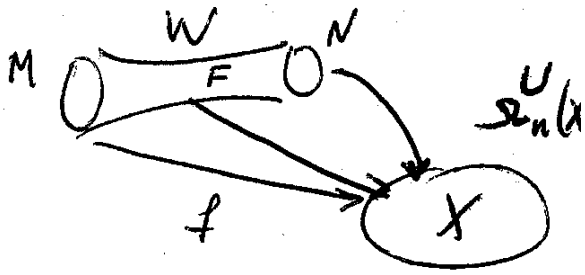


Original source details: GELFAND-P000441

Diagram and typed description. A short slanted stroke leads from the map notation to the caption below.

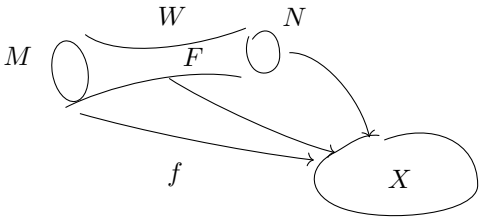
$f : M \longrightarrow X$

\ cont. map

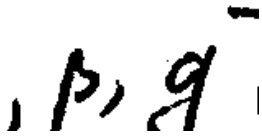


Original source details: GELFAND-P000441

Diagram and typed description. The upper line is separated from the left oval. Both lines stop before the right oval. A separate arc from the right of N points to X, as does the middle arrow. The lower arrow labelled f stops just short of the edge of the oval X.



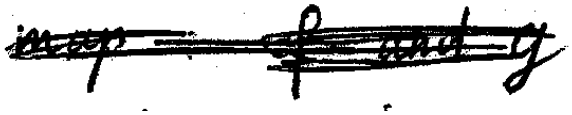
$\Omega_n^U(X)$ — equivalence classes
(group of Bordism of X).
$[M, \xi, \alpha, f] \sim [N, \eta, \rho, g]$



Original source details: GELFAND-P000441

Uncertain reading. The sign before g in the upper line has one loop and is provisionally read as ρ or p. β is possible. The β in the lower line has two bends.

iff ~~map [?] f and g~~ $[M, \xi, \alpha] \sim [N, \eta, \beta]$



Original source details: GELFAND-P000441

Uncertain reading. "map" and "f and g" are visible under the cancellation. The sign between them resembles $=$ or a dash.

and equiv. prolong. over W
$\bigoplus \Omega_n^U(X) = \Omega_*(X)$ — graded group (mult. abs.)
GELFAND-P000442

This is a Ω_*^U-module.



Original source details: GELFAND-P000442

Uncertain reading. A separate dark spot appears below "a". Its function is unresolved.

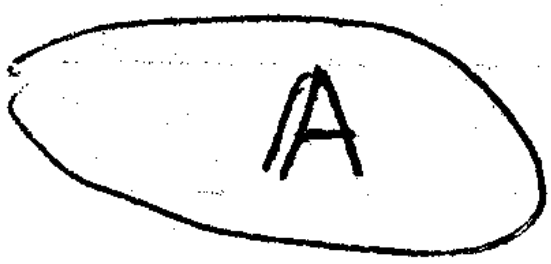
...



Original source details: GELFAND-P000442

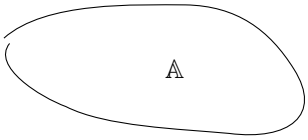
Undeciphered passage. Horizontal and oblique strokes cover the short entry. The letters are unreadable.

Problem: describe Ω_*^U-module structure. (in Ω_*^U)
$\Omega_*^U = \mathbb{Z}[t_0, t_1, \ldots]$ = coordin. ring of a big affine space $\mathbb{A}$.
Module $\Omega_*^U(X)$ — sheaf over $\mathbb{A}$.

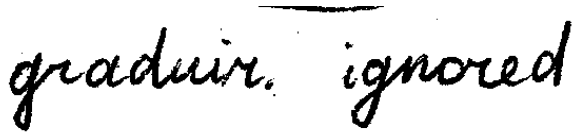


Original source details: GELFAND-P000442

Diagram and typed description. The slanting outline almost closes on the left. Inside it is $\mathbb{A}$, with a double left stroke.



graduir. ignored
$\bigoplus \Omega^U_{2n}(X)$ and $\bigoplus \Omega^U_{2n+1}(X)$ —
— from graduir. (forgets the rest)





Uncertain reading. The abbreviation in both lines reads “graduir.”.

Original source details: GELFAND-P000442

Theor. $\exists$ dens. open set in the Zar. topol. $\mathbb{A}$, such that
Morava and Quillen,
the fibre at $\forall$ point is the gr. $K^\bullet(X)$ $\big(= K^0(X) \oplus K^1(X)\big)$!!

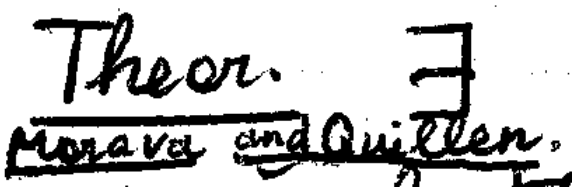


Diagram and typed description. “Morava” and “and Quillen” appear below the heading. There are two separate lines under the names.

Original source details: GELFAND-P000442



Uncertain reading. The Russian is provisionally read as “пл.” and translated as “dens.”. The second letter more closely resembles “л” than the “н” of “мн-во” to the right.

Original source details: GELFAND-P000442



Diagram and typed description. The Russian “в” is written above the line before $\mathbb{A}$, inside a loop-shaped insertion mark.

Original source details: GELFAND-P000442

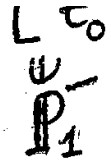


Uncertain reading. The heavily circled sign with intersecting strokes is provisionally set as $\oplus$. $\otimes$ is possible.

Original source details: GELFAND-P000442

$\mathbb{Z}[t_0, \ldots, t_n, \ldots][t_1^{-1}]$ — this is that very open set.

$t_1 = \mathbb{P}_1$ — compl. proj. bundle



Uncertain reading. A separate U-shaped mark with an inner stroke stands above $\mathbb{P}_1$. A detached rising stroke to the right may be a prime. No continuous arrow shaft is visible.

Original source details: GELFAND-P000442

R — commutative; $\Gamma(R) = \left\{ \sum_{i=1}^{\infty} a_i t^i \;\middle|\; a_i \in R \right\}$

$a_1 \in R^*$ — invert.

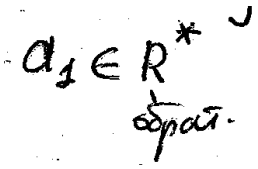


Diagram and typed description. A short oblique stroke lies between R^* and "invert.". No separate arrowhead is visible.

Original source details: GELFAND-P000442

gr. under compos. of maps.

GELFAND-P000443

[source number: 68]

$\Gamma(R)$ — inf. dim alg. group. (compos. — polinomial)

Γ acts on A

Sheaf Ω_*^U is an equivariant sheaf (const on orbits).



Uncertain reading. There is a dark patch of ink at the left of Ω. Any earlier writing underneath is unreadable.

Original source details: GELFAND-P000443

Γ: {Rings} ⟶ {groups}.



Diagram and typed description. The arrowhead at right is clear. At left is a short lower hook whose role is unresolved.

Original source details: GELFAND-P000443

The actual theor. should consist in the descr. of the orbits.

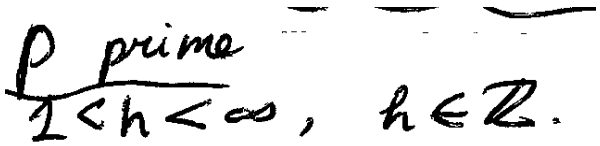
~~Orb.[?]~~ over $\mathcal{D}$ — fibers $K^\bullet(X)$.
$\mathcal{D}_{(p,h)}$ p prime — orbits,
$1 < h < \infty$, $h \in \mathbb{Z}$.
$\mathcal{D}_\infty$ = fibers $H^*(X, \mathbb{Z})$.

orbits.



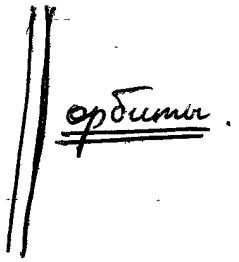
Original source details: GELFAND-P000443

Undeciphered passage. An initial loop, a tall loop and a dot show through the cancellation. “Orb.” is a provisional reading.



Original source details: GELFAND-P000443

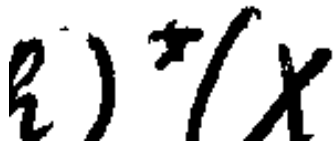
Diagram and typed description. The strict inequality $1 < h < \infty$ is written below “p prime”. The top stroke of 1 meets the wavy underline.



Original source details: GELFAND-P000443

Diagram and typed description. Two near-vertical strokes set off the doubly underlined caption on the right.

$K(h)$ — functor $K(h)^+(X, \mathbb{Z}_p)$ — $\mathbb{Z}_p$-module.



Original source details: GELFAND-P000443

Uncertain reading. The superscript of $K(h)$ has a crossbar and one slanted stroke. It is provisionally set as $+$. $*$ is possible, though the stars in Ω^U_* and H^* show extra arms.

Bott periodicity in dim $2(p^h - 1)$

$\exists$ Adams oper. $\psi^\alpha \colon K(h)$



Original source details: GELFAND-P000443

Diagram and typed description. The arc at the right of $K(h)$ runs from the lower end to the upper end. Its arrowhead points left.

$\alpha \in$ units of p-adic division algebra of ranke h^2.



Original source details: GELFAND-P000443

Uncertain reading. The sign before $\in$ resembles the superscript of ψ. It is provisionally set as α, with λ possible. The final "e" is visible in "ranke".

This is station. subgroup of a point on the orbit.

Next time — continuation



Formal groups



Original source details: GELFAND-P000443

Uncertain reading. An angular loop, a hooked stem and a separate dot lie within the outlined shape. The glyph remains unidentified.

$\mathbb{A}$ — moduli space for 1-dimensional commut. form. groups.

F — form. gr. over a ring R, if an elem. is given $F \in R[[X, Y]]$,



Original source details: GELFAND-P000443

Uncertain reading. The inner opening mark is rounded, almost like a C. Two closing brackets are visible. The double opening bracket is provisional.

1) $F(X,Y) = F(Y,X)$; 2) $F(0,X) = X$; 3) $F(X, F(Y,Z)) = F(F(X,Y),Z)$

(the inv. exists automatically). X, Y — form. var. — letters.

GELFAND-P000444

$f \in \Gamma(R)$ $\Rightarrow$ s'

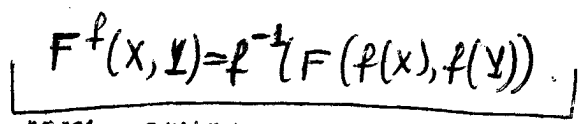


Original source details: GELFAND-P000444

Uncertain reading. A separate slanting stroke sits below the right end of the underline. The small adjacent sign and its raised stroke are provisionally read as s', with $5'$ possible.

$$f = a_1 t + a_2 t^2 + \dots$$
$$a_1 \in R^*$$

Change of var. —

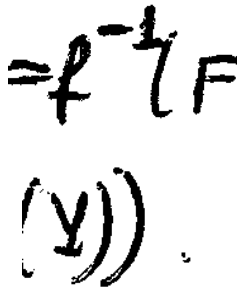


Original source details: GELFAND-P000444

Diagram and typed description. A long line with raised ends encloses the formula from below.

$$F^f(X,Y) = f^{-1}\big[F\big(f(X), f(Y)\big).$$

Uncertain reading. The slanted sign with a short lower hook after f^{-1} is provisionally set as [. Two round parentheses and a stop follow Y. No separate outer closing delimiter is visible.



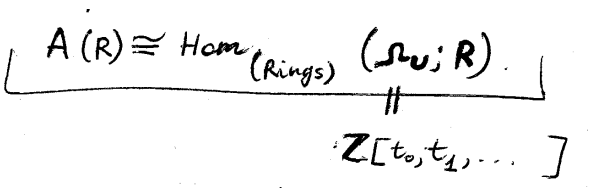
Original source details: GELFAND-P000444

f acts on the set of form. groups.

R — ring $\rightsquigarrow$ {form. groups $A(R)$ — set.}

$$A\colon (\text{Rings}) \xrightarrow{\text{Functor}} (\text{Sets})$$

it is representative



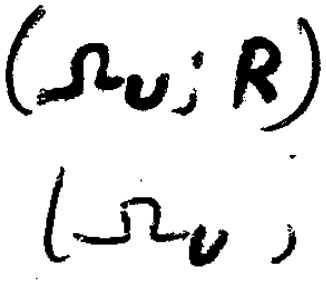
Original source details: GELFAND-P000444

Diagram and typed description. The frame is open at the top. Two vertical strokes below Ω_u cross its lower line. The polynomial ring is written below them.

$$A(R) \cong \text{Hom}_{(\text{Rings})}(\Omega_u; R).$$

$$\mathbb{Z}[t_0, t_1, \dots]$$

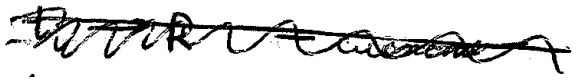
Uncertain reading. The sign after Ω is lowered in both occurrences. It is provisionally set as Ω_u. A capital U in the subscript is possible.



Original source details: GELFAND-P000444

Hom determ. by values on t_i. which form. gr.

~~$\Omega_u \longrightarrow R$[?] ...~~



Original source details: GELFAND-P000444

Undeciphered passage. At left under the cancellation are an omega-like loop and lower sign, followed by a rightward arrow-like stroke and R. The beginning may read $\Omega_u \to R$ or contain $\forall$. A C-like sign is visible farther on. The full inscription is unresolved.

$$\forall R: \quad \Gamma(R) \times A(R) \longrightarrow A(R);$$
$$\Gamma \times \mathbb{A} \longrightarrow \mathbb{A}.$$

(Γ acts on A as an alg. gr. on an alg. set).

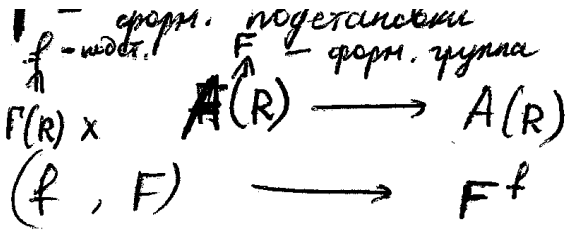
$$A(R) \underset{\text{def}}{=} \text{Hom}(\Omega_u, R).$$

Γ — form. substitutions



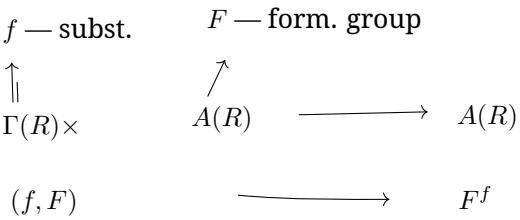
Original source details: GELFAND-P000444

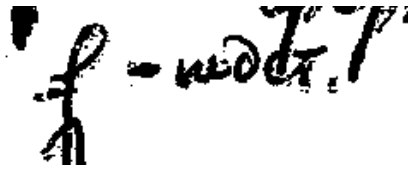
Uncertain reading. The Russian is provisionally read as “подстановки”. Joined strokes between “д” and “т” obscure the reading.



Original source details: GELFAND-P000444

Diagram and typed description. Two short arrows point up from the vicinity of $\Gamma(R)$ to f and from the first $A(R)$ to F. Both long arrows point right.





Original source details: GELFAND-P000444

Uncertain reading. The caption is provisionally read as “подст.”. It has a tall medial loop, a final cross stroke and a stop. The earlier alternatives “индив.” and “модиф.” remain possible.



Original source details: GELFAND-P000444

Uncertain reading. The first lower A is heavily overwritten. The earlier strokes cannot be separated.

GELFAND-P000445

[source number: 69]

Kazhdan.

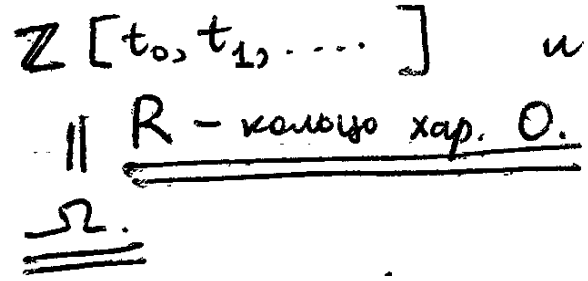


Diagram and typed description. Two vertical strokes stand between the beginning of the line with $\mathbb{Z}$ and Ω. The statement about R is to their right.

$\mathbb{Z}[t_0, t_1, \ldots]$ and form. group ... over R.
$\|$ R — ring of char. 0.
Ω.

Original source details: GELFAND-P000445



Undeciphered passage. The short cancelled inscription before "over" is undeciphered.

Original source details: GELFAND-P000445

Constr. of an inv. diff. on a form. group — uniquely determ.

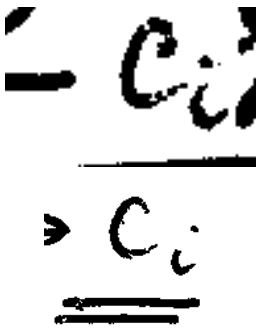
$$\omega = \sum c_i x^i \, dx;$$

Map: $t_i \longrightarrow C_i \in R$.



Uncertain reading. Both variable signs are heavily overwritten. They are provisionally set as x. Their case and earlier strokes are unresolved.

Original source details: GELFAND-P000445



Uncertain reading. The coefficient in the differential is set as c_i. The taller letter in the map is set as C_i. The distinction in case is provisional.

Original source details: GELFAND-P000445

Obtain a map $\Omega \longrightarrow R$.

14/IV I. M. Gelfand.

On Dyson's talk "Unrealized possibilities".

Anecdote about a tunnel.

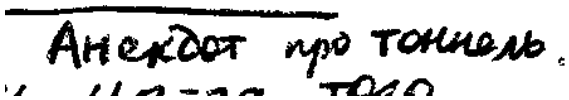


Diagram and typed description. The reference to an anecdote sits in the right margin beneath the title.

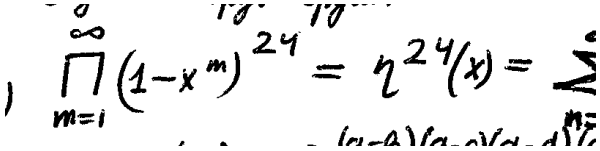
Original source details: GELFAND-P000445

Aim: to convince math. and phys. how much they lost by failing to listen to one another.

Example 1.

(Dyson's amusement)

$$\prod_{m=i}^{\infty}(1-x^m)^{24}=\eta^{24}(x)=\sum_{n=1}^{\infty}\tau(n)x^n \quad \text{(Ramanujan fn.)}$$



Original source details: GELFAND-P000445

Uncertain reading. The lower product index is provisionally read as $m=i$. A short stroke with a small mark above it follows the equals sign. The reading $m=1$ remains possible. No separate factor is visible before the product.

Identity:

$$\tau(n)=\sum_{\substack{a,b,c,d,e\equiv 1,2,3,4,5 \pmod 5,\\ a+b+c+d+e=0\\ a^2+b^2+c^2+d^2+e^2=10n}}\frac{(a-b)(a-c)(a-d)(a-e)(b-c)(b-d)(b-e)(c-d)(c-e)(d-e)}{1!\,2!\,3!\,4!}.$$

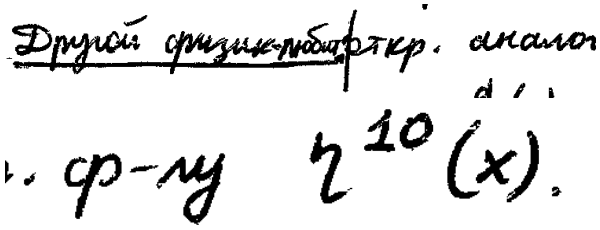


Original source details: GELFAND-P000445

Uncertain reading. The first letter in the penultimate parenthesis is overwritten. The factor is provisionally set as $(c-e)$.

$\eta(x)$ — Dedekind fn..

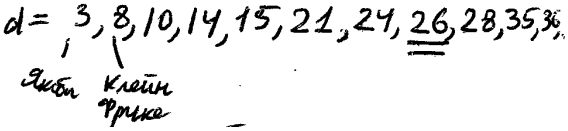
Another amateur physicist discov. an analog. fmla. $\eta^{10}(x)$.



Original source details: GELFAND-P000445

Uncertain reading. The joined writing after the underlined words is provisionally read as "откр.". The abbreviation before the power of η is provisionally read as "ф-лу".

Dyson: for $\eta^d(x)$,



Original source details: GELFAND-P000445

Diagram and typed description. Short slanted strokes lead from 3 and 8 toward the labels. A comma and a small stroke are visible after 36.

$d=$ 3, 8, 10, 14, 15, 21, 24, 26, 28, 35, 36,

Jacobi — 3; Klein, Fricke — 8

These are dimensions of simple Lie algebras.

$$\begin{array}{ccccccc} A_1 & A_2 & Sp(4) & - & - & \cdot & A_4 = SL(5) \\ 3 & 8 & 10 & - & - & \cdot & 24 \end{array}$$

Prod. — over all simple roots (pos.) of gr. $SL(5)$.

Произв. — по всем простым корням (положит.) гр. SL(5)

Original source details: GELFAND-P000445

Uncertain reading. The wording is “простым корням (положит.)”, rendered here as “simple roots (pos.)”.

26 — drops out — unclear

GELFAND-P000446

2. $c \to 0,\ c \to \infty,\ R \to \infty$ — lim. transitions in the kinem. group.

в кинем. группе.

Original source details: GELFAND-P000446

Uncertain reading. The last word reads “группе” (“group”). Joined strokes in the middle obscure the reading slightly.

Morava.

Kirillov (recapitulation).

$X \longrightarrow \Omega_n(X)$ — group, $\underset{[?]}{}$ $\Omega_n(*)$ — module.



Original source details: GELFAND-P000446

Uncertain reading. A separate small sign with two upper strokes is visible beneath the dash after $\Omega_n(X)$. Its reading is unresolved.

~~Triple.~~ (M, ξ, α, f)



Original source details: GELFAND-P000446

Uncertain reading. “Тройка.” (“Triple.”) is the provisional reading beneath the two cancellation strokes.

(M, ξ, α) — weakly compl. set

$$\left(\cdots \quad \underbrace{\tau(M) + [k] = \xi + [k']}_{\alpha}, \quad \xi \text{ — already a compl. bundle} \right)$$



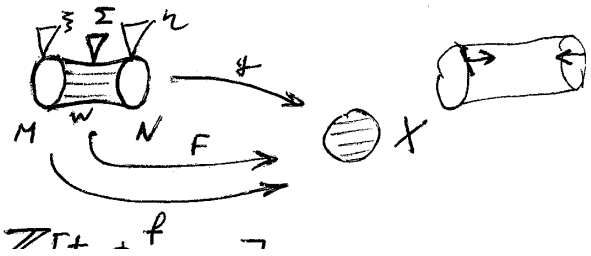
Original source details: GELFAND-P000446

Undeciphered passage. The short sign just inside the left parenthesis is cancelled. The earlier strokes are unresolved.

α — fixed decomp..

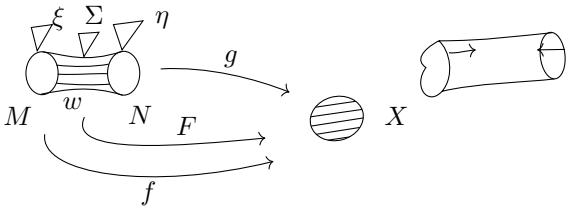
$f\colon M \longrightarrow X$

Equiv. relation.



Original source details: GELFAND-P000446

Diagram and typed description. Three arrows point right toward the hatched region X. The upper label is provisionally g, and the label beneath the middle of the left cylinder is provisionally w. Two arrows in the separate right cylinder point inward from its ends.



Theorem. $\Omega_*(*) = \mathbb{Z}[t_1, t_2, \ldots]$

$$\deg t_i = 2i$$

$t_1 \leftrightarrow CP^1$

A — cobord. ring of a point $\Omega_*(*)$.

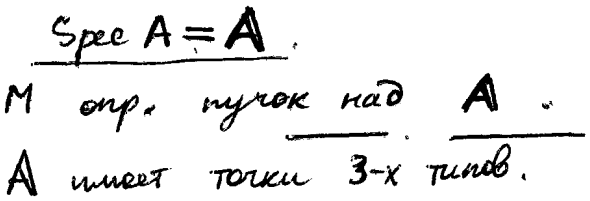
M — module over A $(\Omega_*(X))$.

Spec $A = \mathbb{A}$.

M determ. a sheaf over $\mathbb{A}$.

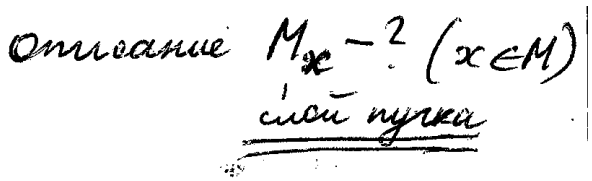
Description of M_x — ? $(x \in M)$
fiber of the sheaf

$\mathbb{A}$ has points of 3 types.



Original source details: GELFAND-P000446

Uncertain reading. The right slanting stroke is doubled in three signs. The sign before the equals sign is written more simply. The three signs are set as $\mathbb{A}$.



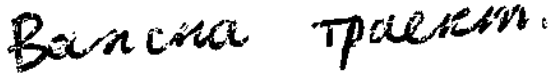
Original source details: GELFAND-P000446

Uncertain reading. The letter following the membership sign in the margin reads M. The heavily overwritten subscript of M_x is provisionally read as x.

GELFAND-P000447

[source number: 70]

What matters is the traj. of a point under the action of an infin. group.



Uncertain reading. The opening reads “Важна траект.”. Some joining strokes are faint.

Original source details: GELFAND-P000447

Orbits:

1) one open dense — gen. pos.
2) one closed
3) count. number of interm. orbits



Uncertain reading. The first letters of the abbreviation are overwritten. The provisional reading is “промежут.” (“interm.”).

Original source details: GELFAND-P000447

1) fiber $K(X)$. (at $\forall$ point) (open dense)
2) p, n (p — prime, n — natur.) $K^{n,p}(n)(X, \mathbb{Z}_p)$.

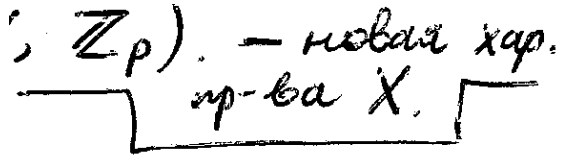
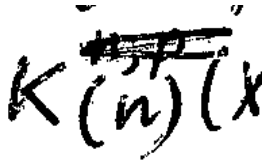


Diagram and typed description. A stepped bracket runs beneath “of the sp. X”.

— new char.
of the sp. X.

Original source details: GELFAND-P000447



Uncertain reading. A short inscription above (n), following K, is cancelled. It is provisionally read as n, p.

Original source details: GELFAND-P000447

3) $H^*(X, \mathbb{Z})$ at $\forall$ point.

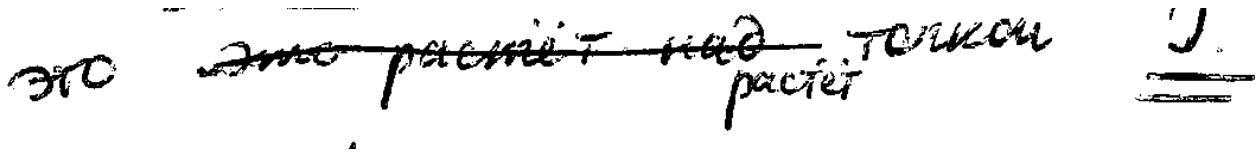
Clarification

$$(a_1, \ldots, a_n, \ldots) \in \mathbb{A}$$

max id.

$$J = (t_1 - a_1, \ldots, t_n - a_n, \ldots)A \subset \mathbb{A}$$

M/MJ — vect. sp. (more precisely A/J — module).

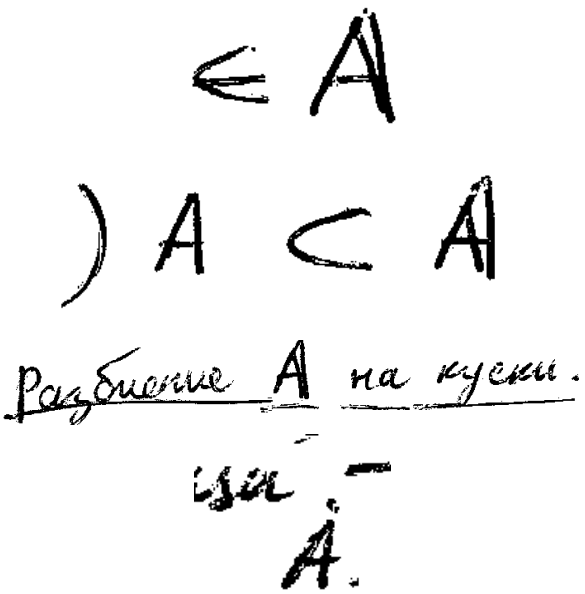


Original source details: GELFAND-P000447

Diagram and typed description. “It” is written separately on the left. “Grows” is inserted beneath the cancelled “over”.

it ~~it grows~~ ~~over~~ grows the point J.

Partition $\mathbb{A}$ into pieces.



Original source details: GELFAND-P000447

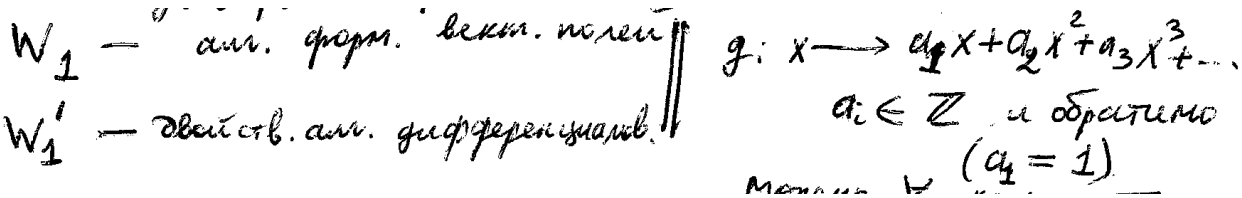
Uncertain reading. The right stroke of A is doubled after $\in$ and to the right of $\subset$. The letter in the “Partition” heading is also set as $\mathbb{A}$. The sign immediately after the closing parenthesis in the J line is plain A.

~~$\mathbb{A}$ [...] one-dimension... form.~~ G — gr. of form. substitutions of variables



Original source details: GELFAND-P000447

Undeciphered passage. Under the wavy cancellation the initial A, the stem “одномерн” (“one-dimension”), and the final “форм.” (“form.”) are visible. The reading of the horizontal stroke and the ending of the first word is unresolved.

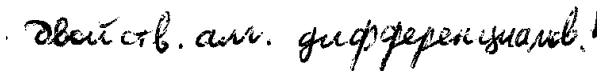


Original source details: GELFAND-P000447

Diagram and typed description. Two vertical strokes separate the W_1, W_1' entries from the substitution g. The subscript on the first a is overwritten and reads as 1.

W_1 — alg. of form. vect. fields
W_1' — dual alg. of differentials.

$g\colon x \longrightarrow$
$a_1 x + a_2 x^2 + a_3 x^3 + \dots$
$a_i \in \mathbb{Z}$ and invertible
$(a_1 = 1)$



Original source details: GELFAND-P000447

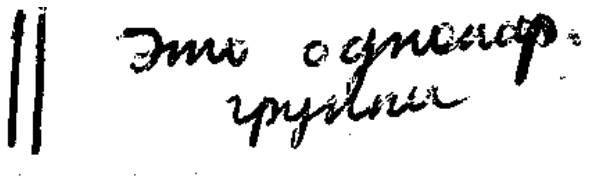
Uncertain reading. The text reads “двойств. алг. дифференциалов” (“dual alg. of differentials”). A separate stroke is visible above “й”.

Possible for $\forall$ rings. —
— in particular from A.
(a_1 — inv. el. of the ring)
Form. 1-dim. group:

$$x * y = x + y + \alpha xy + \beta x^2 y + \delta x y^2 + \dots \qquad (\alpha, \beta \in \text{the ring})$$

associativity. — sole req.
They are equiv. if one law of mult. $\longrightarrow$ the other by a form. subst. of var.

$$\varphi(x) = \lambda_1 x + \lambda_2 x^2 + \dots, \qquad \lambda_1 \text{ — inv. el. of the ring}$$

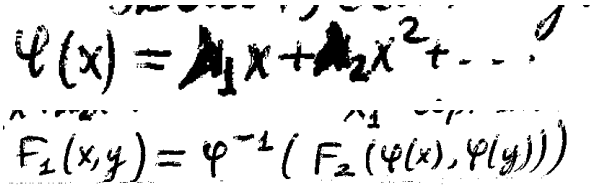


Original source details: GELFAND-P000447

Diagram and typed description. Two vertical strokes separate the right marginal note. Its abbreviation is provisionally read as “одноморф.” (“same-form”).

$F_1(x,y) = \varphi^{-1}\big(F_2(\varphi(x), \varphi(y))\big)$

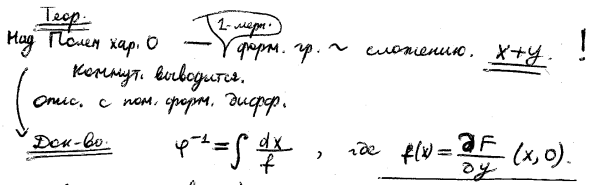
These are same-form groups



Original source details: GELFAND-P000447

Uncertain reading. The coordinate sign is set as φ in all four places. A right loop is visible, especially in the lower formula. The first sign is less clear. The two overwritten coefficients are provisionally read as λ_1, λ_2.

GELFAND-P000448



Original source details: GELFAND-P000448

Diagram and typed description. A loop encloses the insertion “1-dim.”. The left arrow leads down to “Proof”.

Thm.

Over a Field of char. 0 — 1-dim. form. gr. $\sim$ addition. $\underline{x+y}$. !

Commut. follows.

Descr. by means of form. diff.

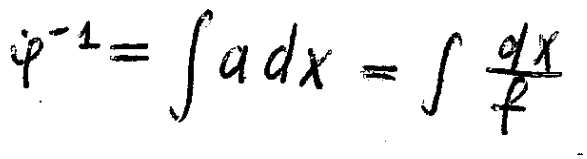
Proof. $\varphi^{-1} = \int \frac{dx}{f}$, where $\underline{f(x) = \frac{\partial F}{\partial y}(x,0)}$.

(simple check).

Outside char. 0 — the integral does not exist.

$\int x^{p-1}\,dx$ does not exist over a field of char. p.

$$\dot{\varphi}^{-1} = \int a\,dx = \int \frac{dx}{f}$$



Uncertain reading. A separate dot is visible above the left part of φ in the second occurrence. It is set as a dot above the letter. It may be a spot.

Original source details: GELFAND-P000448

$\underline{a\,dx}$ — form. diff., ~~reconstr.~~ determ. the group



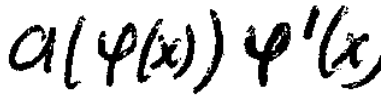
Uncertain reading. The cancelled abbreviation is provisionally read as “восстан.” (“reconstr.”).

Original source details: GELFAND-P000448

$$a(x) = 1 + a_1 x + a_2 x^2 + \dots$$

Action of the group (in diff.)

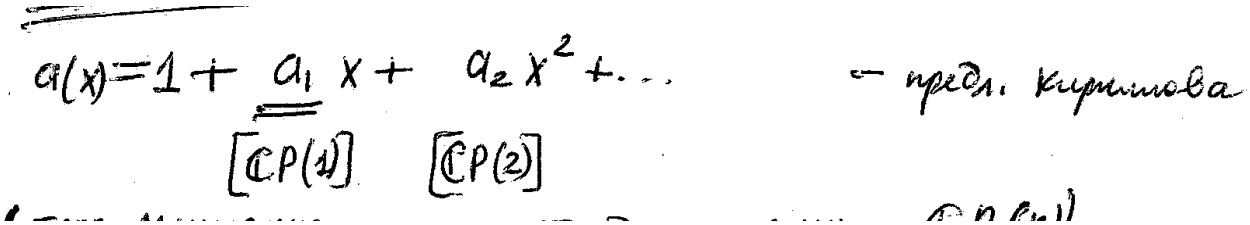
$$a(x) \longrightarrow a(\varphi(x))\varphi'(x)$$



Uncertain reading. Both signs have a right loop. They are set as φ and φ'.

Original source details: GELFAND-P000448

$a(x)$ — any series. To it corresp. a form. group (theorem on existence of the univ. group) Morava.



Original source details: GELFAND-P000448

Diagram and typed description. The labels under a_1 and a_2 are in square brackets. The letter C has an additional stroke inside.

$$a(x) = 1 + \underset{[\mathbb{C}P(1)]}{a_1}\, x + \underset{[\mathbb{C}P(2)]}{a_2}\, x^2 + \dots$$

— Kirillov's prop.

(Mishchenko's thm. — … here one obtains $\mathbb{C}P(n)$).



Undeciphered passage. The short inscription after the dash is heavily cancelled. The letters cannot be separated.

Original source details: GELFAND-P000448

Theorem, G — gr. scheme. $X_{[?]}$ — G-scheme (over $\mathbb{Z}$).



Uncertain reading. A single slanting stroke follows "Theorem". It is provisionally set as a comma.

Original source details: GELFAND-P000448



Uncertain reading. A small looped mark and an adjacent stroke appear below and to the right of X. The reading is unresolved.

Original source details: GELFAND-P000448

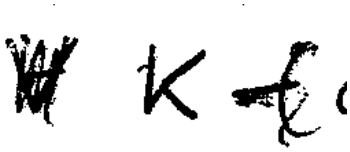
Let X be smooth, G be flat and all be Noetherian



Uncertain reading. Two diagonal strokes overwrite the sign before "smooth". It is provisionally set as X. The earlier loop-like stroke cannot be separated from the overwriting.

Original source details: GELFAND-P000448

… K — (alg. closed field) and $G(K)$ acts transitively on $X(K)$



Uncertain reading. A separate sign with two slanting sides is heavily cancelled before K. One possible reading is $\forall$.

Original source details: GELFAND-P000448

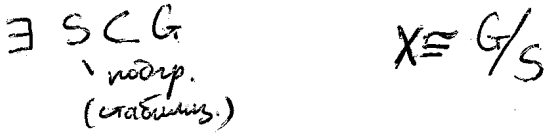


Original source details: GELFAND-P000448

Uncertain reading. The letters in parentheses after G and X are smaller than the standalone K, but have a similar angular form. They are provisionally set as K.

GELFAND-P000449

[source number: 71]



Original source details: GELFAND-P000449

Diagram and typed description. A short slanted stroke links S to its label.

$\exists\ S \subset G \qquad X \cong G/S$

subgr.

(stabiliz.)



Original source details: GELFAND-P000449

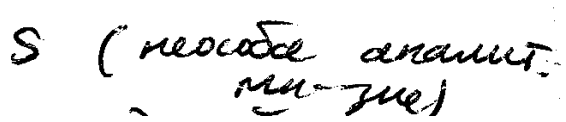
Diagram and typed description. Ten dots form a separator.

. . . . . .

GELFAND-P000450

25/IX-72 yr. Gabrielov.

Fam. of Hodge structures over S (nonsingular analyt. set)



Original source details: GELFAND-P000450

Uncertain reading. Two rounded strokes are joined after “б”. The reading is “неособое” (“nonsingular”).

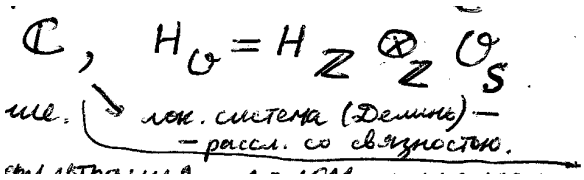
1) $\mathcal{H}_{\mathbb{Z}}$ — loc. free sheaf of $\mathbb{Z}$-modules of fin. type.

~~2)~~ From it constr. a loc. triv. compl. bundle $H_{\mathbb{C}}$.



Original source details: GELFAND-P000450

Uncertain reading. The cancelled beginning of the line is provisionally read as “2)”.



Original source details: GELFAND-P000450

Diagram and typed description. An arrow from the complex scalar extension points to the note. The lower line of the note turns up at its left end.

$H_{\mathbb{C}} = H_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{C}, \quad H_{\mathcal{O}} = H_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathcal{O}_S.$

and $H_{\mathbb{R}}$ — real bundle. loc. system (Deligne) —

— bundle with connection.

2) in $H_{\mathbb{C}}$ a decreasing filtration by holom. bundles

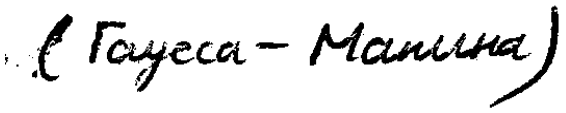
$$H_{\mathbb{C}} = F^0 \supset F^1 \supset \ldots \supset F^n$$



Original source details: GELFAND-P000450

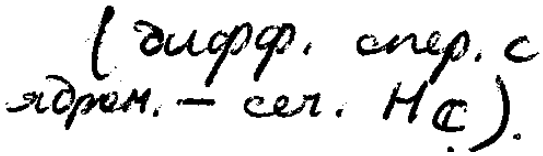
Diagram and typed description. A loop encloses the inserted “есть” (“there is”).

3) in $H_{\mathcal{O}}$ there is nat. connection (Gauss–Manin)



Original source details: GELFAND-P000450

Uncertain reading. The middle letters of the first name are joined. The pair is read as “Гаусса—Манина” (“Gauss–Manin”).

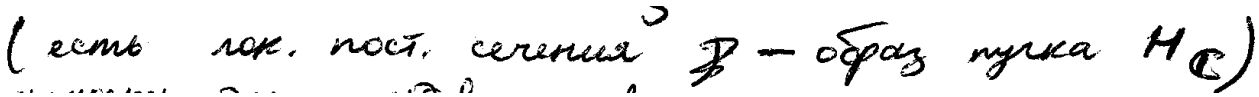


Original source details: GELFAND-P000450

Diagram and typed description. The note on the kernel is to the right of the formula. The mark before the dash is provisionally a comma.

$\nabla\colon H_{\mathcal{O}} \longrightarrow \Omega^1 \otimes_{\mathcal{O}_S} H_{\mathcal{O}}$ (diff. oper. with kernel, — sect. of $H_{\mathbb{C}}$).

(there are loc. const. sections $\mathbb{Z}$ — image of the sheaf $H_{\mathbb{C}}$)



Original source details: GELFAND-P000450

Uncertain reading. The sign between “sections” and the dash is provisionally read as $\mathbb{Z}$. It has a loop and a long lower stroke. The exact reading is unresolved.

The connection must satisfy the cond.:

$$\nabla(\mathcal{F}^p) \subset \Omega^1_S \otimes \mathcal{F}^{p-1} \qquad (\mathcal{F}^p = F^p \otimes_{\mathbb{C}} \mathcal{O}_S)$$

4) ψ — bilin., loc. const. form on $H_{\mathbb{C}}$, giving on restriction to $\forall$ fiber a genuine Hodge struct.

———— • ————

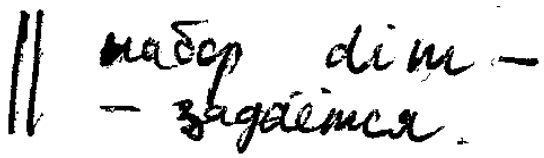
Classifying sp. of Hodge structures.
Consid.

———— • ————

$$F = \{F^0 \supset F^1 \supset \ldots \supset F^n\} \qquad \text{— flags of fixed dim.}$$

in a fixed sp., $\mathbb{C}^n$ (dim can have gaps).

———— • ————

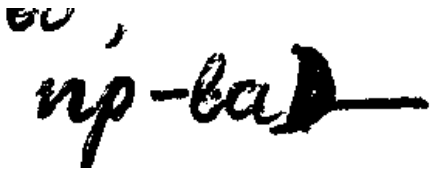


Original source details: GELFAND-P000450

Diagram and typed description. Two vertical strokes separate the right marginal note.

list of dim —
— is given.

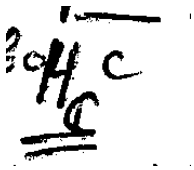
D — sp.;
Points of the sp. ~~D~~ — flags.



Original source details: GELFAND-P000450

Uncertain reading. The heavily overwritten sign before the dash resembles D. The reading of the cancelled letter is provisional.

Take a lin. sp. $H_{\mathbb{C}}$ with a bilin. form $\psi(x, y)$ (nondeg.; skew or symm.)



Original source details: GELFAND-P000450

Diagram and typed description. Two lines under $H_{\mathbb{C}}$ sit below its lowered subscript.

$\bar{D}$ — set of all flags ~~in~~ in $H_{\mathbb{C}}$ with a fixed list of dim



Original source details: GELFAND-P000450

Uncertain reading. A separate stroke is visible above the heavily retraced D.



Original source details: GELFAND-P000450

Uncertain reading. The small cancelled sign before “в” (“in”) is provisionally read as a repeated “в”.

GELFAND-P000451

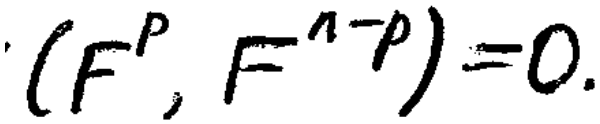
[source number: 72]

... with condition 1) $\psi(F^p, F^{n-p}) = 0$.



Original source details: GELFAND-P000451

Undeciphered passage. The short cancelled sign before “with” is undeciphered.



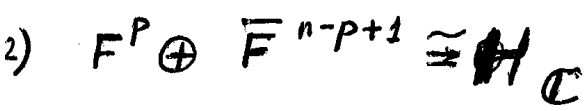
Original source details: GELFAND-P000451

Uncertain reading. The first formula clearly reads $n - p$. The next formula separately reads $n - p + 1$.

This is an algebraic subs. $\bar{D}$.

ψ either symm. or skew-symmetric (def).

$$2) \quad F^p \oplus \bar{F}^{n-p+1} \cong H_{\mathbb{C}}$$



Original source details: GELFAND-P000451

Uncertain reading. The sign before $H_{\mathbb{C}}$ has a wave above two horizontal strokes. A slanted mark joins the lower strokes. The reading $\cong$ is provisional. The letter H is heavily retraced.

$H_{\mathbb{Z}} \subset H_{\mathbb{R}} \subset H_{\mathbb{C}}$ also given

The nondeg. flag determ. the grading and the Weil autom.

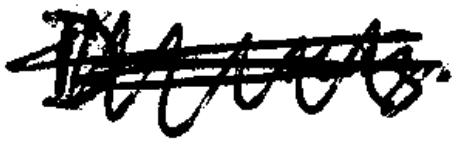
$$H^{p,n-p} = F^p \cap \bar{F}^{n-p}$$

Weil autom.:

$$C\xi = i^{2p-n}\xi \quad \text{when } \xi \in H^{p,n-p}$$

These conds. pick out in $\bar{D}$ the open subs. D.

... Another condition: $\psi(x, C\bar{x}) > 0$ when $x \neq 0$.



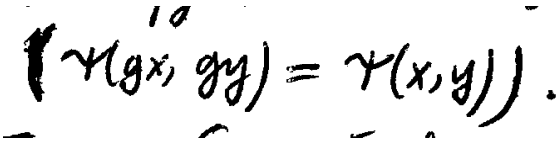
Original source details: GELFAND-P000451

Undeciphered passage. Several horizontal strokes cover the earlier writing before “Another condition”. The ending has a loop and a stop. The text is unresolved.

Define on D the structure of a homog. sp..

G — group of all real transf. of $H_{\mathbb{R}}$, preserv. ψ

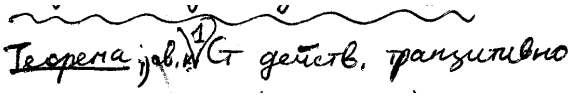
$$\big(\psi(gx, gy) = \psi(x, y)\big).$$



Original source details: GELFAND-P000451

Uncertain reading. The left outer parenthesis is heavily retraced. Two round closing parentheses are distinguishable on the right.

Then G acts on D



Original source details: GELFAND-P000451

Diagram and typed description. Before G is a short inscription provisionally read as “яв.” (“expl.”) and a heavily retraced sign. A 1 is inserted in the loop above it. The beginning of the line provisionally reads 1).

Theorem: 1) expl. ...$\overset{1}{\vee}$$G$ acts transitively

2) stab. Subgroup H (preserv. the flag) is compact.

Proof: from explicit computation of G and H.

$K \subset G$ — max. comp. subgroup

$G/H \longrightarrow G/K$ it is not holomorphic

In G/K there is a compl. structure

GELFAND-P000452

....



Original source details: GELFAND-P000452

Undeciphered passage. Closely spaced wavy strokes cover the underlined heading. A stop is visible to the right. The letters are unresolved.

Manin’s 1st lecture on p-adic numbers. (retelling)

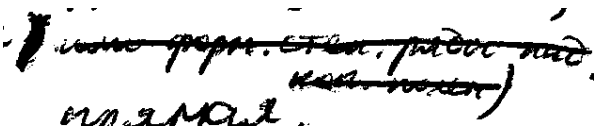
K — loc. field, (...



Original source details: GELFAND-P000452

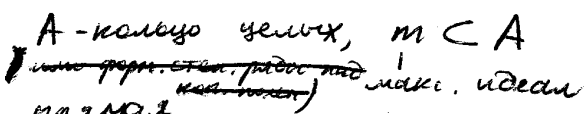
Undeciphered passage. After the comma are a left parenthesis and a short, heavily cancelled inscription. The right parenthesis in this line is unclear.

(fin. ext. of the p-adic numbers) ~~or form. pow. series over a fin. field~~)



Original source details: GELFAND-P000452

Undeciphered passage. The cancelled lines are provisionally read as “or form. pow. series over” and “a fin. field”. The closing parenthesis near “adic numbers” is heavily retraced.

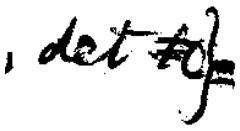


Original source details: GELFAND-P000452

Diagram and typed description. A short vertical stroke connects m with the label below.
A — ring of integers, $m \subset A$
max. ideal

$P^1(K)$ — proj. line.

$$G = \text{its motions} \quad \left\{ z \longrightarrow \frac{az+b}{cz+d} \;\middle|\; a,b,c,d \in K,\ \det \neq 0 \right\} =$$
$$= \mathrm{GL}(2,K)/K^*$$



Original source details: GELFAND-P000452

Uncertain reading. The notation before the right brace reads det $\neq 0$. The sign is heavily retraced. Two horizontal strokes immediately after the brace read as $=$. The next line also starts with $=$.

$U \subset G$; $U = \mathrm{GL}(2,A)/A^*$.

Constr. G/U — graph Δ — analog of the Lobachevsky pl..

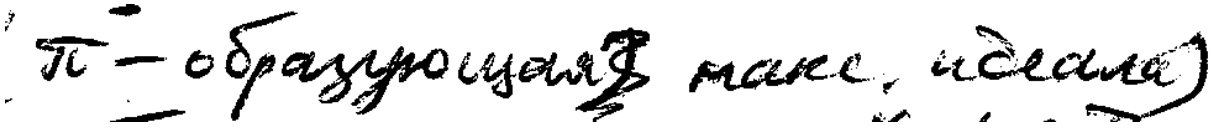
Def.: a lattice in K^2 — a free A-module with 2 gen.

$$L = Ae_1 \oplus Ae_2.$$

Lemma. If L_1 and L_2 — 2 latt., then $\exists$ a basis e_1, e_2 in K^2, such that

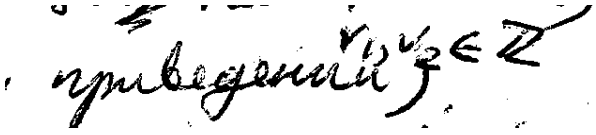
$$L_1 = Ae_1 \oplus Ae_2; \qquad L_2 = m^{\nu_1} e_1 \oplus m^{\nu_2} e_2 =$$
$$= A\pi^{\nu_1} e_1 \oplus A\pi^{\nu_2} e_2$$

(π — generator [?] of the max. ideal)



Original source details: GELFAND-P000452

Uncertain reading. A looped mark with a lengthwise stroke stands between “generator” and “max.” in the Russian. Its reading is unresolved.



Original source details: GELFAND-P000452

Diagram and typed description. The inscription $\nu_1, \nu_2 \in \mathbb{Z}$ sits at the right between the two parenthetical lines.

(analog. to the lin. alg. theor. on simult. reduction) $\nu_1, \nu_2 \in \mathbb{Z}$

Def. $L_1 \sim L_2$, if $L_1 = \alpha L_2$, where $\alpha \in K^*$.

There is an analog of distance — as in symm. sp.. ... — see A. Weil



Undeciphered passage. The small sign before the dash is cancelled. Its reading is unresolved.

Original source details: GELFAND-P000452

Vertices of graph $\Delta^{(0)}$ — equiv. classes of lattices



Undeciphered passage. A small cancelled sign appears below and to the right of the first Δ. Its reading is unresolved. Above it is (0).

Original source details: GELFAND-P000452

G acts on $\Delta^{(0)}$.

GELFAND-P000453

[source number: 73]

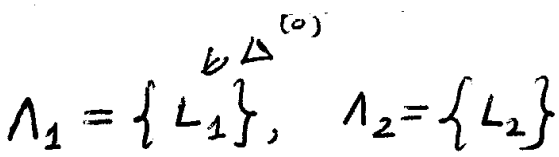


Diagram and typed description. The arrow by $\Delta^{(0)}$ points down and left toward the first entry.

$\Delta^{(0)}$

↙

$\Lambda_1 = \{L_1\}, \quad \Lambda_2 = \{L_2\}$ — classes of lattices.

Original source details: GELFAND-P000453

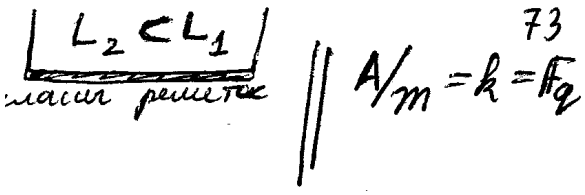


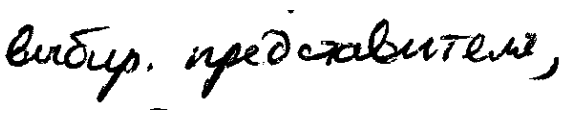
Diagram and typed description. The frame under $L_2 \subset L_1$ is open at the top. Two vertical strokes set off the residue field on the right. The left stroke of F is doubled.

$L_2 \subset L_1 \quad \| \quad A/m = k = \mathbb{F}_q$

Original source details: GELFAND-P000453

$$L_1 = Ae_1 \oplus Ae_2$$
$$L_2 = Ae_1 \oplus A\pi^\nu e_2$$

— thus choose the representatives, $\nu > 0$.



Uncertain reading. The end of the last word is faint. The reading is “представители” (“representatives”).

Original source details: GELFAND-P000453

ν is called distance $\underline{\underline{\nu = d(\Lambda_1, \Lambda_2).}}$

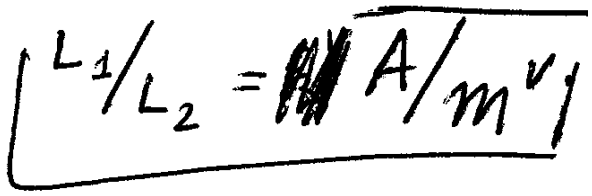


Diagram and typed description. Three strokes frame the formula below and at the sides. The inscription after the equals sign is heavily cancelled. Its reading is unresolved.

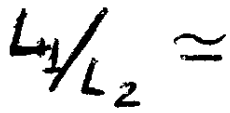
$L_1/L_2 = \dots A/m^\nu$

Original source details: GELFAND-P000453

Thm. This is a corr. def. and is a distance.

Variant:

$$L_1/L_2 \simeq A/\pi^{\nu_1}A \oplus A/\pi^{\nu_2}A$$



Uncertain reading. A single straight stroke lies under the wave. It is set as $\simeq$.

Original source details: GELFAND-P000453

$$d(\Lambda_1, \Lambda_2) = |\nu_1 - \nu_2|$$

Λ_1, Λ_2 — neighbors in Λ^0, if $d(\Lambda', \Lambda^2) = 1$.



Uncertain reading. The set has a Λ with a small raised circle. In the arguments of d, a raised stroke follows the first Λ and a raised 2 follows the second.

Original source details: GELFAND-P000453

Join them by an edge. We obtain a graph Δ.

Theorem: Δ — a tree (conn. graph without cycles), and from $\forall$ vertex there issue exactly $q+1$ edges

Δ is the Lobachevsky pl. — there acts G etc.

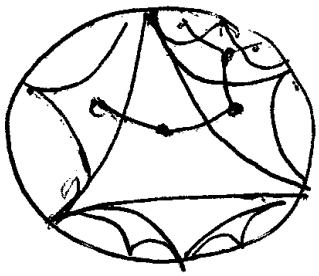
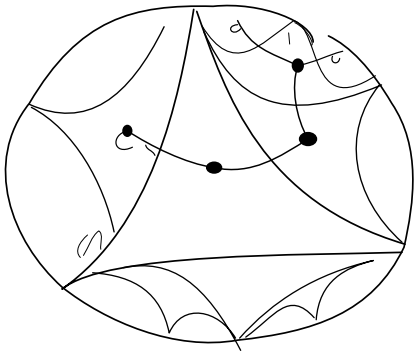


Diagram and typed description. The outer boundary has a gap at the upper right. The inner arcs and chain follow the visible strokes. A fine mark by the lower left arc resembles an oval. Some joins in the upper right remain uncertain.

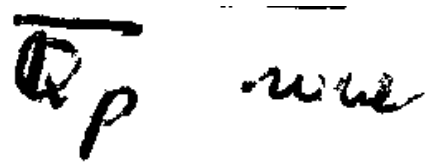


Original source details: GELFAND-P000453

GELFAND-P000454

D. Kazhdan. Tate's art., Matematika, 13 : 2, §3.

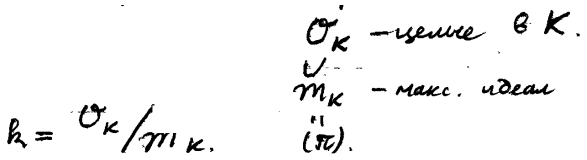
(on the field $\mathbb{C}_p$ — completion of the alg. closure $\overline{\mathbb{Q}}_p$ or $\mathbb{Q}_p$ — p-adic numbers)



Original source details: GELFAND-P000454

Uncertain reading. The connecting word reads "or."

K — finite ext. of the field $\mathbb{Q}_p$.



Original source details: GELFAND-P000454

Diagram and typed description. There is a curved stroke between the two entries. Its meaning is uncertain. Two short vertical strokes sit below the second entry.

$\mathcal{O}_K$ — integers in K.

$\smile$

m_K — max. ideal

$k = \mathcal{O}_K/m_K$. (π).

L — finite ext. of K, $\ldots$ $\mathcal{O}_L$ — integers in it



Original source details: GELFAND-P000454

Uncertain reading. The crossed-out marks before $\mathcal{O}_L$ are illegible.

$m_L = (\tau)$, $\quad \ell = \mathcal{O}_L/m_L$,

$$G = \mathrm{Gal}(L : K)$$

Introduce a filtration in G and prove that G is solvable.

On L and K $\ldots$ define a norm so that $|\pi| = (\mathrm{Card}\, k)^{-1}$



Original source details: GELFAND-P000454

Uncertain reading. A crossed-out sign sits between K and "define." Its reading is uncertain.

Def. $\ldots$.



Original source details: GELFAND-P000454

Undeciphered passage. A short entry is crossed out after "Def." Its reading is uncertain.

$$g \in G_i \Longleftrightarrow \forall x \in \mathcal{O}_L \quad x^g - x \in m_L^i$$

$$G = G_0 \supset G_1 \supset \ldots \supset G_n = \{e\}.$$

G_i — normal subgroup in G.

Theorem: G_i/G_{i+1} — ab. group

1) $i = 0$. $G \xrightarrow{\varphi}$ Gal$(\ell : k)$ (natural map)

Construction of φ: $\bar{x} \in \ell,\ g \in G$

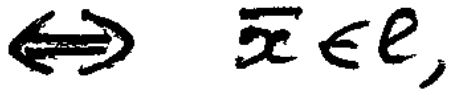
$x \in L$ $\quad \bar{x}^g \overset{\text{def}}{=} \overline{x^g}$



Uncertain reading. The manuscript reads $x \in L$.

Original source details: GELFAND-P000454

$g \in \ker \varphi \iff \bar{x} \in \ell,\ \bar{x}^g = \bar{x}$, i.e. $\bar{x}^g - \bar{x} = 0$, i.e.



Uncertain reading. The equivalence sign is followed by $\bar{x} \in \ell$. No explicit quantifier is visible.

Original source details: GELFAND-P000454

GELFAND-P000455

[source number: 74]

~~i.e. ...~~ $x^g - x \in m_L$, i.e. $g \in G_1$.



Uncertain reading. The start of the crossed-out entry provisionally reads “i.e.”.

Original source details: GELFAND-P000455



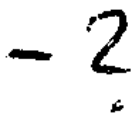
Undeciphered passage. The remainder of the crossed-out entry is illegible.

Original source details: GELFAND-P000455

$$0 \longrightarrow G_0/G_1 \xrightarrow{\varphi} \text{Gal}(\ell : k)$$

Exercise: φ — an isomorphism.

2) G_1/G_2 — ?



Uncertain reading. A question mark follows the dash.

Original source details: GELFAND-P000455

$$G_1 \xrightarrow{\psi_1} \ell^* \qquad g \sim\!\!\!\longrightarrow \overline{\tau^g/\tau} \qquad \text{(indep. of the choice of } \tau\text{)}$$



Uncertain reading. The first arrow is labelled ψ_1.

Original source details: GELFAND-P000455

Its kernel is G_2.



Uncertain reading. The word following "kernel" reads «ero» ("its").

Original source details: GELFAND-P000455

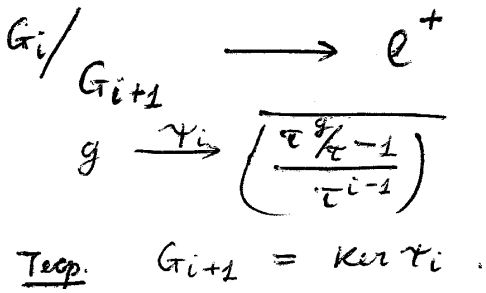
$$G_i/G_{i+1} \longrightarrow \ell^+ \qquad \text{(addit. group of the field } \ell\text{)}.$$
$$g \xrightarrow{\psi_i} \overline{\left(\frac{\tau^g/\tau - 1}{\tau^{i-1}}\right)}$$



Uncertain reading. The arrow from g is labelled ψ_i.

Original source details: GELFAND-P000455

Thm. $G_{i+1} = \operatorname{Ker} \psi_i$.



Uncertain reading. The domain on the upper line is written G_i/G_{i+1}. The line below has $G_{i+1} = \operatorname{Ker} \psi_i$.

Original source details: GELFAND-P000455

G_1 is called the inertia subgroup of the Galois grp.

Unramified extension $\underset{\text{def}}{\iff} G_1 = \{e\}$;

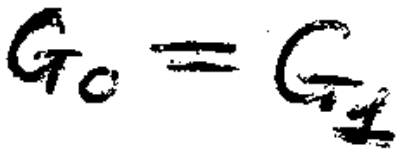
weakly ramif. $G_2 = \{e\}$.



Uncertain reading. The second letter is overwritten. The abbreviation reads «слаборазветвл.» ("weakly ramif.").

Original source details: GELFAND-P000455

purely ramified $G_0 = G_1$



Uncertain reading. The last subscript is overwritten. It provisionally reads 1.

Original source details: GELFAND-P000455

GELFAND-P000456

The seminar will start at $18^{\underline{00}}$.

2/X-72. A. Gabrielov.

S — compl. analytic set

$H_{\mathbb{Z}}$ — loc. free $\mathbb{Z}$-sheaf

($\mathbb{Z}$ — const. sheaf; a section over a conn. region is a const. integer; $H_{\mathbb{Z}}$ — loc. free sheaf of $\mathbb{Z}$-modules.



Uncertain reading. Overlapping strokes follow "modules." Their reading is uncertain.

Original source details: GELFAND-P000456

At $\forall$ point $\exists$ nbhd., in which it is isom. to $\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}$).

~~H...~~



Uncertain reading. The first crossed-out sign provisionally reads H.

Original source details: GELFAND-P000456

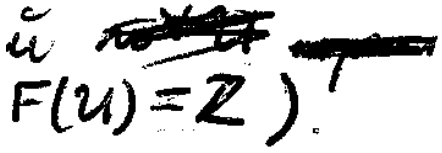


Undeciphered passage. The rest of the short entry before the parenthesis is illegible.

Original source details: GELFAND-P000456

($\mathbb{Z}$-sheaf assoc. with the presheaf which ~~... ...~~

$F(U) = \mathbb{Z}$).



Undeciphered passage. Two entries after "which" are crossed out. Below them is $F(U) = \mathbb{Z}$.

Original source details: GELFAND-P000456

The space is loc. connected $\Rightarrow$ fiber $= \mathbb{Z}$.

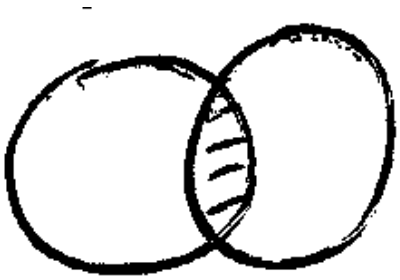
The space of the sheaf $\mathbb{Z}$ is homeom. to $S \times \mathbb{Z}$.

Loc. $H_{\mathbb{Z}}|_U \simeq U \times \mathbb{Z}^n$

$$\Gamma(U, H_{\mathbb{Z}}) \cong \mathbb{Z}^n$$

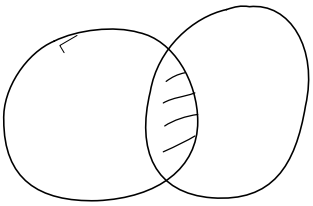
S is connected $\Rightarrow$ n is the same everywhere.

Cover by neighborhoods and fix isomorphisms over the neighborhoods (up to $GL(n, \mathbb{Z})$)



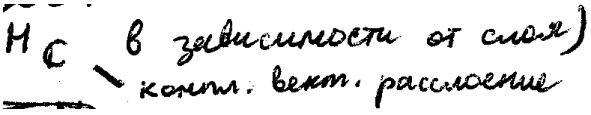
Original source details: GELFAND-P000456

Diagram and typed description. Two overlapping ovals with four strokes in their intersection. There is an angular mark on the left arc.



transition functions from $GL(n, \mathbb{Z})$.

One can construct a loc. triv. bundle with such transition functions.

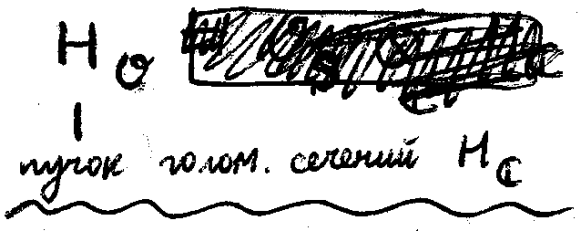


Original source details: GELFAND-P000456

Diagram and typed description. A slanted stroke links $H_{\mathbb{C}}$ to the caption.

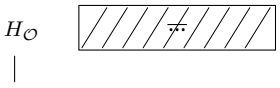
This is $H_{\mathbb{R}}$ (or $H_{\mathbb{C}}$ depending on the fiber)

compl. vector bundle



Original source details: GELFAND-P000456

Diagram and typed description. A vertical stroke leads to the wavy-underlined caption. The rectangle at right is crossed out.

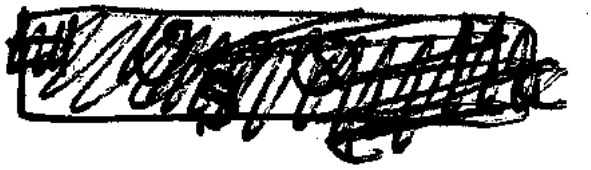


sheaf of holom. sections of $H_{\mathbb{C}}$



Original source details: GELFAND-P000456

Uncertain reading. The crossed-out entry may start with H and a subscript.



Original source details: GELFAND-P000456

Undeciphered passage. The crossed-out entry inside the rectangle is illegible.

$$H_{\mathcal{O}} = H_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathcal{O}_S.$$

There is a natural connection (Gauss–Manin)

$$\nabla \colon H_{\mathcal{O}} \longrightarrow \Omega^1_S \otimes_{\mathcal{O}_S} H_{\mathcal{O}} \qquad (\mathbb{C}\text{-lin. homom.})$$

Ω^1_S — holom. diff. forms of order 1

GELFAND-P000457

[source number: 75]

def. of connection.

1. $\mathbb{C}$-linearity.

2. $\varphi \in \mathcal{O}_{S,s}$ (germs), $f \in H_{\mathcal{O},s} \Rightarrow$

$$\Rightarrow \nabla(\varphi f) = \varphi \nabla f + d\varphi \otimes f$$

Our connection.

Take a loc. trivialization
and a basis of const. sections (bundle $H_{\mathbb{Z}}$) u_i

$$f = \sum \varphi_i u_i, \qquad \varphi_i \in \mathcal{O}_S$$

Then

$$\nabla f = \sum d\varphi_i \otimes u_i$$

This def. is well-defined and satisfies all the axioms.

The curvature of this connection equals 0.



Uncertain reading. The last sign reads 0.

Original source details: GELFAND-P000457

Requirement on the G.-M. connection (or on the ... filtration)

$$\nabla(\mathcal{F}^p) \subset \Omega^1_S \otimes \mathcal{F}^{p-1}.$$

(Griffiths)



Uncertain reading. The crossed-out sign after "or on the" is illegible.

Original source details: GELFAND-P000457

(Given a filtration $H_{\mathbb{C}} = F^0 \supset F^1 \supset \cdots$)
by holom. subbundles

$\mathcal{F}^p$ — sheaves of sections of F^p

There is a loc. constant bilin. form $\psi(x,y)$ on $H_{\mathbb{C}}$



Uncertain reading. The letter before (x,y) reads ψ.

Original source details: GELFAND-P000457

...



Undeciphered passage. The short crossed-out entry before “and” is illegible.

Original source details: GELFAND-P000457

and at $\forall$ point $s \in S$ this is a Hodge structure.

———— . ————

Let $0 \leq p \leq n$, $k_p \geq k_{p+1}$ — a seq. of numbers

$$k_0 \geq k_1 \geq \cdots \geq k_n.$$

Z — space with elem. $F = \{F^0 \supset F^1 \supset \cdots \supset F^n\}$,
where dim $F^p = k_p$.



Uncertain reading. The superscript p on F is overwritten.

Original source details: GELFAND-P000457

This is a proj. alg. set.

GELFAND-P000458

$$H_{\mathbb{C}} = F^0, \qquad H_{\mathbb{Z}} \subset H_{\mathbb{R}} \subset H_{\mathbb{C}}.$$

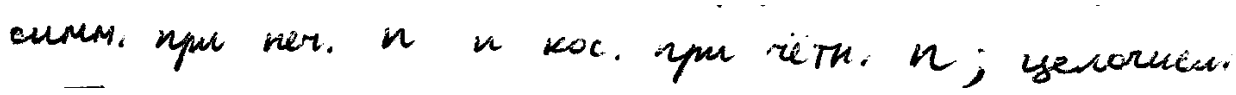
Let there be on $H_{\mathbb{C}}$ a given bilin. form ψ (nondeg.),



Uncertain reading. The sign after “form” reads ψ.

Original source details: GELFAND-P000458

symm. for odd n and skew for even n; integral



Original source details: GELFAND-P000458

Uncertain reading. The manuscript reads “symm. for odd n and skew for even n.”

$\overline{D}$ — subset in Z of those flags for which

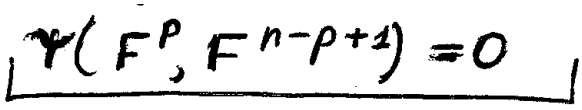
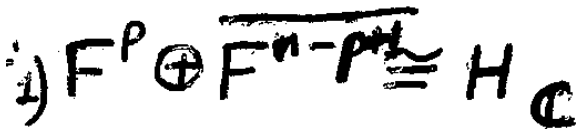


Diagram and typed description. A line encloses the formula from below and at both sides.

$$\psi(F^p, F^{n-p+1}) = 0$$

Original source details: GELFAND-P000458

Add. cond.
on the flags 1) $F^p \oplus \overline{F^{n-p+1}} \cong H_{\mathbb{C}}$



Original source details: GELFAND-P000458

Uncertain reading. The end of the exponent $n-p+1$ is overwritten beside the sign $\cong$.

2) From it are constr. the grading and the Weil autom. C and then

$\psi(x, C\bar{y})$ pos. def. Herm. form on $H_{\mathbb{C}}$.



Original source details: GELFAND-P000458

Uncertain reading. After C the manuscript reads «и тогда» (“and then”).

These conditions single out $\underset{\text{(open).}}{D} \subset \overline{D}$.

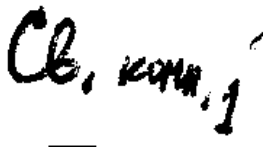
D is called the classify. space of Hodge structures.

Depends on the set of dimensions and the form ψ.

Next time: homogeneity.

grp. of the compl. space $H_{\mathbb{C}}$, preserv. ψ, transit. on $\overline{D}$

Conn. comp. $_1$ grp. of the real space $H_{\mathbb{R}}$, preserv. ψ, transit. on D

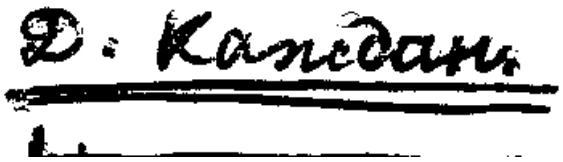


Original source details: GELFAND-P000458

Uncertain reading. The left addition provisionally reads «Св. комп.» (“conn. comp.”) with a lowered sign resembling 1.

Manin's 2nd lecture.

D: Kazhdan. More on Tate's work.



Original source details: GELFAND-P000458

Uncertain reading. Two dots sit one above the other after the initial.



Original source details: GELFAND-P000458

Diagram and typed description. A narrow mark of several strokes crosses the divider. No definite letter or arrow is visible.

